

Chapter 5.

Quantum Markov semigroups

In Chapter 0, we briefly introduced the standard mathematical model for the description of the continuous time evolution of open quantum systems. After taking a suitable limit, we showed that it can be described in terms of a one-parameter semigroup of unital, completely positive maps acting on the von Neumann algebra generated by the observables of the system.

The theory of Markov semigroups acting on abstract operator algebras was initiated by [Albeverio and Høegh-Krohn, 1977] where the bijective correspondence between symmetric Markov semigroups and Dirichlet forms has been established. Their work was followed by [Davies and Lindsay, 1992] which treated the case of semigroups acting on a finite von Neumann algebra that are symmetric with respect to its trace. The general von Neumann algebraic symmetric case was later treated by [Goldstein and Lindsay, 1995].

A large part of the thesis deals with evolutions occurring on matrix algebras. In this chapter, we however decided to introduce the objects with a fair amount of generality, namely weak*-continuous semigroups on the algebra $\mathcal{B}(\mathcal{H})$ of bounded operators on a separable Hilbert space $\mathcal{H}$, which describes dissipative evolutions occurring in quantum optical information processing devices and *continuous variables* (CV) quantum information theory. First of all, this is the amount of generality needed to introduce the quantum heat and Ornstein Uhlenbeck semigroups studied in Chapter 11. Moreover, we believe that the concepts introduced in the next chapters can be extended to this setting, after taking care of the issues arising from the manipulation of unbounded operators that constitute the daily worries of CV information theorists. Quantum optical systems are known to be prominent candidates for implementation of quantum information processing devices. Hence, this can be seen as a series of first steps towards the author's goal to extend the present results to the case of Markovian noise occurring in quantum optical systems. In particular, after an introduction to the non-commutative differential calculus in Section 5.3, we prove a non-commutative, infinite dimensional *Stroock-Varopoulos inequality*. This result is the main ingredient of the proof of the equivalence between the logarithmic Sobolev inequality and hypercontractivity of reversible quantum Markov semigroups.

Layout of the chapter: For sake of comparison with the classical setting, this chapter is organized similarly to Chapter 2: After a brief recapitulation of the theory of semigroups of operators on Banach algebras in Section 5.1. We introduce the notions of invariant states, $\mathbb{L}_p$ quantum Markov semigroups, reversibility and Dirichlet forms in Section 5.2. The notion of a non-commutative differential calculus associated to a QMS is presented in Section 5.3. In Section 5.4, this notion is then used to prove a quantum extension of the $\mathbb{L}_p$ regularity of Dirichlet forms seen in Section 2.4. We end this chapter by introducing examples that will reappear in the sequel.

5.1. Semigroups of operators on Banach algebras

We recall that a *one-parameter semigroup of operators* on a Banach space $\mathcal{X}$ is a map $\mathcal{P} : [0, \infty) \rightarrow \mathcal{B}(\mathcal{X})$ (or equivalently the family $(\mathcal{P}_t)_{t \geq 0}$ of maps on $\mathcal{X}$) such that

- (i) $\mathcal{P}_0 = \text{id}_{\mathcal{X}}$
- (ii) semigroup property: for all $s, t \geq 0$,

$$\mathcal{P}_{s+t} = \mathcal{P}_s \circ \mathcal{P}_t. \quad (5.1)$$

5.1.1. Different notions of continuity:

Such a semigroup is called *uniformly continuous* if the following holds:

$$\|\mathcal{P}_t - \text{id} : \mathcal{X} \rightarrow \mathcal{X}\| \xrightarrow{t \rightarrow 0} 0.$$

The semigroup $(\mathcal{P}_t)_{t \geq 0}$ is called *strongly continuous* if for all $X \in \mathcal{X}$:

$$\|\mathcal{P}_t(X) - X\| \xrightarrow{t \rightarrow 0} 0. \quad (5.2)$$

In the following chapters, we also consider weaker notions of continuity: the semigroup $(\mathcal{P}_t)_{t \geq 0}$ is *weak* continuous* if:

- (i) For all $X \in \mathcal{X}$ and all $\eta \in \mathcal{X}_*$, the map $t \mapsto \eta(\mathcal{P}_t(X))$ is continuous on $\mathbb{R}_+$;
- (ii) For all $t \geq 0$ and any $\eta \in \mathcal{X}_*$, $\eta \circ \mathcal{P}_t \in \mathcal{X}_*$.

Similarly, the semigroup $(\mathcal{P}_t)_{t \geq 0}$ is said to be *weakly continuous* if the above two conditions hold after replacement of the predual $\mathcal{X}_*$ by the dual $\mathcal{X}^*$ of $\mathcal{X}$. In fact, any weakly continuous semigroup is strongly continuous (see Corollary 3.1.8 of [Bratteli and Robinson, 1979]). Therefore, a simple ordering of the topologies implies the following inclusion of the classes introduced:

$$\text{uniform continuous} \subset \text{strongly continuous} = \text{weakly continuous} \subset \text{weak}^* \text{ continuous}.$$

5.1.2. Generators:

Let $(\mathcal{P}_t)_{t \geq 0}$ be a weak* continuous semigroup of operators on the Banach space $\mathcal{X}$. The *weak* generator* $(\mathcal{L}, \text{dom}(\mathcal{L}))$ of $(\mathcal{P}_t)_{t \geq 0}$ is defined as the linear operator $\mathcal{L}$ on $\mathcal{X}$ whose domain is composed of elements $X \in \mathcal{X}$ such that there exists $Y \in \mathcal{X}$ with the property that for all $\eta \in \mathcal{X}_*$, $\eta(Y) = \lim_{t \rightarrow 0} t^{-1} \eta((\mathcal{P}_t - \text{id})(X))$. In this case

$$\mathcal{L}(X) := \text{weak}^* - \lim_{h \rightarrow 0} \frac{\mathcal{P}_h(X) - X}{h}.$$

In the case when $(\mathcal{P}_t)_{t \geq 0}$ is strongly continuous, its *generator* is the linear operator $(\mathcal{L}, \text{dom}(\mathcal{L}))$, where $\text{dom}(\mathcal{L})$ is the subspace of vectors $X \in \mathcal{X}$ for which there exists $Y \in \mathcal{X}$ such that $\lim_{h \rightarrow 0} h^{-1} \|\mathcal{P}_h(X) - X - hY\| \rightarrow 0$. In this case, we set $Y = \mathcal{L}(X)$. In particular, this generator coincides with the weak generator, defined similarly to the weak* generator by simply requiring the limit to exist for any $\eta \in \mathcal{X}^*$.

By the semigroup property (5.1), the domain of $\mathcal{L}$ is then the set of vectors $X \in \mathcal{X}$ for which the map $t \mapsto \mathcal{P}_t(X)$ is continuously differentiable of derivative denoted by $\frac{d}{dt} \mathcal{P}_t(X)$. Moreover, for any

such X ,

$$\frac{d}{dt}\mathcal{P}_t(X) = \mathcal{L} \circ \mathcal{P}_t(X) = \mathcal{P}_t \circ \mathcal{L}(X).$$

A strongly continuous contraction semigroup $(\mathcal{P}_t)_{t \geq 0}$ with generator $(\mathcal{L}, \text{dom}(\mathcal{L}))$ satisfies the following equation (see (10.3.7) of [Hille and Phillips, 1996]): for any $X \in \text{dom}(\mathcal{L})$ and $t \geq 0$,

$$\mathcal{P}_t(X) - X = \int_0^t \mathcal{P}_s \circ \mathcal{L}(X) ds, \quad (5.3)$$

where the integral is defined in the Bochner sense.

The generator $\mathcal{L}$ is bounded, which equivalently means that $\text{dom}(\mathcal{L}) = \mathcal{X}$, if and only if $(\mathcal{P}_t)_{t \geq 0}$ is uniformly continuous. A linear operator on $\mathcal{X}$ is the infinitesimal generator of at most one strongly continuous semigroup. Conversely, the Hille-Yosida-Phillips theorem provides necessary and sufficient conditions under which a linear operator $\mathcal{L}$ is the generator of a semigroup (see [Hille and Phillips, 1996]).

A strongly continuous semigroup $(\mathcal{P}_t)_{t \geq 0}$ is called a *contraction semigroup* if it satisfies $\|\mathcal{P}_t : \mathcal{X} \rightarrow \mathcal{X}\| \leq 1$ for any $t \geq 0$. One can show [Bratteli and Robinson, 1979] that in this case, $\mathcal{L}$ is weak* densely defined. The generator associated to a strongly continuous contraction semigroup is *dissipative*: for all $X \in \mathcal{X}$ there exists $X^* \in \mathcal{X}^*$, the dual of $\mathcal{X}$, such that $\|X^*\|^2 = \|X\|^2 = X^*(X)$, and $\text{Re}(X^*(\mathcal{L}(X))) \leq 0$. More precisely:

Theorem 5.1.1 (Lumer-Phillips theorem for contraction semigroups). *Let $\mathcal{X}$ be a complex Banach space, $\mathcal{L} : \text{dom}(\mathcal{L}) \rightarrow \mathcal{X}$ a densely defined complex linear operator. The following are equivalent:*

(i) $\mathcal{L}$ is the infinitesimal generator of a strongly continuous contraction semigroup.

(ii) For all $\lambda > 0$, $\lambda \text{id} - \mathcal{L} : \text{dom}(\mathcal{L}) \rightarrow \mathcal{X}$ is bijective and

$$\|(\lambda \text{id} - \mathcal{L})^{-1}\| \leq \lambda^{-1}.$$

(iii) For all $\lambda \in \mathbb{C}$ with $\text{Re}(\lambda) > 0$, $\lambda \text{id} - \mathcal{L} : \text{dom}(\mathcal{L}) \rightarrow \mathcal{X}$ is bijective and

$$\|(\lambda \text{id} - \mathcal{L})^{-1}\| \leq \text{Re}(\lambda)^{-1}.$$

(iv) $\mathcal{L}$ is dissipative and there exists $\lambda > 0$ such that $\lambda \text{id} - \mathcal{L} : \text{dom}(\mathcal{L}) \rightarrow \mathcal{X}$ has a dense image.

5.1.3. Quantum Markov semigroups:

As mentioned in Section 0.1.5, we are exclusively interested in the study of weak* continuous, completely positive, unital contraction semigroups $(\mathcal{P}_t)_{t \geq 0}$ acting on a von Neumann algebra of operators $\mathcal{X} := \mathcal{M}$, and refer to them as *quantum Markov semigroup* (QMS). In fact, any completely positive, unital Markov semigroup is necessarily contractive: this is due to the Russo-Dye Theorem (Corollary 2.9 of [Paulsen, 2002]) according to which, for all $t \geq 0$:

$$\|\mathcal{P}_t : \mathcal{M} \rightarrow \mathcal{M}\| = \|\mathcal{P}_t(\mathbb{1}_{\mathcal{M}})\|_{\infty} = 1.$$

Quantum Markov semigroups can be viewed as dual semigroups of strongly continuous semigroups on the predual Banach space $\mathcal{M}_*$, defined by:

$$\mathcal{P}_{*t}(\omega)(X) = \omega(\mathcal{P}_t(X)),$$

for every $X \in \mathcal{M}$ and $\omega \in \mathcal{M}_*$. Indeed, since $t \mapsto \mathcal{P}_t$ is weak* continuous on $\mathcal{M}$, $t \mapsto \mathcal{P}_{*t}$ is weakly continuous on $\mathcal{M}_*$. From our previous discussion, this actually implies that $t \mapsto \mathcal{P}_{*t}$ is strongly continuous on $\mathcal{M}_*$.

Uniformly continuous QMS: In the case of a uniformly continuous completely positive semigroup the following important result was first proved by [Lindblad, 1976, Gorini et al., 1976] for matrix algebras, and extended by [Christensen and Evans, 1979] to the general C*-algebraic setting:

Theorem 5.1.2 (Standard representation of a norm continuous QMS). *Let $(\mathcal{P}_t)_{t \geq 0}$ be a uniformly continuous semigroup of completely positive normal unital maps on a von Neumann algebra $\mathcal{M}$ of operators acting on a Hilbert space $\mathcal{H}$. Then its generator $\mathcal{L}$ may be represented as*

$$\forall X \in \mathcal{M}, \quad \mathcal{L}(X) = i[H, X] - \{\Psi(\mathbb{1}_{\mathcal{A}}), X\} + \Psi(X) = \Psi(X) + K^*X + XK, \quad (5.4)$$

where $K, H = H^* \in \mathcal{M}$ and $\Psi : \mathcal{M} \rightarrow \mathcal{M}$ is a completely positive map.

In this case we obtain the following *Lindblad form* of $\mathcal{L}$ from the Kraus decomposition (0.23) of Ψ ,

$$\mathcal{L}(X) = i[H, X] + \frac{1}{2} \sum_{j \in \mathcal{J}} 2L_j^*XL_j - XL_j^*L_j - L_j^*L_jX, \quad (5.5)$$

where $\{L_j\}_{j \in \mathcal{J}}$ is a family of bounded operators such that $\sum_{j \in \mathcal{J}} L_j^*L_j$ is bounded and $\sum_{j \in \mathcal{J}} L_j^*XL_j \in \mathcal{M}$ for any $X \in \mathcal{M}$. In this case, $\mathcal{L}$ is usually called a *Lindblad generator*.

A physical interpretation for the Lindblad form (5.5) can be given from the Dyson expansion of the solution of the forward Markov master equation

$$\mathcal{P}_t = \hat{\mathcal{P}}_t + \sum_{n=1}^{\infty} \int \dots \int_{0 \leq t_1 \leq \dots \leq t_n \leq t} \hat{\mathcal{P}}_{t_1} \circ \Psi \circ \hat{\mathcal{P}}_{t_2-t_1} \circ \dots \circ \Psi \circ \hat{\mathcal{P}}_{t-t_n} dt_1 \dots dt_n, \quad (5.6)$$

where $\hat{\mathcal{P}}_t(X) := e^{-K^*t} X e^{-Kt}$. In words, the above equation says that $(\mathcal{P}_t)_{t \geq 0}$ can be described by a sequence of “spontaneous jumps” modeled by the map Ψ occurring at times $t_1 \leq \dots \leq t_n$ on the background of a nonunitary reversible evolution given by the semigroup $(\hat{\mathcal{P}}_t)_{t \geq 0}$.

Weak* continuous QMS on $\mathcal{B}(\mathcal{H})$: The non-norm continuous case, while being interesting from both physical and mathematical reasons, is more difficult to handle. The generator of such a QMS may be unbounded, and its domain is not necessarily a *-algebra. Moreover the correspondence between the master equation and the semigroup is not one-to-one, since the solution for the former need not be unique. In a series of articles, Holevo partially introduced the so-called *form generator* associated to a QMS. In words, it corresponds to a differential equation without boundary terms, which makes issues arising from the unboundedness of the weak* generator easier. On the other hand, this formulation introduces other problems such as the one of existence, uniqueness and unitality of the solution.

More precisely, let $(\mathcal{P}_t)_{t \geq 0}$ be a QMS on a von Neumann algebra $\mathcal{M}$ acting on a separable Hilbert space $\mathcal{H}$, and assume there exists a dense domain $\mathcal{V} \subseteq \mathcal{H}$ such that the derivative

$$\left. \frac{d}{dt} \langle \varphi, \mathcal{P}_t(X)\psi \rangle \right|_{t=0} =: \mathcal{L}(X)[\varphi, \psi] \quad (5.7)$$

exists for all $X \in \mathcal{M}$, and any $\varphi, \psi \in \mathcal{V}$. By the semigroup property of $(\mathcal{P}_t)_{t \geq 0}$, this implies that the

following so-called *master equation* holds for any $t \geq 0$, any $X \in \mathcal{M}$ and any $\varphi, \psi \in \mathcal{V}$:

$$\frac{d}{dt} \langle \varphi, \mathcal{P}_t(X)\psi \rangle = \mathcal{L}(\mathcal{P}_t(X))[\varphi, \psi].$$

Moreover, the form $(\mathcal{M}, \mathcal{V}, \mathcal{V}) \ni (X, \varphi, \psi) \mapsto \mathcal{L}(X)[\varphi, \psi]$ satisfies the following properties, due to the definition of the QMS $(\mathcal{P}_t)_{t \geq 0}$:

- Adjoint preservation property: $(X, \varphi, \psi) \mapsto \mathcal{L}(X)[\varphi, \psi]$ is linear in X and ψ , antilinear in φ and

$$\mathcal{L}(X^*)[\varphi, \psi] = \overline{\mathcal{L}(X)[\psi, \varphi]};$$

- Complete positivity: Given any finite subsets $\{\varphi_i\} \subset \mathcal{V}$ and $\{X_i\} \subset \mathcal{M}$ such that $\sum_i X_i \varphi_i = 0$,

$$\sum_{i,j} \mathcal{L}(X_i^* X_j)[\varphi_i, \varphi_j] \geq 0;$$

- Unitality: for any $\varphi, \psi \in \mathcal{V}$,

$$\mathcal{L}(\mathbb{1})[\varphi, \psi] = 0; \quad (5.8)$$

- Weak* continuity: for any $\psi, \varphi \in \mathcal{V}$, the functional $X \mapsto \mathcal{L}(X)[\varphi, \psi]$ is weak* continuous on $\mathcal{M}$.

Any map $(X, \varphi, \psi) \mapsto \mathcal{L}(X)[\varphi, \psi]$ satisfying the above properties is called a *form generator*. The following theorem, proved in [Holevo, 1995a] provides a characterization of form generators in the case when $\mathcal{M} = \mathcal{B}(\mathcal{H})$:

Theorem 5.1.3. *Let $(X, \varphi, \psi) \mapsto \mathcal{L}(X)[\varphi, \psi]$ be a form generator on $\mathcal{B}(\mathcal{H})$. Then there exists an operator $K : \mathcal{V} \rightarrow \mathcal{H}$ and a countable family $\{L_i : \mathcal{V} \rightarrow \mathcal{H}\}_{i \in \mathcal{J}}$ of operators satisfying $\sum_i \|L_i \varphi\|^2 \leq -2 \operatorname{Re} \langle \varphi, K \varphi \rangle$ for any $\varphi \in \mathcal{V}$, and such that*

$$\mathcal{L}(X)[\varphi, \psi] = \sum_{i \in \mathcal{J}} \langle L_i \varphi, X L_i \psi \rangle + \langle K \varphi, X \psi \rangle + \langle \varphi, X K \psi \rangle. \quad (5.9)$$

The operators L_i and K can always be chosen so that the following normalizations hold: for any state $\rho := \sum_j \lambda_j |\psi_j\rangle\langle\psi_j| \in \mathcal{D}(\mathcal{H})$ whose eigenvectors are supposed to be in $\mathcal{V}$, one can without loss of generality assume that for any $i \in \mathcal{J}$:

$$\operatorname{Tr}(\rho L_i) := \sum_j \lambda_j \langle \psi_j, L_i \psi_j \rangle = 0, \quad \operatorname{Im} \operatorname{Tr}(\rho K) := \operatorname{Im} \sum_j \lambda_j \langle \psi_j, K \psi_j \rangle = 0,$$

as long as the associated series converge absolutely: $\sum_j \lambda_j |\langle \psi_j, L_i \psi_j \rangle| < \infty$, $\sum_j \lambda_j |\langle \psi_j, K \psi_j \rangle| < \infty$. Moreover, for any other operators $\tilde{L}_i$ and $\tilde{K}$ satisfying Equation (5.9) with corresponding state $\rho' = V \rho V^*$, $V \in \mathcal{U}(\mathcal{H})$, there exists a unitary operator $U = \{U_{ij}\}_{(i,j) \in \mathcal{J}^2}$, a vector $\{\alpha_i\}_{i \in \mathcal{J}}$ of complex numbers and a real number β such that

$$\tilde{L}_i = \sum_j L_j U_{ij} + \alpha_i \mathbb{1} \quad (5.10)$$

$$\tilde{K} = K - \sum_{i,j \in \mathcal{J}} \overline{U_{ji}} \alpha_j L_i - \frac{1}{2} \left(\sum_{i \in \mathcal{J}} |\alpha_i|^2 - i\beta \right) \mathbb{1}, \quad (5.11)$$

where $\alpha_i \equiv \operatorname{Tr}(\rho \tilde{L}_i)$ and $\beta \equiv \operatorname{Im} \operatorname{Tr}(\rho \tilde{K})$.

The above theorem is important for two main reasons: first, it provides a justification for the search of candidate generators of quantum Markov semigroups in a *generalized GKLS form*, which was already assumed in previous work [Davies, 1977, Chebotarev and Fagnola, 1998, Fagnola, 1993]. Moreover, the equivalence class of representations $(K, \{L_i\}_{i \in \mathcal{J}})$ provided in Equations (5.10) and (5.11) is at the heart of Holevo's characterization of covariant quantum Markov semigroups which lead to a noncommutative extension of the Lévy-Khinchin formula for generators of classical shift-covariant Markov semigroups [Holevo, 1993a, Holevo, 1995a, Holevo, 1993b, Holevo, 1995b, Holevo, 1996, Holevo, 1998]. In Section 5.2.2, we will provide a useful expression for the generator of symmetric quantum Markov semigroups as yet another straightforward consequence of Theorem 5.1.3.

On the other hand, given a form-generator $\mathcal{L}$ in the generalized GKLS form of Equation (5.9), the question of the existence, uniqueness and unitality of a QMS satisfying the Master equation (5.7) has been partially solved under some further conditions on the operators K and L_i , $i \in \mathcal{J}$. In what follows, we provide a brief summary of the main results in that direction. Given a countable set $\mathcal{J}$, let K and L_k , $k \in \mathcal{J}$ be (possibly unbounded) operators on $\mathcal{H}$, and assume the following holds:

Condition 5.1.4 (H-min). The operator K is the generator of a strongly continuous semigroup $(P_t)_{t \geq 0}$ on $\mathcal{H}$, the domain $\text{dom}(L_k)$ of each operator L_k is contained in $\text{dom}(K)$, and for all $\psi, \varphi \in \text{dom}(K)$:

$$\langle K\psi, \varphi \rangle + \sum_{k \in \mathcal{J}} \langle L_k \psi, L_k \varphi \rangle + \langle \psi, K\varphi \rangle = 0.$$

For any $X \in \mathcal{B}(\mathcal{H})$, define the quadratic form $(\psi, \varphi) \mapsto \mathcal{L}(X)[\psi, \varphi]$ as in Equation (5.9) with domain $\text{dom}(K) \times \text{dom}(K)$. Under Condition 5.1.4, one can construct the so-called *minimal semigroup* $(\mathcal{P}_t^{\min})_{t \geq 0}$ on $\mathcal{B}(\mathcal{H})$ associated with the operators K and L_k , $k \in \mathcal{J}$ [Davies, 1977, Chebotarev and Fagnola, 1998, Fagnola, 1999]. This is a weak* continuous semigroup $(\mathcal{P}_t^{\min})_{t \geq 0}$ of completely positive maps satisfying

$$\forall \psi, \varphi \in \text{dom}(K), \quad \langle \psi, \mathcal{P}_t^{\min}(X) \varphi \rangle = \langle \psi, X \varphi \rangle + \int_0^t \mathcal{L}(\mathcal{P}_s^{\min}(X))[\psi, \varphi] ds. \quad (5.12)$$

Moreover, $(\mathcal{P}_t^{\min})_{t \geq 0}$ satisfies the following two properties (Theorem 3.22 of [Fagnola, 1999]):

- $\mathcal{P}_t^{\min}(\mathbb{1}) \leq \mathbb{1}$ for every $t \geq 0$;
- for every other weak* continuous semigroup of positive maps $(\mathcal{P}_t)_{t \geq 0}$ satisfying Equation (5.12), every positive operator $X \in \mathcal{B}(\mathcal{H})$ and all $t \geq 0$:

$$\mathcal{P}_t^{\min}(X) \leq \mathcal{P}_t(X).$$

However, Condition 5.1.4, reminiscent from Equation (5.8), does not insure that the QMS is unital (see e.g. example 3.3 of [Davies, 1977] or example 3.4 of [Fagnola, 1999]). If we further impose the minimal semigroup to be unital, then it becomes the unique weak* continuous semigroup of positive maps on $\mathcal{B}(\mathcal{H})$ that satisfies Equation (5.12). A sufficient condition for unitality was provided in [Chebotarev and Fagnola, 1998] (see also [Holevo, 1995a]). In the statement of the theorems, we will however assume the following weaker condition (see e.g. Proposition 2.3 of [Carbone and Fagnola, 2003]):

Condition 5.1.5 (H-Markov). One of these equivalent conditions holds:

- The minimal semigroup $(\mathcal{P}_t^{\min})_{t \geq 0}$ is unital.
- $(\mathcal{P}_t^{\min})_{t \geq 0}$ is the unique weak* continuous family of positive contractive maps satisfying Equation (5.12) for all $X \in \mathcal{B}(\mathcal{H})$.

- The linear space $\mathcal{D} := \text{span}\{|\psi\rangle\langle\varphi|, \psi, \varphi \in \text{dom}(K)\}$ is a core for $\mathcal{L}_*^{\min}$. In that case, for any $\psi, \varphi \in \mathcal{D}$:

$$\mathcal{L}_*^{\min}(|\psi\rangle\langle\varphi|) = |K\psi\rangle\langle\varphi| + |\psi\rangle\langle K\varphi| + \sum_{r \in \mathcal{J}} |L_r\psi\rangle\langle L_r\varphi| \quad (5.13)$$

The next result provides a characterization of the domain of $\mathcal{L}^{\min}$. It is also due to [Fagnola and Rebolledo, 1998]:

Proposition 5.1.6. *Suppose that Condition 5.1.4 and Condition 5.1.5 hold. Then the domain of the infinitesimal generator $\mathcal{L}^{\min}$ of $(\mathcal{P}_t^{\min})_{t \geq 0}$ is given by all elements $X \in \mathcal{B}(\mathcal{H})$ such that the sesquilinear form defined in Equation (5.9) on $\text{dom}(K) \times \text{dom}(K)$:*

$$(\psi, \varphi) \mapsto \mathcal{L}(X)[\psi, \varphi]$$

is norm continuous.

5.2. $\mathbb{L}_p$ quantum Markov semigroups, reversibility, Dirichlet forms

5.2.1. $\mathbb{L}_p$ quantum Markov semigroups and invariant states

Just as in Section 2.2, we now extend the domain of definition of a QMS $(\mathcal{P}_t)_{t \geq 0}$ defined on $\mathcal{B}(\mathcal{H})$, $\mathcal{H}$ separable, to the larger non-commutative $\mathbb{L}_p$ spaces. Such semigroups were studied by [Albeverio and Høegh-Krohn, 1977] (in the C^* -algebraic setting) and [Davies and Lindsay, 1992] in the case of $\mathbb{L}_p$ spaces defined with respect to a tracial state. The theory was later generalized to semigroups that are symmetric with respect to any faithful normal state by [Goldstein and Lindsay, 1995]. Here, we restrict ourselves to semigroups defined on the algebra $\mathcal{B}(\mathcal{H})$ of bounded operators on a separable Hilbert space $\mathcal{H}$, unless otherwise specified. We refer to the above articles for more details. Let σ be a faithful invariant state on $\mathcal{H}$ and $1 \leq p < \infty$. A strongly continuous, contraction semigroup $(\mathcal{P}_t)_{t \geq 0}$ of positive unital operators on the Banach space $\mathbb{L}_p(\sigma)$ is called an $\mathbb{L}_p(\sigma)$ -quantum Markov semigroup. Let us first consider the case $p = 2$. By the Hille-Yosida theorem, an $\mathbb{L}_2(\sigma)$ -quantum Markov semigroup is fully determined by its generator

$$\mathcal{L}_2 : \begin{cases} \text{dom}(\mathcal{L}_2) \rightarrow \mathbb{L}_2(\sigma) \\ X \mapsto \lim_{\epsilon \rightarrow 0} \frac{\mathcal{P}_\epsilon(X) - X}{\epsilon}, \end{cases} \quad (5.14)$$

the limit being taken in the $\|\cdot\|_{\mathbb{L}_2(\sigma)}$ topology. The state σ is said to be an *invariant state* with respect to $(\mathcal{P}_t)_{t \geq 0}$ if for all $t \geq 0$ and all $X \in \mathbb{L}_\infty(\sigma)$,

$$\text{Tr}(\sigma \mathcal{P}_t(X)) = \text{Tr}(\sigma X). \quad (5.15)$$

Moreover, $(\mathcal{P}_t)_{t \geq 0}$ is called *primitive* if it possesses a unique faithful invariant state. From now on, we assume that the state σ with respect to which the $\mathbb{L}_p(\sigma)$ spaces are being defined is invariant with respect to $(\mathcal{P}_t)_{t \geq 0}$. As in the classical case, the following holds:

Proposition 5.2.1. *Let $(\mathcal{P}_t)_{t \geq 0}$ be a QMS and σ a faithful invariant state. Then $(\mathcal{P}_t)_{t \geq 0}$ can be extended to an $\mathbb{L}_p(\sigma)$ -QMS for any $1 \leq p < \infty$.*

Proof. Since $(\mathcal{P}_t)_{t \geq 1}$ is unital, it satisfies $\mathcal{P}_t(\mathbb{1}) = \mathbb{1}$ for all $t \geq 0$. By positivity of the maps $\mathcal{P}_t$, Russo-Dye theorem (see Corollary 2.9 of [Paulsen, 2002]) implies that $\|\mathcal{P}_t : \mathcal{B}(\mathcal{H}) \rightarrow \mathcal{B}(\mathcal{H})\| = \|\mathcal{P}_t(\mathbb{1})\|_\infty = \|\mathbb{1}\|_\infty = 1$. Moreover, by duality of the $\mathbb{L}_p(\sigma)$ spaces, for any $X \in \mathcal{B}(\mathcal{H})$,

$$\begin{aligned} \|\mathcal{P}_t(X)\|_{\mathbb{L}_1(\sigma)} &= \sup_{\|Y\|_\infty \leq 1} \langle Y, \mathcal{P}_t(X) \rangle_\sigma \\ &= \sup_{\|Y\|_\infty \leq 1} \langle \hat{\mathcal{P}}_t(Y), X \rangle_\sigma \\ &\leq \sup_{\|Y\|_\infty \leq 1} \|\hat{\mathcal{P}}_t(Y)\|_\infty \|X\|_{\mathbb{L}_1(\sigma)} \\ &\leq \|X\|_{\mathbb{L}_1(\sigma)}, \end{aligned}$$

where $\hat{\mathcal{P}}_t$ is the dual of $\mathcal{P}_t$ with respect $\langle \cdot, \cdot \rangle_\sigma$. By density, it then follows that $\|\mathcal{P}_t : \mathbb{L}_1(\sigma) \rightarrow \mathbb{L}_1(\sigma)\| \leq 1$. The result for any $p \geq 1$ follows by interpolation. The strong continuity of the semigroup follows by Proposition 5.8 of [Goldstein and Lindsay, 1995] which we here adapt to our setting: let $\hat{p}$ be the Hölder conjugate of p . Then, for any $X \in \mathcal{B}(\mathcal{H})$ and $Y \in \mathbb{L}_{\hat{p}}(\sigma)$,

$$\langle Y, (\mathcal{P}_t - \text{id})(X) \rangle_\sigma = \text{Tr}(\Gamma_\sigma(Y)^* (\mathcal{P}_t - \text{id})(X)) \xrightarrow{t \rightarrow 0} 0.$$

The limit above holds by weak* continuity of $(\mathcal{P}_t)_{t \geq 0}$ and since

$$\|\Gamma_\sigma(Y)\|_1 = \|Y\|_{\mathbb{L}_1(\sigma)} \leq \|Y\|_{\mathbb{L}_{\hat{p}}(\sigma)} < \infty,$$

so that $\Gamma_\sigma(Y) \in \mathcal{T}_1(\mathcal{H}) \simeq \mathcal{B}(\mathcal{H})_*$. Since the limit holds for any $Y \in \mathbb{L}_{\hat{p}}(\sigma) \simeq \mathbb{L}_p(\sigma)^*$, $t \mapsto \mathcal{P}_t - \text{id}$ converges weakly to 0 as $t \rightarrow 0$. This is equivalent to strong convergence by Proposition 1.23 of [Davies, 1980]. $\square$

For sake of simplicity, we denote the $\mathbb{L}_p(\sigma)$ QMS as $(\mathcal{P}_t)_{t \geq 0}$, and its generator as $(\mathcal{L}_p, \text{dom}(\mathcal{L}_p))$ of dense domain $\text{dom}(\mathcal{L}_p)$ in $\mathbb{L}_p(\sigma)$. We also assume the following condition, similar to Condition 2.2.1, holds true:

Condition 5.2.2. There exists a *-subalgebra $\mathcal{A}_0$ of $\mathcal{B}(\mathcal{H})$ that is $\mathbb{L}_2(\sigma)$ -dense in $\text{dom}(\mathcal{L}_2)$, invariant under $\mathcal{L}_2$ and dense in all the $\mathbb{L}_p(\sigma)$ spaces for $p \in [1, \infty)$. Then, we define $\mathcal{A}_0^+ := \{X + c\mathbb{1}, X \in \mathcal{A}_0, X \geq 0, c > 0\}$.

In practice, infinitesimal functional inequalities such as the logarithmic Sobolev inequality will be defined on the space $\mathcal{A}_0^+$. The density of $\mathcal{A}_0$ will allow us to derive contractivity properties of the semigroup for any initial operator (see Chapter 7). In the case when the $\mathbb{L}_2(\sigma)$ -QMS $(\mathcal{P}_t)_{t \geq 0}$ is the extension of a QMS on $\mathcal{B}(\mathcal{H})$ with associated Lindblad form given in Equation (5.9), the algebra $\mathcal{A}_0$ can be chosen as follows:

Proposition 5.2.3. Assume that $(\mathcal{P}_t)_{t \geq 0}$ is a weak* continuous QMS such that Condition 5.1.4 and Condition 5.1.5 hold, and let $\sigma = \sum_i \lambda_i |\psi_i\rangle\langle\psi_i|$ be a normal faithful invariant state. Assume moreover that, for all $i \in \mathbb{N}$, ψ_i is contained in $\text{dom}(K^*)$, $\text{dom}(L_l^*)$, $l \geq 1$, such that $\sum_l \|L_l^* \psi_i\|^2 < \infty$, and that $K\mathcal{V}, K^*\mathcal{V}, L_j\mathcal{V}, L_j^*\mathcal{V} \subset \mathcal{V}$, where $\mathcal{V} := \text{span}\{\psi_i, i \in \mathbb{N}\}$. Then $\mathcal{A}_0 = \text{span}\{|\psi_i\rangle\langle\psi_j|, i, j \in \mathbb{N}\}$ satisfies Condition 5.2.2.

Proof. We first show that $\mathcal{A}_0$ belongs to $\text{dom}(\mathcal{L}_2)$. By Proposition 5.1.6, we simply need to prove

that for any $i, j \in \mathbb{N}$, $(v, u) \mapsto \mathcal{L}(|\psi_i\rangle\langle\psi_j|)[v, u]$ is norm continuous on $\text{dom}(K) \times \text{dom}(K)$:

$$\begin{aligned} \mathcal{L}(|\psi_i\rangle\langle\psi_j|)[v, u] &= \langle Kv, \psi_i\rangle\langle\psi_j, u\rangle + \sum_l \langle L_l v, \psi_i\rangle\langle\psi_j, L_l u\rangle + \langle v, \psi_i\rangle\langle\psi_j, Ku\rangle \\ &= \langle v, K^* \psi_i\rangle\langle\psi_j, u\rangle + \sum_l \langle v, L_l^* \psi_i\rangle\langle L_l^* \psi_j, u\rangle + \langle v, \psi_i\rangle\langle K^* \psi_j, u\rangle \\ &\leq \|u\| \|v\| \left(\|K^* \psi_i\| \|\psi_j\| + \sum_l \|L_l^* \psi_i\| \|L_l^* \psi_j\| + \|\psi_i\| \|K^* \psi_j\| \right), \end{aligned}$$

and the claim follows by a use of Cauchy Schwartz inequality. This directly implies that $\mathcal{A}_0 \subset \text{dom}(\mathcal{L}_2)$. By definition, $\mathcal{A}_0$ is dense in $\mathcal{T}_p(\mathcal{H})$ for any $p \geq 1$. This implies that for any $p \geq 1$, and any $X \in \mathbb{L}_p(\sigma)$, there exists a sequence $\{X_n\}_{n \in \mathbb{N}}$ of elements in $\mathcal{A}_0$ such that $\|\Gamma_{\sigma^{\frac{1}{p}}}^{-1}(X_n) - X\|_{\mathbb{L}_p(\sigma)} = \|X_n - \Gamma_{\sigma^{\frac{1}{p}}}^{-1}(X)\|_p \rightarrow 0$. This also proves the density of $\mathcal{A}_0$ in $\mathcal{D}(\mathcal{L}_2) \subset \mathbb{L}_2(\sigma)$. The invariance of $\mathcal{A}_0$ under the action of $\mathcal{L}_2$ arises from the condition of invariance of $\mathcal{V}$ under the action of the operators $K, K^*, L_j, L_j^*, j \in \mathcal{J}$. $\square$

5.2.2. KMS symmetry and detailed balance condition

There are various ways of extending the notion of reversibility seen in Section 2.2 to the quantum setting. One possible way is by use of the modular automorphism group associated to the corresponding invariant state (cf. Section 0.1.5).

Definition 5.2.4 (KMS-symmetry). Let $\beta \in \mathbb{R}$, $(\alpha_t)_{t \in \mathbb{R}}$ be a weak* continuous group of *-automorphisms of a von Neumann algebra $\mathcal{M}$ and ω a fixed (α, β) -KMS state. A bounded map $\Phi : \mathcal{M} \rightarrow \mathcal{M}$ is said to be (α, β) -KMS symmetric with respect to ω if

$$\omega(B\Phi(A)) = \omega(\alpha_{-i\beta/2}(A)\Phi(\alpha_{i\beta/2}(B))), \quad (\text{KMS-symmetry})$$

holds for any A, B in a weak* dense $(\alpha_t)_{t \geq 0}$ -invariant *-subalgebra of the set $\mathcal{M}_\alpha$ of analytic elements of $(\alpha_t)_{t \in \mathbb{R}}$. A weak* continuous semigroup $(\mathcal{P}_t)_{t \geq 0}$ on $\mathcal{M}$ is said to be (α, β) -KMS symmetric with respect to ω if $\mathcal{P}_t$ is (α, β) -KMS symmetric with respect to ω for all $t \geq 0$.

In the case when $\beta = 0$ (ω tracial state), the condition of KMS-symmetry simplifies to

$$\langle \Phi(A), B \rangle_{1, \omega} = \langle A, \Phi(B) \rangle_{1, \omega}, \quad A, B \in \mathcal{M}, \quad (\omega\text{-DBC})$$

where $\langle X, Y \rangle_{1, \omega} := \omega(X^*Y)$ is usually referred to as the GNS-*inner product*. This so-called *quantum detailed balance condition* (also known as GNS-symmetry) was introduced by [Kossakowski et al., 1977] and will play an important role in Chapter 12. For the moment, we make the following well-known observation, a proof of which can be found in [Cipriani, 1997]: a bounded linear map $\Phi : \mathcal{M} \rightarrow \mathcal{M}$ is said to commute with a weak* continuous group of automorphisms $(\alpha_s)_{s \in \mathbb{R}}$ if, for any element $X \in \mathcal{M}_\alpha$ and any $s \in \mathbb{R}$:

$$\Phi \circ \alpha_s(X) = \alpha_s \circ \Phi(X).$$

A QMS $(\mathcal{P}_t)_{t \geq 0}$ is said to commute with $(\alpha_s)_{s \in \mathbb{R}}$ if $\mathcal{P}_t$ commutes with $(\alpha_s)_{s \in \mathbb{R}}$ for any $t \geq 0$.

Lemma 5.2.5. *Let $\mathcal{M}$ be a von Neumann algebra, $\beta \in \mathbb{R}$, $(\alpha_t)_{t \in \mathbb{R}}$ a weak* continuous group of automorphisms, ω an (α, β) -KMS state on $\mathcal{M}$ and $\Phi : \mathcal{M} \rightarrow \mathcal{M}$ a bounded map. Then the following holds:*

- (i) *If Φ commutes with $(\alpha_t)_{t \in \mathbb{R}}$, then KMS-symmetry is equivalent to ω -DBC.*

(ii) If $(\alpha_t)_{t \in \mathbb{R}} = (\alpha_t^\omega)_{t \in \mathbb{R}}$ is the modular automorphism group (so that $\beta = -1$), and if Φ satisfies ω -DBC, then Φ commutes with $(\alpha_t^\omega)_{t \in \mathbb{R}}$ and satisfies KMS-symmetry.

Proof. (i) Assume that Φ commutes with $(\alpha_t)_{t \geq 0}$ and that KMS-symmetry holds. Then for all A, B in a norm dense, α -invariant $*$ -subalgebra of $\mathcal{M}_\alpha$,

$$\begin{aligned} \omega(B\Phi(A)) &= \omega(\alpha_{-i\beta/2}(A)\Phi(\alpha_{i\beta/2}(B))) \\ &= \omega(\alpha_{-i\beta/2}(A)\alpha_{i\beta/2}(\Phi(B))) \\ &= \omega(\Phi(B)A), \end{aligned}$$

where the first line follows from KMS-symmetry, the second one by commutation of Φ and $(\alpha_t)_{t \in \mathbb{R}}$ and the fact that $B \in \mathcal{M}_\alpha$, and the last line follows from Equation (0.13). Then ω -DBC follows by weak* density. Since all the steps above are reversible, the equivalence is proved.

(ii) This is done in Proposition 2.1 of [Kossakowski et al., 1977] (see also Lemma 2 of [Bratteli and Robinson, 1976]). $\square$

In the case when $\mathcal{M} = \mathcal{B}(\mathcal{H})$, with $\mathcal{H}$ separable, and ω is normal faithful with associated density operator σ and cyclic vector $\Omega_\omega := \sigma^{\frac{1}{2}}$, a semigroup $(\mathcal{P}_t)_{t \geq 0}$ is $((\alpha_s^\sigma)_{s \in \mathbb{R}}, -1)$ -KMS symmetric with respect to σ , where $(\alpha_s^\sigma)_{s \in \mathbb{R}}$ is the modular automorphism group defined in Section 0.1.5, if and only if for any $t \geq 0$ and all A, B in a weak*-dense $(\alpha_s^\sigma)_{s \in \mathbb{R}}$ -invariant subspace of the algebra $\mathcal{B}(\mathcal{H})_\sigma$ of entire analytic elements of $\mathcal{B}(\mathcal{H})$ for the modular group $(\alpha_s^\sigma)_{s \in \mathbb{R}}$:

$$\text{Tr}(\sigma B \mathcal{P}_t(A)) = \text{Tr}\left(\sigma^{\frac{1}{2}} A \sigma^{\frac{1}{2}} \mathcal{P}_t(\sigma^{\frac{1}{2}} B \sigma^{-\frac{1}{2}})\right). \quad (5.16)$$

However, given an eigenvector decomposition of σ as $\sigma = \sum_{i=1}^\infty \lambda_i |e_i\rangle\langle e_i|$, $\mathcal{A}_0 := \text{span}\{|e_i\rangle\langle e_j|, i, j \in \mathbb{N}\}$ is a weak* dense subalgebra of $\mathcal{B}(\mathcal{H})$ whose norm closure is the algebra $\mathcal{K}$ of compact operators. Moreover, it is easy to see that it is included in the algebra $\mathcal{B}(\mathcal{H})_\sigma$. Therefore, for $A, B \in \mathcal{A}_0$, upon replacement of $\sigma^{\frac{1}{2}} B \sigma^{-\frac{1}{2}}$ by B and A by A^* in Equation (5.16), we get $\langle B, \mathcal{P}_t(A) \rangle_\sigma = \langle A, \mathcal{P}_t(B) \rangle_\sigma$. Since $\mathcal{P}_t$ is continuous and $\mathcal{A}_0$ is dense in $\mathbb{L}_2(\sigma)$, this also holds for any $A, B \in \mathbb{L}_2(\sigma)$. We then proved that in the case of a QMS $(\mathcal{P}_t)_{t \geq 0}$ on $\mathcal{B}(\mathcal{H})$, $((\alpha_s^\sigma)_{s \in \mathbb{R}}, -1)$ -KMS-symmetry with respect to the faithful normal state σ is equivalent to the following: for any $A, B \in \mathbb{L}_2(\sigma)$,

$$\langle B, \mathcal{P}_t(A) \rangle_\sigma = \langle A, \mathcal{P}_t(B) \rangle_\sigma. \quad (\text{KMS-symmetry})$$

That is, the QMS is symmetric with respect to the $\mathbb{L}_2(\sigma)$ inner product $\langle \cdot, \cdot \rangle_\sigma$. This condition then naturally extends the classical detailed balance condition (μ -DBC). Since we will exclusively be working in this setting, from now on, we take the above equation as our definition of KMS-symmetry.

In the case of an $\mathbb{L}_2(\sigma)$ -QMS that satisfies ω -DBC, Theorem 5.1.3 provides an expression for the generator $\mathcal{L}_2$ that will prove very useful in Section 5.4.

Theorem 5.2.6. Assume that $(\mathcal{P}_t)_{t \geq 0}$ is the minimal, unital QMS associated to a form generator in the generalized GKLS form (5.9), with operators L_j, L_j^*, K and K^* densely defined on $\mathcal{V} := \text{span}\{\psi_i, i \in \mathbb{N}\}$, where $\sigma = \sum_i \lambda_i |\psi_i\rangle\langle \psi_i|$ is a normal, faithful state. We further assume that $\sum_j \lambda_j |\langle \psi_j, L_i \psi_j \rangle| < \infty$, $\sum_j \lambda_j |\langle \psi_j, K \psi_j \rangle| < \infty$ and $|\mathcal{J}| < \infty$. Then,

(a) If $(\mathcal{P}_t)_{t \geq 0}$ commutes with the modular automorphism group $(\alpha_s^\sigma)_{s \in \mathbb{R}}$, the operators L_j can be chosen without loss of generality so that the following algebraic conditions hold: for any $j \in \mathcal{J}$, there exist $k, l \in \mathbb{N}$ and constants $\omega_j = \ln \lambda_k - \ln \lambda_l$ such that, for all $z \in \mathbb{C}$, and all $\psi \in \mathcal{V}$:

$$(i) \quad \sigma^{iz} L_j \sigma^{-iz} \psi = e^{iz\omega_j} L_j \psi$$

$$(ii) \quad \sigma K \psi = K \sigma \psi$$

$$(iii) \quad \text{Tr}(\sigma L_i) = \text{Tr}(\sigma L_i^*) = \text{Im Tr}(\sigma K) = \text{Im Tr}(\sigma K^*) = 0.$$

(b) Moreover, if the semigroup satisfies ω -DBC with respect to the state σ , then the operator K can be assumed to satisfy $K = K^*$ on $\mathcal{V}$, and one can choose $\{e^{-\frac{\omega_r}{2}} L_r\}_{r \in \mathcal{J}} = \{L_r^*\}_{r \in \mathcal{J}}$.

Proof. (a) By Theorem 5.1.3, we can assume without loss of generality that $\text{Tr}(\sigma L_r), \text{Tr}(\sigma K) = 0$. Moreover, since $(\mathcal{P}_t)_{t \geq 0}$ commutes with $(\alpha_s^\sigma)_{s \in \mathbb{R}}$, it is covariant with respect to the representation $s \mapsto V(s) := \sigma^{is}$ of $\mathbb{R}$ on $\mathcal{H}$. Adapting the proof of Theorem 2 of [Holevo, 1995a], since $V(s)\mathcal{V} \subset \mathcal{V}$, and since $V(s)\sigma V(s)^* = \sigma$, the operators L_r and K can, without loss of generality, be replaced by $L'_r(s)$ and $K'(s)$ that satisfy Equations (5.10) and (5.11) for any $s \in \mathbb{R}$ and unitary matrices $U(s) \in \mathcal{U}(\mathbb{C}^{\mathcal{J}})$. Moreover, by the covariance property, the operators $L'_r(s)$ and $K'(s)$ can be chosen as follows:

$$L'_r(s) = \sigma^{is} L_r \sigma^{-is} = \sum_j U_{rj}(s) L_j \quad (5.17)$$

$$K'(s) = \sigma^{is} K \sigma^{-is} = K \quad (5.18)$$

(the vectors $(\alpha_i(s))_{i \in \mathcal{J}}$ and numbers $\beta(s)$ appearing in Theorem 5.1.3 can be taken to be null because of the invariance of the normalization condition under the action of the modular group). One can easily verify from Equation (5.17) that the unitaries $U(s)$ form a group $U(s)U(t) = U(s+t)$, $s, t \in \mathbb{R}$. Therefore, it admits a self-adjoint generator H . Without loss of generality, one can then assume that $(L_j)_{j \in \mathcal{J}}$ are the coefficients of a $|\mathcal{J}|$ -dimensional vector $\mathbf{L}$ written in the eigenbasis of H , so that $(U(s)\mathbf{L})_r = e^{i\omega_r s} L_r$ for all $s \in \mathbb{R}$. Therefore $\alpha_s^\sigma(L_r) = e^{i\omega_r s} L_r$ on $\mathcal{V}$ for any $r \in \mathcal{J}$. In particular, for any $j, k \in \mathbb{N}$ and $r \in \mathcal{J}$:

$$\left(\frac{\lambda_j}{\lambda_k}\right)^{is} \langle \psi_j, L_r \psi_k \rangle = \langle \psi_j, \sigma^{is} L_r \sigma^{-is} \psi_k \rangle = e^{i\omega_r s} \langle \psi_j, L_r \psi_k \rangle.$$

This implies that, unless $\langle \psi_j, L_r \psi_k \rangle = 0$, for any $r \in \mathcal{J}$, there exist $j, k \in \mathbb{N}$ such that $\omega_r = \ln(\lambda_j) - \ln(\lambda_k)$. By density of $\mathcal{V}$ in $\mathcal{H}$, for any $z \in \mathbb{C}$, and any $\psi \in \mathcal{V}$:

$$\sigma^{iz} L_r \sigma^{-iz} \psi = e^{i\omega_r z} L_r \psi.$$

(b) This is inspired by [Fagnola and Umanitá, 2007]: we assume now that $(\mathcal{P}_t)_{t \geq 0}$ satisfies ω -DBC with respect to σ . From Equation (5.13), we know that for any $k \in \mathbb{N}$ and $\psi \in \mathcal{V}$:

$$K \psi = \mathcal{L}_*(|\psi\rangle\langle\psi_k|) \psi_k - \sum_r \langle L_r \psi_k, \psi_k \rangle L_r \psi - \langle K \psi_k, \psi_k \rangle \psi.$$

By the adjoint of Equation (5.13), we find similarly the following:

$$K^* \psi = \mathcal{L}_*(|\psi\rangle\langle\psi_k|) \psi_k - \langle \psi_k, K \psi_k \rangle \psi - \sum_r \langle L_r^* \psi_k, \psi_k \rangle L_r^* \psi.$$

Therefore,

$$(K - K^*) \psi = (\langle \psi_k, K \psi_k \rangle - \langle \psi_k, K^* \psi_k \rangle) \psi + \sum_r \langle (L_r^* - L_r) \psi_k, \psi_k \rangle L_r \psi,$$

After multiplication by λ_k and summation over index k , we get

$$(K - K^*) \psi = [\text{Tr}(\sigma K) - \text{Tr}(\sigma K^*)] \psi,$$

where the term involving operators L_r vanishes because of the normalization conditions $\text{Tr}(\sigma L_r) = 0$. Hence, $K - K^*$ acts as an imaginary constant on $\mathcal{V}$, which is equal to 0 by the normalization condition. To prove the last claim, we start from the detailed balance condition $\text{Tr}(\sigma \mathcal{P}_t(A) B) = \text{Tr}(\sigma A \mathcal{P}_t(B))$, and choose $A = |\psi_i\rangle\langle\psi_k|$, $B = |\psi_j\rangle\langle\psi_l|$. After differentiation, we get

$$\mathcal{L}(|\psi_j\rangle\langle\psi_l|)[\psi_k, \sigma\psi_i] = \mathcal{L}(|\psi_i\rangle\langle\psi_k|)[\sigma\psi_l, \psi_j].$$

By a direct computation, we arrive at

$$\begin{aligned} \sum_{r \in \mathcal{J}} \langle \sigma L_r^* \psi_l, \psi_i \rangle \langle \psi_k, L_r^* \psi_j \rangle &= \sum_{r \in \mathcal{J}} \langle L_r \psi_k, \psi_j \rangle \langle \psi_l, L_r \sigma \psi_i \rangle \\ &= \sum_{r \in \mathcal{J}} \langle L_r \sigma \psi_l, \psi_i \rangle \langle \psi_k, L_r \psi_j \rangle \\ &= \sum_{r \in \mathcal{J}} e^{-\omega_r} \langle \sigma L_r \psi_l, \psi_i \rangle \langle \psi_k, L_r \psi_j \rangle. \end{aligned}$$

Then, by weak* density of $\mathcal{A}_0$ in $\mathcal{B}(\mathcal{H})$, we get that for all $\varphi, \psi \in \mathcal{V}$ and any $X \in \mathcal{B}(\mathcal{H})$:

$$\sum_{r \in \mathcal{J}} \langle \sigma L_r^* \psi_l, X L_r^* \psi_j \rangle = \sum_{r \in \mathcal{J}} e^{-\omega_r} \langle \sigma L_r \psi_l, X L_r \psi_j \rangle$$

and therefore $\{e^{-\frac{\omega_r}{2}} L_r\}_{r \in \mathcal{J}} = \{L_r^*\}_{r \in \mathcal{J}}$, without loss of generality. $\square$

The above theorem implies the existence of the following symmetric GKLS form for the form-generator of a QMS satisfying ω -DBC with respect to a normal faithful state σ as well as the domain conditions of Theorem 5.2.6: there exist operators $\{\tilde{L}_r\}_{r \in \mathcal{J}} = \{\tilde{L}_r^*\}_{r \in \mathcal{J}}$ and K defined on $\mathcal{V}$, with $K = K^*$ pointwise in $\mathcal{V}$, satisfying (i)-(ii)-(iii) of Theorem 5.2.6, and such that for any $X \in \mathcal{B}(\mathcal{H})$ and any $\psi, \varphi \in \mathcal{V}$:

$$\mathcal{L}(X)[\psi, \varphi] = \langle \psi, X K \varphi \rangle + \langle K \psi, X \varphi \rangle + \sum_{r \in \mathcal{J}} e^{-\frac{\omega_r}{2}} \langle \tilde{L}_r \psi, X \tilde{L}_r \varphi \rangle. \quad (5.19)$$

Indeed, assuming that $L_r^* = e^{-\frac{\omega_r'}{2}} \tilde{L}_{r'}$, one simply needs to pick $\tilde{L}_{r'} := e^{\frac{\omega_r'}{4}} L_r$. This form reduces to the one that was recently given by [Carlen and Maas, 2017] in the finite dimensional case: since $\mathcal{L}(1) = 0$, $K = -\frac{1}{2} \sum_{r \in \mathcal{J}} e^{-\frac{\omega_r}{2}} L_r^* L_r$, so that

$$\mathcal{L}(X) = \sum_{r \in \mathcal{J}} e^{-\omega_r/2} (\tilde{L}_r^* [X, \tilde{L}_r] + [\tilde{L}_r^*, X] \tilde{L}_r). \quad (5.20)$$

5.2.3. Dirichlet forms

The analysis of QMS and their contractivity properties is greatly simplified by the introduction of a Dirichlet form¹: let $(\mathcal{P}_t)_{t \geq 0}$ be an $\mathbb{L}_2(\sigma)$ QMS with associated generator $(\mathcal{L}_2, \text{dom}(\mathcal{L}_2))$. Then, for $X, Y \in \text{dom}(\mathcal{L}_2)$, the mapping

$$\mathcal{E}_{2, \mathcal{L}}(X, Y) := -\langle X, \mathcal{L}_2(Y) \rangle_\sigma$$

¹For a thorough study of Dirichlet forms on general von Neumann algebras, we refer to [Cipriani, 1997].

is called the *Dirichlet form* associated to $(\mathcal{P}_t)_{t \geq 0}$. In fact, symmetric, weak* continuous positive contraction semigroups are fully characterized by their Dirichlet forms. This noncommutative generalisation of the so-called Beurling–Deny characterisation of the form generators of classical symmetric Markov semigroups was studied in the general von Neumann algebraic setting in [Cipriani, 1997, Goldstein and Lindsay, 1995, Goldstein and Lindsay, 1999].

In Section 3.2 we introduced the logarithmic Sobolev inequality $\text{LSI}_p(c, d)$ where the right hand side involves a classical Dirichlet form $\mathcal{E}(f^{p-1}, f)$. We will see in Part IV that, in order to define a quantum logarithmic Sobolev inequality as the infinitesimal version of the hypercontractivity property of a QMS, one needs to introduce a parametrized family of such forms: For any $p > 1$, the *Dirichlet form of order p* associated to the QMS $(\mathcal{P}_t)_{t \geq 0}$ is defined as follows for all $X \in \text{dom}(\mathcal{L}_p)$:²

$$\mathcal{E}_{p,\mathcal{L}}(X) := -\frac{p\hat{p}}{4} \text{Re}(\langle \mathbb{I}_{\hat{p},p}(X), \mathcal{L}_p(X) \rangle_\sigma). \quad (5.21)$$

Since $(\mathcal{P}_t)_{t \geq 0}$ is contractive in each $\mathbb{L}_p(\sigma)$ space, the Lumer-Phillips theorem 5.1.1 directly implies the following:

Proposition 5.2.7. *For any $p > 1$ and any positive, semidefinite operator $X \in \text{dom}(\mathcal{L}_p)$:*

$$\mathcal{E}_{p,\mathcal{L}}(X) \geq 0. \quad (5.22)$$

Proof. Let $p > 1$. Then, by Proposition 5.2.1 the semigroup $(\mathcal{P}_t)_{t \geq 0}$ is $\mathbb{L}_p(\sigma)$ contractive. Hence, by part (iv) of Theorem 5.1.1, $\mathcal{L}_p$ is dissipative, which means that there exists $X_p^* \in \mathbb{L}_p(\sigma)^* \simeq \mathbb{L}_{\hat{p}}(\sigma)$ such that $X_p^*(X) = \|X\|_{\mathbb{L}_p(\sigma)}^2$ and $\text{Re}(X_p^*(\mathcal{L}_p(X))) \leq 0$. However, by Proposition 1.1.4, the only operator X_p^* that satisfies $X_p^*(X) = \|X\|_{\mathbb{L}_p(\sigma)}^2$ is $X_p^* \equiv \|X\|_{\mathbb{L}_p(\sigma)}^{2-p} \mathbb{I}_{\hat{p},p}(X)$. Therefore:

$$\text{Re}(X_p^*(\mathcal{L}_p(X))) = \|X\|_{\mathbb{L}_p(\sigma)}^{2-p} \text{Re}(\langle \mathbb{I}_{\hat{p},p}(X), \mathcal{L}_p(X) \rangle_\sigma) = -\frac{4 \|X\|_{\mathbb{L}_p(\sigma)}^{2-p}}{\hat{p}p} \mathcal{E}_{p,\mathcal{L}}(X) \leq 0 \Rightarrow \mathcal{E}_{p,\mathcal{L}}(X) \geq 0.$$

□

In Section 7.4, we will need to extend the definition of $\mathcal{E}_{p,\mathcal{L}}$ to $p \in \mathbb{R} \setminus \{0, 1\}$. The same definition will be of use, after replacing $\mathcal{L}_p$ by the generator $\mathcal{L}_2$ of the QMS: assume that there exists an algebra $\mathcal{A}_0$ that satisfies Condition 5.2.2. Then, for all $X \in \mathcal{A}_0^+$ and $p \in \mathbb{R} \setminus \{0, 1\}$,

$$\mathcal{E}_{p,\mathcal{L}}(X) := -\frac{p\hat{p}}{4} \langle \mathbb{I}_{\hat{p},p}(X), \mathcal{L}(X) \rangle_\sigma. \quad (5.23)$$

This definition coincides with the one given in Equation (5.21) for $X \in \mathcal{A}_0^+$ and $p > 1$. The verification of the following properties of the Dirichlet form is easy.

Lemma 5.2.8. *Let $X \in \mathcal{A}_0^+$:*

- (i) $\mathcal{E}_{\hat{p},\mathcal{L}}(\mathbb{I}_{\hat{p},2}(X)) = \mathcal{E}_{p,\mathcal{L}}(\mathbb{I}_{p,2}(X))$ for all $p \in \mathbb{R} \setminus \{0, 1\}$.
- (ii) $\mathcal{E}_{p,\mathcal{L}}(cX) = c^p \mathcal{E}_{p,\mathcal{L}}(X)$ for any $p \in \mathbb{R} \setminus \{0, 1\}$ and $c \geq 0$.

²The normalization chosen here differs from the one of [Kastoryano and Temme, 2013]. The advantage of our normalization resides in that the quantity stays positive even for $p < 1$, which will prove useful in Section 7.4.

5.3. Non-commutative differential calculus

5.3.1. Derivations and differential operators

In Part II, we claimed that the proof of the equivalence between hypercontractivity and the logarithmic Sobolev inequality mainly consists in the use of a differential calculus that is based on the existence of a certain chain rule. This is due to the fact that the property of $\mathbb{L}_p$ -regularity of Dirichlet forms depends on it.

From this observation, the question of the existence of a noncommutative differential calculus becomes central. In this section, we recall the concept of a derivation on the algebra $\mathcal{B}(\mathcal{H})$ of bounded operators on a separable Hilbert space, and explain how in this framework, the generator of a QMS should be interpreted as a noncommutative second order differential operator. We then describe a noncommutative chain rule due to [Birman and Solomyak, 1993] that is the main ingredient in the proof of $\mathbb{L}_p$ -regularity of QMS satisfying ω -DBC. This chain rule will also prove to be very useful in the development of a quantum theory of Ricci curvature (see Chapter 12).

The study of noncommutative differential calculus on abstract C^* -algebras can be traced back to [Guido et al., 1996, Cipriani and Sauvageot, 2003]. Given a C^* -algebra $\mathcal{A}$, an $\mathcal{A}$ – $\mathcal{A}$ bimodule $\mathcal{K}$ is a Hilbert space together with commuting left and *right* representations of $\mathcal{A}$.

Definition 5.3.1 (Derivations). A linear map $\nabla : \text{dom}(\nabla) \rightarrow \mathcal{K}$ defined on a subspace $\text{dom}(\nabla)$ of $\mathcal{A}$ is then called a *derivation* if

- $\text{dom}(\nabla)$ is a subalgebra of $\mathcal{A}$;
- The *Leibniz rule* holds: for any $X, Y \in \text{dom}(\nabla)$,

$$\nabla(XY) = \nabla(X)Y + X \nabla(Y),$$

where, given $X \in \mathcal{A}$ and $\mathbf{V} \in \mathcal{K}$, $X \mathbf{V}$, resp. $\mathbf{V} X$, stands for the operation of left-multiplication, resp. right-multiplication, of $\mathbf{V}$ by X .

Example 5.3.2 (Gradients on $\mathbb{R}^n$). Let $\mathcal{A} = C^0(\mathbb{R}^n)$ be the algebra of continuous functions $f : \mathbb{R}^n \rightarrow \mathbb{R}$ that vanish at infinity, and let $\mathcal{S}(\mathbb{R}^n)$ be the Schwartz space of smooth, rapidly decaying functions:

$$\mathcal{S}(\mathbb{R}^n) = \left\{ f \in C^\infty(\mathbb{R}^n) : \sup_{x \in \mathbb{R}^n} |x^\alpha D^\beta f(x)| < \infty \ \forall \alpha, \beta \in \mathbb{N}^n \right\},$$

where, given $x \in \mathbb{R}^n$ and $\alpha = (\alpha_1, \dots, \alpha_n)$, $\beta = (\beta_1, \dots, \beta_n) \in \mathbb{N}^n$, $x^\alpha := x_1^{\alpha_1} \dots x_n^{\alpha_n}$, and $D^\beta f := \partial_{x_1}^{\beta_1} \dots \partial_{x_n}^{\beta_n} f$. The usual gradient

$$\nabla : \mathcal{S}(\mathbb{R}^n) \rightarrow \mathbb{L}_2(\mu_{\text{Leb}}(\mathbb{R}^n), \mathbb{R}^n) := \left\{ v : \mathbb{R}^n \rightarrow \mathbb{R}^n, \int_{\mathbb{R}^n} \|v(x)\|_{\mathbb{R}^n}^2 \mu_{\text{Leb}}(dx) < \infty \right\}$$

is a derivation in the sense of Definition 5.3.1.

Example 5.3.3 (Discrete derivation on finite sample spaces). Let $\mathcal{A} = F(I)$ the algebra of functions defined on a finite set I and let $\pi : \mathcal{M}(I \times I)$ be a positive measure on the cartesian product $I \times I$. Then, the map $\nabla : F(I) \rightarrow \mathbb{L}_2(I \times I, \pi)$ defined as

$$\forall f \in F(I), \ \forall x, y \in I, \ \nabla(f)(x, y) := f(x) - f(y).$$

is a derivation in the sense of Definition 5.3.1.

Example 5.3.4 (Quantum commutations). In the next section, we will need to manipulate commutators $[A, B]$ between two possibly unbounded operators A and B . Let $\mathcal{A} = \mathcal{B}(\mathcal{H})$ be the algebra of bounded linear operators on a separable Hilbert space $\mathcal{H}$, and let $L : \text{dom}(L) \rightarrow \mathcal{H}$ be a (possibly unbounded) closed, densely defined linear operator on $\mathcal{H}$. Let X, Y two closed operators on $\mathcal{H}$ with dense domains. Assume there exists a dense subspace $\mathcal{V}$ of $\mathcal{H}$ such that $\mathcal{V} \subset \text{dom}(X), \text{dom}(X^*), \text{dom}(Y), \text{dom}(L), \text{dom}(L^*)$. Moreover, assume that the following sesquilinear form defined, for any $\psi, \varphi \in \mathcal{V}$, as

$$(\psi, \varphi) \mapsto \tilde{C}_L^{X,Y}[\psi, \varphi] := \langle L^* \psi, Y \varphi \rangle - \langle X^* \psi, L \varphi \rangle \quad (5.24)$$

is continuous on $\mathcal{H} \times \mathcal{H}$. To this form one can then associate a bounded operator, which we denote by $C_L^{X,Y}$, such that $\tilde{C}_L^{X,Y}[\psi, \varphi] = \langle \psi, C_L^{X,Y} \varphi \rangle$, $\psi, \varphi \in \mathcal{V}$. Formally, “ $C_L^{X,Y} = LY - XL$ ”.

In the case when $X = Y$, we will denote $C_L^{X,X}$ by $[L, X]$. If X is bounded, the commutator $[L, X]$ can be interpreted as a derivation in the sense of Definition 5.3.1: assume that $\mathcal{V} := \text{span}\{\psi_i\}$ for a given orthonormal basis $\{\psi_i\}_{i \in \mathbb{N}}$ of $\mathcal{H}$, and let $\sigma := \sum_i \lambda_i |\psi_i\rangle \langle \psi_i|$ be a faithful normal state in $\mathcal{D}(\mathcal{H})$. Since $\mathbb{L}_2(\sigma)$ is invariant under left and right multiplication by $\mathcal{B}(\mathcal{H})$, it defines a $\mathcal{B}(\mathcal{H}) - \mathcal{B}(\mathcal{H})$ bimodule. Then, define the operator $\nabla_L : \text{dom}(\nabla_L) \rightarrow \mathbb{L}_2(\sigma)$ as follows:

$$\text{dom}(\nabla_L) = \{X \in \mathcal{B}(\mathcal{H}) : X\mathcal{V} \subset \mathcal{V}, \tilde{C}_L^{X,X} \text{ continuous in } \mathcal{H} \times \mathcal{H}, [L, X] \in \mathbb{L}_2(\sigma)\}, \quad \nabla_L(X) := [L, X].$$

This set is an algebra: for any $X, Y \in \text{dom}(\nabla_L)$, $XY\mathcal{V} \subset \mathcal{V}$ and for all $\psi, \varphi \in \mathcal{V}$:

$$\begin{aligned} \tilde{C}_L^{XY,XY}[\psi, \varphi] &= \langle L^* \psi, XY \varphi \rangle - \langle (XY)^* \psi, L \varphi \rangle \\ &= \langle L^* \psi, X(Y \varphi) \rangle - \langle X^* \psi, L(Y \varphi) \rangle \\ &\quad + \langle L^*(X^* \psi), Y \varphi \rangle - \langle Y^*(X^* \psi), L \varphi \rangle. \end{aligned}$$

The above also justifies the Leibnitz rule

$$\nabla_L(XY) = \nabla_L(X)Y + X\nabla_L(Y),$$

and that $\nabla_L(XY) \in \mathbb{L}_2(\sigma)$. Therefore, $\text{dom}(\nabla_L)$ is an algebra.

For instance, let $\mathcal{H} = \mathbb{L}_2(\mathbb{R})$, and let Q, P denote the position and momentum operators defined in Section 0.2. Then, for any smooth, bounded function φ on $\mathbb{R}$, $\varphi(Q) \in \text{dom}(\nabla_P)$ and, on $\mathcal{S}(\mathbb{R})$,

$$\nabla_P(\varphi(Q)) = \frac{1}{i} \varphi'(Q).$$

More generally, given a vector $\mathbf{L} := (L_j)_{j \in \mathcal{J}}$ of closed, densely defined operators L_j such that L_j and L_j^* all have common dense domain $\mathcal{V}$, we introduce the *noncommutative gradient* as follows:

$$\nabla_{\mathbf{L}} : \begin{cases} \bigcap_{j \in \mathcal{J}} \text{dom}(\nabla_{L_j}) \rightarrow \bigoplus_{j \in \mathcal{J}} \mathbb{L}_2(\sigma) \\ X \mapsto (\nabla_{L_1}(X), \dots, \nabla_{L_n}(X)) \end{cases}. \quad (5.25)$$

The *noncommutative divergence* is then defined in the weak sense as follows: for any $\mathbf{V} \in \bigoplus_{j \in \mathcal{J}} \mathcal{T}_2(\mathcal{H})$ and $X \in \mathcal{A}_0 \equiv \text{span}\{|\psi_i\rangle \langle \psi_j|\}$,

$$\langle X, \text{div}_{\mathbf{L}}(\mathbf{V}) \rangle_{\text{HS}} = - \sum_{j \in \mathcal{J}} \langle \nabla_{L_j}(X), V_j \rangle_{\text{HS}}.$$

Finally, the *noncommutative Laplacian* $\Delta_{\mathbf{L}}$ is weakly defined as follows:

$$\Delta_{\mathbf{L}} := \operatorname{div}_{\mathbf{L}} \circ \nabla_{\mathbf{L}} = - \sum_{j \in \mathcal{J}} \nabla_{L_j}^* \circ \nabla_{L_j}.$$

5.3.2. Double operator integrals and Noncommutative chain rule:

In order to extend some of the calculations of Part II involving differentiations to our noncommutative setting, we need the notion of a noncommutative chain rule. Such a chain rule exists and is connected to the abstract theory of double integrals [Daleckiĭ and Krein, 1951, Daletskiĭ and Krein, 1965, Birman and Solomyak, 1967, De Pagter et al., 2002, de Pagter and Sukochev, 2004, Ptapov and Sukochev, 2008, Potapov, D. and Sukochev, F., 2010]. Here, we follow the treatment of [Birman and Solomyak, 1993]. In spirit, given any two spectral projections $\mathcal{B}(\mathbb{R}) \ni A \mapsto E_0(A), E_1(A)$ and a function $f : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{C}$, one wants to make sense of the following integral:

$$T_f : X \mapsto \int_{\mathbb{R}^2} f(x, y) E_0(dx) X E_1(dy).$$

More precisely, let A_0, A_1 be (possibly unbounded) self-adjoint operators in a separable Hilbert space $\mathcal{H}$, of respective domains $\operatorname{dom}(A_k)$ and associated spectral measures E_k , $k = 0, 1$. The measures E_k induce new spectral measures acting on $\mathcal{T}_2(\mathcal{H})$: for $T \in \mathcal{T}_2(\mathcal{H})$, set $\mathcal{E}_0(\cdot)T = E_0(\cdot)T$ and $\mathcal{E}_1(\cdot)T = TE_1(\cdot)$. The spectral measures $\mathcal{E}_k$ commute, and their product $\tilde{\mathcal{E}}$ is a spectral measure on $\mathcal{T}_2(\mathcal{H})$ on the Borel subsets of $\mathbb{R}^2$. Now, given an arbitrary $\tilde{\mathcal{E}}$ -measurable and $\tilde{\mathcal{E}}$ -almost everywhere finite function h defined on $\operatorname{sp}(A_0) \times \operatorname{sp}(A_1)$, we associate the integral

$$T_h = \int h d\tilde{\mathcal{E}}.$$

T_h is called the *transformer* of h . The transformer T_h defines a bounded linear operator on $\mathcal{T}_2(\mathcal{H})$, since $\|T_h : \mathcal{T}_2(\mathcal{H}) \rightarrow \mathcal{T}_2(\mathcal{H})\| = \|h\|_{\mathbb{L}_\infty(\mathbb{R}^2)}$. More generally, given $1 \leq p \leq \infty$, we are interested in the class $[T_p(\mathcal{H})]$ of transformers T_h that are bounded in $\mathcal{T}_p(\mathcal{H})$ (with $\mathcal{T}_\infty(\mathcal{H}) := \mathcal{B}(\mathcal{H})$), i.e. such that $\|T_h : \mathcal{T}_p(\mathcal{H}) \rightarrow \mathcal{T}_p(\mathcal{H})\| < \infty$. On any of these ideals, the map $f \mapsto T_f$ is

- (i) linear: $T_{\lambda f + g} = \lambda T_f + T_g$;
- (ii) multiplicative: $T_{fg} = T_f \circ T_g$;
- (iii) involution: when $p = 2$, $T_f^* = T_{\bar{f}}$, where the adjoint is taken with respect to the Hilbert-Schmidt inner product;
- (iv) positivity preserving: $f \geq 0$ implies that T_f is a positive operator on $\mathcal{T}_2(\mathcal{H})$.

In particular, given a complex-valued Borel function $f : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{C}$, we are interested in functions $\tilde{f}$ of the following form

$$\tilde{f}(x, y) := \begin{cases} \frac{f(x) - f(y)}{x - y} & \text{if } (x, y) \in \operatorname{sp}(H_0) \times \operatorname{sp}(H_1) \\ 0 & \text{else} \end{cases}. \quad (5.26)$$

Now, let X, Y and L be defined as in Example 5.3.4 on a dense subspace $\mathcal{V}$. Assume moreover that X and Y are self-adjoint, and that for any $g \in C_c^\infty(\mathbb{R})$, $g(X)\mathcal{V}, g(Y)\mathcal{V} \subset \mathcal{V}$. The following theorem from [Birman and Solomyak, 1993] provides a *noncommutative chain rule* for the derivation ∇_L :

Theorem 5.3.5. *Given $1 \leq p \leq \infty$, assume that the operator $C_L^{X,Y}$ defined through Equation (5.24) is in $\mathcal{T}_p(\mathcal{H})$ and let $f : \mathbb{R} \rightarrow \mathbb{C}$ be a Borel function such that the transformer $T_{\bar{f}}^{X,Y}$ associated to the spectral measures of X and Y is in $[\mathcal{T}_p(\mathcal{H})]$. Then, the form $\tilde{C}_{L,f}^{X,Y}$ defined for any $\psi, \varphi \in \mathcal{V}$ as*

$$\tilde{C}_{L,f}^{X,Y}[\psi, \varphi] := \langle L^* \psi, f(Y)\varphi \rangle - \langle \bar{f}(X)\psi, L\varphi \rangle$$

is continuous on $\mathcal{H} \times \mathcal{H}$. Moreover, the associated bounded operator $C_{L,f}^{X,Y}$ can be represented as follows:

$$C_{L,f}^{X,Y} = T_{\bar{f}}^{X,Y}(C_L^{X,Y}) \in \mathcal{T}_p(\mathcal{H}). \quad (\text{n-c chain rule})$$

Remark 5.3.6. In the case when $X = Y$ the n-c chain rule simplifies to

$$\nabla_L f(X) = T_{\bar{f}}^{X,X}(\nabla_L X),$$

where $\nabla_L X := [L, X]$. This should be interpreted as a noncommutative extension of the usual chain rule $(f \circ g)' = f' \circ g \cdot g'$ for functions.

5.4. Stroock Varopoulos inequality and regularity of Dirichlet forms

In Section 2.3 and Section 2.4, we introduced regularity properties of the Dirichlet form of a classical Markov semigroup that play a key role in the proof of the equivalence between the logarithmic Sobolev inequality and hypercontractivity. In particular, we saw that if $(P_t)_{t \geq 0}$ is an $\mathbb{L}_2(\mu)$ -Markov semigroup defined on a measure space $(E, \mathcal{F}, \mu)$ that is symmetric with respect to the measure μ , then, for any $p > 0$ and $f \geq 0$:

$$\mathcal{E}(f^{p-1}, f) \geq \frac{4}{p\hat{p}} \mathcal{E}(f^{p/2}, f^{p/2}). \quad (5.27)$$

When $(P_t)_{t \geq 0}$ is a reversible Markov semigroup defined on a finite sample space, inequality (5.27) is actually a simple consequence of the so-called Stroock Varopoulos inequality (first established by [Carlen et al., 1987, Varopoulos, 1985] in the case $p = 2$ and $q \in (1, 2]$, see [Mossel et al., 2013] for the more general result provided here):

Theorem 5.4.1 (Stroock-Varopoulos inequality). *Let $p, q \in (0, 2] \setminus \{1\}$ and $p > q$. Then for every function $g > 0$:*

$$q\hat{q} \mathcal{E}(g^{1/q}, g^{1/\hat{q}}) \geq p\hat{p} \mathcal{E}(g^{1/p}, g^{1/\hat{p}}). \quad (5.28)$$

When q spans $(1, 2]$, $\hat{q}$ spans $[2, \infty)$, so that one recovers (5.27) after taking $p = 2$ and $g = f^{\hat{q}}$ in (5.28).

A fully quantum generalization of these results is currently lacking. Here we provide a partial result in this direction: let $(\mathcal{P}_t)_{t \geq 0}$ be a quantum Markov semigroup on the space $\mathcal{B}(\mathcal{H})$, $\mathcal{H}$ separable, with faithful normal invariant state $\sigma = \sum_i \lambda_i |\psi_i\rangle\langle\psi_i|$. In this section, we assume that the QMS $(\mathcal{P}_t)_{t \geq 0}$ satisfies the conditions of Proposition 5.2.3, so that its form generator takes the form of Equation (5.20), and choose $\mathcal{A}_0 := \text{span}\{|\psi_i\rangle\langle\psi_j|, i, j \in \mathbb{N}\}$. By Proposition 5.2.3, the algebra $\mathcal{A}_0$ satisfies Condition 5.2.2,

and for any $X \in \mathcal{A}_0^+$:

$$\begin{aligned}\mathcal{E}_{p,\mathcal{L}}(X) &= -\frac{p\hat{p}}{4} \langle \mathbf{I}_{\hat{p},p}(X), \mathcal{L}(X) \rangle_\sigma \\ &= -\frac{p\hat{p}}{4} \sum_{r \in \mathcal{J}} e^{-\frac{\omega_r}{2}} \langle \mathbf{I}_{\hat{p},p}(X), \nabla_{\tilde{L}_r^*}(X) \tilde{L}_r - \tilde{L}_r^* \nabla_{\tilde{L}_r}(X) \rangle_\sigma,\end{aligned}$$

where $\nabla_{\tilde{L}_r} : \text{dom}(\nabla_{\tilde{L}_r}) \rightarrow \mathbb{L}_2(\sigma)$, $\nabla_{\tilde{L}_r^*} : \text{dom}(\nabla_{\tilde{L}_r^*}) \rightarrow \mathbb{L}_2(\sigma)$ are defined as in Example 5.3.4, with $\mathcal{V} := \text{span}\{\psi_i, i \in \mathbb{N}\}$. In particular, for $p = 2$, $\mathcal{E}_{2,\mathcal{L}}$ satisfies the following *integration by parts formula* analogous to IBP: for any $X, Y \in \mathcal{A}_0$:

$$\mathcal{E}_{2,\mathcal{L}}(X, Y) = -\langle X, \mathcal{L}(Y) \rangle_\sigma = \sum_{r \in \mathcal{J}} \langle \nabla_{\tilde{L}_r} X, \nabla_{\tilde{L}_r} Y \rangle_\sigma. \quad (\text{n-cIBP})$$

Theorem 5.4.2 (Quantum Stroock-Varopoulos inequality). *Let $(\mathcal{P}_t)_{t \geq 0}$ be a quantum Markov semigroup satisfying the conditions of Theorem 5.2.6(b). Then, for all $X \in \mathcal{A}_0^+$ and any $0 \leq p \leq q \leq 2$:*

$$\mathcal{E}_{p,\mathcal{L}}(\mathbf{I}_{p,2}(X)) \geq \mathcal{E}_{q,\mathcal{L}}(\mathbf{I}_{q,2}(X)).$$

Remark 5.4.3. We should point out that a quantum Stroock-Varopoulos inequality in the special case when σ is the completely mixed state of a QMS defined on a finite dimensional Hilbert space was proven in [Cubitt et al., 2015]. Moreover, sufficient conditions for a finite dimensional QMS to satisfy *strong $\mathbb{L}_p$ -regularity* were previously provided in [Kastoryano and Temme, 2013]. The strong $\mathbb{L}_p$ -regularity was later proved in [Bardet, 2017] under the condition of ω -DBC.

In the next two subsections, we list two proofs for this theorem. The first one builds on the proof of strong regularity of [Kastoryano and Temme, 2013] and only works in finite dimensions. The second one is more similar to the proof of [Bardet, 2017] and extends to the case of a separable Hilbert space. We present the two proofs since they are different in nature and we hope that the ideas used in both can be useful elsewhere.

5.4.1. First proof of Theorem 5.4.2 in the finite dimensional case

In this section, we assume that the QMS $(\mathcal{P}_t)_{t \geq 0}$ is defined on $\mathcal{B}(\mathcal{H})$, where $d_{\mathcal{H}} < \infty$, and satisfies σ -DBC.

Lemma 5.4.4. *For every $t \geq 0$ there are operators $R_k \in \mathcal{B}(\mathcal{H})$ and $\nu_k > 0$ such that $\Delta_\sigma(R_k) = \nu_k R_k$,*

$$\mathcal{P}_t(X) = \sum_k R_k X R_k^*, \quad (5.29)$$

and $\sum_k R_k R_k^ = \mathbb{1}$.*

Proof. By Lemma 5.2.5 the Lindblad generator $\mathcal{L}$ and then $\mathcal{P}_t = e^{t\mathcal{L}}$ commute with Δ_σ , i.e.,

$$\mathcal{P}_t \circ \Delta_\sigma = \Delta_\sigma \circ \mathcal{P}_t. \quad (5.30)$$

Fix an orthonormal basis $\{e_i\}_{i=1}^d$ for the underlying Hilbert space $\mathcal{H} = \mathcal{H}_A$ and define

$$\Upsilon := \sum_{i=1}^d e_i \otimes e_i \in \mathcal{H}_{AB},$$

where $\mathcal{H}_B$ is isomorphic to $\mathcal{H}_A$. It is not hard to verify that for any matrix M we have

$$(M_A \otimes \mathbb{1}_B)|\Upsilon\rangle = \mathbb{1}_A \otimes M_B^T |\Upsilon\rangle, \quad (5.31)$$

where the transpose is with respect to the basis $\{e_i\}_{i=1}^d$. The Choi-Jamiolkowski representation of $\mathcal{P}_t$ is

$$J_{AB} := (\mathcal{P}_t \otimes \mathbb{1}_B)(|\Upsilon\rangle\langle\Upsilon|).$$

Then using (5.31) it is not hard to verify that (5.30) translates to

$$(\sigma_A^{-1} \otimes \sigma_B^T) J_{AB} = J_{AB} (\sigma_A^{-1} \otimes \sigma_B^T).$$

That is, J_{AB} and $\sigma_A^{-1} \otimes \sigma_B^T$ commute. On the other hand, J_{AB} is positive semidefinite since it is the Choi-Jamiolkowski representation of a completely positive map. Therefore, J_{AB} and $\sigma_A^{-1} \otimes \sigma_B^T$ can be simultaneously diagonalized in an orthonormal basis, i.e., there exists an orthonormal basis $\{v_k\}_{k=1}^{d^2}$ of $\mathcal{H}_{AB}$ such that

$$J_{AB} v_k = \lambda_k v_k \quad (5.32)$$

$$\sigma_A^{-1} \otimes \sigma_B^T v_k = \nu_k^{-1} v_k, \quad (5.33)$$

where $\lambda_k \geq 0$, $\nu_k > 0$. Define the operator V_k by

$$(V_k \otimes \mathbb{1}_B) \Upsilon = v_k.$$

Then again using (5.31), equation (5.33) translates to

$$\sigma^{-1} V_k \sigma = \nu_k^{-1} V_k.$$

Moreover, equation (5.32) means that

$$(\mathcal{P}_t \otimes \mathbb{1}_B)(|\Upsilon\rangle\langle\Upsilon|) = J_{AB} = \sum_k \lambda_k |v_k\rangle\langle v_k| = \sum_k \lambda_k (V_k \otimes \mathbb{1}_B) |\Upsilon\rangle\langle\Upsilon| (V_k^* \otimes \mathbb{1}_B),$$

which gives

$$\mathcal{P}_t(X) := \sum_k \lambda_k V_k X V_k^*.$$

Then letting $R_k := \sqrt{\lambda_k} V_k$ we have $\sigma R_k = \nu_k R_k \sigma$ and (5.29) holds. The other equation comes from $\mathcal{P}_t(\mathbb{1}) = \mathbb{1}$.

Proof of Theorem 5.4.2 ($d_{\mathcal{H}} < \infty$) : For any $t \geq 0$ define the function $h_t : (0, \infty) \rightarrow \mathbb{R}$ by

$$h_t(s) := \langle \mathbb{I}_{2/(2-s), 2}(X), \mathcal{P}_t \circ \mathbb{I}_{2/s, 2}(X) \rangle_{\sigma}.$$

Since by part (ii) of Lemma 5.2.5, $\mathcal{P}_t = e^{t\mathcal{L}}$ is self-adjoint with respect to the inner product $\langle \cdot, \cdot \rangle_{\sigma}$, we have $h_t(2-s) = h_t(s)$ and h_t is symmetric about $s = 1$. Therefore, all the odd-order derivatives of h_t at $s = 1$ vanish, and we have

$$h_t(s) = h_t(1) + \sum_{j=1}^{\infty} \frac{c_j}{(2j)!} (s-1)^{2j}, \quad (5.34)$$

where

$$c_j = \frac{d^{2j}}{ds^{2j}} h_t(s) \Big|_{s=1}.$$

We claim that all the even-order derivatives of h_t at $s = 1$ are non-negative, i.e., $c_j \geq 0$. We use Lemma 5.4.4 to verify this. Let R_k 's be operators such that

$$\sigma R_k \sigma^{-1} = \nu_k R_k, \quad (5.35)$$

with $\nu_k > 0$ and (5.29) holds. Then letting $Y := \Gamma_\sigma^{1/2}(X)$ and using (5.35) we compute

$$\begin{aligned} h_t(s) &= \text{Tr} \left[\Gamma_\sigma^{\frac{s}{2}}(Y^{2-s}) \cdot \mathcal{P}_t(\Gamma_\sigma^{-\frac{s}{2}}(Y^s)) \right] \\ &= \sum_k \text{Tr} \left[Y^{2-s} \sigma^{\frac{s}{4}} R_k \sigma^{-\frac{s}{4}} Y^s \sigma^{-\frac{s}{4}} R_k^* \sigma^{\frac{s}{4}} \right] \\ &= \sum_k \nu_k^{\frac{s}{2}} \text{Tr} \left[Y^{2-s} R_k Y^s R_k^* \right]. \end{aligned}$$

Now diagonalizing Y in its eigenbasis: $Y = \sum_\ell \mu_\ell |e_\ell\rangle\langle e_\ell|$, we find that

$$h_t(s) = \sum_{k, \ell, \ell'} \mu_\ell^2 |\langle e_\ell, R_k e_{\ell'} \rangle|^2 \left(\frac{\sqrt{\nu_k} \mu_{\ell'}}{\mu_\ell} \right)^s.$$

Therefore, $h_t(s)$ is a sum of exponential functions with positive coefficients. From this expression it is clear that c_j 's are all non-negative. Let us define

$$g_t(s) := \frac{h_t(s) - h_t(0)}{(s-1)^2 - 1} = \sum_{j=1}^{\infty} \frac{c_j}{(2j)!} \left(\sum_{i=0}^{j-1} (s-1)^{2i} \right).$$

From this expression it is clear that $g_t(s)$ is non-decreasing on $[1, +\infty)$. Therefore, $\lim_{t \rightarrow 0^+} g_t(s)/t$ is non-decreasing on $[1, +\infty)$. On the other hand, we have $h_t(0) = \text{Tr}(Y^2) = h_0(s)$. We thus can compute

$$\begin{aligned} \lim_{t \rightarrow 0^+} \frac{g_t(s)}{t} &= \frac{1}{(s-1)^2 - 1} \lim_{t \rightarrow 0^+} \frac{h_t(s) - h_t(0)}{t} \\ &= \frac{1}{(s-1)^2 - 1} \lim_{t \rightarrow 0^+} \frac{h_t(s) - h_0(s)}{t} \\ &= \frac{1}{(s-1)^2 - 1} \frac{\partial}{\partial t} h_t(s) \Big|_{t=0} \\ &= \frac{1}{(s-1)^2 - 1} \langle \text{I}_{2/(2-s), 2}(X), \mathcal{L} \circ \text{I}_{2/s, 2}(X) \rangle_\sigma. \end{aligned}$$

Therefore

$$s \mapsto \frac{1}{(s-1)^2 - 1} \langle \text{I}_{2/(2-s), 2}(X), \mathcal{L} \circ \text{I}_{2/s, 2}(X) \rangle_\sigma,$$

is non-decreasing on $[1, +\infty)$. Now the desired result follows once we identify $2/s$ with p (and $2/(2-s)$ with $\hat{p}$, its Hölder conjugate). □

5.4.2. Second proof of Theorem 5.4.2 in the infinite dimensional case

The proof of Theorem 5.4.2 presented in Section 5.4.1 has the disadvantage that it only holds in finite dimensions. This is mostly due to the fact that it relies on the introduction of the unnormalized maximally entangled state Υ that becomes ill defined when $\dim(\mathcal{H}) = \infty$. Here we provide a more

algebraic proof based on the noncommutative chain rule introduced in Theorem 5.3.5. First, for any $r, s \in \mathbb{R} \setminus \{0, 1\}$ and $X \in \mathcal{A}_0^+$, $\mathbf{I}_{r,s}(X) \in \text{dom } \nabla_{\tilde{L}_j}$ and the following holds on $\mathcal{V} = \text{span}\{\psi_i, i \in \mathbb{N}\}$:

$$\begin{aligned} \nabla_{\tilde{L}_j} \mathbf{I}_{r,s}(X) &= \tilde{L}_j \sigma^{-\frac{1}{2r}} \left(\sigma^{\frac{1}{2s}} X \sigma^{\frac{1}{2s}} \right)^{\frac{s}{r}} \sigma^{-\frac{1}{2r}} - \sigma^{-\frac{1}{2r}} \left(\sigma^{\frac{1}{2s}} X \sigma^{\frac{1}{2s}} \right)^{\frac{s}{r}} \sigma^{-\frac{1}{2r}} \tilde{L}_j \\ &= e^{-\frac{\omega_j}{2r}} \sigma^{-\frac{1}{2r}} \tilde{L}_j \left(\sigma^{\frac{1}{2s}} X \sigma^{\frac{1}{2s}} \right)^{\frac{s}{r}} \sigma^{-\frac{1}{2r}} - e^{\frac{\omega_j}{2r}} \sigma^{-\frac{1}{2r}} \left(\sigma^{\frac{1}{2s}} X \sigma^{\frac{1}{2s}} \right)^{\frac{s}{r}} \tilde{L}_j \sigma^{-\frac{1}{2r}} \end{aligned} \quad (5.36)$$

Next, define $Y_j := e^{-\frac{\omega_j}{4}} \Gamma_{\sigma}^{\frac{1}{2}}(X)$ and $Z_j := e^{\frac{\omega_j}{4}} \Gamma_{\sigma}^{\frac{1}{2}}(X)$. Using Equation (5.36) we compute

$$\begin{aligned} \mathcal{E}_{q,\mathcal{L}}(\mathbf{I}_{q,2}(X)) &= -\frac{q\hat{q}}{4} \langle \mathbf{I}_{\hat{q},q}(\mathbf{I}_{q,2}(X)), \mathcal{L}(\mathbf{I}_{q,2}(X)) \rangle_{\sigma} \\ &= -\frac{q\hat{q}}{4} \langle \mathbf{I}_{\hat{q},2}(X), \mathcal{L}(\mathbf{I}_{q,2}(X)) \rangle_{\sigma} \\ &= \frac{q\hat{q}}{4} \sum_j \langle \nabla_{\tilde{L}_j} \mathbf{I}_{\hat{q},2}(X), \nabla_{\tilde{L}_j} \mathbf{I}_{q,2}(X) \rangle_{\sigma} \end{aligned} \quad (5.37)$$

$$= \frac{q\hat{q}}{4} \sum_j \left\langle \Gamma_{\sigma}^{-\frac{1}{\hat{q}}} \left(\tilde{L}_j Y_j^{\frac{2}{\hat{q}}} - Z_j^{\frac{2}{\hat{q}}} \tilde{L}_j \right), \Gamma_{\sigma}^{-\frac{1}{q}} \left(\tilde{L}_j Y_j^{\frac{2}{q}} - Z_j^{\frac{2}{q}} \tilde{L}_j \right) \right\rangle_{\sigma} \quad (5.38)$$

$$= \frac{q\hat{q}}{4} \sum_j \left\langle \tilde{L}_j Y_j^{\frac{2}{\hat{q}}} - Z_j^{\frac{2}{\hat{q}}} \tilde{L}_j, \tilde{L}_j Y_j^{\frac{2}{q}} - Z_j^{\frac{2}{q}} \tilde{L}_j \right\rangle_{\text{HS}} \quad (5.39)$$

$$= \frac{q\hat{q}}{4} \sum_j \left\langle T_{\tilde{f}_2^{\frac{2}{\hat{q}}}}^{Z_j, Y_j} (\tilde{L}_j Y_j - Z_j \tilde{L}_j), T_{\tilde{f}_2^{\frac{2}{q}}}^{Z_j, Y_j} (\tilde{L}_j Y_j - Z_j \tilde{L}_j) \right\rangle_{\text{HS}} \quad (5.40)$$

$$\begin{aligned} &= \frac{q\hat{q}}{4} \sum_j \left\langle \tilde{L}_j Y_j - Z_j \tilde{L}_j, \left(T_{\tilde{f}_2^{\frac{2}{\hat{q}}}}^{Z_j, Y_j} \right)^* T_{\tilde{f}_2^{\frac{2}{q}}}^{Z_j, Y_j} (\tilde{L}_j Y_j - Z_j \tilde{L}_j) \right\rangle_{\text{HS}} \\ &= \frac{q\hat{q}}{4} \sum_j \left\langle \tilde{L}_j Y_j - Z_j \tilde{L}_j, T_{\tilde{f}_2^{\frac{2}{\hat{q}}} \tilde{f}_2^{\frac{2}{q}}}^{Z_j, Y_j} (\tilde{L}_j Y_j - Z_j \tilde{L}_j) \right\rangle_{\text{HS}}, \end{aligned} \quad (5.41)$$

where in (5.37) we used n-cIBP, in (5.38) we used (5.36), and in (5.40) we used the chain rule formula of Theorem 5.3.5 for the functions $f_{\alpha} : x \mapsto x^{\alpha}$. Finally, in (5.41) we used the fact that $f \mapsto T_f$ is multiplicative and involutive (cf. Section 5.3.2). Now, using the proofs of Theorem 2.1 and Lemma 2.4 of [Mossel et al., 2013], for any $x, y \geq 0$ and $0 \leq p \leq q \leq 2$ we have

$$q\hat{q} (x^{1/\hat{q}} - y^{1/\hat{q}}) (x^{1/q} - y^{1/q}) \leq p\hat{p} (x^{1/\hat{p}} - y^{1/\hat{p}}) (x^{1/p} - y^{1/p}).$$

This means that for all x, y we have

$$q\hat{q} \tilde{f}_2^{\frac{2}{\hat{q}}}(x, y) \tilde{f}_2^{\frac{2}{q}}(x, y) \leq p\hat{p} \tilde{f}_2^{\frac{2}{\hat{p}}}(x, y) \tilde{f}_2^{\frac{2}{p}}(x, y).$$

Hence, by positivity we have

$$\begin{aligned} \mathcal{E}_{q,\mathcal{L}}(\mathbf{I}_{q,2}(X)) &\leq \frac{p\hat{p}}{4} \sum_j \left\langle \tilde{L}_j Y_j - Z_j \tilde{L}_j, T_{\tilde{f}_2^{\frac{2}{\hat{p}}} \tilde{f}_2^{\frac{2}{p}}}^{Z_j, Y_j} (\tilde{L}_j Y_j - Z_j \tilde{L}_j) \right\rangle_{\text{HS}} \\ &= \mathcal{E}_{p,\mathcal{L}}(\mathbf{I}_{p,2}(X)). \end{aligned}$$

□

5.5. Examples

5.5.1. Generalized depolarizing semigroup

Perhaps the simplest example of a quantum channel is the *depolarizing channel*: given a state $\rho \in \mathcal{D}(\mathcal{H})$ and $0 \leq p \leq 1$:

$$\mathcal{P}_*^{\text{depol}}(\rho) := (1-p)\rho + p \frac{\mathbb{1}}{d_{\mathcal{H}}}.$$

The depolarizing channel corresponds to the case when the information on the input state is completely lost (i.e. replaced by the completely mixed state $\mathbb{1}/d_{\mathcal{H}}$) with some probability p . More generally, consider the channel that replaces the input state with a full-rank fixed state σ :

$$\mathcal{P}_*^{\text{depol},\sigma}(\rho) = (1-p)\rho + p\sigma.$$

The *generalized depolarizing semigroup* $(\mathcal{P}_t^{\text{depol},\sigma})$ constitutes a simple continuous time version of the above channel. It is defined as follows: for all $t \geq 0$ and any $X \in \mathcal{B}(\mathcal{H})$, $\rho \in \mathcal{D}(\mathcal{H})$:

$$\mathcal{P}_t^{\text{depol},\sigma}(X) = e^{-t} X + (1 - e^{-t}) \text{Tr}(\sigma X) \mathbb{1}_{\mathcal{H}} \Leftrightarrow \mathcal{P}_{t*}^{\text{depol},\sigma}(\rho) = e^{-t} \rho + (1 - e^{-t}) \sigma.$$

The generator of $(\mathcal{P}_t^{\text{depol},\sigma})_{t \geq 0}$ is then given by

$$\mathcal{L}_{\sigma}(X) = \text{Tr}(\sigma X) \mathbb{1}_{\mathcal{H}} - X.$$

$(\mathcal{P}_t^{\text{depol}})_{t \geq 0}$ is obviously primitive with respect to σ . It is easy to also see that it satisfies ω -DBC: for any $X, Y \in \mathcal{B}(\mathcal{H})$,

$$\text{Tr}(\sigma X^* \mathcal{P}_t^{\text{depol},\sigma}(Y)) = e^{-t} \text{Tr}(\sigma X^* Y) + (1 - e^{-t}) \text{Tr}(\sigma X^*) \text{Tr}(\sigma Y) = \text{Tr}(\sigma \mathcal{P}_t^{\text{depol},\sigma}(X)^* Y).$$

Therefore, $\mathcal{L}_{\sigma}$ takes the form of Equation (5.20): given the following eigenvalue decomposition $\sigma := \sum_{i=1}^{d_{\mathcal{H}}} \lambda_i |\psi_i\rangle\langle\psi_i|$, define the operators $L_{ij} := \sqrt{\lambda_i} |\psi_i\rangle\langle\psi_j|$. Hence for any $X \in \mathcal{B}(\mathcal{H})$, $\text{Tr}(\sigma X) \mathbb{1}_{\mathcal{H}} = \sum_{i,j=1}^d L_{ij}^* X L_{ij}$, so that

$$\mathcal{L}_{\sigma}(X) = -\frac{1}{2} \sum_{i,j=1}^{d_{\mathcal{H}}} L_{ij}^* L_{ij} X - 2L_{ij}^* X L_{ij} + X L_{ij}^* L_{ij}. \quad (5.42)$$

Moreover, $\Delta_{\sigma}(L_{ij}) = \lambda_i/\lambda_j L_{ij}$, so that $\omega_{ij} \equiv \log \lambda_j - \log \lambda_i$. Therefore, for any $X \in \mathcal{B}(\mathcal{H})$, Equation (5.42) can be rewritten in the form of Equation (5.20) by taking $\tilde{L}_{ij} = (\lambda_i \lambda_j / 4)^{1/4} |\psi_i\rangle\langle\psi_j|$.

5.5.2. Quantum diffusions on phase space

Quantum Ornstein Uhlenbeck semigroups In this section, we introduce the so-called *quantum Ornstein Uhlenbeck semigroup* (also known as the *damped harmonic oscillator*) on the representation of the CCR algebra over a two-dimensional phase space (cf. Section 0.2), that is, the algebra $\mathcal{B}(\mathcal{H})$ of bounded linear operators over the Hilbert space $\mathcal{H} := \mathbb{L}_2(\mathbb{R})$. This Markovian dynamics models the evolution of a two-energy levels atom which traverses a photonic cavity. The dynamics then results from the interaction of photons, modeled as a quantized radiation field, with the incident atom. The Master equation can be obtained by different standard approximation procedures (weak coupling limit, coarse graining, see e.g. [Strunz, 2002, Fagnola et al., 1994, Weidlich and Haake, 1965]).

First, let $0 < \lambda < \mu$, $\nu := \lambda^2/\mu^2$, and defined the state:

$$\sigma_\nu := (1 - \nu) \sum_{n \in \mathbb{N}} \nu^n |\psi_n\rangle\langle\psi_n| = \frac{e^{\beta N}}{Z}, \quad (5.43)$$

where $\{\psi_n, n \in \mathbb{N}\}$ corresponds to the eigenbasis of the number operator N , and $\beta = \ln \nu$ can be interpreted as an inverse temperature. σ_ν can be shown to be a Gaussian state. Then, on the $*$ -subalgebra $\mathcal{A}_0 := \text{span}\{|\psi_n\rangle\langle\psi_m|, m, n \in \mathbb{N}\}$, the generator of the quantum Ornstein Uhlenbeck semigroup takes the following Lindblad form:

$$\mathcal{L}^{\text{qOU}}(X) := -\frac{\mu^2}{2}(a^*aX - 2a^*Xa + Xa^*a) - \frac{\lambda^2}{2}(aa^*X - 2aXa^* + Xaa^*).$$

In Theorem 4.2 and Corollary 4.3 of [Cipriani et al., 2000], the authors proved the existence of a weak* continuous, positive contraction semigroup $(\mathcal{P}_t^{\text{qOU}})_{t \geq 0}$ on $\mathcal{B}(\mathcal{H})$ that is KMS-symmetric with respect to ρ_ν , through the careful analysis of its associated Dirichlet form (see also [Ko and Park, 2004]). They also showed that $(\mathcal{P}_t^{\text{qOU}})_{t \geq 0}$ is a Feller semigroup with respect to the algebra $\mathcal{K}(\mathcal{H})$ of compact operators on $\mathcal{H}$ (see also [Carbone and Fagnola, 2003]). Moreover, they showed that $\mathcal{A}_0$ is a core for the strong generator $\mathcal{L}_{\mathcal{K}(\mathcal{H})}^{\text{qOU}}$ of $(\mathcal{P}_t^{\text{qOU}}|_{\mathcal{K}(\mathcal{H})})_{t \geq 0}$: for any $X \in \text{dom}(\mathcal{L}_{\mathcal{K}(\mathcal{H})}^{\text{qOU}})$, there exists a sequence $\{X_n\}_{n \in \mathbb{N}}$ of operators in $\mathcal{A}_0$ such that

$$\|\mathcal{L}_{\mathcal{K}(\mathcal{H})}^{\text{qOU}}(X_n) - \mathcal{L}_{\mathcal{K}(\mathcal{H})}^{\text{qOU}}(X)\|_\infty + \|X_n - X\|_\infty \xrightarrow{n \rightarrow \infty} 0.$$

$\mathcal{A}_0$ is also a weak* core for the weak* generator $\mathcal{L}^{\text{qOU}}$, that is, for any $X \in \text{dom}(\mathcal{L}^{\text{qOU}})$, there exists a sequence $\{X_n\}_{n \in \mathbb{N}}$ of operators on $\mathcal{A}_0$ such that for any $T \in \mathcal{T}_1(\mathcal{H}) \simeq \mathcal{B}(\mathcal{H})_*$,

$$\text{Tr}(T(X_n - X)) \xrightarrow{n \rightarrow \infty} 0 \quad \text{and} \quad \text{Tr}(T(\mathcal{L}(X_n) - \mathcal{L}(X))) \xrightarrow{n \rightarrow \infty} 0.$$

Another approach is via the form-generator formalism introduced in Section 5.1. In particular, the form-generator associated with $L_1 = a$ and $L_2 = a^*$ satisfies Conditions 5.1.4 and 5.1.5. Hence, it generates a unique weak* continuous QMS $(\mathcal{P}_t^{\text{qOU}})_{t \geq 0}$ on $\mathcal{B}(\mathcal{H})$. Finally, $(\mathcal{P}_t^{\text{qOU}})_{t \geq 0}$ is primitive with respect to ρ_ν (cf. Theorem 6.2 of [Cipriani et al., 2000]). The reason behind the name of the semigroup resides in the following theorem which one can find in [Cipriani et al., 2000]:

Theorem 5.5.1. *For any polynomial p , $p(Q) \in \text{dom}(\mathcal{L}^{\text{qOU}})$ and*

$$\mathcal{L}^{\text{qOU}}(p(Q)) = L^{\text{cOU}}(p)(Q),$$

where L^{cOU} is the generator of the classical Ornstein Uhlenbeck semigroup which is defined, for example on $C^2(\mathbb{R})$, as:

$$L^{\text{cOU}}(f)(x) := \left(\frac{\mu^2 + \lambda^2}{4}\right) f''(x) - \left(\frac{\mu^2 - \lambda^2}{2}\right) x f'(x).$$

On the other hand, for any polynomial q ,

$$\mathcal{L}^{\text{qOU}}(q(N)) = L^{\text{BD}}(q)(N),$$

where L^{BD} is the generator of a birth and death process with birth rates $\{\lambda^2(k+1)\}_{k \in \mathbb{N}}$ and death rates

$\{\mu^2 k\}_{k \in \mathbb{N}}$. That is

$$L^{\text{BD}}(f)(k) = \mu^2 k (f(k-1) - f(k)) + \lambda^2 (k+1) (f(k+1) - f(k)),$$

Remark 5.5.2. The last theorem provides an example of how from a single quantum Markov semigroup, one can recover different classical Markov semigroups of very different nature, depending on the invariant commutative subalgebra on which the semigroup acts. In this case, the action of the semigroup on the algebra of functions of the position operator leads to a diffusion semigroup, namely the Ornstein Uhlenbeck semigroup, whereas its action on the algebra generated by the number operator leads to a jump semigroup, namely a birth and death process.

Quantum Brownian motion In the infinite temperature limit, that is when $\mu = \nu (= 1/\sqrt{2})$, the generator of the quantum Ornstein Uhlenbeck semigroup formally converges to the one of the *quantum heat semigroup*: on the subalgebra $\mathcal{A}_0 := \text{span}\{|\varphi\rangle\langle\psi| : |\varphi\rangle, |\psi\rangle \in \bigcap_{j=1}^n (\text{dom}(P_j^2) \cap \text{dom}(Q_j^2))\}$ of the CCR algebra $\mathcal{B}(\mathbb{L}_2(\mathbb{R}^n))$ over the $2n$ -dimensional phase space $\mathcal{Z} = \mathbb{R}^{2n}$, consider the following generator

$$\mathcal{L}^{\text{qheat}}(\cdot) := -\frac{1}{4} \sum_{j=1}^{2n} [R_j, [R_j, \cdot]], \quad (5.44)$$

with R_j being the j^{th} element of the vector $\mathbf{R} = (Q_1, P_1, \dots, Q_n, P_n)$ defined through Equation (0.30). By Section 2 of [Holevo, 1996], $\mathcal{L}^{\text{qheat}}$ extends to an unbounded operator on $\mathcal{T}_1(\mathcal{H})$ such that, if an operator X has finite moments of order 2, then $X \in \text{dom}(\mathcal{L}^{\text{qheat}})$; this in particular is true if X is a Gaussian state or a Schwartz operator (cf. Section 0.2). The following theorem lists some important properties of the quantum heat semigroup. Some of these can already be found in [Koenig and Smith, 2014].

Proposition 5.5.3. *The generator defined in Equation (5.44) is the weak* generator of a unique unital quantum Markov semigroup $(\mathcal{P}_t^{\text{qheat}})_{t \geq 0}$ on $\mathcal{B}(\mathbb{L}_2(\mathbb{R}^n))$ called the quantum heat semigroup. Moreover,*

1. *The quantum heat semigroup is reversible with respect to the trace:*

$$\langle A, \mathcal{P}_t^{\text{qheat}}(B) \rangle_{\text{HS}} = \langle \mathcal{P}_t^{\text{qheat}}(A), B \rangle_{\text{HS}}, \quad \forall A, B \in \mathcal{T}_2(\mathcal{H}). \quad (5.45)$$

Denoting by $\mathcal{L}_2^{\text{qheat}}$ the $\mathbb{L}_2(\text{Tr})$ generator of the above QMS, this implies that, for all $A, B \in \text{dom}(\mathcal{L}_2^{\text{qheat}})$,

$$\langle A, \mathcal{L}_2^{\text{qheat}}(B) \rangle_{\text{HS}} = \langle \mathcal{L}_2^{\text{qheat}}(A), B \rangle_{\text{HS}}. \quad (5.46)$$

2. *Let the state $\rho_t \equiv \mathcal{P}_t^{\text{qheat}}(\rho)$ denote the solution of the quantum diffusion equation defined through Equation (5.44). Then for each $t \geq 0$, the characteristic function of ρ_t is given by*

$$\mathcal{F}_{\rho_t}^{\text{q}}(z) = \mathcal{F}_{\rho}^{\text{q}}(z) e^{-|z|^2 t/4}, \quad \forall z \in \mathcal{Z}, \quad (5.47)$$

and we have

$$\rho_t \equiv \rho * g_{t/2} := \int_{\mathcal{Z}} W_z \rho W_{-z} g_{t/2}(z) dz, \quad (5.48)$$

where $g_{t/2}$ denotes the probability density function (pdf) of a Gaussian random variable on $\mathbb{R}^{2n}$ with zero mean and variance equal to $t/2$, i.e.

$$g_{t/2}(z) = \frac{1}{(\pi t)^n} e^{-|z|^2/t}, \quad (5.49)$$

and W_z denotes the Weyl operator defined in Equation (0.28). In particular, if ρ is a Gaussian state, then so is ρ_t .

Proof. Equation (5.47) can be found in [Koenig and Smith, 2014]. In fact, Equation (5.46) can be directly verified by using Equation (5.44). The proof of Equation (5.48) is obtained as follows: Note that $e^{-|z|^2 t/4}$ is the characteristic function of a Gaussian random variable of associated pdf $g_{t/2}$. It is well-known that if f and g are two pdfs then the characteristic function of their convolution $f * g$ is equal to the product of their characteristic functions:

$$\mathcal{F}_{f*g}(z) = \mathcal{F}_f(z) \mathcal{F}_g(z).$$

This property also holds when one of the pdfs is replaced by a quantum state and the standard definition of convolution is replaced by the one of *quantum convolution* (given by Equation (5.48)), as shown by [Werner, 1984] (see also [Kossakowski, 1972]) in his generalization of harmonic analysis to the quantum framework. Hence, (5.47) allows us to express the state ρ_t as a quantum convolution of the initial state ρ and the pdf $g_{t/2}$, as given by Equation (5.48). This provides an easy proof of self-duality (5.45): the operator $\mathcal{P}_t(B)$ is given by the right hand side of Equation (5.48). Hence,

$$\begin{aligned} \text{Tr}(A^* \mathcal{P}_t(B)) &= \text{Tr}\left(A^* \int_{\mathcal{Z}} W_z B W_{-z} g_{t/2}(z) dz\right) \\ &= \text{Tr}\left(A^* \int_{\mathcal{Z}} W_{-z} B W_z g_{t/2}(z) dz\right) \\ &= \text{Tr}\left(\int_{\mathcal{Z}} W_z A^* W_{-z} B g_{t/2}(z) dz\right) \\ &= \text{Tr}\left(\left(\int_{\mathcal{Z}} W_z A W_{-z}\right)^* B g_{t/2}(z) dz\right) \\ &= \text{Tr}((\mathcal{P}_t(A))^* B), \end{aligned} \quad (5.50)$$

where we have used the symmetry of the Gaussian pdf, the cyclicity of the trace, the fact that $W_z^* = W_{-z}$, and Equation (5.48). □

Remark 5.5.4. The semigroup does not possess an invariant state: if there was an invariant state ρ , then its characteristic function would satisfy $\mathcal{F}_{\rho_t}^q = \mathcal{F}_\rho^q$ for all $z \in \mathcal{Z}$, which is impossible by Equation (5.47). The semifinite trace Tr is, however, invariant under $(\mathcal{P}_t^{\text{qheat}})_{t \geq 0}$. This is in complete analogy with the classical heat semigroup $(P_t^{\text{heat}})_{t \geq 0}$ which leaves invariant the Lebesgue measure. Moreover, for any smooth function f on $\mathbb{R}^{2n}$:

$$L^{\text{heat}}(f)(Q) = \mathcal{L}^{\text{qheat}}(f(Q)),$$

where $L^{\text{heat}} \equiv \frac{1}{4} \Delta$ is the generator of the classical heat semigroup.

5.5.3. Quantum convolution semigroups

More generally, [Kossakowski, 1972] introduced a technique to define quantum Markov semigroups on the space $\mathcal{T}_1(\mathcal{H})$, $\mathcal{H}$ separable, starting from a classical one parameter convolution semigroup (we

refer to Section 2.5 for the definition of such semigroups):

Theorem 5.5.5. *Let $(G, \mathcal{B}(G))$ be a locally compact separable group together with its Borel algebra $\mathcal{B}(G)$, and let U be a unitary representation of G on $\mathcal{H}$. Any one-parameter convolution semigroup $(\mu_t)_{t \geq 0}$ on $\mathcal{M}(G)$ induces a strongly continuous semigroup $(\mathcal{P}_{t*}^\mu)_{t \geq 0}$ on $\mathcal{T}_1(\mathcal{H})$ defined as follows: for any $\rho \in \mathcal{T}_1(\mathcal{H})$:*

$$\mathcal{P}_{t*}^\mu(\rho) := \int_G U_g \rho U_{g^{-1}} \mu_t(dg), \quad \rho \in \mathcal{T}_1(\mathcal{H}).$$

In finite dimensions, a characterization of the Heisenberg dual of the quantum convolution semigroups defined by Kossakowski was given by [Kümmerer and Maassen, 1987] in terms of their dilation properties: We mentioned in Section 2.1 that to any classical Markov semigroup $(P_t)_{t \geq 0}$ one can associated a unique minimal Markov process. In the noncommutative setting, the situation is more complicated and is known as the *dilation problem*. Here, by a quantum Markov process (or Markov dilation), we mean a group of $*$ -automorphisms occurring on a larger system from which the local irreversible evolution is recovered by coarse graining (see [Hudson and Parthasarathy, 1984, Parthasarathy, 1992, Meyer, 1993, Kümmerer and Maassen, 1987, Rajarama Bhat and Parthasarathy, 1995] and references therein). In finite dimensions, [Kümmerer and Maassen, 1987] showed that the existence of a certain type of dilation characterizes those unital semigroups that can be written as a quantum convolution semigroup.

More precisely, let $(\mathcal{P}_t)_{t \geq 0}$ be a QMS defined a von Neumann algebra $\mathcal{M}$, and assume that it has a faithful normal invariant state ω . If there exists a group $(\alpha_t)_{t \in \mathbb{R}}$ of $*$ -automorphisms on a von Neumann $\mathcal{M}'$ together with faithful normal invariant state ω' on $\mathcal{M}'$, and completely positive, unital operators $\pi: \mathcal{M} \rightarrow \mathcal{M}'$, $E_{\mathcal{M}}: \mathcal{M}' \rightarrow \mathcal{M}$ such that $\omega \circ E_{\mathcal{M}} = \omega'$ and $\omega' \circ \pi = \omega$, and such that the following diagram commutes for all $t \geq 0$:

$$\begin{array}{ccc} (\mathcal{M}, \omega) & \xrightarrow{\mathcal{P}_t} & (\mathcal{M}, \omega) \\ \downarrow \pi & & \uparrow E_{\mathcal{M}} \\ (\mathcal{M}', \omega') & \xrightarrow{\alpha_t} & (\mathcal{M}', \omega') \end{array}$$

then we call the tuplet $(\mathcal{M}', \omega', (\alpha_t)_{t \geq 0})$ a *dilation* of the *dynamical system* $(\mathcal{M}, \omega, (\mathcal{P}_t)_{t \geq 0})$. Such dilation is called *essentially commutative* if the relative commutant of $\pi(\mathcal{M})$ in $\mathcal{M}'$ is commutative.

Following [Kümmerer and Maassen, 1987], we now restrict our attention to the case when $(\mathcal{P}_t)_{t \geq 0}$ is defined on the algebra $\mathcal{B}(\mathcal{H})$ of bounded operators on the finite dimensional Hilbert space $\mathcal{H}$, and assume that the state ρ is invariant. In this case, any commutative dilation is characterized by a commutative von Neumann algebra $\mathcal{C}$ with faithful normal state $\mathbb{E}_\mu$ such that $\mathcal{M}' = \mathcal{B}(\mathcal{H}) \otimes \mathcal{C}$ and $\omega' = \omega \otimes \mathbb{E}_\mu$. Moreover, for any $X \in \mathcal{B}(\mathcal{H})$, $\pi(X) = X \otimes \mathbb{1}_{\mathcal{C}}$ and $E_{\mathcal{B}(\mathcal{H})}$ is uniquely defined by its action on tensor products: $E_{\mathcal{B}(\mathcal{H})}(X \otimes f) = \mathbb{E}_\mu(f) X$. Since any automorphism of $\mathcal{B}(\mathcal{H})$ leaves the trace invariant, it follows that one can take $\omega \equiv d_{\mathcal{H}}^{-1} \text{Tr}$, without loss of generality. Then (see Theorem 1.1.1 of [Kümmerer and Maassen, 1987]):

Theorem 5.5.6. *The following are equivalent:*

- (i) *The dynamical system $(\mathcal{B}(\mathcal{H}), \frac{\mathbb{1}_{\mathcal{H}}}{d_{\mathcal{H}}}, (\mathcal{P}_t)_{t \geq 0})$ admits an essentially commutative Markov dilation.*
- (ii) *There exists a weak*-continuous convolution semigroup $(\mu_t)_{t \geq 0}$ on the space $\mathcal{M}(\text{Aut}(\mathcal{B}(\mathcal{H})))$ of probability measures on the group of automorphisms of $\mathcal{B}(\mathcal{H})$ such that for all $X \in \mathcal{B}(\mathcal{H})$ and $t \geq 0$:*

$$\mathcal{P}_t(X) = \int_{\alpha \in \text{Aut}(\mathcal{B}(\mathcal{H}))} \alpha(X) \mu_t(d\alpha)$$

(iii) The generator $\mathcal{L}$ of $(\mathcal{P}_t)_{t \geq 0}$ is of the form:

$$\mathcal{L}(X) = i[H, X] + \frac{1}{2} \sum_{j=1}^k 2A_j X A_j - A_j^2 X - X A_j^2 + \sum_{i=1}^l c_i (U_i^* X U_i - X), \quad (5.51)$$

where H and A_j , $j \in \{1, \dots, k\}$ are self-adjoint elements of $\mathcal{B}(\mathcal{H})$, U_i , $i \in \{1, \dots, l\}$, are unitaries in $\mathcal{B}(\mathcal{H})$ and c_i , $i \in \{1, \dots, l\}$ are positive numbers.

(iv) The generator $\mathcal{L}$ of $(\mathcal{P}_t)_{t \geq 0}$ is in the closure of the cone generated by $\{\alpha - \text{id}; \alpha \in \text{Aut}(\mathcal{B}(\mathcal{H}))\}$

(v) For all $t \geq 0$, $\mathcal{P}_t$ lies in the convex hull of $\text{Aut}(\mathcal{B}(\mathcal{H}))$.

We come back to this decomposition in Chapter 9, where a simpler version of the theorem is presented.

Remark 5.5.7. Not every unital QMS on $\mathcal{B}(\mathcal{H})$ can be seen as a quantum convolution semigroup, as shown in [Kümmerer and Maassen, 1987]. This becomes the case when the QMS satisfies ω -DBC with respect to the completely mixed state (see e.g. [Frigerio and Gorini, 1984]). However, there exist QMS that are not $d_{\mathcal{H}}^{-1} \mathbb{1}_{\mathcal{H}}$ -DBC and still admit a Markov dilation [Kümmerer and Maassen, 1987].

As we will see in Chapter 9, there may be several convolution semigroups of probability measures leading to the same QMS $(\mathcal{P}_t)_{t \geq 0}$ on $\mathcal{B}(\mathcal{H})$. However, Theorem 2.5.1 allows us to write a canonical form for the generator of such a semigroup in terms of the form (2.33) of the generator of $(\mu_t)_{t \geq 0}$. This is the content of Theorem 1.5.1 of [Kümmerer and Maassen, 1987]:

Theorem 5.5.8. *Let $X_1, \dots, X_{n^2-1}$ be a fixed basis of the Lie algebra $\text{aut}(\mathcal{B}(\mathcal{H}))$ of $\text{Aut}(\mathcal{B}(\mathcal{H}))$ that is orthonormal with respect to $\langle \cdot, \cdot \rangle_{\text{HS}}$. Choose functions $g_1, \dots, g_{n^2-1}$ on $C_2(\text{Aut}(\mathcal{B}(\mathcal{H})))$ with the properties $g_i(\text{id}) = 0$, $D_i g_j(\text{id}) = \delta_{ij}$ where D_i is the derivation on $C^2(\text{Aut}(\mathcal{B}(\mathcal{H})))$ induced by X_i . Moreover, let $(\mathcal{P}_t)_{t \geq 0}$ be a unital QMS on $\mathcal{B}(\mathcal{H})$. Then there exists a one-to-one correspondence between:*

(i) essentially commutative minimal Markov dilations of $(\mathcal{B}(\mathcal{H}), d_{\mathcal{H}}^{-1} \mathbb{1}_{\mathcal{H}}, (\mathcal{P}_t)_{t \geq 0})$;

(ii) convolution semigroups $(\mu_t)_{t \geq 0}$ of measures on $\text{Aut}(\mathcal{B}(\mathcal{H}))$ such that

$$\mathcal{P}_t(X) = \int_{\text{Aut}(\mathcal{B}(\mathcal{H}))} \alpha(X) \mu_t(d\alpha);$$

(iii) triples $(H, \{b_{ij}\}_{1 \leq i, j \leq n^2-1}, \nu)$ where H is a self-adjoint element of $\mathcal{B}(\mathcal{H})$, $\{b_{ij}\}$ a real positive semidefinite matrix and ν a Lévy measure such that for all $X \in \mathcal{B}(\mathcal{H})$,

$$\mathcal{L}(X) = i[H, X] + \sum_{i,j=1}^{n^2-1} b_{ij} [X_i, [X_j, X]] + \int_{\text{Aut}(\mathcal{B}(\mathcal{H}))} \left(\alpha(X) - X + \sum_{i=1}^{n^2-1} g_i(\alpha) [X_i, X] \right) \nu(d\alpha).$$

Chapter 6.

Decoherence

Quantum and classical systems exhibit very different behaviors, and it is not clear in practice how the former should be thought of as generalizations of the latter. For example, there is no real parameter governing the smooth transition between systems that behave according to the superposition principle and those who don't. This is in total contrast with the other major extension of classical physics, namely special relativity, whose laws converge to Newtonian kinematics and dynamics when taking the speed of light to infinity. This simple observation lead Bohr to formulate the Copenhagen interpretation of quantum mechanics according to which classical and quantum descriptions of a physical system are complementary to each other. The Copenhagen interpretation is somewhat unsatisfactory for two main reasons: Firstly, it imposes the necessary cohabitation of two very distinct physical theories with apparently little logical relation. Secondly, it does not provide any way to decide when one should use either one or the other theory to describe a physical system.

The program of environment induced decoherence was first proposed as a way to solve this issue. Roughly speaking, it states that the universe is fundamentally quantum, but most macroscopic systems acquire classical properties, and hence effectively behave like classical systems, due to unavoidable interactions with their environment. The measurement-like interactions between a system and its environment lead to a strong entanglement between them. This in turn results in the dynamical destruction of certain initial quantum superpositions between vectors belonging to different subspaces in the state of the system under consideration.

An early contribution along these lines is the paper [Zeh, 1970]. Decoherence was popularized by Zurek's articles [Zurek, 1981, Zurek, 1982] (see also [Joos et al., 2003, Blanchard et al., 2000] and the references therein). Perhaps the simplest example of environment induced decoherence is the one treated by [Joos and Zeh, 1985]. There, they showed that the “non-diagonal” elements (in the position basis) of the reduced density matrix of a particle subject to scattering particles of its environment vanish exponentially fast. Similarly, in our Markovian setting, typical decoherent evolutions will be non-primitive. In Chapter 8, we introduce a new set of functional inequalities adapted to the study of the asymptotic properties of such QMS.

Layout of the chapter: We briefly review some general features of the mathematical theory of decoherence in Section 6.1. Section 6.2 deals with the finite dimensional case. We end the chapter by mentioning some important examples of QMS that display decoherence in Section 6.4.

6.1. General theory

One of the first steps towards a rigorous algebraic approach to the study of decoherence was made by [Blanchard and Olkiewicz, 2003] (the slightly different definition given here comes from [Hellmich, 2011], see also [Olkiewicz, 1999, Olkiewicz, 2000] for former special cases previously studied, and [Rebolledo, 2005] for an alternative definition)¹: consider a von Neumann algebra $\mathcal{M}$ together with an irreversible (reduced) dynamics $(\mathcal{P}_t)_{t \geq 0}$, that is a family of positive, normal unital maps such that $t \mapsto \mathcal{P}_t(X)$ is weak* continuous for every $X \in \mathcal{M}^2$. $(\mathcal{P}_t)_{t \geq 0}$ is said to display *decoherence* if the following holds: there exists a von Neumann subalgebra $\mathcal{N}_0 \subseteq \mathcal{M}$ and a weak* continuous group of *-automorphisms $(\alpha_t)_{t \geq 0}$ on $\mathcal{N}_0$ such that $\mathcal{P}_t(\mathcal{N}_0) \subseteq \mathcal{N}_0$, $\mathcal{P}_t|_{\mathcal{N}_0} = \alpha_t$ for any $t \geq 0$, as well as a *-invariant, weak* closed subspace $\mathcal{V}_0 \subset \mathcal{M}$ with $\mathcal{P}_t(\mathcal{V}_0) \subseteq \mathcal{V}_0$ for all $t \geq 0$ such that

$$\mathcal{M} = \mathcal{N}_0 \oplus \mathcal{V}_0, \quad (\text{EID})$$

and such that for all $X \in \mathcal{V}_0$ and any $\omega \in \mathcal{M}_*$,

$$\lim_{t \rightarrow \infty} \omega(\mathcal{P}_t(X)) = 0.$$

The algebra $\mathcal{N}_0$ is chosen as the maximal von Neumann subalgebra on which $(\mathcal{P}_t)_{t \geq 0}$ reduces to a group of *-automorphisms, and is referred to as the *algebra of effective observables*. On the other hand, the subspace $\mathcal{V}_0$ is known as the space of *non-detectable observables*. In other words, decoherence as defined here demands that the non-automorphic part of the quantum evolution vanishes in time. Hence, after a sufficiently long time, the system is effectively characterized by observables belonging to the algebra $\mathcal{N}_0$ and the reversible time evolution is given by the group $(\alpha_t)_{t \geq 0}$. This means that it effectively evolves like a closed system whose properties may differ from the ones of the original system. Note that the above definition makes no assumptions about the rate at which decoherence occurs. In Chapters 8 and 9 we devise some functional analytic techniques to estimate such rates for finite dimensional systems.

The problem of finding minimal conditions for the occurrence of environment induced decoherence is a difficult one. Even when assuming that decoherence occurs, uniqueness of the EID decomposition is not known in general. Here, we briefly review some of the mains results in the general von Neumann algebraic setting. In the next section, we will specialize to the case of QMS defined on the algebra $\mathcal{B}(\mathcal{H})$ of linear operators on a finite dimensional Hilbert space.

First of all, we introduce the *decoherence free algebra* $\mathcal{N}_0 \subseteq \mathcal{N}$ and the space $\mathcal{V}_0 \subseteq \mathcal{V}$ of weakly vanishing operators in $\mathcal{M}$:

$$\begin{aligned} \mathcal{N} &:= \{X \in \mathcal{M} \mid \forall t \geq 0, \mathcal{P}_t(X^*X) = \mathcal{P}_t(X)^* \mathcal{P}_t(X), \mathcal{P}_t(XX^*) = \mathcal{P}_t(X) \mathcal{P}_t(X)^*\}, \\ \mathcal{V} &:= \{X \in \mathcal{M}; \text{ weak}^* \lim_{t \rightarrow \infty} \mathcal{P}_t(X) = 0\}. \end{aligned}$$

$\mathcal{N}$ and $\mathcal{V}$ typically constitute natural candidates for $\mathcal{N}_0$ and $\mathcal{V}_0$ when EID occurs. The inclusion $\mathcal{N}_0 \subseteq \mathcal{N}$ follows from the following result, the proof of which can be found in Proposition 1 of [Carbone et al., 2011]:

Proposition 6.1.1. *$\mathcal{N}$ is the largest von Neumann subalgebra of $\mathcal{M}$ for which $\mathcal{P}_t|_{\mathcal{N}}$ acts as a*

¹A similar definition of decoherence in the C^* -algebraic setting was also considered in [Blanchard et al., 2007].

²Remark that we do not yet require $(\mathcal{P}_t)_{t \geq 0}$ to form a QMS at this stage.

**-homomorphism. In particular, for all $X, Y \in \mathcal{M}$, if either X or Y belongs to $\mathcal{N}$, then for all $t \geq 0$:*

$$\mathcal{P}_t(X^*Y) = \mathcal{P}_t(X^*)\mathcal{P}_t(Y). \quad (6.1)$$

Proof. Obviously, $\mathcal{N}^* = \mathcal{N}$. Moreover, since $\mathcal{P}_t$ is completely positive unital for any $t \geq 0$, it satisfies the Schwarz inequality (1.55):

$$\mathcal{P}_t(X^*X) \geq \mathcal{P}_t(X^*)\mathcal{P}_t(X). \quad (6.2)$$

Therefore the map $D_t : (X, Y) \mapsto \mathcal{P}_t(X^*Y) - \mathcal{P}_t(X^*)\mathcal{P}_t(Y)$ is positive and sesquilinear, so that $D_t(X, X) = 0$ is equivalent to $D_t(X, Y) = 0$ for all Y . Now, since $D_t(X, X) = 0$ for all $X \in \mathcal{N}$, Equation (6.1) follows. Next, we prove that $\mathcal{N}$ is an algebra. Linearity is obvious. Now, let $X, Y \in \mathcal{N}$:

$$\begin{aligned} \mathcal{P}_t((XY)^*(XY)) &= \mathcal{P}_t(Y^*X^*XY) \\ &= \mathcal{P}_t(Y^*)\mathcal{P}_t(X^*XY) \\ &= \mathcal{P}_t(Y^*)\mathcal{P}_t(X^*)\mathcal{P}_t(XY) \\ &= \mathcal{P}_t(Y^*X^*)\mathcal{P}_t(XY) \\ &= \mathcal{P}_t((XY)^*)\mathcal{P}_t(XY), \end{aligned}$$

and $\mathcal{P}_t((XY)(XY)^*) = \mathcal{P}_t(XY)\mathcal{P}_t((XY)^*)$ follows similarly. Therefore, $XY \in \mathcal{N}$. $\mathcal{P}_t$ invariance follows from

$$\mathcal{P}_t(\mathcal{P}_s(X)^*\mathcal{P}_s(X)) = \mathcal{P}_t(\mathcal{P}_s(X^*X)) = \mathcal{P}_{t+s}(X^*X) = \mathcal{P}_t(\mathcal{P}_s(X^*))\mathcal{P}_t(\mathcal{P}_s(X)),$$

for any $X \in \mathcal{N}$, and $s, t \geq 0$. Finally, the weak* closure comes from the following observation: define for any $Y \in \mathcal{M}$:

$$\varphi_Y : \mathcal{M} \ni X \mapsto \mathcal{P}_t(Y^*X) - \mathcal{P}_t(Y^*)\mathcal{P}_t(X) \in \mathcal{M}, \quad (6.3)$$

Then $\mathcal{N} = \cap_{Y \in \mathcal{M}} \varphi_Y^{-1}(\{0\})$. The weak* closure follows from the fact that $\mathcal{P}_t$ is normal and $X \mapsto YX$ is weak* continuous. $\square$

It is further known that $\mathcal{N} = \mathcal{N}_0$ (i.e. $\mathcal{N}$ is the largest von Neumann subalgebra of $\mathcal{M}$ on which the maps $\mathcal{P}_t$ act as *-automorphisms) if any of the following holds [Robinson, 1982] (see also Proposition 3 of [Carbone et al., 2015]):

- $(\mathcal{P}_t)_{t \geq 0}$ possesses a faithful normal invariant state;
- $(\mathcal{P}_t)_{t \geq 0}$ is uniformly continuous on $\mathcal{M} := \mathcal{B}(\mathcal{H})$, $\mathcal{H}$ separable;
- $\mathcal{N} \subseteq \mathcal{F}$, where $\mathcal{F}$ is the set of fixed points of $(\mathcal{P}_t)_{t \geq 0}$.³

$$\mathcal{F} := \{X \in \mathcal{M}; \mathcal{P}_t(X) = X \forall t \geq 0\}.$$

In the uniformly continuous case, this trivially holds since the generator $\mathcal{L}$ is bounded, so that $\mathcal{P}_t$ is invertible for any $t \geq 0$, of inverse $\mathcal{P}_{-t} := e^{-t\mathcal{L}}$. Moreover, $\mathcal{N} \cap \mathcal{V} = \{0\}$, so that if EID holds one necessarily has $\mathcal{N} = \mathcal{N}_0$ and $\mathcal{V} = \mathcal{V}_0$ (see Proposition 2 of [Carbone et al., 2013], Proposition 6

³In fact, the link between $\mathcal{N}$ and $\mathcal{F}$ was already investigated in the 70s and used to study ergodic properties of quantum Markov semigroups (see [Evans, 1977, Frigerio, 1978, Frigerio and Verri, 1982, Robinson, 1982, Fagnola and Rebolledo, 2008]).

of [Carbone et al., 2015]). This fact is no longer true in general (see Example 9 of [Carbone et al., 2015]).

One natural condition that one can require is the existence of a faithful normal invariant state $\omega \in \mathcal{M}_*$.⁴ In this case, the set $\mathcal{F}$ of fixed points is a von Neumann subalgebra (see [Spohn, 1977, Frigerio, 1978], Theorem 6.12 of [Wolf, 2012]). In fact, the following was proved in [Frigerio and Verri, 1982]:

Theorem 6.1.2. *Given a QMS $(\mathcal{P}_t)_{t \geq 0}$ with faithful normal invariant state ω , the limit*

$$E_{\mathcal{F}}[X] := \text{weak}^* \lim_{t \rightarrow 0} \frac{1}{t} \int_0^t \mathcal{P}_s(X) ds$$

exists for all $X \in \mathcal{B}(\mathcal{H})$ and defines a $(\mathcal{P}_t)_{t \geq 0}$ invariant normal conditional expectation $E_{\mathcal{F}}$ onto the algebra $\mathcal{F}$ of fixed points of $(\mathcal{P}_t)_{t \geq 0}$.

Under the condition of existence of a faithful normal invariant state ω , EID was moreover proved to hold in the following cases:

- $\mathcal{N} \subseteq \mathcal{F}$ (and in fact $\mathcal{N} = \mathcal{F}$);
- $\mathcal{M} = \mathcal{B}(\mathcal{H})$, $\mathcal{H}$ finite dimensional;
- $\mathcal{M} = \mathcal{B}(\mathcal{H})$, $\mathcal{H}$ separable, $\mathcal{N}$ atomic, $(\mathcal{P}_t)_{t \geq 0}$ uniformly continuous.

with $\mathcal{V}_0$ being the orthogonal complement $\mathcal{N}^\perp$ of $\mathcal{N}$ under the GNS inner product $\langle A, B \rangle_{1, \omega} := \omega(A^* B)$. Moreover, in these cases, there exists a conditional expectation $E_{\mathcal{N}}$ onto $\mathcal{N}$, compatible with ω , with $\ker(E_{\mathcal{N}}) = \mathcal{N}^\perp$ (see Theorem 1.17 of [Carbone et al., 2014], Theorem 9 of [Carbone et al., 2013], Theorem 22 of [Deschamps et al., 2016] respectively). In general, $\mathcal{M}$ might not be complete with respect to $\langle \cdot, \cdot \rangle_{1, \omega}$, so that $\mathcal{N}$ might not be closed with respect to the corresponding norm and one cannot define $E_{\mathcal{N}}$ as the projection onto $\mathcal{N}$. In this case, [Carbone et al., 2015] considered the completion $\overline{\mathcal{M}}^{\langle \cdot, \cdot \rangle_\omega}$ together with the splitting

$$\overline{\mathcal{M}}^{\langle \cdot, \cdot \rangle_\omega} = \overline{\mathcal{N}}^{\langle \cdot, \cdot \rangle_\omega} \oplus (\overline{\mathcal{N}}^{\langle \cdot, \cdot \rangle_\omega})^\perp.$$

In general, it is not true that $\mathcal{M} = \mathcal{N} \oplus (\overline{\mathcal{N}}^{\langle \cdot, \cdot \rangle_\omega})^\perp \cap \mathcal{M}$. The situation simplifies if one further assumes the existence of a conditional expectation $E_{\mathcal{N}}$ onto $\mathcal{N}$ (cf. Theorem 19 of [Carbone et al., 2015]). A proof of it in the case when $\mathcal{M} = \mathcal{B}(\mathcal{H})$, $\mathcal{H}$ finite, is provided at the end of Section 6.2 for sake of completeness.

Theorem 6.1.3. *Assume there exists a faithful normal invariant state ω and a conditional expectation $E_{\mathcal{N}}$ compatible with ω . Then EID occurs with $\mathcal{N}_0 = \mathcal{N}$ and $\mathcal{V}_0 = \ker E_{\mathcal{N}} = \overline{\mathcal{N}}^{\langle \cdot, \cdot \rangle_\omega}^\perp \cap \mathcal{M}$ being $\mathcal{P}_t$ -invariant for all $t \geq 0$. Moreover, $\mathcal{P}_t$ commutes with $E_{\mathcal{N}}$.*

Remark 6.1.4. The sufficient condition given in the above theorem is equivalent to the invariance of $\mathcal{N}$ under the modular automorphism group $(\sigma_s^\omega)_{s \in \mathbb{R}}$ (cf. Theorem 0.1.4). This in particular happens when each $\mathcal{P}_t$ commutes with $(\sigma_s^\omega)_{s \in \mathbb{R}}$ (see Remark 20 of [Carbone et al., 2015]). The former fact always holds in finite dimensions, as we see in the next section.

Other splittings of the algebra $\mathcal{M}$ into a unitarily evolving part and a decaying part have been extensively studied and compared to EID (see [Carbone et al., 2015, Hellmich, 2009] and references therein for more details).

⁴This assumption was actually part of the original definition of EID by [Blanchard and Olkiewicz, 2003]. It is however not necessary for EID to hold (see Proposition 2.5 of [Carbone et al., 2014]).

Primitive quantum Markov semigroups: In the case when $\mathcal{N} = \mathcal{F}$, the semigroup is said to be *ergodic*. In this case, one can use Frigerio and Verri's sufficient condition for the convergence of a QMS towards a steady state⁵: under the condition of existence of a faithful normal invariant state, the following convergence holds:

$$E_{\mathcal{F}}[X] = \text{weak}^* \lim_{t \rightarrow \infty} \mathcal{P}_t(X).$$

where $E_{\mathcal{F}}$ is the conditional expectation of Theorem 6.1.2.

In the case when $\mathcal{N} = \mathcal{F} = \mathbb{C} \mathbb{1}$, the conditional expectation simply reduces to the average with respect to the unique invariant state ω :

$$\text{weak}^* \lim_{t \rightarrow \infty} \mathcal{P}_t(X) = \omega(X) \mathbb{1} \quad \Leftrightarrow \quad \text{strong} \lim_{t \rightarrow \infty} \mathcal{P}_{t*} = \omega.$$

In this case, the semigroup is said to be *primitive*. A sufficient condition for primitivity is then the existence of a unique faithful invariant state (see also [Burgarth et al., 2013, Theorem 14]).

6.2. The case $\mathcal{M} = \mathcal{B}(\mathcal{H})$, $\mathcal{H}$ finite

In this section, we restrict our attention to the case of norm continuous QMS defined on the algebra $\mathcal{B}(\mathcal{H})$ of linear operators acting on a finite dimensional Hilbert space $\mathcal{H}$ (see [Carbone et al., 2011, Deschamps et al., 2016, Carbone et al., 2013]). The following result can be found in [Fagnola and Rebolledo, 2008] (for an extension to minimal QMS whose generator is represented by a generalised GKLS form, see [Dhahri et al., 2010]):

Theorem 6.2.1. *Given a QMS $(\mathcal{P}_t)_{t \geq 0}$ with GKLS form given by Equation (5.5), the decoherence free algebra $\mathcal{N}$ is characterized by*

$$\{\nabla_H^k(L_j), \nabla_H^k(L_j^*); k \in \mathbb{N}, j \in I\}',$$

where $\nabla_H(L) := [H, L]$. This is independent of the choice of Lindblad operators.

As already discussed in Section 0.1.4, a basic result from the theory of $*$ -algebras on finite dimensional Hilbert spaces states that $\mathcal{N}$ can always be decomposed into a direct sum of subparts where it restricts to a factor [Kadison and Ringrose, 2015]. More precisely, up to a unitary transformation, the Hilbert space $\mathcal{H}$ admits the following decomposition

$$\mathcal{H} = \bigoplus_{i \in I} \mathcal{H}_i \otimes \mathcal{K}_i, \quad (6.4)$$

such that $\mathcal{N}$ is unitarily isomorphic to a matrix algebra: there exists a unitary $U \in \mathcal{U}(\mathcal{H})$ such that⁶

$$\mathcal{N} = U \bigoplus_{i \in I} \mathcal{B}(\mathcal{H}_i) \otimes \mathbb{1}_{\mathcal{K}_i} U^* \equiv U \sum_{i \in I} P_i \mathcal{N} P_i U^*, \quad (6.5)$$

where the projections P_i are mutually orthogonal projections in $\mathcal{N}$ onto $\bigoplus_{j=1}^{\dim(\mathcal{K}_i)} \mathcal{H}_i$, $\sum_{i \in I} P_i = \mathbb{1}_{\mathcal{H}}$, and clearly commute with any elements of $\mathcal{N}$. They are minimal in the sense that for any other

⁵This condition turns out to be necessary in the finite dimensional setting, see [Frigerio and Verri, 1982].

⁶From now on, we forget the rotation induced by the unitary operator U for sake of simplicity, this can be done after redefining the semigroup as $(U \mathcal{P}_t(U^* (\cdot) U) U^*)_{t \geq 0}$.

projection $P \in \mathcal{N}$,

$$P \leq P_i \Rightarrow P = 0 \text{ or } P = P_i. \quad (6.6)$$

In the primitive case, we recall that $\mathcal{N} = \mathbb{C} \mathbb{1}_{\mathcal{H}}$ so that there exists a unique minimal projection which is equal to $\mathbb{1}_{\mathcal{H}}$. More generally, for each $i \in I$, P_i is in the centralizer $\mathcal{Z}(\mathcal{N})$ of $\mathcal{N}$ ⁷.

Since we are in finite dimensions, we know from last section that EID occurs under the condition of existence of a full-rank invariant state σ , and that, in this case, $\mathcal{N}_0 = \mathcal{N}$ and $\mathcal{V}_0 = \mathcal{V}$. Moreover, there exists a conditional expectation $E_{\mathcal{N}} : \mathcal{B}(\mathcal{H}) \rightarrow \mathcal{N}$ (see end of this section). We call such semigroups *faithful*. In order to get an expression for $E_{\mathcal{N}}$, we first need to study the structure of invariant states of $(\mathcal{P}_t)_{t \geq 0}$. This is taken from Theorem 21 of [Deschamps et al., 2016]:

Theorem 6.2.2. *Assume that $(\mathcal{P}_t)_{t \geq 0}$ admits a full-rank invariant state σ . Then, there exists a family $\{\tau_i\}_{i \in I}$ of full-rank invariant states on each $\mathcal{K}_i$ such that any other $(\mathcal{P}_t)_{t \geq 0}$ -invariant state ρ can be written as*

$$\rho = \sum_{i \in I} \text{Tr}_{\mathcal{K}_i}(P_i \rho P_i) \otimes \tau_i. \quad (6.7)$$

The proof consists in a reduction to the evolution in each factor $P_i \mathcal{N} P_i = \mathcal{B}(\mathcal{H}_i) \otimes \mathbb{1}_{\mathcal{K}_i}$. We first state the following technical lemma that is also of independent interest:

Lemma 6.2.3. *The projections P_i belong to the set of fixed points $\mathcal{F}$. Moreover, for any $i, j \in I$, any $X \in \mathcal{B}(\mathcal{H})$ and any $t \geq 0$:*

$$\mathcal{P}_t(P_j X P_i) = P_j \mathcal{P}_t(X) P_i.$$

In particular, each factor $P_i \mathcal{N} P_i = \mathcal{B}(\mathcal{H}_i) \otimes \mathbb{1}_{\mathcal{K}_i}$ of $\mathcal{N}$ is invariant under the action of $(\mathcal{P}_t)_{t \geq 0}$.

Proof. The proof proceeds in two steps. First, we show that any projection P_i is contained in the space $\mathcal{F}$ of fixed points of $(\mathcal{P}_t)_{t \geq 0}$. This is due to the fact that $(\mathcal{P}_t)_{t \geq 0}$ acts as a $*$ -automorphism on $\mathcal{N}$ (cf. Section 6.1). Therefore, since each $P_i \in \mathcal{Z}(\mathcal{N})$, and since for any $X \in \mathcal{N}$, there exists $Y \in \mathcal{N}$ such that $X = \mathcal{P}_t(Y)$:

$$X \mathcal{P}_t(P_i) = \mathcal{P}_t(Y) \mathcal{P}_t(P_i) = \mathcal{P}_t(Y P_i) = \mathcal{P}_t(P_i Y) = \mathcal{P}_t(P_i) \mathcal{P}_t(Y) = \mathcal{P}_t(P_i) X,$$

i.e. $\mathcal{P}_t(P_i) \in \mathcal{Z}(\mathcal{N})$. Hence, $\{\mathcal{P}_t(P_i)\}_{i \in I}$ is a family of mutually orthogonal projections in $\mathcal{Z}(\mathcal{N})$. Similarly, for any $i, j \in I$, it is easy to show that $P_j \mathcal{P}_t(P_i) P_j$ is a projection in $\mathcal{Z}(\mathcal{N})$, that is $(P_j \mathcal{P}_t(P_i) P_j)^2 = P_j \mathcal{P}_t(P_i) P_j$. Moreover, since $P_j \mathcal{P}_t(P_i) P_j \leq P_j$, $P_j \mathcal{P}_t(P_i) P_j = 0$ or $P_j \mathcal{P}_t(P_i) P_j = P_j$ (cf. (6.6)). By continuity of $t \rightarrow \mathcal{P}_t(X)$ for all $X \in \mathcal{B}(\mathcal{H})$, we actually have that $P_j \mathcal{P}_t(P_i) P_j = 0$ for $i \neq j$ and $P_i \mathcal{P}_t(P_i) P_i = P_i$. Therefore

$$P_i \mathcal{P}_t(\mathbb{1} - P_i) P_i = \sum_{j \neq i} P_i \mathcal{P}_t(P_j) P_i = 0 \Rightarrow \mathcal{P}_t(P_i) = P_i.$$

We proved that for any $i \in I$, $P_i \in \mathcal{F}$. This implies that, for all $t \geq 0$ and any $X \in \mathcal{N}$:

$$\mathcal{P}_t(P_i X P_j) = \mathcal{P}_t(P_i) \mathcal{P}_t(X) \mathcal{P}_t(P_j) = P_i \mathcal{P}_t(X) P_j,$$

⁷We recall that the centralizer of a von Neumann algebra $\mathcal{M}$ is the von Neumann algebra that consists in the elements X of $\mathcal{M}$ that commute with all $Y \in \mathcal{M}$.

where the first identity above comes from the fact that, under the condition of existence of an invariant full-rank state, $\mathcal{F}$ is an algebra (and hence a subalgebra of $\mathcal{N}$), so that Equation (6.1) holds.⁸ $\square$

The next lemma is key in the reduction of the problem to the case of a factor algebra $\mathcal{N}$, i.e. when $|I| = 1$. It can be found in Proposition 14 of [Deschamps et al., 2016].

Lemma 6.2.4. *Under the assumption of existence of a full-rank invariant state, $P_i \rho P_j = 0$ for any $i \neq j$ and any invariant state ρ .*

Proof. We showed in Lemma 6.2.3 that for any $i, j \in I$ and any $t \geq 0$, $\mathcal{P}_t(P_j X P_i) = P_j \mathcal{P}_t(X) P_i$. By duality, this implies that for any trace-class operator T , $\mathcal{P}_{t*}(P_j T P_i) = P_j \mathcal{P}_{t*}(T) P_i$. Hence, for $T \equiv \rho$ being an invariant state, for any $s \geq 0$:

$$\mathrm{Tr}(P_i \rho P_j X) = \mathrm{Tr}(P_i \mathcal{P}_{s*}(\rho) P_j X) = \mathrm{Tr}(\rho \mathcal{P}_s(P_j X P_i)) = \mathrm{Tr}(\rho P_j \mathcal{P}_s(X) P_i).$$

Integrating the above from 0 to $t \geq 0$, we get

$$\mathrm{Tr}(P_i \rho P_j X) = \mathrm{Tr}\left(\rho P_j \left(t^{-1} \int_0^t \mathcal{P}_s(X) ds\right) P_i\right).$$

Then, it follows from Theorem 6.1.2 that, after taking the limit $t \rightarrow 0$ in the above right hand side:

$$\mathrm{Tr}(P_i \rho P_j X) = \mathrm{Tr}(\rho P_j E_{\mathcal{F}}[X] P_i).$$

Since in this case $\mathcal{F} \subseteq \mathcal{N}$, and since $P_i, P_j \in \mathcal{Z}(\mathcal{N})$, $P_j E_{\mathcal{F}}[X] P_i = P_j P_i E_{\mathcal{F}}[X] = 0$, which implies that, for any $i \neq j$:

$$\mathrm{Tr}(P_i \rho P_j X) = 0.$$

Since this is true for any $X \in \mathcal{B}(\mathcal{H})$, the result follows. $\square$

From Lemma 6.2.4, it is enough to prove that, for any QMS $(\mathcal{P}_t)_{t \geq 0}$ on $\mathcal{B}(\mathcal{H}_0 \otimes \mathcal{K}_0)$ of associated decoherence-free algebra $\mathcal{N} = \mathcal{B}(\mathcal{H}_0) \otimes \mathbb{1}_{\mathcal{K}_0}$ possessing a full-rank invariant state σ , any other invariant state ρ can be written as

$$\rho = \mathrm{Tr}_{\mathcal{K}_0}(\rho) \otimes \tau, \tag{6.8}$$

for some given full-rank state τ on $\mathcal{K}_0$. This is done by means of the following structure theorem (see Theorem 11 of [Deschamps et al., 2016]):

Theorem 6.2.5. *In the case when $\mathcal{N}(\mathcal{P}) = \mathcal{B}(\mathcal{H}_0) \otimes \mathbb{1}_{\mathcal{K}_0}$, the semigroup $(\mathcal{P}_t)_{t \geq 0}$ can be decomposed as follows:*

$$(\mathcal{P}_t)_{t \geq 0} = (\mathcal{P}_t^{\mathcal{H}_0} \otimes \mathcal{P}_t^{\mathcal{K}_0})_{t \geq 0},$$

where $(\mathcal{P}_t^{\mathcal{H}_0})_{t \geq 0}$ is a semigroup of automorphisms of associated Hamiltonian H_W , and $(\mathcal{P}_t^{\mathcal{K}_0})_{t \geq 0}$ is primitive.

Proof. By Theorem 6.2.1, we know that L_k, L_k^* have to commute with the algebra $\mathcal{N} = \mathcal{B}(\mathcal{H}_0) \otimes \mathbb{1}_{\mathcal{K}_0}$. This implies the existence of operators M_k on $\mathcal{K}_0$, such that $L_k = \mathbb{1}_{\mathcal{H}_0} \otimes M_k$. Next, given $X \equiv X_0 \otimes \mathbb{1}_{\mathcal{K}_0} \in$

⁸In fact, the identity holds even without the assumption that $\mathcal{F}$ forms a $*$ -algebra by showing that any projection belonging to $\mathcal{F}$ commutes with the Lindblad operators of any GKLS form of the generator $\mathcal{L}$, so that $\mathcal{L}^n(P_i X P_j) = P_i \mathcal{L}^n(X) P_j$ for any $n \geq 0$, see Lemma 7 of [Deschamps et al., 2016].

$\mathcal{N}$, since $\mathcal{P}_t(X) = e^{itH}(X_0 \otimes \mathbb{1}_{\mathcal{K}_0})e^{-itH} \in \mathcal{N}$, it must be of the form $(W_t \otimes \mathbb{1}_{\mathcal{K}_0})(X_0 \otimes \mathbb{1}_{\mathcal{K}_0})(W_t \otimes \mathbb{1}_{\mathcal{K}_0})^*$ for some one parameter group $(W_t)_{t \geq 0}$ of unitary operators on $\mathcal{H}_0$. Therefore,

$$(W_t \otimes \mathbb{1}_{\mathcal{K}_0})^* e^{itH}(X_0 \otimes \mathbb{1}_{\mathcal{K}_0}) = (X_0 \otimes \mathbb{1}_{\mathcal{K}_0})(W_t \otimes \mathbb{1}_{\mathcal{K}_0})^* e^{itH}.$$

From the above identity, $(W_t \otimes \mathbb{1}_{\mathcal{K}_0})^* e^{itH}$ commutes with $\mathcal{B}(\mathcal{H}_0) \otimes \mathbb{1}_{\mathcal{K}_0}$, and therefore needs to be of the form $\mathbb{1}_{\mathcal{H}_0} \otimes R_t$ for each $t \geq 0$. Moreover, since $(e^{itH})_{t \geq 0} = (W_t \otimes R_t)_{t \geq 0}$ is a group of unitaries, so must be $(R_t)_{t \geq 0}$. Differentiating, we end up with $H = H_W \otimes \mathbb{1}_{\mathcal{K}_0} + \mathbb{1}_{\mathcal{H}_0} \otimes H_R$, where H_W , resp. H_R , denotes the generator of $(W_t)_{t \geq 0}$, resp. $(R_t)_{t \geq 0}$. Therefore, we showed that

$$\mathcal{L} = \mathcal{L}_{\mathcal{H}_0} \otimes \text{id}_{\mathcal{K}_0} + \text{id}_{\mathcal{H}_0} \otimes \mathcal{L}_{\mathcal{K}_0},$$

where for any $X \in \mathcal{B}(\mathcal{H}_0)$, $Y \in \mathcal{B}(\mathcal{K}_0)$:

$$\begin{aligned} \mathcal{L}_{\mathcal{H}_0}(X) &:= i[H_W, X], \\ \mathcal{L}_{\mathcal{K}_0}(Y) &:= i[H_R, Y] - \frac{1}{2} \sum_k M_k^* M_k Y + Y M_k^* M_k - 2M_k^* Y M_k. \end{aligned}$$

Hence, $(\mathcal{P}_t)_{t \geq 0} = (\mathcal{P}_t^{\mathcal{H}_0} \otimes \mathcal{P}_t^{\mathcal{K}_0})_{t \geq 0}$, where $(\mathcal{P}_t^{\mathcal{H}_0})_{t \geq 0}$, resp. $(\mathcal{P}_t^{\mathcal{K}_0})_{t \geq 0}$, is the semigroup corresponding to $\mathcal{L}_{\mathcal{H}_0}$, resp. $\mathcal{L}_{\mathcal{K}_0}$. Since for each $t \geq 0$, $\mathcal{P}_t^{\mathcal{H}_0}$ is an automorphism on $\mathcal{B}(\mathcal{H}_0)$, its decoherence free algebra $\mathcal{N}(\mathcal{P}^{\mathcal{H}_0})$ is $\mathcal{B}(\mathcal{H}_0)$. On the other hand, the decoherence free algebra $\mathcal{N}(\mathcal{P}^{\mathcal{K}_0})$ is equal to the commutant of $\{\nabla_{H_R}^k(M_j), \nabla_{H_R}^k(M_j^*); k \in \mathbb{N}, j \in I\}$ by Theorem 6.2.1. This in particular implies that for any $X \in \mathcal{N}(\mathcal{P}^{\mathcal{K}_0})$, $\mathbb{1}_{\mathcal{H}_0} \otimes X$ belongs to the commutant of $\{\nabla_H^k(L_j), \nabla_H^k(L_j^*); k \in \mathbb{N}, j \in I\}$, that is $\mathcal{N}(\mathcal{P}) = \mathcal{B}(\mathcal{H}) \otimes \mathbb{1}_{\mathcal{K}_0}$. Hence, $\mathcal{N}(\mathcal{P}^{\mathcal{K}_0}) = \mathbb{C} \mathbb{1}_{\mathcal{K}_0}$ and $(\mathcal{P}_t^{\mathcal{K}_0})_{t \geq 0}$ is primitive $\square$

Proof of Theorem 6.2.2 : It remains to prove Equation (6.8). The proof, that can be found in Theorem 20 of [Deschamps et al., 2016], is restated here for sake of completeness. Let $\{e_j\}_{j \geq 1}$ be a basis of eigenvectors of the Hamiltonian H_W (cf. Theorem 6.2.5) so that $H_W e_j = \kappa_j e_j$ for some eigenvalues $\kappa_j \in \mathbb{R}$. Now, let ρ be an invariant state that we write as follows,

$$\rho = \sum_{jk} |e_j\rangle\langle e_k| \otimes \rho_{jk}$$

where $\rho_{jk} = \text{Tr}_{\mathcal{H}_0}(|e_j\rangle\langle e_k| \otimes \mathbb{1}_{\mathcal{K}_0})\rho$. Then, since ρ is invariant:

$$\rho = \mathcal{P}_{t*}(\rho) = \sum_{kj} e^{i(\kappa_k - \kappa_j)t} |e_j\rangle\langle e_k| \otimes \mathcal{P}_{t*}^{\mathcal{K}_0}(\rho_{jk}).$$

Since the operators $|e_j\rangle\langle e_k|$ are linearly independent, $e^{i(\kappa_j - \kappa_k)t} \rho_{jk} = \mathcal{P}_{t*}^{\mathcal{K}_0}(\rho_{jk})$ for all j, k . Since $(\mathcal{P}_t^{\mathcal{K}_0})_{t \geq 0}$ is primitive, with unique full-rank invariant state τ , $\mathcal{P}_{t*}^{\mathcal{K}_0}(\rho_{jk}) \rightarrow \text{Tr}(\rho_{jk})\tau$. Therefore, if $\kappa_j \neq \kappa_k$, we have $\text{Tr}(\rho_{jk}) = 0$, while if $\kappa_j = \kappa_k$, $\text{Tr}(\rho_{jk})\tau = \rho_{jk}$. It follows that

$$\rho = \sum_{jk} (\text{Tr}(\rho_{jk}) |e_j\rangle\langle e_k|) \otimes \tau = \text{Tr}_{\mathcal{K}_0}(\rho) \otimes \tau.$$

$\square$

Since $\text{Tr}(\rho_{jk}) = 0$ if $\kappa_j \neq \kappa_k$, one can easily derive the following corollary:

Corollary 6.2.6. *The Hamiltonian H_W associated to the asymptotic evolution $(\mathcal{P}_t|_{\mathcal{N}})_{t \geq 0}$ admits the decomposition $U H_W U^* = \sum_i H_i \otimes \mathbb{1}_{\mathcal{K}_i}$, where for each $i \in I$, $H_i \in \mathcal{B}_{sa}(\mathcal{H}_i)$. Moreover given any invariant state ρ of decomposition $U \rho U^* = \sum_i \text{Tr}_{\mathcal{K}_i}(P_i \rho P_i) \otimes \tau_i$, $[H_i, \text{Tr}_{\mathcal{K}_i}(P_i \rho P_i)] = 0$.*

We are now ready to prove that EID occurs under the unique assumption of existence of a full-rank invariant state σ . To do so, we construct a conditional expectation $E_{\mathcal{N}}$ onto $\mathcal{N}$ as the orthogonal projection onto $\mathcal{N}$ under the GNS inner product associated to the (and in fact, any) invariant state σ . Theorem 6.1.3, for which we provide a proof in our special finite dimensional case below, allows us to conclude.

Construction and properties of the conditional expectation: Define $E_{\mathcal{N}} : \mathcal{B}(\mathcal{H}) \rightarrow \mathcal{N}$ as follows:

$$E_{\mathcal{N}}[X] := \sum_{i \in I} \text{Tr}_{\mathcal{K}_i}((\mathbb{1}_{\mathcal{H}_i} \otimes \tau_i) P_i X P_i) \otimes \mathbb{1}_{\mathcal{K}_i}, \quad (6.9)$$

One can simply verify from the definition of $E_{\mathcal{N}}$, Lemma 6.2.3 and Theorem 6.2.5 that $E_{\mathcal{N}}$ commutes with $\mathcal{P}_t$ for any $t \geq 0$. Moreover, by simple duality, we also get the following expression for $E_{\mathcal{N}*}$:

$$\rho_{\mathcal{N}} \equiv E_{\mathcal{N}*}(\rho) = \sum_{i \in I} \text{Tr}_{\mathcal{K}_i}(P_i \rho P_i) \otimes \tau_i. \quad (6.10)$$

We show that this superoperator is a valid conditional expectation as well as the orthogonal projection onto $\mathcal{N}$ with respect to $\langle \cdot, \cdot \rangle_{1, \rho}$. Indeed,

- For any invariant state ρ , $E_{\mathcal{N}*}(\rho) = \rho$ (simply compare Equation (6.10) with Equation (6.7));
- For any $X \in \mathcal{N}$, $E_{\mathcal{N}}[X] = X$;
- For any $X \in \mathcal{B}(\mathcal{H})$,

$$\begin{aligned} \|E_{\mathcal{N}}[X]\|_{\infty} &= \sum_{i \in I} \|\text{Tr}_{\mathcal{K}_i}((\mathbb{1}_{\mathcal{H}_i} \otimes \tau_i) P_i X P_i) \otimes \mathbb{1}_{\mathcal{K}_i}\|_{\infty} \\ &= \sum_{i \in I} \|\text{Tr}_{\mathcal{K}_i}((\mathbb{1}_{\mathcal{H}_i} \otimes \tau_i) P_i X P_i)\|_{\infty} \\ &\leq \sum_{i \in I} \|P_i X P_i\|_{\infty} \\ &= \|X\|_{\infty}. \end{aligned}$$

By Theorem 0.1.4, $E_{\mathcal{N}}$ is the unique conditional expectation with respect to any full-rank invariant state ρ , and the modular automorphism group associated to any such state leaves the algebra $\mathcal{N}$ invariant (we remark that this could directly be verified from the structure (6.7) of these states). In other words:

$$\Delta_{\rho}(\mathcal{N}) = \rho \mathcal{N} \rho^{-1} = \mathcal{N}. \quad (6.11)$$

As claimed above, the conditional expectation $E_{\mathcal{N}}$ is the orthogonal projection onto $\mathcal{N}$ with respect to $\langle \cdot, \cdot \rangle_{1, \rho}$, for any full-rank invariant state ρ . The only property left to be proved is the self-adjointness of $E_{\mathcal{N}}$ with respect to this inner product: for any $X, Y \in \mathcal{B}(\mathcal{H})$,

$$\begin{aligned} \text{Tr}(\rho E_{\mathcal{N}}[X] Y) &= \sum_{i \in I} \text{Tr}(\rho [\text{Tr}_{\mathcal{K}_i}((\mathbb{1}_{\mathcal{H}_i} \otimes \tau_i) P_i X P_i) \otimes \mathbb{1}_{\mathcal{K}_i}] Y) \\ &= \sum_{i \in I} \text{Tr}(\text{Tr}_{\mathcal{K}_i}(\rho) \text{Tr}_{\mathcal{K}_i}((\mathbb{1}_{\mathcal{H}_i} \otimes \tau_i) P_i X P_i) \text{Tr}_{\mathcal{K}_i}((\mathbb{1}_{\mathcal{H}_i} \otimes \tau_i) P_i Y P_i)) \\ &= \sum_{i \in I} \text{Tr}((\text{Tr}_{\mathcal{K}_i}(\rho) \otimes \tau_i) (\mathbb{1}_{\mathcal{H}_i} \otimes \tau_i) P_i X P_i [\text{Tr}_{\mathcal{K}_i}((\mathbb{1}_{\mathcal{H}_i} \otimes \tau_i) P_i Y P_i) \otimes \mathbb{1}_{\mathcal{K}_i}]) \\ &= \text{Tr}(\rho X E_{\mathcal{N}}[Y]). \end{aligned}$$

From Lemma 5.2.5, we directly get for free that $E_{\mathcal{N}}$ is also self-adjoint with respect to the KMS inner product $\langle \cdot, \cdot \rangle_{\rho}$. Equivalently:

$$\Gamma_{\rho} \circ E_{\mathcal{N}} = E_{\mathcal{N}*} \circ \Gamma_{\rho}. \quad (6.12)$$

The following theorem is then a direct consequence of the existence of $E_{\mathcal{N}}$ together with Theorem 6.1.3:

Theorem 6.2.7. *When $\mathcal{M} = \mathcal{B}(\mathcal{H})$, $\mathcal{H}$ finite dimensional, and under the condition of existence of a full-rank invariant state, EID occurs with $\mathcal{N}_0 = \mathcal{N}$ and $\mathcal{V}_0 = \mathcal{V} = \mathcal{N}^{\perp, \rho}$.*

Proof of Theorem 6.1.3 for $\mathcal{M} = \mathcal{B}(\mathcal{H})$, $\mathcal{H}$ finite dimensional : Given a full-rank invariant state ρ , decompose $\mathcal{B}(\mathcal{H})$ into $\mathcal{N} \oplus \mathcal{N}^{\perp, \rho} = \text{im}(E_{\mathcal{N}}) \oplus \text{ker}(E_{\mathcal{N}})$. Now, it is obvious that $\mathcal{N} \subset \mathcal{N}_0$, and equality holds since $\mathcal{N}$ is the largest subalgebra of $\mathcal{B}(\mathcal{H})$ on which $\mathcal{P}_t$ acts as a $*$ -homomorphism for any $t \geq 0$, and since $\mathcal{P}_t|_{\mathcal{N}}$ is an automorphism. Moreover, $\mathcal{V} \cap \mathcal{N} = \{0\}$ and $\mathcal{V} \subseteq \mathcal{N}^{\perp, \rho}$. Indeed, for any $X \in \mathcal{N}$ and $Y \in \mathcal{V}$, by Equation (6.1):

$$\text{Tr}(\rho XY) = \text{Tr}(\rho \mathcal{P}_t(XY)) = \text{Tr}(\rho \mathcal{P}_t(X) \mathcal{P}_t(Y)) \xrightarrow{t \rightarrow \infty} 0.$$

As for the reverse inclusion $\mathcal{N}^{\perp, \rho} \subset \mathcal{V}$, let $X \in \mathcal{N}^{\perp, \rho}$. Since $[E_{\mathcal{N}}, \mathcal{P}_t] = 0$ for any $t \geq 0$, $\mathcal{P}_t(X) \in \mathcal{N}^{\perp, \rho}$ for all time. The net $(\mathcal{P}_t(X))_{t \geq 0}$ being bounded, it admits a cluster point X_{∞} by Bolzano–Weierstrass theorem. However, any cluster point of $(\mathcal{P}_t(X))_{t \geq 0}$ belongs to $\mathcal{N}$ (see the proof of Theorem 3.1 of [Frigerio, 1978]). Hence $X_{\infty} \in \mathcal{N} \cap \mathcal{N}^{\perp, \rho} = \{0\}$. Therefore, $\mathcal{V}_0 = \mathcal{V} = \mathcal{N}^{\perp, \rho}$, and the proof follows. $\square$

An invariant state that will play an important role in Chapter 8 is

$$\sigma_{\text{Tr}} := E_{\mathcal{N}*} \left(\frac{\mathbb{1}}{d_{\mathcal{H}}} \right) = \frac{1}{d_{\mathcal{H}}} \sum_{i \in I} d_{\mathcal{K}_i} \mathbb{1}_{\mathcal{H}_i} \otimes \tau_i. \quad (6.13)$$

The relevance of σ_{Tr} comes from the fact that σ_{Tr} is *tracial* on $\mathcal{N}$, that is, for all $X \in \mathcal{N}$ and all $Y \in \mathcal{B}(\mathcal{H})$,

$$\text{Tr}(\sigma_{\text{Tr}} XY) = \text{Tr}(\sigma_{\text{Tr}} YX).$$

We close this section with a discussion of the relevance of the assumption of existence of a full-rank invariant state. In [Carbone et al., 2011], decoherence was shown to occur for any QMS when $\dim(\mathcal{H}) = 2$. However, this is not the case in general, even in finite dimensions, as it was showed by a counterexample in [Carbone et al., 2013] even when there exists a (non full-rank) invariant state. In Theorem 6 of the same article, the authors showed some necessary and sufficient conditions: EID holds if and only if the algebra $\mathcal{N}$ contains all the eigenvectors of the generator $\mathcal{L}$ associated with eigenvalues with null real part. This is equivalent to the fact that $\mathcal{N}$ coincides with the linear space generated by these eigenvectors, which always holds true under the existence of a full-rank invariant state.

6.3. Quantum Markovian evolutions in discrete time

Similarly to the classical setting, the theory of quantum Markov chains in discrete time is well-established and shares some similarities with its continuous time counterpart described above. Here, we regroup some basic facts concerning quantum Markov chains on finite dimensional Hilbert spaces that are going to be useful in Chapter 14. For more details on these and other basic facts of quantum information theory, we refer e.g. to [Wolf, 2012].

First, we recall that given a linear map $\Phi : \mathcal{B}(\mathcal{H}) \rightarrow \mathcal{B}(\mathcal{H})$, $\mathcal{H}$ finite dimensional, its spectrum $\text{sp}(\Phi)$ coincides with the set of λ 's such that there exists $X \in \mathcal{B}(\mathcal{H})$ for which

$$\Phi(X) = \lambda X. \quad (6.14)$$

Moreover, $\text{sp}(\Phi) = \text{sp}(\Phi^*)$, where Φ^* corresponds to the dual map with respect to the Hilbert Schmidt inner product $\langle A, B \rangle_{\text{HS}} := \text{Tr}(A^* B)$. Now, like any linear operator, Φ admits a Jordan decomposition:

$$\Phi = \Phi_P + \Phi_Q, \quad \Phi_P = \sum_{k: \lambda_k \text{ peripheral}} \lambda_k P_k + N_k, \quad \Phi_Q = \sum_{k: \lambda_k \text{ not peripheral}} \lambda_k P_k + N_k, \quad (6.15)$$

where λ_k are the eigenvalues of Φ , P_k the associated (not necessarily orthogonal) eigenprojections, and $N_k^{d_k} = 0$, where $d_k := \text{Tr}(P_k)$, so that $\sum_k P_k = \mathbb{1}$. For any k , $\lambda_k P_k + N_k$ constitutes the k -th Jordan block of Φ . The linear span $\tilde{\mathcal{N}}(\Phi)$ of the peripheral points is called *phase subspace*, and we denote by $\mathcal{P}_{\tilde{\mathcal{N}}(\Phi)}$ the projection onto it. As we will see in Section 6.3.1, the phase subspace is what takes the role of the decoherence-free subalgebra for discrete time evolutions.

In particular, if Φ is hermiticity preserving, (6.14) implies that the eigenvalues of Φ either are real, or come in conjugate pairs. If, moreover, Φ is positive unital ($\Phi(\mathbb{1}) = \mathbb{1}$) or trace preserving ($\text{Tr} \Phi(A) = \text{Tr}(A)$ for all $A \in \mathcal{B}(\mathcal{H})$), $1 \in \text{sp}(\Phi)$ and all the other eigenvalues of Φ lie in the unit disc of the complex plane, and the eigenvalues lying on the peripheral spectrum are associated to one-dimensional Jordan blocks, so that $|\lambda_k| = 1 \Rightarrow N_k = 0$.

6.3.1. Decoherence for discrete time quantum Markov chains

Given a general quantum channel $\Phi : \mathcal{B}(\mathcal{H}) \rightarrow \mathcal{B}(\mathcal{H})$, its phase subspace introduced in Section 6.3 is known to possess the following structure (Theorem 6.16 of [Wolf, 2012], Theorem 8 of [Wolf and Perez-Garcia, 2010]): there exists a decomposition of $\mathcal{H}$ as $\mathcal{H} = \bigoplus_{j \in \mathcal{J}} \mathcal{H}_j \otimes \mathcal{K}_j \oplus \mathcal{K}_0$ such that

$$\tilde{\mathcal{N}}(\Phi) := \bigoplus_{i \in \mathcal{J}} \mathcal{B}(\mathcal{H}_i) \otimes \tau_i \oplus 0_{\mathcal{K}_0}, \quad \mathcal{P}_{\tilde{\mathcal{N}}(\Phi)}(\rho \oplus 0_{\mathcal{K}_0}) = \sum_{i \in \mathcal{J}} \text{Tr}_{\mathcal{K}_i}(p_i \rho p_i) \otimes \tau_i, \quad (6.16)$$

where p_i is the orthogonal projector onto the i -th subspace, for some fixed full-rank states $\tau_i \in \mathcal{D}(\mathcal{K}_i)$. Assuming that $\mathcal{K}_0 = \{0\}$, the range of the projector $\mathcal{P}_{\tilde{\mathcal{N}}(\Phi)}^*$, which is called the *decoherence-free subalgebra* of $\{\Phi^n\}_{n \in \mathbb{N}}$ and denoted by $\mathcal{N}(\Phi^*)$, has the form

$$\mathcal{N}(\Phi^*) := \bigoplus_{i \in \mathcal{J}} \mathcal{B}(\mathcal{H}_i) \otimes \mathbb{1}_{\mathcal{K}_i}. \quad (6.17)$$

This occurs when Φ is *faithful*, which means that it possesses a full-rank invariant state. For any such quantum Markov chain, the following convergence result is known: for any $\rho \in \mathcal{D}(\mathcal{H})$,

$$\Phi^n(\rho) - \Phi^n \circ \mathcal{P}_{\tilde{\mathcal{N}}(\Phi)}(\rho) \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

This is identical to the continuous time case. Here, $\mathcal{P}_{\tilde{\mathcal{N}}(\Phi)}$ plays the role of the (dual) conditional expectation $E_{\mathcal{N}^*}$ of Section 6.2. Coming back to the evolution of states, the map $\mathcal{P}_{\tilde{\mathcal{N}}(\Phi)}$ projecting states onto the peripheral subspace of Φ is such that $\sigma_{\text{Tr}} := d_{\mathcal{H}}^{-1} \mathcal{P}_{\tilde{\mathcal{N}}(\Phi)}(\mathbb{1})$ commutes with $\mathcal{N}(\Phi^*)$. The map $\mathcal{P}_{\tilde{\mathcal{N}}(\Phi)}^*$ is a conditional expectation and hence satisfies the following properties:

Lemma 6.3.1.

- (i) $\mathcal{P}_{\tilde{\mathcal{N}}(\Phi)}^*$ is adjoint preserving: for any $X \in \mathcal{B}(\mathcal{H})$, $\mathcal{P}_{\tilde{\mathcal{N}}(\Phi)}^*(X^*) = \mathcal{P}_{\tilde{\mathcal{N}}(\Phi)}^*(X)^*$;
- (ii) for any $X \in \mathcal{B}(\mathcal{H})$ and $Y, Z \in \mathcal{N}(\Phi^*)$, $\mathcal{P}_{\tilde{\mathcal{N}}(\Phi)}^*(Y X Z) = Y \mathcal{P}_{\tilde{\mathcal{N}}(\Phi)}^*(X) Z$;

- (iii) for any $X \in \mathcal{B}(\mathcal{H})$, $\text{Tr}(\sigma_{\text{Tr}} X) = \text{Tr}(\sigma_{\text{Tr}} \mathcal{P}_{\tilde{\mathcal{N}}(\Phi)}^*(X))$;
 (iv) $\mathcal{P}_{\tilde{\mathcal{N}}(\Phi)}^*$ is self-adjoint with respect to σ_{Tr} : for any $X, Y \in \mathcal{B}(\mathcal{H})$.

$$\langle X, \mathcal{P}_{\tilde{\mathcal{N}}(\Phi)}^*(Y) \rangle_{\sigma_{\text{Tr}}} = \langle \mathcal{P}_{\tilde{\mathcal{N}}(\Phi)}^*(X), Y \rangle_{\sigma_{\text{Tr}}} = \langle \mathcal{P}_{\tilde{\mathcal{N}}(\Phi)}^*(X), \mathcal{P}_{\tilde{\mathcal{N}}(\Phi)}^*(Y) \rangle_{\sigma_{\text{Tr}}}.$$

Proof.

- (i) is obvious since $\mathcal{P}_{\tilde{\mathcal{N}}(\Phi)}$ itself is adjoint-preserving (cf. Proposition 6.3 of [Wolf, 2012]).
 (ii) For $Y, Z \in \mathcal{N}(\Phi^*)$, we have the decomposition $Y = \sum_i Y_i \otimes \mathbb{1}_{\mathcal{K}_i}$, where $p_i(Y_i \otimes \mathbb{1}_{\mathcal{K}_i})p_i = Y_i \otimes \mathbb{1}_{\mathcal{K}_i}$, and similarly with Z . Then,

$$\begin{aligned} \mathcal{P}_{\tilde{\mathcal{N}}(\Phi)}^*(YXZ) &= \sum_i \text{Tr}_{\mathcal{K}_i}((\mathbb{1}_{\mathcal{H}_i} \otimes \tau_i)p_i YXZ p_i) \otimes \mathbb{1}_{\mathcal{K}_i} \\ &= \sum_i \text{Tr}_{\mathcal{K}_i}((\mathbb{1}_{\mathcal{H}_i} \otimes \tau_i)(Y_i \otimes \mathbb{1}_{\mathcal{K}_i})p_i X p_i(Z_i \otimes \mathbb{1}_{\mathcal{K}_i})) \otimes \mathbb{1}_{\mathcal{K}_i} \\ &= \sum_i [\text{Tr}_{\mathcal{K}_i}(Y_i(\mathbb{1}_{\mathcal{H}_i} \otimes \tau_i)p_i X p_i)Z_i] \otimes \mathbb{1}_{\mathcal{K}_i} \\ &= \sum_j Y_j \otimes \mathbb{1}_{\mathcal{K}_j} \left(\sum_i \text{Tr}_{\mathcal{K}_i}((\mathbb{1}_{\mathcal{H}_i} \otimes \tau_i)p_i X p_i) \otimes \mathbb{1}_{\mathcal{K}_i} \right) \sum_j Z_j \otimes \mathbb{1}_{\mathcal{K}_j} \\ &= Y \mathcal{P}_{\tilde{\mathcal{N}}(\Phi)}^*(X) Z. \end{aligned}$$

- (iii) follows from a simple computation: since Φ is trace preserving, so is $\mathcal{P}_{\tilde{\mathcal{N}}(\Phi)}$ (Proposition 6.3 of [Wolf, 2012]), and hence

$$\text{Tr}(\sigma_{\text{Tr}} \mathcal{P}_{\tilde{\mathcal{N}}(\Phi)}^*(X)) = \text{Tr}(\mathcal{P}_{\tilde{\mathcal{N}}(\Phi)} \circ \Gamma_{\sigma_{\text{Tr}}}(X)) = \text{Tr}(\Gamma_{\sigma_{\text{Tr}}}(X)) = \text{Tr}(\sigma_{\text{Tr}} X).$$

- (iv) is a consequence of (i)–(iii):

$$\begin{aligned} \langle X, \mathcal{P}_{\tilde{\mathcal{N}}(\Phi)}^*(Y) \rangle_{\sigma_{\text{Tr}}} &= \text{Tr}(\sigma_{\text{Tr}}^{1/2} X^* \sigma_{\text{Tr}}^{1/2} \mathcal{P}_{\tilde{\mathcal{N}}(\Phi)}^*(Y)) \\ &= \text{Tr}(\sigma_{\text{Tr}} X^* \mathcal{P}_{\tilde{\mathcal{N}}(\Phi)}^*(Y)) \\ &= \text{Tr}(\sigma_{\text{Tr}} \mathcal{P}_{\tilde{\mathcal{N}}(\Phi)}^*(X^* \mathcal{P}_{\tilde{\mathcal{N}}(\Phi)}^*(Y))) \\ &= \text{Tr}(\sigma_{\text{Tr}} \mathcal{P}_{\tilde{\mathcal{N}}(\Phi)}^*(X^*) \mathcal{P}_{\tilde{\mathcal{N}}(\Phi)}^*(Y)) \\ &= \text{Tr}(\sigma_{\text{Tr}} \mathcal{P}_{\tilde{\mathcal{N}}(\Phi)}^*(X)^* \mathcal{P}_{\tilde{\mathcal{N}}(\Phi)}^*(Y)) \\ &= \text{Tr}(\sigma_{\text{Tr}}^{1/2} \mathcal{P}_{\tilde{\mathcal{N}}(\Phi)}^*(X)^* \sigma_{\text{Tr}}^{1/2} \mathcal{P}_{\tilde{\mathcal{N}}(\Phi)}^*(Y)) \\ &= \langle \mathcal{P}_{\tilde{\mathcal{N}}(\Phi)}^*(X), \mathcal{P}_{\tilde{\mathcal{N}}(\Phi)}^*(Y) \rangle_{\sigma_{\text{Tr}}} \end{aligned}$$

where we used the commutativity of σ_{Tr} with $\mathcal{N}(\Phi^*)$ in the second and sixth lines, (i) in the fifth line, (ii) in the fourth line and (iii) in the third line. The first identity in (iv) follows by symmetry. $\square$

6.4. Examples

6.4.1. Simple semigroups associated to a conditional expectation

The depolarizing semigroup introduced in Section 5.5.1 corresponds to a very symmetric situation where the noise added to the quantum system is assumed to be isotropic. Assuming that $\mathcal{H} = \mathbb{C}^2$, this

is clear from the following reformulation:

$$\begin{aligned}\mathcal{P}_*^{\text{depol}}(\rho) &= (1-p)\rho + p\frac{\mathbb{1}}{2} \\ &= (1-p)\rho + \frac{p}{4} \sum_{i=0}^3 \sigma_i \rho \sigma_i,\end{aligned}$$

where $\sigma_0 = \frac{\mathbb{1}}{2}$ and

$$\sigma_1 := \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 := \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 := \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

are the Pauli noise operators in each direction of a computational basis that is assumed to be fixed beforehand. Hence,

$$\mathcal{P}_*^{\text{depol}}(\rho) = \left(1 - \frac{3p}{4}\right)\rho + \frac{p}{4} (\sigma_x \rho \sigma_x + \sigma_y \rho \sigma_y + \sigma_z \rho \sigma_z).$$

Hence, the depolarizing channel can be interpreted as follows: with probability $\frac{p}{4}$, the noise σ_i in the direction i is applied to the state uniformly at random. Otherwise, the state remains untouched.

On the contrary, the quantum system might get affected in a single direction, say along the z axis. The resulting quantum channel, known as *dephasing channel* (or *phase flip channel*), takes the following simpler form:

$$\mathcal{P}_*^{\text{deph}}(\rho) = (1-p)\rho + p\sigma_z \rho \sigma_z.$$

More generally, one considers the so-called *Pauli channel*:

$$\mathcal{P}_*^{\text{Pauli}}(\rho) = \sum_{i=0}^3 p(i) \sigma_i \rho \sigma_i$$

for a given probability mass function p on $\mathbb{Z}_4$. Assuming that the probability that an error occurs is uniform among the subset of the Pauli matrices, with associated index set I , the Pauli channel reduces to the following:

$$\mathcal{P}_*^I(\rho) = (1-p)\rho + \frac{p}{|I|} \sum_{i \in I} \sigma_i \rho \sigma_i.$$

As in the case of the depolarizing semigroup, this quantum channel has a simple semigroup analogue, when assuming that the probability $1-p$ vanishes exponentially fast: for all $X \in \mathcal{B}(\mathbb{C}^2)$:

$$\mathcal{P}_t^I(X) = e^{-t} X + \frac{1-e^{-t}}{|I|} \sum_{i \in I} \sigma_i X \sigma_i \quad \Rightarrow \quad \mathcal{L}^I(X) = \frac{1}{|I|} \sum_{i \in I} \sigma_i X \sigma_i - X.$$

The QMS $(\mathcal{P}_t^I)_{t \geq 0}$ is a perfect example of a non primitive semigroup. First of all, one easily reads its associated Lindblad operators (cf. Equation (5.5)) from the above expression of its generator: $H = 0$ and $L_j = |I|^{-\frac{1}{2}} \sigma_i$. Theorem 6.2.1 leads to

$$\mathcal{N}(\mathcal{P}^I) = \{\sigma_i, i \in I\}'.$$

In fact, the completely positive map $X \mapsto |I|^{-1} \sum_{i \in I} \sigma_i X \sigma_i$ is a conditional expectation associated to the algebra $\mathcal{N}(\mathcal{P}^I)$ with respect to the completely mixed state $\frac{\mathbb{1}}{2}$.

More generally, let $\mathcal{N}$ be a $*$ -subalgebra of $\mathcal{B}(\mathcal{H})$, $\mathcal{H}$ being finite dimensional. As we saw in Section 0.1.4, $\mathcal{N}$ can be written as

$$\mathcal{N} := \bigoplus_{i \in I} \mathcal{B}(\mathcal{H}_i) \otimes \mathbb{1}_{\mathcal{K}_i}, \quad \mathcal{H} := \bigoplus_{i \in I} \mathcal{H}_i \otimes \mathcal{K}_i.$$

Choosing σ_{Tr} as in Equation (6.13) and letting $E_{\mathcal{N}} : \mathcal{B}(\mathcal{H}) \rightarrow \mathcal{N}$ be its associated conditional expectation, the following is a valid generator of a QMS of associated decoherence free subalgebra $\mathcal{N}$:

$$\mathcal{L}^{\mathcal{N}}(X) = E_{\mathcal{N}}[X] - X.$$

We call the associated QMS $(\mathcal{P}_t^{\mathcal{N}})_{t \geq 0}$ the *simple QMS associated to $\mathcal{N}$* .

Lemma 6.4.1. *For any $*$ -subalgebra $\mathcal{N}$ of $\mathcal{B}(\mathcal{H})$, and any conditional expectation $E_{\mathcal{N}}$ with respect to the state σ_{Tr} , the simple QMS $(\mathcal{P}_t^{\mathcal{N}})_{t \geq 0}$ satisfies σ_{Tr} -DBC. Moreover,*

$$E_{\mathcal{N}} \circ \Delta_{\sigma_{\text{Tr}}} = \Delta_{\sigma_{\text{Tr}}} \circ E_{\mathcal{N}} \quad \text{and} \quad \Gamma_{\sigma_{\text{Tr}}} \circ E_{\mathcal{N}} = E_{\mathcal{N}*} \circ \Gamma_{\sigma_{\text{Tr}}}. \quad (6.18)$$

Proof. In view of Lemma 5.2.5, it suffices to prove that $\text{Tr}(\sigma_{\text{Tr}} E_{\mathcal{N}}[X]Y) = \text{Tr}(\sigma_{\text{Tr}} X E_{\mathcal{N}}[Y])$ for any $X, Y \in \mathcal{B}(\mathcal{H})$. But this follows by a direct computation using the expression (6.9) for $E_{\mathcal{N}}$. $\square$

6.4.2. Collective decoherence

Here, we describe an important type of non primitive quantum Markovian evolution that has found application in quantum error correction, since it is believed to be a good candidate for the implementation of fault tolerant universal quantum computation devices [Lidar et al., 1998]. Here, the system under consideration is constituted of n qubits, that is $\mathcal{H} = (\mathbb{C}^2)^{\otimes n}$, which experience the exact same noise. This leads to an overall invariance of the evolution under permutation of the qubits that is referred to as *collective decoherence*. In the case when the system is subject to a collective dephasing along the z -axis, the generator of the resulting *weak collective decoherence* is given by

$$\mathcal{L}_n^{\text{wcd}}(X) := \Sigma_z^n X \Sigma_z^n - \frac{1}{2}((\Sigma_z^n)^2 X + X(\Sigma_z^n)^2), \quad \text{where} \quad \Sigma_z^n := \sum_{i=1}^n \mathbb{1}_{\mathbb{C}^2}^{\otimes i-1} \otimes \sigma_z \otimes \mathbb{1}_{\mathbb{C}^2}^{n-i}.$$

More generally, if any collective Pauli noise is allowed, with associated index set $I \subset \{1, 2, 3\}$:

$$\mathcal{L}_n^{\text{cd}, I}(X) := \sum_{i \in I} \Sigma_i^n X \Sigma_i^n - \frac{1}{2}((\Sigma_i^n)^2 X + X(\Sigma_i^n)^2), \quad \text{where} \quad \Sigma_i^n := \sum_{k=1}^n \mathbb{1}_{\mathbb{C}^2}^{\otimes k-1} \otimes \sigma_i \otimes \mathbb{1}_{\mathbb{C}^2}^{n-k}.$$

The case when $I = \{1, 2, 3\}$ is commonly referred to as *strong collective decoherence*, and in this case we denote the generator by $\mathcal{L}_n^{\text{scd}}$.

Decoherence free subsystems and quantum error protection As yet another approach to decoherence, [Lidar et al., 1998] (see also [Ticozzi and Viola, 2008, Lidar and Whaley, 2003, Lidar, 2014, Knill et al., 2000, Viola et al., 2001, Kempe et al., 2001]) looked at the evolution in the Schrödinger picture. In particular, they identified parts of the Hilbert space where the evolution is unitary. These so-called *decoherence-free subsystems* have been proposed as possible candidates to encode quantum information that is protected from the environmental noise. More precisely, given a decomposition of $\mathcal{H}$ as follows:

$$\mathcal{H} = \mathcal{H}_s \otimes \mathcal{H}_f \oplus \mathcal{H}_r.$$

the space $\mathcal{H}_s$ is said to support a decoherence-free/noiseless subsystem for a given QMS $(\mathcal{P}_t)_{t \geq 0}$ if for any initial state $\rho = \rho_s \otimes \rho_f$ supported on $\mathcal{H}_s \otimes \mathcal{H}_f$, and any $t \geq 0$:

$$\mathcal{P}_t(\rho) = U_t \rho_s U_t^* \otimes \mathcal{P}_t^f(\rho_f),$$

where U_t is a unitary operator on $\mathcal{H}_s$ and $(\mathcal{P}_t^f)_{t \geq 0}$ is a QMS on $\mathcal{B}(\mathcal{H}_f)$. In the case when $\dim(\mathcal{H}_f) = 1$, $\mathcal{H}_s$ is said to support a *decoherence-free subspace*.

The following theorem, a proof of which can be found in [Deschamps et al., 2016] in the uniformly continuous case, provides an identification of decoherence-free subsystems of a QMS $(\mathcal{P}_t)_{t \geq 0}$ in terms of the structure of its decoherence-free algebra (in the case of unbounded generators, see [Agredo et al., 2014]):

Theorem 6.4.2. *Given the decomposition (6.5) of the decoherence-free algebra of a QMS $(\mathcal{P}_t)_{t \geq 0}$, each $\mathcal{H}_i$ supports a decoherence-free subsystem. Moreover, any direct sum $\bigoplus_{i \in I'} \mathcal{H}_i$, where $I' \subset I$ is such that for all $i \in I'$, $\dim(\mathcal{K}_i) = 1$, supports a decoherence-free subspace.*

Part IV.

Quantum functional inequalities

Chapter 7.

Primitive quantum Markov semigroups

In Part II, we saw that functional inequalities represent a powerful tool to estimate mixing times of classical Markov semigroups. Generalizations of these tools were initiated in [Gross, 1975a, Lindsay, 1990, Carlen and Loss, 1993, Carlen and Lieb, 1993, Biane, 1997] for specific semigroups. The notion of hypercontractivity and logarithmic Sobolev inequalities was fully extended in [Olkiewicz and Zegarlinski, 1999] to arbitrary reversible primitive semigroups on non-commutative $\mathbb{L}_p$ -spaces defined over approximately finite C^* -algebras. The exploration of the asymptotic behavior of a QMS satisfying a functional inequality has since become an area of focus in the quantum information and quantum probability communities [Carbone and Sasso, 2008, Temme et al., 2010, Montanaro, 2012, Carbone and Martinelli, 2015, Kastoryano and Temme, 2013, Temme et al., 2014, Cubitt et al., 2015, Kastoryano and Temme, 2016, Müller-Hermes and França, 2018, Delgosha and Beigi, 2014, Beigi and King, 2016, Junge et al., 2015, Carbone, 2004, Carbone and Fagnola, 2000].

Layout of the chapter: In this chapter, we survey the quantum versions of the functional inequalities of Chapter 3, namely Poincaré inequality (Section 7.1), logarithmic Sobolev inequalities and hypercontractivity (Section 7.2), Nash inequalities and ultracontractivity (Section 7.5), in the case of a primitive quantum Markov semigroup. Apart from Theorem 7.2.4 which provides the exact logarithmic Sobolev constant of order 2 for the generalized quantum depolarizing semigroup, the results discussed in Sections 7.1 to 7.3 and 7.5.1 are not new. In contrast, the notion of quantum reverse hypercontractivity for non doubly-stochastic QMS as discussed in Section 7.4 extends the one of [Cubitt et al., 2015]. We end this chapter with the statement and proof of a quantum Nash inequality for the quantum heat semigroup (cf. Section 5.5.2) in Section 7.5.2. The functional inequalities described in this chapter, and in particular Section 7.2, will be extended to the case of non-primitive QMS in Chapter 8.

7.1. Quantum Poincaré inequality

Let $\mathcal{H}$ be a separable Hilbert space, and fix a faithful normal state σ on $\mathcal{H}$. An $\mathbb{L}_2(\sigma)$ QMS $(\mathcal{P}_t)_{t \geq 0}$ satisfies a *quantum Poincaré inequality* with constant $\lambda > 0$, if for all $X \in \text{dom}(\mathcal{L}_2)$,¹

$$\lambda \text{Var}_\sigma(X) \leq \mathcal{E}_{2,\mathcal{L}}(X), \quad (\text{PI}(\lambda))$$

¹The following discussion is very general and would apply to any strongly continuous contraction semigroup $(\mathcal{P}_t)_{t \geq 0}$ on a Hilbert space $\mathcal{K}$ with an invariant vector ψ .

where $\text{Var}_\sigma(X) := \|X\|_{\mathbb{L}_2(\sigma)}^2 - \text{Tr}(\sigma X)^2 = \|X - \text{Tr}(\sigma X) \mathbb{1}\|_{\mathbb{L}_2(\sigma)}^2$. The *spectral gap* of $\mathcal{L}$ is defined by the following nonnegative number:

$$\lambda(\mathcal{L}) = \inf_{X \in \text{dom}(\mathcal{L}_2)} -\frac{\text{Re}\langle X, \mathcal{L}_2 X \rangle_\sigma}{\|X - \langle \mathbb{1}, X \rangle_\sigma \mathbb{1}\|_{\mathbb{L}_2(\sigma)}^2} \equiv \inf_{X \in \text{dom}(\mathcal{L}_2)} \frac{\mathcal{E}_{2,\mathcal{L}}(X)}{\text{Var}_\sigma(X)},$$

The next proposition, whose proof consists in a simple differentiation of (7.1), is a straightforward extension of Proposition 3.1.1. It can be found e.g. in [Carbone and Fagnola, 2000]:

Proposition 7.1.1. *Under the above conditions, the spectral gap of $\mathcal{L}_2$ is the maximum positive value λ such that the following exponential decay occurs in $\mathbb{L}_2(\sigma)$: for any $X \in \mathbb{L}_2(\sigma)$ and all $t \geq 0$,*

$$\|\mathcal{P}_t(X) - \text{Tr}(\sigma X) \mathbb{1}\|_{\mathbb{L}_2(\sigma)} \leq e^{-\lambda t} \|X - \text{Tr}(\sigma X) \mathbb{1}\|_{\mathbb{L}_2(\sigma)}. \quad (7.1)$$

In other words, the spectral gap corresponds to the maximal positive value λ such that

$$\text{Var}_\sigma(\mathcal{P}_t(X)) \leq e^{-2\lambda t} \text{Var}_\sigma(X).$$

Finding the spectral gap of quantum Markov semigroups is already a challenging problem (see [Kastoryano and Brandão, 2016]). However, they usually don't provide the tightest bounds on the mixing time of a QMS. In the next section, we review the more refined noncommutative theory of hypercontractivity.

7.2. Quantum logarithmic Sobolev inequality, hypercontractivity

As mentioned in the preamble of the chapter, the theory of hypercontractivity for primitive QMS was fully formalized in [Olkiewicz and Zegarlinski, 1999], using Kosaki's theory of non-commutative interpolating weighted $\mathbb{L}_p$ spaces [Kosaki, 1984, Majewski and Zegarlinski, 1996], where the weights here are given in terms of a faithful invariant state of the evolution. This study was further pursued by different authors [Temme et al., 2014, Carbone and Martinelli, 2015] and applied to the problem of estimating mixing times in [Kastoryano and Temme, 2013]. We recall that, given a subalgebra $\mathcal{A}_0$ satisfying Condition 5.2.2, $\mathcal{A}_0^+$ denotes the subset $\{X + c \mathbb{1}, X \in \mathcal{A}_0, X \geq 0, c > 0\}$ of positive definite operators whose spectrum is uniformly bounded away from 0. Then, for $p \in \mathbb{R} \setminus \{0\}$, define the *entropy function of order p* as follows: for all $X \in \mathcal{A}_0^+$,

$$\text{Ent}_{p,\sigma}(X) := \text{Tr} \left(\Gamma_{\sigma}^{\frac{1}{p}}(X) \right)^p \ln \left(\Gamma_{\sigma}^{\frac{1}{p}}(X) \right)^p - \text{Tr} \left(\Gamma_{\sigma}^{\frac{1}{p}}(X) \right)^p \ln \sigma - \|X\|_{\mathbb{L}_p(\sigma)}^p \ln \|X\|_{\mathbb{L}_p(\sigma)}, \quad (7.2)$$

where the family $\|\cdot\|_p$ can be naturally extended on $\mathcal{A}_0^+$ for $p \leq 1$ (see Section 7.4). Similarly, extending the definition of the maps $I_{q,p}$ on $\mathcal{A}_0^+$ to any $q, p \in \mathbb{R} \setminus \{0\}$, the following well-known properties of the entropy function are easy to verify.

Lemma 7.2.1. *Let $X \in \mathcal{A}_0^+$. Then, for any $p, q \in \mathbb{R} \setminus \{0\}$:*

- (i) $\text{Ent}_{p,\sigma}(I_{p,2}(X)) = \text{Ent}_{q,\sigma}(I_{q,2}(X))$;
- (ii) $\text{Ent}_{p,\sigma}(cX) = c^p \text{Ent}_{p,\sigma}(X)$ for any $c > 0$;
- (iii) $\text{Ent}_{p,\sigma}(X) = D(\rho \parallel \sigma) - \text{Tr} \rho \ln \text{Tr} \rho$, where $\rho = \left(\Gamma_{\sigma}^{\frac{1}{p}}(X) \right)^p$, and $D(\rho \parallel \sigma) = \text{Tr}(\rho \ln \rho) - \text{Tr}(\rho \ln \sigma)$ is Umegaki's relative entropy.

We are now ready to define quantum logarithmic Sobolev inequalities:

Definition 7.2.2. Given $p > 1$, $(\mathcal{P}_t)_{t \geq 0}$ is said to satisfy a *quantum logarithmic Sobolev inequality* of order p if for any $X \in \mathcal{A}_0^+$:

$$\text{Ent}_{p,\sigma}(X) \leq c \mathcal{E}_{p,\mathcal{L}}(X) + 2d \|X\|_{\mathbb{L}_p(\sigma)}^p. \quad (\text{qLSI}_p(c, d))$$

In the case when $d \neq 0$, the logarithmic Sobolev inequality is said to be *defective*. In fact, one can always reduce to the case $d = 0$ in the primitive, finite dimensional setting². This legitimates the definition of the *p-logarithmic Sobolev constant* as

$$\alpha_p(\mathcal{L}) := \inf_{X \in \mathcal{A}_0^+} \frac{\mathcal{E}_{p,\mathcal{L}}(X)}{\text{Ent}_{p,\sigma}(X)}. \quad (7.3)$$

The $\mathbb{L}_2(\sigma)$ -QMS $(\mathcal{P}_t)_{t \geq 0}$ is hypercontractive if for all $p \leq q \leq 1 + (p-1)e^{\frac{4t}{c}}$, the following holds:

$$\|\mathcal{P}_t : \mathbb{L}_p(\sigma) \rightarrow \mathbb{L}_q(\sigma)\| \leq \exp \left\{ 2d \left(\frac{1}{p} - \frac{1}{q} \right) \right\}. \quad (\text{HC}_p(c, d))$$

Quantum logarithmic Sobolev inequalities reduce to $\text{LSI}_p(c, d)$ in the commutative case. In fact, $\text{qLSI}_p(c, d)$ is also well-defined for $p \in \mathbb{R} \setminus \{0, 1\}$. The range $p \in (-\infty, 0) \cup (0, 1)$ will be studied in Section 7.4, whereas the limiting case $p = 1$ gives the modified logarithmic Sobolev inequality introduced in Section 7.3.

The following theorem was first proved by [Olkiewicz and Zegarlinski, 1999]:

Theorem 7.2.3. Let $(\mathcal{P}_t)_{t \geq 0}$ be an $\mathbb{L}_2(\sigma)$ -quantum Markov semigroup, $p > 1$, $c > 0$ and $d \geq 0$. Then

- (i) If $\text{HC}_p(c, d)$ holds, then $\text{qLSI}_p(c, d)$ holds.
- (ii) If $\text{qLSI}_q(c, d)$ holds for all $q \geq p$, then $\text{HC}_p(c, d)$ holds.

Moreover, if $(\mathcal{P}_t)_{t \geq 0}$ satisfies ω -DBC with respect to σ , then $\text{HC}_p(c, d)$ holds for any $p > 1$ with $d = 0$ and $c = \alpha_2(\mathcal{L})^{-1}$.

Proof. Items (i) and (ii) were proved in [Olkiewicz and Zegarlinski, 1999]. The equivalence under the condition of ω -DBC is a easy consequence of the quantum Stroock-Varopoulos inequality as well as part (i) of Proposition 7.2.1, from which we get that $p \mapsto \alpha_p(\mathcal{L})$ is non-increasing on $[1, 2]$ and that $\alpha_p(\mathcal{L}) = \alpha_{\bar{p}}(\mathcal{L})$ (see Proposition 7.4.6). $\square$

Example: the generalized depolarizing semigroup We now give the exact expression of the 2-logarithmic Sobolev constant of the generalized depolarizing semigroup introduced in Section 5.5.1. To prove Theorem 7.2.4 we need to show that a certain function of qubit density matrices is optimized over diagonal ones. Once we show this, the explicit expression for the 2-logarithmic Sobolev constant is obtained from the associated classical constant derived in [Diaconis and Saloff-Coste, 1996a].

Theorem 7.2.4. Let $\sigma \in \mathcal{D}_+(\mathcal{H})$ be arbitrary and let $\mathcal{L}_\sigma(X) = \text{Tr}(\sigma X) \mathbb{1} - X$. Then we have

$$\alpha_2(\mathcal{L}_\sigma) = \frac{1 - 2\|\sigma^{-1}\|_\infty^{-1}}{\ln(\|\sigma^{-1}\|_\infty - 1)}. \quad (7.4)$$

Proof. Since both $\text{Ent}_{2,\sigma}(X)$ and $\mathcal{E}_{2,\mathcal{L}_\sigma}(X)$ are homogenous of degree two, to prove a log-Sobolev inequality, without loss of generality we can assume that X is of the form $X = \Gamma_\sigma^{-1/2}(\sqrt{\rho})$ where ρ is a

²This is in sharp contrast with the non primitive case that we introduce in Chapter 8.

density matrix. In this case

$$\text{Ent}_{2,\sigma}(X) = D(\rho\|\sigma), \quad -\langle X, \mathcal{L}_\sigma X \rangle_\sigma = 1 - [\text{Tr}(\sqrt{\sigma}\sqrt{\rho})]^2.$$

Let $\sigma = \sum_{i=1}^d s_i |i\rangle\langle i|$ and $\rho = \sum_{k=1}^d r_k |\tilde{k}\rangle\langle \tilde{k}|$ be the eigen-decompositions of σ and ρ . Then

$$\text{Ent}_{2,\sigma}(X) = \sum_{k=1}^d r_k \ln r_k - \sum_{i,k=1}^d |\langle i|\tilde{k}\rangle|^2 r_k \ln s_i,$$

and

$$-\langle X, \mathcal{L}_\sigma X \rangle_\sigma = 1 - \left(\sum_{i,k=1}^d |\langle i|\tilde{k}\rangle|^2 \sqrt{s_i r_k} \right)^2.$$

Let $A = (a_{ik})_{d \times d}$ be a $d \times d$ matrix whose entries are given by

$$a_{ik} = |\langle i|\tilde{k}\rangle|^2.$$

Observe that, fixing the eigenvalues s_i 's and r_k 's, the entropy $\text{Ent}_{2,\sigma}(X)$ is a linear function of A and $\mathcal{E}_{2,\mathcal{L}_\sigma}(X)$ is a concave function of A . On the other hand, since both $\{|1\rangle, \dots, |d\rangle\}$ and $\{|\tilde{1}\rangle, \dots, |\tilde{d}\rangle\}$ form orthonormal bases, A is a doubly stochastic matrix. Then by Birkhoff's theorem, A can be written as a convex combination of permutations matrices. We conclude that if an inequality of the form

$$\alpha \left(\sum_{k=1}^d r_k \ln r_k - \sum_{i,k=1}^d a_{ik} r_k \ln s_i \right) \leq 1 - \left(\sum_{i,k=1}^d a_{ik} \sqrt{s_i r_k} \right)^2,$$

holds for all permutation matrices A , then it holds for all doubly stochastic A , and then for all σ, ρ with the given eigenvalues. We note that A is a permutation matrix when $\{|1\rangle, \dots, |d\rangle\}$ and $\{|\tilde{1}\rangle, \dots, |\tilde{d}\rangle\}$ are the same bases (under some permutation) which means that σ and ρ commute. Therefore, a logarithmic Sobolev inequality of the form

$$\alpha \text{Ent}_{2,\sigma}(\Gamma_\sigma^{-1/2}(\rho)) \leq \mathcal{E}_{2,\mathcal{L}_\sigma}(\Gamma_\sigma^{-1/2}(\rho))$$

holds for all ρ if and only if it holds for all ρ that commute with σ . That is, to find the log-Sobolev constant

$$\alpha_2(\mathcal{L}_\sigma) = \inf_{\rho} \frac{\mathcal{E}_{2,\mathcal{L}_\sigma}(\Gamma_\sigma^{-1/2}(\rho))}{\text{Ent}_{2,\sigma}(\Gamma_\sigma^{-1/2}(\rho))},$$

we may restrict to those ρ that commute with σ . This optimization problem over such ρ is equivalent to computing the 2-log-Sobolev constant of the *classical* depolarizing Lindblad generator, and has been solved in Theorem A.1 of [Diaconis and Saloff-Coste, 1996a]. $\square$

Remark 7.2.5. The above result should be compared with the expression for the MLSI constant derived in [Müller-Hermes et al., 2016] (see Section 7.3):

$$\alpha_1(\mathcal{L}_\sigma) = \min_{x \in [0,1]} \frac{1}{4} \left\{ 1 + \frac{D_{\text{bin}}(\|\sigma^{-1}\|_\infty^{-1} \|x\|)}{D_{\text{bin}}(x \| \|\sigma^{-1}\|_\infty^{-1})} \right\} \geq \frac{1}{4}, \quad (7.5)$$

where $D_{\text{bin}}(x\|y) := x \ln \frac{x}{y} + (1-x) \ln \frac{1-x}{1-y}$ denotes the binary relative entropy between the distributions $\{x, 1-x\}$ and $\{y, 1-y\}$.

7.3. Quantum modified logarithmic Sobolev inequality

Here, we restrict ourselves to a finite dimensional Hilbert space $\mathcal{H}$.

Theorem 7.3.1. *Let $(\mathcal{P}_t)_{t \geq 0}$ be a primitive QMS on $\mathcal{B}(\mathcal{H})$ with invariant state σ . Then, for any $\rho \in \mathcal{D}(\mathcal{H})$, the following holds:*

$$\frac{d}{dt} D(\mathcal{P}_{*t}(\rho) \| \sigma) = \text{Tr}(\mathcal{L}_*(\rho_t)(\ln \rho_t - \ln \sigma)) .$$

The result follows by a simple application of the integral representation for the logarithm (see [Kastoryano and Temme, 2013]).

Theorem 7.3.1 provides a justification for the following definition:

Definition 7.3.2. For any $\rho = \Gamma_\sigma(X)$, $X > 0$, the *quantum entropy production* of ρ along the semigroup $(\mathcal{P}_t)_{t \geq 0}$ is defined as:

$$\text{EP}_\sigma(\rho) = - \left. \frac{d}{dt} \right|_{t=0} D(\rho_t \| \sigma) = - \text{Tr}(\mathcal{L}_*(\rho)(\ln \rho - \ln \sigma)) .$$

Then, the QMS $(\mathcal{P}_t)_{t \geq 0}$ is said to satisfy a *quantum modified log-Sobolev inequality* with constant $\alpha_1 > 0$ if for all $\rho \in \mathcal{D}_+(\mathcal{H})$,

$$4\alpha_1 D(\rho \| \sigma) \leq \text{EP}_\sigma(\rho) . \quad (\text{MLSI}(\alpha_1))$$

which implies the following convergence:

$$\|\rho_t - \sigma\|_1 \leq \sqrt{2 \ln \|\sigma^{-1}\|_\infty} e^{-2\alpha_1(\mathcal{L})t} . \quad (7.6)$$

7.4. Quantum reverse hypercontractivity

Until now, we restricted our analysis to p -logarithmic Sobolev inequalities and hypercontractivity in the range $p > 1$. It turns out that, for $p < 1$, the contractivity properties of the maps $\mathcal{P}_t$ are flipped. This was first observed in the classical literature in [Borell, 1982, Mossel et al., 2006, Mossel et al., 2013], and extended to the quantum case for finite dimensional primitive semigroups whose invariant state is completely mixed in [Cubitt et al., 2015]. In this section we develop the theory of *quantum reverse hypercontractivity* beyond this so-called doubly stochastic case. The analysis carried out here is analogous to Section 7.2.

First, given a normal faithful state on $\mathcal{B}(\mathcal{H})$, $\mathcal{H}$ finite dimensional, we extend the definition of $\mathbb{L}_p(\sigma)$ norms to the range $p < 1$, $p \neq 0$: given $X \in \mathcal{B}(\mathcal{H})^+ := \{Y \in \mathcal{B}(\mathcal{H}), Y = Z + c \mathbb{1}, Z \geq 0, c > 0\}$:

$$\|X\|_{\mathbb{L}_p(\sigma)} := \text{Tr} \left(|\Gamma_\sigma^{\frac{1}{p}}(X)|^p \right)^{\frac{1}{p}} .$$

Similarly to the classical case, $\|\cdot\|_{\mathbb{L}_p(\sigma)}$ is simply a pseudo-norm for $p < 1$.

Lemma 7.4.1 (Reverse Hölder inequality). *Let $X, Y \in \mathcal{B}(\mathcal{H})^+$. Then, for any $0 < p < 1$ of Hölder conjugate $\hat{p}$, we have*

$$\langle X, Y \rangle_\sigma \geq \|X\|_{\mathbb{L}_p(\sigma)} \|Y\|_{\mathbb{L}_{\hat{p}}(\sigma)} .$$

Proof. The proof is a direct generalization of equation (32) of [Tomamichel et al., 2014]: for any two positive, definite operators $A, B \in \mathcal{B}(\mathcal{H})$:

$$\mathrm{Tr}(AB) \geq \|B\|_{\hat{p}} \|A\|_p. \quad (7.7)$$

From there, choosing $A := \Gamma_{\sigma}^{\frac{1}{p}}(X)$ and $B := \Gamma_{\sigma}^{\frac{1}{\hat{p}}}(Y)$,

$$\langle X, Y \rangle_{\sigma} := \mathrm{Tr} \left(\sigma^{\frac{1}{2p}} X \sigma^{\frac{1}{2p}} \sigma^{\frac{1}{2\hat{p}}} Y \sigma^{\frac{1}{2\hat{p}}} \right) = \mathrm{Tr}(AB) \geq \|A\|_p \|B\|_{\hat{p}} = \|X\|_{\mathbb{L}_p(\sigma)} \|Y\|_{\mathbb{L}_{\hat{p}}(\sigma)}.$$

□

Lemma 7.4.2 (Reverse Minkowski inequality). *For $p < 1$, $p \neq 0$, and $X, Y \in \mathcal{B}(\mathcal{H})^+$,*

$$\|X\|_{\mathbb{L}_p(\sigma)} + \|Y\|_{\mathbb{L}_p(\sigma)} \leq \|X + Y\|_{\mathbb{L}_p(\sigma)}.$$

Proof. This inequality in the special case of σ being the completely mixed state is proven in [Cubitt et al., 2015] but the generalization to arbitrary σ is trivial and follows from the reverse Hölder inequality:

$$\begin{aligned} \|X + Y\|_{\mathbb{L}_p(\sigma)}^p &= \left\| \left(\Gamma_{\sigma}^{\frac{1}{p}}(X + Y) \right)^p \right\|_1 \\ &= \left\| \Gamma_{\sigma}^{\frac{1}{p}}(X + Y) \left(\Gamma_{\sigma}^{\frac{1}{p}}(X + Y) \right)^{p-1} \right\|_1 \\ &= \left\| \Gamma_{\sigma}^{\frac{1}{p}}(X) \left(\Gamma_{\sigma}^{\frac{1}{p}}(X + Y) \right)^{p-1} \right\|_1 + \left\| \Gamma_{\sigma}^{\frac{1}{p}}(Y) \left(\Gamma_{\sigma}^{\frac{1}{p}}(X + Y) \right)^{p-1} \right\|_1 \\ &\geq \left(\|X\|_{\mathbb{L}_p(\sigma)} + \|Y\|_{\mathbb{L}_p(\sigma)} \right) \|X + Y\|_{\mathbb{L}_p(\sigma)}^{p-1}. \end{aligned}$$

□

Now, given a full-rank state σ , a quantum Markov semigroup $(\mathcal{P}_t)_{t \geq 0}$ for which σ is invariant is called *reverse $\mathbb{L}_p(\sigma)$ -contractive* if for any $p < 1$, all $t \geq 0$ and $X \in \mathcal{B}(\mathcal{H})^+$:

$$\|\mathcal{P}_t(X)\|_{\mathbb{L}_p(\sigma)} \geq \|X\|_{\mathbb{L}_p(\sigma)}.$$

We already saw in Proposition 5.2.1 that any quantum Markov semigroup is $\mathbb{L}_p(\sigma)$ -contractive for any $p \geq 1$ whenever σ is an invariant state. Next theorem is an extension of this to the range $p < 1$:

Proposition 7.4.3. (i) *Any quantum Markov semigroup $(\mathcal{P}_t)_{t \geq 0}$ with invariant state σ is reverse $\mathbb{L}_p(\sigma)$ -contractive for $p \in (-\infty, -1] \cup [1/2, 1)$.*

(ii) *A doubly stochastic quantum Markov semigroup is reverse $\mathbb{L}_p(\mathbb{1}/\dim(\mathcal{H}))$ -contractive for all $p < 1$.*

Proof. (i) First let $p = -q \in (-\infty, -1]$, and $X > 0$. We note that

$$\|\mathcal{P}_t(X)\|_{\mathbb{L}_p(\sigma)} = \|\mathcal{P}_t(X)^{-1}\|_{\mathbb{L}_q(\sigma)}^{-1}.$$

On the other hand, $\mathcal{P}_t$ is completely positive and unital, and $z \mapsto z^{-1}$ is operator convex. Therefore, $\mathcal{P}_t(X^{-1}) \geq \mathcal{P}_t(X)^{-1}$ and $\|\mathcal{P}_t(X)^{-1}\|_{\mathbb{L}_q(\sigma)} \leq \|\mathcal{P}_t(X^{-1})\|_{\mathbb{L}_q(\sigma)}$. We conclude that

$$\|\mathcal{P}_t(X)\|_{\mathbb{L}_p(\sigma)} \geq \|\mathcal{P}_t(X^{-1})\|_{\mathbb{L}_q(\sigma)}^{-1} \geq \|X^{-1}\|_{\mathbb{L}_q(\sigma)}^{-1} = \|X\|_{\mathbb{L}_p(\sigma)},$$

where for the second inequality we use q -contractivity of $\mathcal{P}_t$ for $q \geq 1$. Now suppose that $p \in [1/2, 1)$. We note that $\hat{p} \in (-\infty, -1]$, and that the conjugate $\hat{\mathcal{P}}_t$ of $\mathcal{P}_t$ with respect to $\langle \cdot, \cdot \rangle_{\sigma}$ is reverse $\mathbb{L}_{\hat{p}}$ -contractive.

Then using Hölder's duality, for $X > 0$ we have

$$\begin{aligned}
\|\mathcal{P}_t(X)\|_{\mathbb{L}_p(\sigma)} &= \inf_{Y>0: \|Y\|_{\mathbb{L}_{\hat{p}}(\sigma)} \geq 1} |\langle Y, \mathcal{P}_t(X) \rangle_\sigma| \\
&= \inf_{Y>0: \|Y\|_{\mathbb{L}_{\hat{p}}(\sigma)} \geq 1} |\langle \hat{\mathcal{P}}_t(Y), X \rangle_\sigma| \\
&\geq \inf_{Z>0: \|Z\|_{\mathbb{L}_{\hat{p}}(\sigma)} \geq 1} |\langle Z, X \rangle_\sigma| \\
&= \|X\|_{\mathbb{L}_p(\sigma)}.
\end{aligned}$$

Here the inequality follows from the reverse $\mathbb{L}_{\hat{p}}$ -contractivity of $\hat{\mathcal{P}}_t$, i.e., $\|\hat{\mathcal{P}}_t(Y)\|_{\mathbb{L}_{\hat{p}}(\sigma)} \geq \|Y\|_{\mathbb{L}_{\hat{p}}(\sigma)} \geq 1$.

(ii) As worked out in Lemma 8 of [Cubitt et al., 2015] this is an immediate consequence of the operator Jensen inequality. $\square$

Remark 7.4.4. The (reverse) contractivity for the range of parameters $p \geq 1/2$ can be shown to hold from the data processing inequality of sandwiched p -Rényi relative entropy, which is known to hold [Frank and Lieb, 2013, Beigi, 2013, Müller-Lennert et al., 2013] for $p \geq 1/2$. Here, we gave a proof of part (i) for the range $p \in (-\infty, -1] \cup [1/2, 1)$ based on new ideas which may be of independent interest.

By (ii) and (iii) of Lemma 7.2.1, for any $p \in \mathbb{R} \setminus \{0, 1\}$ and $X > 0$,

$$\text{Ent}_{p,\sigma}(X) \geq 0.$$

In the case when $p > 1$, we already saw that $\mathcal{E}_{p,\mathcal{L}}(X) \geq 0$ for any $X > 0$. This can be extended to the case $p \in \mathbb{R} \setminus \{0, 1\}$ so long as $(\mathcal{P}_t)_{t \geq 0}$ is reverse $\mathbb{L}_p(\sigma)$ -contractive:

Proposition 7.4.5. *Given $p < 1$, $p \neq 0$, suppose that $\mathcal{L}$ is the generator of a reverse $\mathbb{L}_p(\sigma)$ -contractive QMS $(\mathcal{P}_t)_{t \geq 0}$. Then for any $X > 0$,*

$$\mathcal{E}_{p,\mathcal{L}}(X) \geq 0.$$

Proof. From Lemma 5.2.8 (i), it is enough to consider the range $0 < p < 1$. Then, define

$$g(t) := \hat{p} \|\mathcal{P}_t(X)\|_{\mathbb{L}_p(\sigma)}^p - \hat{p} \|X\|_{\mathbb{L}_p(\sigma)}^p.$$

By assumption of reverse $\mathbb{L}_p(\sigma)$ -contractivity, for all $t \geq 0$ we have $g(t) \leq 0$. We note that $g(0) = 0$. Therefore, $g'(0) \leq 0$. We compute

$$\begin{aligned}
0 &\geq g'(0) \\
&= \frac{d}{dt} \hat{p} \|\mathcal{P}_t(X)\|_{p,\sigma}^p \Big|_{t=0} \\
&= \frac{d}{dt} \hat{p} \text{Tr} \left(\Gamma_\sigma^{\frac{1}{p}} \circ \mathcal{P}_t(X)^p \right) \Big|_{t=0} \\
&= p\hat{p} \text{Tr} \left(\Gamma_\sigma^{\frac{1}{p}} \circ \mathcal{L}(X) \cdot \Gamma_\sigma^{\frac{1}{p}}(X)^{p-1} \right) \\
&= p\hat{p} \text{Tr} \left(\mathcal{L}(X) \cdot \Gamma_\sigma^{\frac{1}{p}} \left(\Gamma_\sigma^{\frac{1}{p}}(X)^{p-1} \right) \right) \\
&= p\hat{p} \langle I_{\hat{p},p}(X), \mathcal{L}(X) \rangle_\sigma \\
&= -4\mathcal{E}_{p,\mathcal{L}}(X).
\end{aligned}$$

$\square$

Extending the notion of a *p-logarithmic Sobolev constant* as defined in (Equation (7.3)) to the range $p \in \mathbb{R} \setminus \{0, 1\}$, the following proposition is easily verified from the properties of the entropy function and of the Dirichlet form:

Proposition 7.4.6. *Let $(\mathcal{P}_t)_{t \geq 0}$ be a QMS on the algebra $\mathcal{B}(\mathcal{H})$ of bounded operators over a finite dimensional Hilbert space $\mathcal{H}$. Then, for any $p \in \mathbb{R} \setminus \{0, 1\}$ of Hölder conjugate $\hat{p}$,*

$$\alpha_p(\mathcal{L}) = \alpha_{\hat{p}}(\mathcal{L}).$$

Proof. Identifying X with $I_{p,2}(Y)$ for some $Y \in \mathcal{B}(\mathcal{H})^+$, this is an immediate consequence of part (i) of Lemma 7.2.1 and part (i) of Lemma 5.2.8. $\square$

The above proposition allows us to restrict ourselves to log-Sobolev constants for values of $p \in [0, 2]$. The proof of next proposition is standard [Olkiewicz and Zegarlinski, 1999, Kastoryano and Temme, 2013] and will be extended to weighted amalgamated norms in Chapter 8:

Proposition 7.4.7. *For a differentiable operator valued function $p \mapsto X_p$ we have, for any $p \in \mathbb{R} \setminus \{0\}$:*

$$\frac{d}{dp} \|X_p\|_{\mathbb{L}_p(\sigma)} = \frac{1}{p^2} \|X_p\|_{\mathbb{L}_p(\sigma)}^{1-p} \cdot \left(\frac{1}{2} \text{Ent}_{p,\sigma}(I_{p,p}(X_p)) + \frac{1}{2} \text{Ent}_{p,\sigma}(I_{p,p}(X_p^*)) + \gamma \right).$$

Here γ is given by

$$\gamma = \frac{p^2}{2} \left(\text{Tr} \left[\Gamma_{\sigma}^{\frac{1}{p}}(Z_p^*) \cdot \Gamma_{\sigma}^{\frac{1}{p}}(X_p) \cdot |\Gamma_{\sigma}^{\frac{1}{p}}(X_p)|^{p-2} \right] + \text{Tr} \left[\Gamma_{\sigma}^{\frac{1}{p}}(X_p^*) \cdot \Gamma_{\sigma}^{\frac{1}{p}}(Z_p) \cdot |\Gamma_{\sigma}^{\frac{1}{p}}(X_p)|^{p-2} \right] \right),$$

where $Z_p := \frac{d}{dp} X_p$.

Theorem 7.4.8. *Let $(\mathcal{P}_t)_{t \geq 0}$ be a primitive QMS and σ a full-rank invariant state. Then the QMS is reverse hypercontractive: Suppose that $\tilde{\alpha}_1 = \inf_{p \in [0,1]} \alpha_p(\mathcal{L}) > 0$. Then for $p \leq q < 1$ and $t \geq \frac{1}{4\tilde{\alpha}_1} \ln \frac{p-1}{q-1}$, and all $X > 0$:*

$$\|\mathcal{P}_t(X)\|_{\mathbb{L}_p(\sigma)} \geq \|X\|_{\mathbb{L}_q(\sigma)}. \quad (7.8)$$

Proof. It suffices to prove the theorem when $t = t(p) \equiv \frac{1}{4\tilde{\alpha}_1} \ln \frac{p-1}{q-1}$, for a given fixed q . Define

$$f(p) := \|\mathcal{P}_{t(p)}(X)\|_{\mathbb{L}_p(\sigma)} - \|X\|_{\mathbb{L}_q(\sigma)} = \|X_p\|_{\mathbb{L}_p(\sigma)} - \|X\|_{\mathbb{L}_q(\sigma)},$$

where $X_p := \mathcal{P}_{t(p)}(X) > 0$. To continue the proof we compute the derivative of $f(p)$ using Proposition 7.4.7.

$$f'(p) = \frac{d}{dp} \|X_p\|_{\mathbb{L}_p(\sigma)} = \frac{1}{p^2} \|X_p\|_{\mathbb{L}_p(\sigma)}^{1-p} \cdot \left(\text{Ent}_{p,\sigma}(X_p) + p^2 \text{Tr} \left[\Gamma_{\sigma}^{\frac{1}{p}}(Z_p) \cdot \Gamma_{\sigma}^{\frac{1}{p}}(X_p)^{p-1} \right] \right),$$

where

$$Z_p = \frac{d}{dp} X_p = -t'(p) \mathcal{L}(X_p) = -\frac{1}{4\tilde{\alpha}_1(p-1)} \mathcal{L}(X_p).$$

Therefore,

$$f'(p) = \frac{1}{p^2} \|X_p\|_{\mathbb{L}_p(\sigma)}^{1-p} \cdot \left(\text{Ent}_{p,\sigma}(X_p) - \frac{1}{\tilde{\alpha}_1} \mathcal{E}_{p,\mathcal{L}}(X_p) \right).$$

Now, assume that $q < 1$ and $\tilde{\alpha}_1 \leq \alpha_p(\mathcal{L})$ for all $p \in [0, 1]$. Then for $p \leq q$ we have

$$\text{Ent}_{p,\sigma}(X_p) \leq \frac{1}{\alpha_p(\mathcal{L})} \mathcal{E}_{p,\mathcal{L}}(X_p) \leq \frac{1}{\tilde{\alpha}_1} \mathcal{E}_{p,\mathcal{L}}(X_p),$$

where the second inequality holds since $p < 1$, so either p or its Hölder conjugate belongs to $[0, 1]$. Therefore, $f'(p) \leq 0$ for all $p \leq q < 1$, and since $f(q) = 0$, $f(p) \geq 0$ for all $p < q$. $\square$

In a similar vein as in the setting of hypercontractivity, one gets a stronger result under the condition that $(\mathcal{P}_t)_{t \geq 0}$ satisfies σ -DBC, due to the Stroock Varopoulos inequality of Theorem 5.4.2. First, we get the reverse $\mathbb{L}_p(\sigma)$ -contractivity for any value of $p < 1$:

Corollary 7.4.9. *Let the QMS $(\mathcal{P}_t)_{t \geq 0}$ satisfy σ -DBC. Then the followings hold:*

- (i) *For all $p < 1$ and $X > 0$, $\mathcal{E}_{p,\mathcal{L}}(X) \geq 0$.*
- (ii) *The associated QMS is reverse $\mathbb{L}_p(\sigma)$ -contractive for all $p < 1$.*

As mentioned before, (reverse) $\mathbb{L}_p(\sigma)$ -contractivity of $\mathcal{P}_t$ implies that Sandwiched $\mathbb{L}_p(\sigma)$ -Rényi relative entropy is monotone under $\mathcal{P}_t$ [Frank and Lieb, 2013, Beigi, 2013, Müller-Lennert et al., 2013]. Therefore, when $\mathcal{P}_t$ comes from a QMS satisfying ω -DBC, the p -Rényi relative entropy is monotone under $\mathcal{P}_t$ not only for $p \geq 1/2$ but *all* values of p .

Proof. (i) By Theorem 5.4.2 (and part (i) of Lemma 5.2.8) for every p we have

$$\mathcal{E}_{p,\mathcal{L}}(\mathbb{I}_{p,2}(X)) \geq \mathcal{E}_{2,\mathcal{L}}(X).$$

On the other hand, by Proposition 7.4.5 and Proposition 5.22, we have $\mathcal{E}_{2,\mathcal{L}}(X) \geq 0$. Therefore, $\mathcal{E}_{p,\mathcal{L}}(\mathbb{I}_{p,2}(X)) \geq 0$.

(ii) Define $g(t)$ as in the proof of Proposition 7.4.5. By part (i) we have $g'(t) \leq 0$ for all $t \geq 0$ and $g(0) = 0$. Therefore, $g(t) \geq 0$ for all $t \geq 0$. $\square$

The following corollary is an immediate consequence of the quantum Stroock-Varopoulos inequality as well as part (i) of Proposition 7.2.1:

Corollary 7.4.10. *Let $\mathcal{L}$ be the generator of a primitive QMS $(\mathcal{P}_t)_{t \geq 0}$ that satisfies σ -DBC. Then the QMS is reverse hypercontractive: For $p \leq q < 1$, $t \geq \frac{1}{4\alpha_1(\mathcal{L})} \ln \frac{p-1}{q-1}$ and all $X > 0$,*

$$\|\mathcal{P}_t(X)\|_{\mathbb{L}_p(\sigma)} \geq \|X\|_{\mathbb{L}_q(\sigma)}. \quad (7.9)$$

7.5. Quantum Nash inequality

7.5.1. Nash inequalities in finite dimensions

Nash inequalities were extended to the finite dimensional quantum setting in [Kastoryano and Temme, 2016] in the spirit of [Diaconis and Saloff-Coste, 1996b]. Here, we simply recall their main result showing the equivalence between a Nash inequality and ultracontractivity of the semigroup. Let $(\mathcal{P}_t = e^{t\mathcal{L}})_{t \geq 0}$ be a primitive QMS on $\mathcal{B}(\mathcal{H})$, $\mathcal{H}$ finite dimensional, with full-rank invariant state σ . The QMS is said to satisfy a *quantum Nash inequality* if there exist constants $C > 0$, $n \geq 2$ and $T \geq 0$ such that for any $X \in \mathcal{B}(\mathcal{H})$:

$$\|X\|_{\mathbb{L}_2(\sigma)}^{2+4/n} \leq C \left(\mathcal{E}_{2,\mathcal{L}}(X) + \frac{1}{T} \|X\|_{\mathbb{L}_2(\sigma)}^2 \right) \|X\|_{\mathbb{L}_1(\sigma)}^{4/n}. \quad (\text{qNash}(C, n, T))$$

The following result was then proved in [Kastoryano and Temme, 2016]:

Theorem 7.5.1. *Assume that $\text{qNash}(C, n, T)$ holds. Then, for all $t \leq T$:*

$$\|\mathcal{P}_t : \mathbb{L}_1(\sigma) \rightarrow \mathbb{L}_2(\sigma)\| \leq e \left(\frac{nC}{4t} \right)^{n/4}.$$

On the other hand, assume that $(\mathcal{P}_t)_{t \geq 0}$ is KMS-symmetric with respect to σ . Then $\|\mathcal{P}_t : \mathbb{L}_1(\sigma) \rightarrow \mathbb{L}_2(\sigma)\| \leq \left(\frac{C_0}{4t} \right)^{n/4}$ for any $t \leq T$ implies $\text{qNash}(C, n, T)$ with $C \equiv 2C_0(1 + 2/n)(1 + n/2)^{2/n}$.

7.5.2. Nash inequalities on the quantum phase space

As mentioned in Section 3.4, the original Nash inequality on $\mathbb{R}^n$ has the following form: for any smooth enough function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ (e.g. a Schwartz function),

$$\|f\|_{\mathbb{L}_2(\mathbb{R}^n)}^{2+4/n} \leq C_n \|f\|_{\mathbb{L}_1(\mathbb{R}^n)}^{4/n} \|\nabla f\|_{\mathbb{L}_2(\mathbb{R}^n)}^2, \quad (7.10)$$

where the sharp constant $C_n := \frac{2((n+2)/2)^{(n+2)/n}}{n \lambda(B^n) |B^n|^{2/n}}$ was found in [Carlen and Loss, 1993], with the constant $|B^n|$ denoting the volume of the unit ball B^n on $\mathbb{R}^n$, whereas $\lambda(B^n)$ is the spectral gap of the Laplacian on B^n . After a simple integration by parts, this inequality can be re-expressed in terms of the Dirichlet form of the heat semigroup:

$$\|f\|_{\mathbb{L}_2(\mathbb{R}^n)}^{2+4/n} \leq C_n \|f\|_{\mathbb{L}_1(\mathbb{R}^n)}^{4/n} \mathcal{E}(f), \quad \mathcal{E}(f) := - \int_{\mathbb{R}^n} f \Delta f dx. \quad (7.11)$$

In this section, we prove that the quantum heat semigroup $(\mathcal{P}_t)_{t \geq 0}$ defined in Section 5.5.2 also satisfies a Nash inequality, at least on Schwartz operators of positive Wigner function. In the following we denote $\rho_t = \mathcal{P}_t^{\text{qheat}}(\rho) = \mathcal{P}_{t*}^{\text{qheat}}(\rho)$.

Theorem 7.5.2 (A non-commutative Nash inequality). *If ρ is a Schwartz state of positive Wigner function, then $\rho \in \text{dom}(\mathcal{L}^{\text{qheat}})$ and the following non-commutative Nash inequality holds:*

$$\|\rho\|_2^{2+2/n} \leq C_{2n} \mathcal{E}_{2, \mathcal{L}^{\text{qheat}}}(\rho). \quad (7.12)$$

Remark 7.5.3. Upon replacing $\mathbb{L}_p(\mathbb{R}^n)$ norms by Schatten norms and n by $2n$, the similarity between (7.12) and (7.11) is obvious and unsurprising in view of Parseval's relation 0.2.3. The doubling of the dimension is also expected since the quantum heat semigroup is defined on a representation of a $2n$ -dimensional phase space.

Proof. We prove that (7.12) is satisfied for any Schwartz state ρ with positive Wigner function. For such states $\mathcal{L}^{\text{qheat}}(\rho)$ is well-defined and trace class (see e.g. Propositions 3.14 and 3.15 of [Keyl et al., 2016]), and using the non-commutative Parseval relation, Theorem 0.2.3, and the Dirichlet form $\mathcal{E}_{2, \mathcal{L}^{\text{qheat}}}$ takes the following form:

$$\mathcal{E}_{2, \mathcal{L}^{\text{qheat}}}(\rho) := -\text{Tr}(\rho \mathcal{L}^{\text{qheat}}(\rho)) = -(2\pi)^{-n} \int_{\mathcal{Z}} \overline{\mathcal{F}_\rho^{\text{q}}(z)} \mathcal{F}_{\mathcal{L}^{\text{qheat}}(\rho)}^{\text{q}}(z) dz. \quad (7.13)$$

Now, using Equation (0.29), we find that $\mathcal{F}_{\mathcal{L}^{\text{qheat}}(\rho)}^{\text{q}}(z) = -\frac{1}{4}|z|^2 \mathcal{F}_\rho^{\text{q}}(z)$. Substituting this into the right hand side of Equation (7.13), we get,

$$\mathcal{E}_{2, \mathcal{L}^{\text{qheat}}}(\rho) = \frac{1}{4} (2\pi)^{-n} \int_{\mathcal{Z}} |z|^2 |\mathcal{F}_\rho^{\text{q}}(z)|^2 dz. \quad (7.14)$$

Now, $z \mapsto \mathcal{F}_\rho^{\text{q}}(z)$ is a Schwartz function, and therefore its inverse Fourier transform is also a Schwartz

function. Let us denote by $\mathcal{F}_f$ the Fourier transform of an integrable function f on $\mathbb{R}^{2n}$:

$$\mathcal{F}_f(z) := (2\pi)^{-n} \int_{\mathbb{R}^{2n}} f(x) e^{i(x,z)} dx. \quad (7.15)$$

We make use of the following basic facts. Firstly, the Fourier transform satisfies the following useful identity

$$\mathcal{F}_{x \mapsto \partial_{x_j} f(x)}(z) = -iz_j \mathcal{F}_f(z), \quad (7.16)$$

for any integrable, continuously differentiable function f , which has an integrable partial derivative $\partial_{x_j} f$, where z_j denotes the j^{th} component of the vector $z \in \mathcal{Z}$. Secondly, for any square integrable function h on $\mathbb{R}^{2n}$, we have that

$$\|\mathcal{F}_h\|_2 = \|h\|_2, \quad (7.17)$$

which is the classical Plancherel identity. We also employ the well-known polarization identity:

$$\int_{\mathbb{R}^{2n}} \overline{g(x)} h(x) dx = \frac{1}{4} (\|g+h\|_2^2 - \|g-h\|_2^2 + i\|g+ih\|_2^2 - i\|g-ih\|_2^2), \quad (7.18)$$

for any two square integrable functions g and h , and the fact that the characteristic function $\mathcal{F}_\rho^q$ is equal to $\mathcal{F}_{f_\rho}$, where f_ρ denotes the Wigner function of ρ and is a Schwartz function.

$$\begin{aligned} \mathcal{E}_{2, \mathcal{L}^{\text{qheat}}}(\rho) &= \frac{1}{4} (2\pi)^{-n} \int_{\mathcal{Z}} \overline{\mathcal{F}_\rho^q(z)} (|z|^2 \mathcal{F}_\rho^q(z)) dz \\ &= -\sum_{i=1}^{2n} \frac{1}{4} (2\pi)^{-n} \int_{\mathcal{Z}} \overline{\mathcal{F}_{f_\rho}(z)} (-iz_j)^2 \mathcal{F}_{f_\rho}(z) dz \\ &= -\sum_{i=1}^{2n} \frac{1}{4} (2\pi)^{-n} \int_{\mathcal{Z}} \overline{\mathcal{F}_{f_\rho}(z)} \mathcal{F}_{x \mapsto \frac{\partial^2}{\partial x_i^2} f_\rho(x)}(z) dz \\ &= -\frac{1}{4} \sum_{i=1}^{2n} (2\pi)^{-n} \int_{\mathcal{Z}} f_\rho(x) \partial_{x_i^2}^2 f_\rho(x) dx \\ &= -(2\pi)^{-n} \int_{\mathcal{Z}} f_\rho(x) \frac{1}{4} \Delta f_\rho(x) dx \equiv \frac{1}{4(2\pi)^n} \mathcal{E}(f_\rho), \end{aligned} \quad (7.19)$$

where $\mathcal{E}(\cdot)$ is the Dirichlet form of the classical heat semigroup as given in Equation (7.11). The third line follows from two uses of Equation (7.16) and the fourth line follows from Equation (7.17) and Equation (7.18).

Moreover, by the non-commutative Parseval relation, Theorem 0.2.3, and Equation (7.17), we have

$$\|\rho\|_2^2 = \text{Tr}(\rho^2) = (2\pi)^{-n} \int_{\mathcal{Z}} |\mathcal{F}_\rho^q(z)|^2 dz = (2\pi)^{-n} \|\mathcal{F}_\rho^q\|_2^2 = (2\pi)^{-n} \|f_\rho\|_{\mathbb{L}_2(\mathbb{R}^n)}^2. \quad (7.20)$$

Further,

$$(2\pi)^n \|\rho\|_1 = (2\pi)^n = \|f_\rho\|_{\mathbb{L}_1(\mathbb{R}^n)}, \quad (7.21)$$

which follows from the fact that the Wigner function of a state (as defined through Equation (0.35)) has integral equal to $(2\pi)^n$, and the assumption that f_ρ is positive. Hence,

$$\|\rho\|_2^{2+2/n} = (2\pi)^{-(n+1)} \|f_\rho\|_{\mathbb{L}_2(\mathbb{R}^n)}^{2+2/n} \leq (2\pi)^{-n+1} C_{2n} \mathcal{E}(f_\rho) = 8\pi C_{2n} \mathcal{E}_{2, \mathcal{L}^{\text{qheat}}}(\rho), \quad (7.22)$$

where we made use of the classical Nash inequality (7.12), with n replaced by $2n$. $\square$

As expected, (7.12) easily implies ultracontractivity of $(\mathcal{P}_t)_{t \geq 0}$.

Theorem 7.5.4 (Ultracontractivity of the quantum heat semigroup). *If ρ is a Schwartz state of positive Wigner function, then there exists a positive constant κ_n such that for any $t > 0$*

$$\|\rho_t\|_\infty \leq \|\rho_t\|_2 \leq \left(\frac{nC_{2n}}{2t} \right)^{\frac{n}{2}}. \quad (7.23)$$

Moreover, for any such initial state ρ evolving under the action of the semigroup $(\mathcal{P}_t^{\text{qheat}})_{t \geq 0}$, the following bounds give the rate of decay of its purity and the rate of increase of its von Neumann entropy, respectively:

$$\text{Tr } \rho_t^2 \equiv \|\rho_t\|_2^2 \leq \left(\frac{nC_{2n}}{2t} \right)^n; \quad (7.24)$$

$$S(\rho_t) \geq \frac{n}{2} \ln \left(\frac{2t}{nC_{2n}} \right). \quad (7.25)$$

Proof. The proof is identical to the one given in [Kastoryano and Temme, 2016] in the case of finite dimensional QMS, which in turn closely follows the proof in the classical case studied in [Diaconis and Saloff-Coste, 1996b]. Let ρ be a Schwartz state with a positive Wigner function. By Proposition 5.5.3, $\rho_t := \mathcal{P}_t(\rho)$ is also such a state, and Theorem 7.5.2 applies. Let $u(t) := \|\rho_t\|_2^2$; using Theorem 0.2.3 and Equation (5.47), one can verify that the function u is differentiable and $\dot{u}(t) = -2\mathcal{E}_{2, \mathcal{L}^{\text{qheat}}}(\rho_t)$. Theorem 7.5.2 implies that

$$u^{1+1/n}(t) = \|\rho_t\|_2^{2+2/n} \leq C_{2n} \mathcal{E}_{2, \mathcal{L}^{\text{qheat}}}(\rho_t) = -\frac{C_{2n}}{2} \dot{u}(t)$$

so that

$$\frac{d}{dt} \frac{1}{u^{1/n}(t)} \frac{nC_{2n}}{2} = -\frac{C_{2n}}{2} \frac{\dot{u}(t)}{u(t)^{1+1/n}} \geq 1,$$

and by integrating both sides of the above inequality from 0 to t , one gets:

$$\frac{1}{u^{1/n}(t)} \geq \frac{2t}{nC_{2n}} + \frac{1}{u^{1/n}(0)} \geq \frac{2t}{nC_{2n}},$$

so that

$$\|\rho_t\|_2^2 \equiv u(t) \leq \left(\frac{nC_{2n}}{2t} \right)^n.$$

In particular, this implies that

$$\|\rho_t\|_\infty \leq \|\rho_t\|_2 \leq \left(\frac{nC_{2n}}{2t} \right)^{\frac{n}{2}}.$$

Therefore,

$$S(\rho_t) = -\text{Tr}(\rho_t \ln \rho_t) \geq -\ln \|\rho_t\|_\infty \geq \frac{n}{2} \ln \left(\frac{2t}{nC_{2n}} \right).$$

$\square$

Chapter 8.

Non primitive functional inequalities for the study of decoherence

Largely inspired by [Diaconis and Saloff-Coste, 1996a], our goal in this chapter is to develop the theory of LSI and HC for non-primitive QMS and its use in proving rapid decoherence. This line of research started with [Bardet, 2017] who introduced non-primitive generalizations of the Poincaré and the modified logarithmic Sobolev inequality. Note that, due to the non-commutativity of quantum systems, typical quantum features arise in this situation that are absent from the classical theory.

Of course, the hypercontractive property depends highly on the choice of the interpolating family of $\mathbb{L}_p$ norms. In particular, a QMS which is hypercontractive for Kosaki's $\mathbb{L}_p$ norms will be primitive. The main contribution of the present work is to study hypercontractivity with respect to a generalisation of Kosaki's norms, called the amalgamated norms and defined by [Junge and Parcet, 2010]. We refer to Section 1.1.1 for a detailed analysis of amalgamated $\mathbb{L}_p$ norms. Using these norms, we will be able to reproduce the steps of [Diaconis and Saloff-Coste, 1996a] for non-primitive QMS.

Throughout this chapter, we consider quantum Markov semigroups defined on the algebra $\mathcal{B}(\mathcal{H})$ of linear operators acting on a finite dimensional Hilbert space $\mathcal{H}$. Following ideas from [Beigi and King, 2016], we derive a formula for the differential of the amalgamated $\mathbb{L}_p$ norms (see Theorem 8.3.3), with respect to the index p . This leads to the definition of the *weak decoherence-free logarithmic Sobolev inequality* (DF-wLSI) and the *weak decoherence-free hypercontractivity* (DF-wHC), and allows us to extend Gross' integration lemma to this setting (see Theorem 8.2.2). A first difference compared to the primitive case is that LSI implies HC but with a larger weak constant which depends on the structure of the DF-algebra.

In the primitive case, the uniform convexity of the $\mathbb{L}_p$ norms was used in [Olkiewicz and Zegarlinski, 1999] to show that wLSI together with PI give rise to a strong logarithmic Sobolev inequality (sLSI) (cf. Proposition 3.2.6). We show that a similar analysis can be performed in our extended framework, in order to derive universal upper bounds on the log-Sobolev constants (see Definition 8.2.1 and Corollary 8.2.6). We also prove that, except in the primitive case, the strong LSI does not hold and therefore neither does the related notion of strong hypercontractivity. This implies that the uniform convexity no longer holds for the amalgamated $\mathbb{L}_p$ -norms.

Layout of the chapter: In Section 8.1, we sketch a succinct overview of the framework of [Bardet, 2017] and relate his decoherence-free Poincaré inequality for a quantum Markov semigroup to the spectral gap of a corresponding discrete-time quantum Markov chain. Next, we define and study the notions of a non-primitive logarithmic Sobolev inequality and hypercontractivity: our main results are stated in Section 8.2. The notions of decoherence-free log-Sobolev inequality and hypercontractivity

are studied in Section 8.4, where we prove Gross' integration Lemma as well as a universal upper bound on the constants. In Section 8.5 we prove that the strong LSI fails for non-trivially primitive QMS. Some applications of our framework to the derivation of decoherence rates are provided in Section 8.6.

8.1. Decoherence-free spectral gap and MLSI

Assume given a faithful QMS $(\mathcal{P}_t)_{t \geq 0}$, that is one possessing a full-rank invariant state. We are in the setting of Section 6.2: the semigroup converges to its decoherence-free algebra $\mathcal{N}$ which we may decompose as in Equation (6.5). Moreover, there exists a conditional expectation $E_{\mathcal{N}}$ onto $\mathcal{N}$ and the state $\sigma_{\text{Tr}} := E_{\mathcal{N}*}(d_{\mathcal{H}}^{-1} \mathbb{1})$ is left invariant under the evolution induced by $(\mathcal{P}_t)_{t \geq 0}$. We are interested in estimating the speed of convergence to the following limit by means of non-commutative functional inequalities:

$$\mathcal{P}_t(X) \underset{t \rightarrow \infty}{\sim} \mathcal{P}_t \circ E_{\mathcal{N}}[X]. \quad (8.1)$$

We first briefly review the results of [Bardet, 2017], where the Poincaré and modified logarithmic Sobolev inequalities were extended to the present non-primitive scenario.

Decoherence-free Poincaré inequality : The Poincaré inequality can be simply modified in order to take into account $\mathbb{L}_2$ convergence of non-primitive evolutions: we say that a *DF-Poincaré inequality* holds if for all $X \in \mathcal{B}_{sa}(\mathcal{H})$:

$$\lambda \text{Var}_{\mathcal{N}}(X) \leq \mathcal{E}_{2,\mathcal{L}}(X). \quad (\text{PI}_{\mathcal{N}}(\lambda))$$

where $\text{Var}_{\mathcal{N}}(X) := \|X - E_{\mathcal{N}}[X]\|_{\mathbb{L}_2(\sigma_{\text{Tr}})}^2$ is the *DF-variance* of X . Then, define the *decoherence-free Poincaré constant* as follows:

$$\lambda(\mathcal{L}) := \inf_{X \in \mathcal{B}_{sa}(\mathcal{H})} \frac{\mathcal{E}_{2,\mathcal{L}}(X)}{\text{Var}_{\mathcal{N}}(X)}. \quad (8.2)$$

The Poincaré constant turns out to be the spectral gap of the operator $\frac{\mathcal{L} + \hat{\mathcal{L}}}{2}$, where $\hat{\mathcal{L}}$ is the adjoint of $\mathcal{L}$ with respect to $\langle \cdot, \cdot \rangle_{\sigma_{\text{Tr}}}$, that is minus its second (negative) largest eigenvalue. In complete analogy with Proposition 7.1.1, we obtain the following $\mathbb{L}_2(\sigma_{\text{Tr}})$ exponential decay:

$$\|\mathcal{P}_t(X - E_{\mathcal{N}}(X))\|_{\mathbb{L}_2(\sigma_{\text{Tr}})} \leq e^{-\lambda(\mathcal{L})t} \|X - E_{\mathcal{N}}[X]\|_{\mathbb{L}_2(\sigma_{\text{Tr}})}. \quad (8.3)$$

From this, bounds on the convergence in trace distance of the corresponding evolution of quantum states can be derived using

$$\|\mathcal{P}_{t*} \circ (\rho - E_{\mathcal{N}*}(\rho))\|_1 = \|\hat{\mathcal{P}}_t(X) - \hat{\mathcal{P}}_t \circ E_{\mathcal{N}}[X]\|_{\mathbb{L}_1(\sigma_{\text{Tr}})} \leq \|\hat{\mathcal{P}}_t(X) - \hat{\mathcal{P}}_t \circ E_{\mathcal{N}}[X]\|_{\mathbb{L}_2(\sigma_{\text{Tr}})}, \quad (8.4)$$

where $(\hat{\mathcal{P}}_t)_{t \geq 0}$ denotes the semigroup associated with $\hat{\mathcal{L}}$.

In the case of a quantum Markov chain $\{\Phi^n\}_{n \in \mathbb{N}}$, we define the *absolute spectral gap* $\lambda(\Phi)$ as $1 - \max_{|\lambda_k| \neq 1} |\lambda_k|$, where the maximization is taken over all the eigenvalues of Φ of absolute value less than 1. Convergence bounds in terms of the absolute spectral gap, as well as the Jordan structure of the map Φ were found in [Szehr et al., 2015].

Decoherence-free MLSI As expected, one can derive better bounds on the convergence (8.1) using more powerful methods. In [Bardet, 2017], the modified logarithmic Sobolev constant $\alpha_1(\mathcal{L})$ was defined as follows:

$$\alpha_1(\mathcal{L}) := \inf_{\rho \in \mathcal{D}(\mathcal{H})} \frac{\text{EP}_{\sigma_{\text{Tr}}}(\rho)}{4 D(\rho \| E_{\mathcal{N}^*}(\rho))}.$$

The following stronger exponential decay in relative entropy is a straightforward consequence of the above definition: denoting $\rho_t := \mathcal{P}_{*t}(\rho)$ and $\rho_{\mathcal{N}} := E_{\mathcal{N}}(\rho)$:

$$D(\rho_t \| \mathcal{P}_{*t}(\rho_{\mathcal{N}})) \leq e^{-4\alpha_1(\mathcal{L})t} D(\rho \| \rho_{\mathcal{N}}).$$

As opposed to the primitive case, there exists no lower bound on the constant $\alpha_1(\mathcal{L})$ for a generic evolution for finite dimensional systems to this day. This is always the case for primitive evolutions satisfying a detailed balance condition with respect to their unique invariant state. This is due to the fact that, in this case, the constant $\alpha_1(\mathcal{L})$ can be lower bounded by $\alpha_2(\mathcal{L})$ using regularity of the Dirichlet forms. On the other hand, general lower bounds on $\alpha_2(\mathcal{L})$ were found in [Temme et al., 2014] by standard interpolation techniques.

This constitutes a good reason for introducing a decoherence-free analogue of $\alpha_2(\mathcal{L})$ in the context of non-primitive evolutions. As it turns out, we will show in Section 8.5 that the constant $\alpha_2(\mathcal{L})$ is positive only in the case of a primitive QMS. Therefore, the problem of finding a lower bound to $\alpha_1(\mathcal{L})$ for non-primitive semigroups remains unanswered. Fortunately, the introduction and study of a weak decoherence-free logarithmic Sobolev inequality, which is the main topic of this chapter, remains interesting since it will allow us to derive similar decoherence times via hypercontractivity.

8.2. Decoherence-free hypercontractivity: main results

By Corollary 6 of [Temme et al., 2014], in finite dimensions, hypercontractivity of a QMS with respect to the $\mathbb{L}_p(\sigma_{\text{Tr}})$ norms is equivalent to the primitivity of the QMS. In order to deal with non-primitive QMS, a possible choice of norms are the so-called *amalgamated norms* introduced in [Junge and Parcet, 2010] (cf. Section 1.1.1). We recall that these norms are defined as follows: for $1 \leq q \leq p \leq +\infty$ and $\frac{1}{r} = \frac{1}{q} - \frac{1}{p}$, define

$$\|X\|_{\mathbb{L}_q(\mathcal{N}, \mathbb{L}_p(\sigma_{\text{Tr}}))} := \inf_{\substack{A, B \in \mathcal{N}, Y \in \mathcal{B}(\mathcal{H}) \\ X = AYB}} \|A\|_{\mathbb{L}_{2r}(\sigma_{\text{Tr}})} \|B\|_{\mathbb{L}_{2r}(\sigma_{\text{Tr}})} \|Y\|_{\mathbb{L}_p(\sigma_{\text{Tr}})}, \quad (8.5)$$

$$\|Y\|_{\mathbb{L}_p(\mathcal{N}, \mathbb{L}_q(\sigma_{\text{Tr}}))} := \sup_{A, B \in \mathcal{N}} \frac{\|AYB\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}}{\|A\|_{\mathbb{L}_{2r}(\sigma_{\text{Tr}})} \|B\|_{\mathbb{L}_{2r}(\sigma_{\text{Tr}})}}. \quad (8.6)$$

We shall prove that they are particularly well-suited to study the hypercontractivity of the QMS, since:

- (i) they reduce to the $\mathbb{L}_p(\sigma)$ norms when the QMS is primitive with unique invariant state σ ;
- (ii) they reduce to the $\mathbb{L}_q(\sigma_{\text{Tr}})$ norms when evaluated on $\mathcal{N}$;
- (iii) the QMS is contractive with respect to these norms for all $p, q \geq 1$.

When differentiating an amalgamated norm with respect to p , some natural quantities will appear that we will connect with entropic ones in Lemma 8.3.2. Similarly to [Olkiewicz and Zegarlinski, 1999], we

thus introduce a decoherence-free generalisation of the $\mathbb{L}_p$ relative entropies as follows: define the map

$$S_p(X) = -p \partial_s I_{p+s,p}(X)|_{s=0},$$

referred to as *operator valued relative entropy*, where $I_{q,p}$ is defined in Equation (1.12)¹. It can be computed explicitly: when $X \geq 0$,

$$S_p(X) = \Gamma_{\sigma_{\text{Tr}}}^{-\frac{1}{p}} [\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{p}}(X) \ln \Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{p}}(X)] - \frac{1}{2p} \{X, \ln \sigma_{\text{Tr}}\}.$$

We then define the DF- $\mathbb{L}_p$ *relative entropy* associated with the algebra $\mathcal{N} \equiv \mathcal{N}(\mathcal{P})$ as follows: for $\frac{1}{p} + \frac{1}{\hat{p}} = 1$,

$$\text{Ent}_{p,\mathcal{N}}(X) := p \langle I_{\hat{p},p}(X), S_p(X) \rangle_{\sigma_{\text{Tr}}} - \text{Tr} \left[\left(\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{p}}(X) \right)^p \ln E_{\mathcal{N}} \left[\Gamma_{\sigma_{\text{Tr}}}^{-1} \left(\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{p}}(X) \right)^p \right] \right]. \quad (8.7)$$

In the case of a primitive QMS where σ_{Tr} is the unique invariant state of the evolution, $E_{\mathcal{N}}[\cdot] := \text{Tr}(\sigma_{\text{Tr}} \cdot) \mathbb{1}$ and we get back the original definition of [Kastoryano and Temme, 2013], which we denoted by $\text{Ent}_{p,\sigma_{\text{Tr}}}(X)$ in Equation (7.2). In general, $\text{Ent}_{p,\mathcal{N}}(X) \leq \text{Ent}_{p,\sigma_{\text{Tr}}}(X)$. In the important cases $p = 1$ and $p = 2$, Equation (8.7) reduces to

$$\begin{aligned} \text{Ent}_{1,\mathcal{N}}(X) &:= \text{Tr} \left[\Gamma_{\sigma_{\text{Tr}}}(X) \left(\ln \frac{\Gamma_{\sigma_{\text{Tr}}}(X)}{\text{Tr}(\Gamma_{\sigma_{\text{Tr}}}(X))} - \ln \sigma_{\text{Tr}} \right) \right] - \text{Tr} \left[\Gamma_{\sigma_{\text{Tr}}}(X) \ln \frac{E_{\mathcal{N}}[X]}{\text{Tr}(\Gamma_{\sigma_{\text{Tr}}}(X))} \right], \\ \text{Ent}_{2,\mathcal{N}}(X) &:= \text{Tr} \left(\left[\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{2}}(X) \right]^2 \left(\ln \left[\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{2}}(X) \right]^2 - \ln E_{\mathcal{N}} \left[\Gamma_{\sigma_{\text{Tr}}}^{-1} \left(\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{2}}(X) \right)^2 \right] - \ln \sigma_{\text{Tr}} \right) \right). \end{aligned} \quad (8.8)$$

Definition 8.2.1. We say that the QMS $(\mathcal{P}_t)_{t \geq 0}$ of generator $\mathcal{L}$

- (i) satisfies a *weak DF- q -log-Sobolev inequality* with positive *strong DF- q -log-Sobolev constant* $c > 0$ and *weak DF- q -log-Sobolev constant* $d \geq 0$ if for all $X > 0$,

$$\text{Ent}_{q,\mathcal{N}}(X) \leq c \mathcal{E}_{q,\mathcal{L}}(X) + 2d \|X\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^q; \quad (\text{LSI}_{q,\mathcal{N}}(c, d))$$

- (ii) is *weakly q -DF-hypercontractive* for positive constants $c > 0$ and $d \geq 0$ if for any function $p: [0, +\infty) \rightarrow \mathbb{R}$ such that for any $t \geq 0$, $q \leq p(t) \leq 1 + (q-1) e^{\frac{4t}{c}}$,

$$\|\mathcal{P}_t: \mathbb{L}_q(\sigma_{\text{Tr}}) \rightarrow \mathbb{L}_q(\mathcal{N}, \mathbb{L}_{p(t)}(\sigma_{\text{Tr}}))\| \leq \exp \left\{ 2d \left(\frac{1}{q} - \frac{1}{p(t)} \right) \right\}. \quad (\text{HC}_{q,\mathcal{N}}(c, d))$$

The first main result of this chapter is the following generalisation of Gross' integration lemma that establishes the equivalence between hypercontractivity and the log-Sobolev inequality for a faithful QMS:

Theorem 8.2.2. *Let $(\mathcal{P}_t)_{t \geq 0}$ be a faithful QMS on $\mathcal{B}(\mathcal{H})$ and let $q \geq 1$, $c > 0$ and $d \geq 0$.*

- (i) *If $\text{HC}_{q,\mathcal{N}}(c, d)$ holds, then $\text{LSI}_{q,\mathcal{N}}(c, d)$ holds.*
- (ii) *If $\text{LSI}_{\tilde{q},\mathcal{N}}(c, d)$ holds for all $\tilde{q} \geq q$, then $\text{HC}_{q,\mathcal{N}}(c, d + \ln \sqrt{|I|})$ holds, where $|I|$ denotes the number of blocks of $\mathcal{N}$ in (0.9).*

This theorem is quite surprising compared to the (classical and quantum) primitive case, where there is an exact equivalence between hypercontractivity and the logarithmic Sobolev inequality (i.e.

¹Here, the operators $I_{q,p}$ are defined with respect to the reference state σ_{Tr} .

with the same constant). In general, the term $\ln \sqrt{|I|}$ appearing in the theorem is not optimal (see end of Section 8.4.1).

The case where $\mathcal{N}$ is a factor and where the QMS is unital and trace-preserving was proved in [Beigi and King, 2016], but only in the case $d = 0$. However, the authors failed to give an example where the constant c is finite. We shall actually prove in Section 8.5 that this is impossible. More generally, we prove that, as soon as the QMS is truly non-primitive and non-invertible (that is, not a unitary evolution), necessarily $c < +\infty$ implies $d > 0$.

Remark also that the last statement is weaker than in the classical case, when one only needs to assume that the weak LSI holds for $\tilde{q} = q$. We recall that this is due to the fact that the regularity conditions seen in Lemma 2.4.2 always hold in the commutative setting, which ensures that $\text{LSI}_2(c, d)$ implies $\text{LSI}_{q, \mathcal{N}}(2c, d)$ for all $q \geq 2$, and even $\text{LSI}_{q, \mathcal{N}}(c, d)$ in the reversible case. This condition needs to be assumed in the general quantum setting, even in the primitive case. A generator $\mathcal{L}$ of a QMS $(\mathcal{P}_t)_{t \geq 0}$ is said to be *strongly $\mathbb{L}_p$ -regular* if there exists $d_0 \geq 0$ such that for all $p \geq 1$ and all $X \in \mathcal{B}_{sa}(\mathcal{H})$,

$$d_0 \|X\|_{\mathbb{L}_p(\sigma_{\text{Tr}})}^p + \mathcal{E}_{p, \mathcal{L}}(X) \geq \mathcal{E}_{2, \mathcal{L}}(\text{I}_{2, p}(X)). \quad (\text{s-}\mathbb{L}_p(d_0))$$

With these definitions, we can prove the following theorem.

Theorem 8.2.3. *Assume that $\text{LSI}_{2, \mathcal{N}}(c, d)$ holds. If the generator $\mathcal{L}$ is strongly $\mathbb{L}_p$ -regular for some $d_0 \geq 0$, then $\text{LSI}_{q, \mathcal{N}}(c, d + c d_0/2)$ holds for all $q \geq 1$, so that $\text{HC}_{2, \mathcal{N}}(c, d + \ln \sqrt{|I|} + c d_0/2)$ holds.*

The last two theorems generalise Theorem 3.8 of [Olkiewicz and Zegarlinski, 1999] as well as Theorem 15 of [Kastoryano and Temme, 2013]. Moreover, it was conjectured in [Kastoryano and Temme, 2013] that KMS-symmetric QMS are strongly $\mathbb{L}_p$ -regular, again with $d_0 = 0$. This was shown to hold in [Bardet, 2017] under the condition of σ_{Tr} -DBC and without the primitive assumption (cf. Section 5.4). For a KMS-symmetric QMS, a straightforward extension of the proof of Proposition 5.2 of [Olkiewicz and Zegarlinski, 1999] implies that the strong regularity of $\mathcal{L}$ always holds, with $d_0 = \|\mathcal{L} : \mathbb{L}_2(\sigma_{\text{Tr}}) \rightarrow \mathbb{L}_2(\sigma_{\text{Tr}})\| + 1$. These remarks motivate the following corollary of Theorem 8.2.3:

Corollary 8.2.4. *Assume that $\text{LSI}_{2, \mathcal{N}}(c, d)$ holds. Then:*

- (i) *If $(\mathcal{P}_t)_{t \geq 0}$ is KMS-symmetric, then $\text{HC}_{2, \mathcal{N}}(c, d + \ln \sqrt{|I|} + c(\|\mathcal{L} : \mathbb{L}_2(\sigma_{\text{Tr}}) \rightarrow \mathbb{L}_2(\sigma_{\text{Tr}})\| + 1))$ holds.*
- (ii) *If $\mathcal{L}$ satisfies σ_{Tr} -DBC, then $\text{HC}_{2, \mathcal{N}}(c, d + \ln \sqrt{|I|})$ holds.*

We also prove that it is always possible to get a weak DF-2-log-Sobolev inequality with a universal weak DF-2-log-Sobolev constant from any weak DF-2-log-Sobolev inequality, hence extending Theorem 4.2 of [Olkiewicz and Zegarlinski, 1999] to the non-primitive case.

Theorem 8.2.5. *Assume that $\text{LSI}_{2, \mathcal{N}}(c, d)$ holds and denote by $\lambda(\mathcal{L})$ the spectral gap of $(\mathcal{L} + \hat{\mathcal{L}})/2$. Then $\text{LSI}_{2, \mathcal{N}}(c + 2 \frac{d+1}{\lambda(\mathcal{L})}, d' = \ln \sqrt{2})$ holds.*

Finally, using the DF-hypercontractivity and complex interpolation methods, we derive the following universal DF-2-log-Sobolev constants:

Corollary 8.2.6. *Given a KMS-symmetric QMS $(\mathcal{P}_t)_{t \geq 0}$ with spectral gap $\lambda(\mathcal{L})$, $\text{LSI}_{2, \mathcal{N}}(c, \ln \sqrt{2})$ holds, with*

$$c \leq \frac{\ln(\|\sigma_{\text{Tr}}^{-1}\|_{\infty}) + 2}{\lambda(\mathcal{L})}.$$

8.2.1. Application to decoherence rates

Given a QMS $(\mathcal{P}_t)_{t \geq 0}$, its *decoherence time* is defined as:

$$\tau_{\text{deco}}(\varepsilon) := \inf \{t \geq 0 : \|\mathcal{P}_{*t}(\rho - E_{\mathcal{N}*}(\rho))\|_1 \leq \varepsilon, \forall \rho \in \mathcal{D}(\mathcal{H})\}.$$

The standard method to obtain estimates for $\tau(\varepsilon)$ in the primitive case is to use Pinsker's inequality in order to upper bound the trace distance in terms of the relative entropy, which decays exponentially fast with decay rate provided by $\alpha_1(\mathcal{L})$ [Diaconis and Saloff-Coste, 1996a, Kastoryano and Temme, 2013] (cf. Chapters 3 and 7). An estimate on the constant $\alpha_1(\mathcal{L})$ can then be found in terms of $\alpha_2(\mathcal{L})$, under the condition that the corresponding weak LSI₂ constant is null. However we prove in Section 8.5 that this condition holds only for primitive and unitary evolution. In all the other cases when there is only access to a weak DF-log-Sobolev inequality, we can fortunately still derive bounds on the decoherence times by extending a technique already used in the classical case in [Zegarlinski, 1995, Diaconis and Saloff-Coste, 1996a], by combining Poincaré's inequality and the weak DF-hypercontractivity property of the semigroup (cf. Section 3.2).

Proposition 8.2.7. *Assume that a QMS $(\mathcal{P}_t)_{t \geq 0}$ satisfies $\text{HC}_{2,\mathcal{N}}(c, d)$, and that $\|\sigma_{\text{Tr}}^{-1}\|_{\infty} \geq e$. Then, given $t = \frac{c}{4} \ln \ln \|\sigma_{\text{Tr}}^{-1}\|_{\infty} + \frac{\kappa}{\lambda(\mathcal{L})}$, $\kappa > 0$:*

$$\forall \rho \in \mathcal{D}(\mathcal{H}), \quad \|\mathcal{P}_{*t}(\rho - E_{\mathcal{N}*}(\rho))\|_1 \leq \max_{i \in I} \sqrt{d_{\mathcal{H}_i}} e^{1+d-\kappa}, \quad (8.9)$$

where the coefficients $d_{\mathcal{H}_i}$ are the dimensions of the spaces $\mathcal{H}_i$ arising in the decomposition of $\mathcal{N}$ given by (0.9). The above inequality provides the following bound on the decoherence time of the QMS:

$$\tau_{\text{deco}}(\varepsilon) \leq \frac{\ln(\max_{i \in I} \sqrt{d_{\mathcal{H}_i}} \varepsilon^{-1}) + 1 + d}{\lambda(\mathcal{L})} + \frac{c}{4} \ln \ln \|\sigma_{\text{Tr}}^{-1}\|_{\infty}.$$

Remark that the assumption on $\|\sigma_{\text{Tr}}^{-1}\|_{\infty}$ is not restrictive: it means that the lowest eigenvalue of σ_{Tr} has to be smaller than $1/e$. In particular, it always holds when $d_{\mathcal{H}} \geq 3$.

We see that having a weak constant $d = \sqrt{2}$ has in practice no effect on the decoherence-time. Remark also that the constant $\max_{i \in I} \sqrt{d_{\mathcal{H}_i}}$ is again a signature of the non-primitive case.

8.3. Advanced properties of DF norms

In the following proposition, we gather properties of $\mathbb{L}_p(\mathcal{N}, \mathbb{L}_q(\sigma_{\text{Tr}}))$, when $\mathcal{N} \equiv \mathcal{N}(\mathcal{P})$ is the decoherence-free algebra of a faithful QMS $(\mathcal{P}_t)_{t \geq 0}$, that will be particularly useful throughout this chapter:

Proposition 8.3.1. *Fix $1 \leq q \leq p \leq +\infty$ and let $(\mathcal{P}_t)_{t \geq 0}$ be a faithful QMS, with $\mathcal{N} \equiv \mathcal{N}(\mathcal{P})$. Then the following properties hold:*

- (i) $(\mathcal{P}_t)_{t \geq 0}$ is contractive with respect to $\|\cdot\|_{\mathbb{L}_q(\mathcal{N}, \mathbb{L}_p(\sigma_{\text{Tr}}))}$ for all $1 \leq q, p \leq +\infty$.
- (ii) For all $X \in \mathcal{N}(\mathcal{P})$, $\|X\|_{\mathbb{L}_q(\mathcal{N}, \mathbb{L}_p(\sigma_{\text{Tr}}))} = \|X\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}$.
- (iii) Ordering of the norms: for fixed $q \geq 1$ and for $q \leq p_1 \leq p_2$, $\|\cdot\|_{\mathbb{L}_q(\mathcal{N}, \mathbb{L}_{p_1}(\sigma_{\text{Tr}}))} \leq \|\cdot\|_{\mathbb{L}_q(\mathcal{N}, \mathbb{L}_{p_2}(\sigma_{\text{Tr}}))}$.
- (iv) In the case when $\mathcal{N} = \mathcal{N}(\mathcal{P}) \equiv \mathbb{C} \mathbb{1}$ and $\sigma_{\text{Tr}} \equiv \sigma$ is the unique invariant state, equality holds in the second inequality of (1.20) as well as in the first inequality of (1.21).

Proof.

- (i) In Proposition 5.2.1, we showed that for any $p \geq 1$, $t \geq 0$ and any invariant state σ , $(\mathcal{P}_t)_{t \geq 0}$ is an $\mathbb{L}_p(\sigma)$ contraction. In particular, it means for $\sigma = \sigma_{\text{Tr}}$ that for all $X \in \mathcal{B}(\mathcal{H})$,

$$\|\mathcal{P}_t(X)\|_{\mathbb{L}_p(\sigma_{\text{Tr}})} \leq \|X\|_{\mathbb{L}_p(\sigma_{\text{Tr}})}.$$

Assume now that $1 \leq q \leq p \leq +\infty$. We first prove that $(\mathcal{P}_t)_{t \geq 0}$ is contractive for the $\|\cdot\|_{\mathbb{L}_p(\mathcal{N}, \mathbb{L}_q(\sigma_{\text{Tr}}))}$ norm. By definition,

$$\begin{aligned} \|\mathcal{P}_t(X)\|_{\mathbb{L}_p(\mathcal{N}, \mathbb{L}_q(\sigma_{\text{Tr}}))} &= \sup_{\substack{A, B \in \mathcal{N}(\mathcal{P}), \\ \|A\|_{\mathbb{L}_{2r}(\sigma_{\text{Tr}})}, \|B\|_{\mathbb{L}_{2r}(\sigma_{\text{Tr}})}=1}} \|\mathcal{P}_t(A)\mathcal{P}_t(X)\mathcal{P}_t(B)\|_{\mathbb{L}_q(\sigma_{\text{Tr}})} \\ &= \sup_{\substack{A, B \in \mathcal{N}(\mathcal{P}), \\ \|A\|_{\mathbb{L}_{2r}(\sigma_{\text{Tr}})}, \|B\|_{\mathbb{L}_{2r}(\sigma_{\text{Tr}})}=1}} \|\mathcal{P}_t(AXB)\|_{\mathbb{L}_q(\sigma_{\text{Tr}})} \\ &\leq \sup_{\substack{A, B \in \mathcal{N}(\mathcal{P}), \\ \|A\|_{\mathbb{L}_{2r}(\sigma_{\text{Tr}})}, \|B\|_{\mathbb{L}_{2r}(\sigma_{\text{Tr}})}=1}} \|AXB\|_{\mathbb{L}_q(\sigma_{\text{Tr}})} = \|X\|_{\mathbb{L}_p(\mathcal{N}, \mathbb{L}_q(\sigma_{\text{Tr}}))}, \end{aligned}$$

where $\frac{1}{r} = \frac{1}{q} - \frac{1}{p}$. Here the first line follows from the fact that $(\mathcal{P}_t)_{t \geq 0}$ acts unitarily on $\mathcal{N}$, with associated Hamiltonian H_W that commutes with σ_{Tr} (cf. Corollary 6.2.6). The second line follows by definition of a decoherence-free algebra, and the third one from the contractivity of $\mathcal{P}_t$ as a map from $\mathbb{L}_q(\sigma_{\text{Tr}})$ to $\mathbb{L}_q(\sigma_{\text{Tr}})$. The case of $\|\cdot\|_{\mathbb{L}_q(\mathcal{N}, \mathbb{L}_p(\sigma_{\text{Tr}}))}$ follows by the duality of Proposition 1.1.8 (ii) and Hölder's inequality (Proposition 1.1.8 (i)):

$$\begin{aligned} \|\mathcal{P}_t(X)\|_{\mathbb{L}_q(\mathcal{N}, \mathbb{L}_p(\sigma_{\text{Tr}}))} &= \sup_{\|Y\|_{\mathbb{L}_{\hat{q}}(\mathcal{N}, \mathbb{L}_{\hat{p}}(\sigma_{\text{Tr}}))} \leq 1} |\langle Y, \mathcal{P}_t(X) \rangle_{\sigma_{\text{Tr}}}| \\ &= \sup_{\|Y\|_{\mathbb{L}_{\hat{q}}(\mathcal{N}, \mathbb{L}_{\hat{p}}(\sigma_{\text{Tr}}))} \leq 1} |\langle \hat{\mathcal{P}}_t(Y), X \rangle_{\sigma_{\text{Tr}}}| \\ &\leq \sup_{\|Y\|_{\mathbb{L}_{\hat{q}}(\mathcal{N}, \mathbb{L}_{\hat{p}}(\sigma_{\text{Tr}}))} \leq 1} \|\hat{\mathcal{P}}_t(Y)\|_{\mathbb{L}_{\hat{q}}(\mathcal{N}, \mathbb{L}_{\hat{p}}(\sigma_{\text{Tr}}))} \|X\|_{\mathbb{L}_q(\mathcal{N}, \mathbb{L}_p(\sigma_{\text{Tr}}))}, \end{aligned}$$

where $\frac{1}{p} + \frac{1}{\hat{p}} = 1$ and $\frac{1}{q} + \frac{1}{\hat{q}} = 1$. We conclude by using the above proof of DF-contractivity for $1 \leq \hat{p} \leq \hat{q} \leq +\infty$, applied to the dual QMS $(\hat{\mathcal{P}}_t)_{t \geq 0}$.

- (ii) This is point (viii) of Proposition 1.1.8 for $\mathcal{N} \equiv \mathcal{N}(\mathcal{P})$.
- (iii) This is point (iv) of Proposition 1.1.8 for $\mathcal{N} \equiv \mathcal{N}(\mathcal{P})$.
- (iv) This is obvious by definition.

□

8.3.1. Differentiation of the decoherence-free norms

As in the primitive case, the equivalence between hypercontractivity and the log-Sobolev inequality relies on a formula for the differentiation of the decoherence-free norms, commonly called Gross' integration Lemma. In the bipartite case, where $\mathcal{N}(\mathcal{P}) = \mathcal{B}(\mathcal{H}_A) \otimes \mathbb{1}_{\mathcal{H}_B}$ and the invariant state is the maximally mixed state, this differentiation was done in [Beigi and King, 2016]. Here we generalise this result to the case of the amalgamated $\mathbb{L}_p$ norms associated to a faithful QMS. The next lemma, which extends Lemma 5 of [Kastoryano and Temme, 2013], provides a physical interpretation of the DF- $\mathbb{L}_p$ relative entropies in terms of the *quantum relative entropy* of a state and its projection onto the decoherence-free subalgebra. Recall that the quantum relative entropy $D(\rho\|\sigma)$ of two states

$\rho, \sigma \in \mathcal{D}(\mathcal{H})$ is given by

$$D(\rho\|\sigma) := \begin{cases} \text{Tr}(\rho(\ln \rho - \ln \sigma)) & \text{supp}(\rho) \subset \text{supp}(\sigma), \\ +\infty & \text{otherwise.} \end{cases} \quad (8.10)$$

Lemma 8.3.2. *Let $\rho \in \mathcal{D}_+(\mathcal{H})$ and $X \in \mathcal{B}(\mathcal{H})$ positive definite, then*

- (i) $\text{Ent}_{q,\mathcal{N}}(\Gamma_{\sigma_{\text{Tr}}}^{-\frac{1}{q}}(\rho^{\frac{1}{q}})) = D(\rho\|\rho_{\mathcal{N}})$ for any $q \geq 1$.
- (ii) If $X \in \mathcal{N}(\mathcal{P})$, then for any $q \geq 1$, $\text{Ent}_{q,\mathcal{N}}(X) = 0$.
- (iii) $\text{Ent}_{p,\mathcal{N}}(X) = \text{Ent}_{2,\mathcal{N}}(\text{I}_{2,p}(X))$ for any $p \geq 1$.

Proof.

- (i) For $X = \Gamma_{\sigma_{\text{Tr}}}^{-\frac{1}{q}}(\rho^{\frac{1}{q}})$, Equation (8.7) reduces to

$$\text{Ent}_{q,\mathcal{N}}(X) = \text{Tr}(\rho \ln \rho) - \text{Tr}(\rho \ln E_{\mathcal{N}}[\Gamma_{\sigma_{\text{Tr}}}^{-1}(\rho)]) - \text{Tr}(\rho \ln \sigma_{\text{Tr}}). \quad (8.11)$$

Now, $\text{Tr}(\rho \ln E_{\mathcal{N}}[\Gamma_{\sigma_{\text{Tr}}}^{-1}(\rho)]) = \text{Tr}(\rho_{\mathcal{N}} \ln E_{\mathcal{N}}[\Gamma_{\sigma_{\text{Tr}}}^{-1}(\rho)])$. Using Equation (1.17) together with $[\sigma_{\text{Tr}}, E_{\mathcal{N}*}(\rho)] = 0$, we arrive at

$$\text{Tr}(\rho_{\mathcal{N}} \ln E_{\mathcal{N}}[\Gamma_{\sigma_{\text{Tr}}}^{-1}(\rho)]) = D(\rho_{\mathcal{N}}\|\sigma_{\text{Tr}}).$$

Substituting the above right hand side into (8.11), we finally arrive at (cf. [Bardet, 2017])

$$\text{Ent}_{q,\mathcal{N}}(Y) = D(\rho\|\sigma_{\text{Tr}}) - D(\rho_{\mathcal{N}}\|\sigma_{\text{Tr}}) = D(\rho\|\rho_{\mathcal{N}}). \quad (8.12)$$

- (ii) This is a simple consequence of (i) together with the fact that if $X \in \mathcal{N}(\mathcal{P})$, $[X, \sigma_{\text{Tr}}] = 0$ and

$$\rho_{\mathcal{N}} := E_{\mathcal{N}*}[\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(X)^q] = E_{\mathcal{N}*}[\Gamma_{\sigma_{\text{Tr}}}(X^q)] = \Gamma_{\sigma_{\text{Tr}}}(E_{\mathcal{N}}[X^q]) = \Gamma_{\sigma_{\text{Tr}}}(X^q) = (\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(X))^q \equiv \rho.$$

- (iii) follows by direct computation.

□

The proof of next theorem follows closely the one of Theorem 7 of [Beigi and King, 2016], and is discussed in Appendix 8.A for sake of clarity. It can be seen as both a generalisation of the differentiation done in the primitive case in [Olkiewicz and Zegarlinski, 1999] to non-primitive QMS (see also Lemma 14 of [Kastoryano and Temme, 2013]), and the one carried out for the CB-norm in [Beigi and King, 2016] to the non-unital case.

Theorem 8.3.3. *Let $t \mapsto p(t)$ be a twice continuously differentiable increasing function in a neighborhood of 0, with $p(0) = q \geq 1$. Also let $t \mapsto Y(t) \in \mathcal{B}(\mathcal{H})$ be an operator-valued twice continuously differentiable function, where $Y(t)$ is positive definite in a neighborhood of 0, and denote $Y \equiv Y(0)$. Then*

$$\left. \frac{d}{dt} \|Y(t)\|_{\mathbb{L}_q(\mathcal{N}, \mathbb{L}_{p(t)}(\sigma_{\text{Tr}}))} \right|_{t=0} = \frac{p'(0)}{q^2 \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^{q-1}} \left(\text{Ent}_{q,\mathcal{N}}(Y) + \frac{q^2}{p'(0)} \text{Tr} \left(\left[\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y) \right]^{q-1} \Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y'(0)) \right) \right).$$

We shall apply this theorem to different situations. Perhaps the most relevant one is when $Y(t)$ models the evolution of an initial observable $X \in \mathcal{B}(\mathcal{H})$ under the QMS $(\mathcal{P}_t)_{t \geq 0}$. We state it as a corollary.

Corollary 8.3.4. *For any positive definite $X \in \mathcal{B}(\mathcal{H})$,*

$$\left. \frac{d}{dt} \|\mathcal{P}_t(X)\|_{\mathbb{L}_q(\mathcal{N}, \mathbb{L}_{p(t)}(\sigma_{\text{Tr}}))} \right|_{t=0} = \frac{p'(0)}{q^2 \|X\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^{q-1}} \left(\text{Ent}_{q, \mathcal{N}}(X) - \frac{4(q-1)}{p'(0)} \mathcal{E}_{q, \mathcal{L}}(X) \right). \quad (8.13)$$

8.3.2. Almost uniform convexity

In this subsection, we study an analogue of the well-known uniform convexity of the Schatten norms proved in [Ball et al., 1994]. This analogue was proved in the context of weighted $\mathbb{L}_p(\sigma)$ norms by [Olkiewicz and Zegarlinski, 1999]. This will be an essential tool when proving universal lower bounds on the weak DF-log-Sobolev constants. This inequality states that for all X positive semidefinite, any full-rank state σ , and all $p \in [1, 2]$,

$$\|X\|_{\mathbb{L}_p(\sigma)}^2 \geq (p-1) \|X - \text{Tr}(\sigma X)\|_{\mathbb{L}_p(\sigma)}^2 + \text{Tr}(\sigma X)^2. \quad (8.14)$$

For the Schatten norms, this inequality can be seen as a consequence of Clarkson inequalities (see [Pisier and Xu, 2003] for a discussion of this fact). It has many important applications in the theory of non-commutative $\mathbb{L}_p$ spaces, such as yielding the optimal constant for Fermionic hypercontractivity [Carlen and Lieb, 1993]. We shall prove however in Section 8.5 that the inequality does not hold for the amalgamated $\mathbb{L}_p$ spaces. Instead, in this section we prove a weaker form of it.

It will be useful to denote by $\mathcal{S}_{\mathbb{L}_1(\sigma_{\text{Tr}})}^+$ the set of positive definite operators on the sphere of radius 1 in $\mathbb{L}_1(\text{Tr})$. Then, for $X \in \mathcal{B}(\mathcal{H})$, $A \in \mathcal{N}(\mathcal{P}) \cap \mathcal{S}_{\mathbb{L}_1(\sigma_{\text{Tr}})}^+$ and $p \geq 1$, we define

$$\Phi(X, A, p) := \left\| \Gamma_A^{-1/r}(X) \right\|_p = \text{Tr} \left[\left| A^{-\frac{1}{2r}} X A^{-\frac{1}{2r}} \right|^p \right]^{\frac{1}{p}}, \quad (8.15)$$

where we recall that $1/r = |1/2 - 1/p|$. Remark that for all positive semidefinite $X \in \mathcal{B}(\mathcal{H})$ and all $A \in \mathcal{N}(\mathcal{P}) \cap \mathcal{S}_{\mathbb{L}_1(\sigma_{\text{Tr}})}^+$, $\Phi(\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{2}}(X), A, 2) = \|X\|_{2, \sigma_{\text{Tr}}}$. We shall prove that a similar inequality as (8.14) holds for Φ , which we subsequently refer to as *almost uniform convexity*.

Lemma 8.3.5. *The two following properties hold:*

(i) *For all $X \in \mathcal{B}_{sa}^+(\mathcal{H})$, $A \in \mathcal{N}(\mathcal{P}) \cap \mathcal{S}_{\mathbb{L}_1(\sigma_{\text{Tr}})}^+$ and all $1 \leq p \leq 2$,*

$$\Phi(\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{p}}(X), A, p)^2 \geq (p-1) \Phi(\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{p}}(X - E_{\mathcal{N}}[X]), A, p)^2 + \Phi(\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{p}}(E_{\mathcal{N}}[X]), A, p)^2. \quad (8.16)$$

(ii) *For all $X \in \mathcal{B}_{sa}^+(\mathcal{H})$ and $A \in \mathcal{N}(\mathcal{P}) \cap \mathcal{S}_{\mathbb{L}_1(\sigma_{\text{Tr}})}^+$,*

$$\begin{aligned} \left. \frac{\partial}{\partial p} \Phi(\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{p}}(X), A, p)^2 \right|_{p=2} &\leq \left. \frac{\partial}{\partial p} \Phi(\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{p}}(X - E_{\mathcal{N}}[X]), A, p)^2 \right|_{p=2} \\ &\quad + \left. \frac{\partial}{\partial p} \Phi(\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{p}}(E_{\mathcal{N}}[X]), A, p)^2 \right|_{p=2} \\ &\quad + \|X - E_{\mathcal{N}}[X]\|_{\mathbb{L}_2(\sigma_{\text{Tr}})}^2. \end{aligned} \quad (8.17)$$

Proof. We follow the proof of Lemma 2.9 in [Olkiewicz and Zegarlinski, 1999] in order to prove the

first claim. We adopt the following notations. For $0 \leq t \leq 1$, define

$$\begin{aligned} X(t) &= E_{\mathcal{N}}[X] + t(X - E_{\mathcal{N}}[X]), \\ \varphi(t) &= \Phi(\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{p}}(X(t)), A, p)^2, \\ h &= \Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{p}} \circ \Gamma_A^{-\frac{1}{r}}(X - E_{\mathcal{N}}[X]). \end{aligned}$$

Then, Equation (8.16) reduces to:

$$\varphi(1) \geq (p-1)\|h\|_p^2 + \varphi(0). \quad (8.18)$$

This inequality follows directly from:

1. $\varphi'(0) = 0$;
2. $\varphi''(t) \geq 2(p-1)\|h\|_p^2$ for all $0 \leq t \leq 1$.

We start by computing $\varphi'(t)$. Writting $Z(t) = \Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{p}} \circ \Gamma_A^{-\frac{1}{r}}(X(t))$, we have by integral representation that for all $0 \leq t \leq 1$

$$\varphi'(t) = 2 \text{Tr}[h Z(t)^{p-1}] \text{Tr}[Z(t)^p]^{2/p-1}, \quad (8.19)$$

$$\varphi''(t) \geq 2 \frac{\partial}{\partial t} (\text{Tr}[h Z(t)^{p-1}]) \text{Tr}[Z(t)^p]^{2/p-1}. \quad (8.20)$$

We start by proving claim 1. First remark that, since elements of $\mathcal{N}$ commute with σ_{Tr} ,

$$Z(0)^{p-1} = \sigma_{\text{Tr}}^{\frac{1}{2} - \frac{1}{2p}} E_{\mathcal{N}} \left[\Gamma_A^{-\frac{1}{r}}(X) \right]^{p-1} \sigma_{\text{Tr}}^{\frac{1}{2} - \frac{1}{2p}}.$$

Therefore $\text{Tr}[h Z(0)^{p-1}] = \langle X - E_{\mathcal{N}}[X], B \rangle_{\sigma_{\text{Tr}}}$ where $B = \Gamma_A^{-\frac{1}{r}} \left(E_{\mathcal{N}} \left[\Gamma_A^{-\frac{1}{r}}(X) \right]^{p-1} \right) \in \mathcal{N}$. By Equation (1.16) we get that $\text{Tr}[h Z^{p-1}] = 0$ which results in $\varphi'(0) = 0$. The proof of claim 2 is a direct copy of the proof of Lemma 2.9 in [Olkiewicz and Zegarliniski, 1999] and we omit it. Hence, (8.16) holds.

In order to prove (ii), we rearrange the terms in (8.16) to get

$$\begin{aligned} (2-p)\Phi(\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{p}}(X - E_{\mathcal{N}}[X]), A, p)^2 \\ \geq \left(\Phi(\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{2}}(X), A, 2) + \Phi(\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{p}}(X), A, p) \right) \left(\Phi(\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{2}}(X), A, 2) - \Phi(\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{p}}(X), A, p) \right) \\ - \left(\Phi(\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{2}}(X - E_{\mathcal{N}}[X]), A, 2) + \Phi(\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{p}}(X - E_{\mathcal{N}}[X]), A, p) \right) \times \\ \times \left(\Phi(\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{2}}(X - E_{\mathcal{N}}[X]), A, 2) - \Phi(\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{p}}(X - E_{\mathcal{N}}[X]), A, p) \right) \\ - \left(\Phi(\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{2}}(E_{\mathcal{N}}[X]), A, 2) + \Phi(\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{p}}(E_{\mathcal{N}}[X]), A, p) \right) \times \\ \times \left(\Phi(\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{2}}(E_{\mathcal{N}}[X]), A, 2) - \Phi(\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{p}}(E_{\mathcal{N}}[X]), A, p) \right), \end{aligned}$$

where we used that

$$\Phi(\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{2}}(X), A, 2)^2 = \Phi(\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{2}}(X - E_{\mathcal{N}}[X]), A, 2)^2 + \Phi(\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{2}}(E_{\mathcal{N}}[X]), A, 2)^2.$$

(8.17) follows by dividing this inequality by $2-p$ and taking the limit $p \rightarrow 2$. $\square$

8.4. DF-hypercontractivity and the log-Sobolev inequality

In this section we state and prove the main results of this article. In Section 8.4.1, we prove the equivalence between hypercontractivity for the amalgamated norms and the DF-log-Sobolev inequality. In Section 8.4.2, we prove that the weak constant in the DF-log-Sobolev inequality can always be upper bounded by a universal constant, namely $\ln \sqrt{2}$. In Section 8.4.3, we show how to derive estimates on the log-Sobolev constants using interpolation techniques. Finally, we combine these two last results in order to obtain generic bounds on both constants.

8.4.1. Equivalence between HC and LSI

Here, we state and prove the main result of this section, that is, the equivalence between the DF-log-Sobolev inequality and DF-hypercontractivity.

Theorem 8.4.1. *Let $(\mathcal{P}_t)_{t \geq 0}$ be a faithful QMS on $\mathcal{B}(\mathcal{H})$ with associated generator $\mathcal{L}$, and let $q \geq 1$, $d \geq 0$ and $p(t) = 1 + (q-1)e^{\frac{4t}{c}}$ for some constant $c > 0$. Then*

- (i) *If $\text{HC}_{q,\mathcal{N}}(c,d)$ holds, then $\text{LSI}_{q,\mathcal{N}}(c,d)$ holds.*
- (ii) *If $\text{LSI}_{p(t),\mathcal{N}}(c,d)$ holds for all $t \geq 0$, then $\text{HC}_{q,\mathcal{N}}(c, d + \ln \sqrt{|I|})$ holds, where $|I|$ denotes the number of blocks in the decomposition of $\mathcal{N}$ as given in Equation (0.9).*

Remark 8.4.2. For primitive evolution, $|I| = 1$ and we recover the equivalence between hypercontractivity and the logarithmic Sobolev inequality of [Olkiewicz and Zegarlinski, 1999]. The equivalence is also achieved in the more general situation where $\mathcal{N}$ is a factor, that is, in the situation of Section 10.2. Below, we discuss why the term $\ln \sqrt{|I|}$ may not be optimal.

Proof. We first prove (i). For $X > 0$, define the function

$$F : [0, +\infty) \ni t \mapsto \exp \left\{ -2d \left(\frac{1}{q} - \frac{1}{p(t)} \right) \right\} \|\mathcal{P}_t(X)\|_{\mathbb{L}_q(\mathcal{N}, \mathbb{L}_{p(t)}(\sigma_{\text{Tr}}))},$$

where $p(t) := 1 + (q-1)e^{\frac{4t}{c}}$. $\text{HC}_{q,\mathcal{N}}(c,d)$ implies that $\ln F(t) \leq \ln F(0)$ for all $t \geq 0$, with equality at $t = 0$. Therefore,

$$\left. \frac{d \ln F(t)}{dt} \right|_{t=0^+} = -2d \frac{p'(0)}{q^2} + \frac{d}{dt} \ln \|\mathcal{P}_t(X)\|_{\mathbb{L}_q(\mathcal{N}, \mathbb{L}_{p(t)}(\sigma_{\text{Tr}}))} \Big|_{t=0^+} \leq 0.$$

Using Equation (8.13), the above inequality reduces to

$$-2d + \frac{1}{\|X\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^q} (\text{Ent}_{q,\mathcal{N}}(X) - c \mathcal{E}_{q,\mathcal{L}}(X)) \leq 0,$$

which yields $\text{LSI}_{q,\mathcal{N}}(c,d)$.

To prove (ii), we proceed by contradiction, similarly to [Beigi and King, 2016]. The main difference resides in the replacement of the norm by an auxiliary quantity that allows to control a remainder term that does not appear in the case where $\mathcal{N}$ is a factor. Without loss of generality, we assume that the evolution on elements of the decoherence-free algebra is trivial, and refer to Lemma 8.4.3 for a justification of this fact.

Next, assume that there exists an $X \in \mathcal{B}(\mathcal{H})$ such that hypercontractivity fails for this X . Following the same proof as Theorem 12 of [Devetak et al., 2006], we can show that it is sufficient to

consider that X is a positive definite operator. Indeed, for fixed $q \leq p$, if there exists $C > 0$ such that for any X positive definite,

$$\|\mathcal{P}_t(X)\|_{\mathbb{L}_q(\mathcal{N}, \mathbb{L}_p(\sigma_{\text{Tr}}))} \leq C \|X\|_{\mathbb{L}_q(\sigma_{\text{Tr}})},$$

then the inequality remains true for any $X \in \mathcal{B}(\mathcal{H})$. Without loss of generality, we also assume that $\|X\|_{\mathbb{L}_q(\sigma_{\text{Tr}})} = 1$. Then, suppose that there exists some time $t_0 > 0$ such that

$$\|\mathcal{P}_{t_0}(X)\|_{\mathbb{L}_q(\mathcal{N}, \mathbb{L}_{q(t_0)}(\sigma_{\text{Tr}}))} > \exp \left\{ 2(d + \ln \sqrt{|I|}) \left(\frac{1}{q} - \frac{1}{q(t_0)} \right) \right\}.$$

Define, for $\varepsilon > 0$,

$$\tilde{\varphi}(t) := \|\mathcal{P}_t(X)\|_{\mathbb{L}_q(\mathcal{N}, \mathbb{L}_{p(t)}(\sigma_{\text{Tr}}))} \exp \left\{ -2(d + \ln \sqrt{|I|}) \left(\frac{1}{q} - \frac{1}{p(t)} \right) \right\} - \varepsilon t,$$

where $\|\cdot\|_{\mathbb{L}_q(\mathcal{N}, \mathbb{L}_p(\sigma_{\text{Tr}}))}$ is defined in Section 8.B. By definition, $\|\mathcal{P}_{t_0}(X)\|_{\mathbb{L}_q(\mathcal{N}, \mathbb{L}_{p(t_0)}(\sigma_{\text{Tr}}))} \geq \|\mathcal{P}_{t_0}(X)\|_{\mathbb{L}_q(\mathcal{N}, \mathbb{L}_{p(t_0)}(\sigma_{\text{Tr}}))}$ so that $\tilde{\varphi}(t_0) > 1$ for ε small enough. Define the set $U := \{t \in [0, t_0] : \tilde{\varphi}(t) \leq 1\}$. Since $\mathcal{P}_0 = \text{id}$ and $p(0) = q$, we have $\tilde{\varphi}(0) = \|X\|_{\mathbb{L}_q(\sigma_{\text{Tr}})} = 1$, so that $U \neq \emptyset$. Let u be the supremum of the set U . By continuity of $t \mapsto \|\mathcal{P}_t(X)\|_{\mathbb{L}_q(\mathcal{N}, \mathbb{L}_{p(t)}(\sigma_{\text{Tr}}))}$ (cf. Lemma 8.B.4), $\tilde{\varphi}$ is continuous and therefore $u \in U$ and $u < t_0$. Now, by definition of u , for all $t \in (u, t_0]$, $\tilde{\varphi}(t) > 1 = \tilde{\varphi}(u)$. For $t > 0$, let $\tilde{A}(t)$ be the unique minimiser of

$$\mathcal{N}(\mathcal{P}) \cap \tilde{\mathcal{S}}_{\mathbb{L}_1}^+(\sigma_{\text{Tr}}) \ni A \mapsto \|A^{-s(t)/2} \mathcal{P}_t(X) A^{-s(t)/2}\|_{\mathbb{L}_{p(t)}(\sigma_{\text{Tr}})},$$

as characterised in Lemma 8.B.3, where $\tilde{\mathcal{S}}_{\mathbb{L}_1}^+(\sigma_{\text{Tr}})$ is defined in Section 8.B and $s(t) = \frac{1}{q} - \frac{1}{p(t)}$. Define

$$\mu(t) := \|\tilde{A}(u)^{-s(t)/2} \mathcal{P}_t(X) \tilde{A}(u)^{-s(t)/2}\|_{\mathbb{L}_{p(t)}(\sigma_{\text{Tr}})} \exp \left\{ -2(d + \ln \sqrt{|I|}) \left(\frac{1}{q} - \frac{1}{p(t)} \right) \right\} - \varepsilon t.$$

Therefore, for all $t \geq u$,

$$\mu(t) \geq \inf_{A \in \mathcal{N}(\mathcal{P}) \cap \tilde{\mathcal{S}}_{\mathbb{L}_1}^+(\sigma_{\text{Tr}})} \|A^{-s(t)/2} \mathcal{P}_t(X) A^{-s(t)/2}\|_{\mathbb{L}_{p(t)}(\sigma_{\text{Tr}})} \exp \left\{ -2(d + \ln \sqrt{|I|}) \left(\frac{1}{q} - \frac{1}{p(t)} \right) \right\} - \varepsilon t = \tilde{\varphi}(t)$$

and $\tilde{\varphi}(u) = \mu(u)$. Now, the derivative of $\mu(t)$ at $t = u$ can be computed using Equation (8.40) with $X(t) = \Gamma_{\sigma_{\text{Tr}}}^{1/p(t)} \circ \mathcal{P}_t(X)$ and $A = \tilde{A}(u)$. Given $M(t) := \tilde{A}(u)^{-s(t)/2} \mathcal{P}_t(X) \tilde{A}(u)^{-s(t)/2}$, one finds

$$\left. \frac{\partial}{\partial t} \right|_{t=u} \|\tilde{A}(u)^{-s(t)/2} \mathcal{P}_t(X) \tilde{A}(u)^{-s(t)/2}\|_{\mathbb{L}_{p(t)}(\sigma_{\text{Tr}})} \quad (8.21)$$

$$\begin{aligned} &= \frac{1}{\|M(u)\|_{p(u), \sigma_{\text{Tr}}}^{p(u)-1}} \left(-\frac{p'(u)}{p(u)^2} \text{Tr} \left(\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{p(u)}} (M(u))^{p(u)} \right) \ln \text{Tr} \left[\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{p(u)}} (M(u))^{p(u)} \right] \right. \\ &+ \frac{p'(u)}{p(u)^2} \text{Tr} \left[\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{p(u)}} (M(u))^{p(u)} \ln \Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{p(u)}} (M(u))^{p(u)} \right] - \frac{p'(u)}{p(u)^2} \text{Tr} \left[\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{p(u)}} (M(u))^{p(u)} \ln \tilde{A}(u) \right] \\ &\left. + \text{Tr} \left[\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{p(u)}} (M(u))^{p(u)-1} \tilde{A}(u)^{-\frac{s(u)}{2}} \left\{ \Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{p(u)}} \mathcal{L}(\mathcal{P}_u(X)) - \frac{p'(u) \{ \ln \sigma_{\text{Tr}}, \Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{p(u)}} (\mathcal{P}_u(X)) \}}{2p(u)^2} \right\} \tilde{A}(u)^{-\frac{s(u)}{2}} \right] \right). \end{aligned} \quad (8.22)$$

Define $\rho(u) := \Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{p(u)}} (M(u))^{p(u)}$. Then, Equation (8.22) simplifies into

$$\begin{aligned}
& \left. \frac{\partial}{\partial t} \right|_{t=u} \left\| \tilde{A}(u)^{-s(t)/2} \mathcal{P}_t(X) \tilde{A}(u)^{-s(t)/2} \right\|_{\mathbb{L}_{p(t)}(\sigma_{\text{Tr}})} \\
&= \frac{p'(u)}{p(u)^2 \|M(u)\|_{\mathbb{L}_{p(u)}(\sigma_{\text{Tr}})}^{p(u)-1}} \left(D(\rho(u) \| E_{\mathcal{N}*}(\rho(u))) - c \mathcal{E}_{p(u), \mathcal{L}}(M(u)) \right. \\
&\quad \left. - \text{Tr}(\rho(u)) \ln \text{Tr} \rho(u) + \text{Tr}(\rho(u) \ln E_{\mathcal{N}*}(\rho(u))) - \text{Tr}(\rho(u) \ln \tilde{A}(u)) - \text{Tr}(\rho(u) \ln \sigma_{\text{Tr}}) \right) \\
&= \frac{p'(u)}{p(u)^2 \|M(u)\|_{\mathbb{L}_{p(u)}(\sigma_{\text{Tr}})}^{p(u)-1}} \left\{ \text{Ent}_{p(u), \mathcal{N}}(M(u)) - c \mathcal{E}_{p(u), \mathcal{L}}(M(u)) \right. \\
&\quad \left. + (-\text{Tr}(\rho(u)) \ln \text{Tr} \rho(u) + \text{Tr}(\rho(u) \ln E_{\mathcal{N}*}(\rho(u))) - \text{Tr}(\rho(u) \ln \tilde{A}(u)) - \text{Tr}(\rho(u) \ln \sigma_{\text{Tr}})) \right\},
\end{aligned} \tag{8.23}$$

where, in order to get the Dirichlet form, we also used that for any $A \in \mathcal{N}$ and $X \in \mathcal{B}(\mathcal{H})$, $\mathcal{P}_t(AXA) = A\mathcal{P}_t(X)A \rightarrow \mathcal{L}(AXA) = A\mathcal{L}(X)A$, since we assume the evolution to be trivial on $\mathcal{N}$. Using the expression for $\tilde{A}(u)$ derived in Equation (8.62),

$$\begin{aligned}
\ln \tilde{A}(u) &= \ln \sum_{i \in I} P_i \tilde{A}(u) P_i \\
&= \sum_{i \in I} \ln P_i \tilde{A}(u) P_i \\
&= \sum_{i \in I} -\ln |I| P_i - \ln \text{Tr}(P_i \rho(u) P_i) P_i - \ln P_i \sigma_{\text{Tr}} P_i + \ln P_i E_{\mathcal{N}*}(\rho(u)) P_i \\
&\geq -\ln |I| - \ln \sigma_{\text{Tr}} - \ln \text{Tr}(\rho(u)) + \ln E_{\mathcal{N}*}(\rho(u)).
\end{aligned}$$

Using this expression, Equation (8.23) leads to:

$$\begin{aligned}
& \left. \frac{\partial}{\partial t} \right|_{t=u} \left\| \tilde{A}(u)^{-s(t)/2} \mathcal{P}_t(X) \tilde{A}(u)^{-s(t)/2} \right\|_{\mathbb{L}_{p(t)}(\sigma_{\text{Tr}})} \\
&\leq \frac{p'(u)}{p(u)^2 \|M(u)\|_{\mathbb{L}_{p(u)}(\sigma_{\text{Tr}})}^{p(u)-1}} \left\{ \text{Ent}_{p(u), \mathcal{N}}(M(u)) - c \mathcal{E}_{p(u), \mathcal{L}}(M(u)) + \ln |I| \|M(u)\|_{\mathbb{L}_{p(u)}(\sigma_{\text{Tr}})}^{p(u)} \right\}.
\end{aligned}$$

Then, using the assumption that $\text{LSI}_{p(u), \mathcal{N}}(c, d)$ holds, we find that

$$\mu'(u) \leq -\varepsilon.$$

Therefore, there exists $\delta > 0$ such that $u + \delta \leq t_0$ and $\mu(u + \delta) \leq \mu(u)$. We then have

$$\tilde{\varphi}(u + \delta) \leq \mu(u + \delta) \leq \mu(u) = \tilde{\varphi}(u) \leq 1,$$

which is in contradiction with the very definition of u . □

Lemma 8.4.3. *Let $(\mathcal{P}_t)_{t \geq 0}$ be a faithful QMS of generator $\mathcal{L}$ and associated decoherence-free Hamiltonian H_W (cf. Corollary 6.2.6). Then $\text{HC}_{q, \mathcal{N}}(c, d)$ holds for $(\mathcal{P}_t)_{t \geq 0}$ if and only if it holds for the semigroup $(\mathcal{P}'_t)$ constructed from $(\mathcal{P}_t)_{t \geq 0}$ by removing the unitary evolution associated to H_W : $\mathcal{L}' = \mathcal{L} - i[H_W, \cdot]$. Consequently, $\text{LSI}_{q, \mathcal{N}}(c, d)$ holds for $(\mathcal{P}_t)_{t \geq 0}$ if and only if it holds for $(\mathcal{P}'_t)_{t \geq 0}$.*

Proof. From Theorem 6.2.5, the following holds for any $X \in \mathcal{B}(\mathcal{H})$:

$$\mathcal{P}'_t(X) = e^{-iH_W t} \mathcal{P}_t(X) e^{iH_W t}.$$

Therefore, for any $1 \leq q \leq p \leq \infty$

$$\begin{aligned}
 \|\mathcal{P}_t(X)\|_{\mathbb{L}_q(\mathcal{N}, \mathbb{L}_p(\sigma_{\text{Tr}}))} &= \inf_{\substack{A, B \in \mathcal{N}, Y \in \mathcal{B}(\mathcal{H}) \\ \mathcal{P}_t(X) = AYB}} \|A\|_{\mathbb{L}_{2r}(\sigma_{\text{Tr}})} \|B\|_{\mathbb{L}_{2r}(\sigma_{\text{Tr}})} \|Y\|_{\mathbb{L}_p(\sigma_{\text{Tr}})} \\
 &= \inf_{\substack{A, B \in \mathcal{N}, Y \in \mathcal{B}(\mathcal{H}) \\ \mathcal{P}'_t(X) = e^{-iH_W t} AYB e^{iH_W t}}} \|A\|_{\mathbb{L}_{2r}(\sigma_{\text{Tr}})} \|B\|_{\mathbb{L}_{2r}(\sigma_{\text{Tr}})} \|Y\|_{\mathbb{L}_p(\sigma_{\text{Tr}})} \\
 &= \inf_{\substack{A, B \in \mathcal{N}, Y \in \mathcal{B}(\mathcal{H}) \\ \mathcal{P}'_t(X) = AYB}} \|A\|_{\mathbb{L}_{2r}(\sigma_{\text{Tr}})} \|B\|_{\mathbb{L}_{2r}(\sigma_{\text{Tr}})} \|Y\|_{\mathbb{L}_p(\sigma_{\text{Tr}})} \\
 &= \|\mathcal{P}'_t(X)\|_{\mathbb{L}_q(\mathcal{N}, \mathbb{L}_p(\sigma_{\text{Tr}}))}.
 \end{aligned}$$

where the third line follows from the commutation of H_W with σ_{Tr} and the unitary invariance of Schatten norms. From this, the equivalence between the hypercontractivity properties of $(\mathcal{P}_t)_{t \geq 0}$ and $(\mathcal{P}'_t)_{t \geq 0}$ follows directly. From Theorem 8.4.1 (i) we also get the correspondence of the logarithmic Sobolev constants between the two semigroups. $\square$

In the above theorem, one needs $\text{LSI}_{\tilde{q}, \mathcal{N}}(c, d)$ to hold for any $\tilde{q} \geq q$ in order to conclude that $\text{HC}_{q, \mathcal{N}}(c, d + \ln \sqrt{|I|})$ holds. Under the assumption of regularity of the Dirichlet forms, it is enough to assume that it holds for $q = 2$ only.

Theorem 8.4.4. *Assume that $\text{LSI}_{2, \mathcal{N}}(c, d)$ holds. If the generator $\mathcal{L}$ is strongly $\mathbb{L}_p$ -regular for some $d_0 \geq 0$, then $\text{LSI}_{q, \mathcal{N}}(c, d + c d_0/2)$ holds for all $q \geq 1$, so that $\text{HC}_{2, \mathcal{N}}(c, d + \ln \sqrt{|I|} + c d_0/2)$ holds.*

Proof. From Lemma 8.3.2 (iii),

$$\begin{aligned}
 \text{Ent}_{q, \mathcal{N}}(X) &= \text{Ent}_{2, \mathcal{N}}(\text{I}_{2, q}(X)) \\
 &\leq (c \mathcal{E}_{2, \mathcal{L}}(\text{I}_{2, q}(X)) + 2d \|\text{I}_{2, q}(X)\|_{\mathbb{L}_2(\sigma_{\text{Tr}})}^2) \\
 &\leq c \mathcal{E}_{q, \mathcal{L}}(X) + 2 \left(d + \frac{c d_0}{2} \right) \|X\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^q,
 \end{aligned}$$

where in the last line we used that $\mathcal{E}_{2, \mathcal{L}}(\text{I}_{2, q}(X)) \leq \mathcal{E}_{q, \mathcal{L}}(X) + d_0 \|X\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^q$ by strong $\mathbb{L}_p$ -regularity. $\square$

It was shown in [Bardet, 2017] that any generator satisfying σ_{Tr} -DBC is strongly regular with constant $d_0 = 0$ (cf. Section 5.4). Furthermore, under the condition of KMS-symmetry, a straightforward extension of the proof of Proposition 5.2 of [Olkiewicz and Zegarlinski, 1999] to the case of a non-primitive QMS implies that the strong $\mathbb{L}_p$ -regularity of $\mathcal{L}$ always holds, with $d_0 = \|\mathcal{L} : \mathbb{L}_2(\sigma_{\text{Tr}}) \rightarrow \mathbb{L}_2(\sigma_{\text{Tr}})\| + 1$. The following corollary is a straightforward consequence of these two facts.

Corollary 8.4.5. *Assume that $\text{LSI}_{2, \mathcal{N}}(c, d)$ holds. Then:*

(i) *If $(\mathcal{P}_t)_{t \geq 0}$ is KMS-symmetric, then $\text{HC}_{2, \mathcal{N}}(c, d + \ln \sqrt{|I|} + c(\|\mathcal{L} : \mathbb{L}_2(\sigma_{\text{Tr}}) \rightarrow \mathbb{L}_2(\sigma_{\text{Tr}})\| + 1)/2)$ holds.*

(ii) *If $\mathcal{L}$ satisfies σ_{Tr} -DBC, then $\text{HC}_{2, \mathcal{N}}(c, d + \ln \sqrt{|I|})$ holds.*

Bounds on the amalgamated $\mathbb{L}_p$ norms of the identity: One can potentially hope to improve the bound (1.28) by applying Theorem 8.4.1 to the trivial QMS $(\text{id})_{t \geq 0}$:

Proposition 8.4.6. *Let $\mathcal{N}$ be a matrix algebra on $\mathcal{H}$ with decomposition (0.9), $E_{\mathcal{N}} : \mathcal{B}(\mathcal{H}) \rightarrow \mathcal{N}$ a conditional expectation onto $\mathcal{N}$ and $\sigma_{\text{Tr}} = E_{\mathcal{N}*}(d_{\mathcal{H}}^{-1} \mathbb{1})$. Then:*

- (i) Assume that $\|\text{id} : \mathbb{L}_2(\sigma_{\text{Tr}}) \rightarrow \mathbb{L}_2(\mathcal{N}, \mathbb{L}_p(\sigma_{\text{Tr}}))\| \leq C^{\frac{1}{2} - \frac{1}{p}}$ for some $C > 0$ and for all $p \geq 2$. Then $D(\rho \| E_{\mathcal{N}*}(\rho)) \leq \ln C$ for any density matrix $\rho \in \mathcal{D}(\mathcal{H})$.
- (ii) Conversely, assume that there exists a constant $C > 0$ such that $D(\rho \| E_{\mathcal{N}*}(\rho)) \leq \ln C$ for all density matrix $\rho \in \mathcal{D}(\mathcal{H})$. Then for any $p \geq 2$

$$\|\text{id} : \mathbb{L}_2(\sigma_{\text{Tr}}) \rightarrow \mathbb{L}_2(\mathcal{N}, \mathbb{L}_p(\sigma_{\text{Tr}}))\| \leq (|I|C)^{\frac{1}{2} - \frac{1}{p}}$$

where $|I|$ is the number of blocks in the decomposition (0.9) of $\mathcal{N}$.

Remark 8.4.7. In the proposition we ask that $\|\text{id} : \mathbb{L}_2(\sigma_{\text{Tr}}) \rightarrow \mathbb{L}_2(\mathcal{N}, \mathbb{L}_p(\sigma_{\text{Tr}}))\| \leq C^{\frac{1}{2} - \frac{1}{p}}$ for all $p \geq 2$. This is actually not needed, since by Riesz-Thorin interpolation Theorem this is equivalent to

$$\|\text{id} : \mathbb{L}_1(\sigma_{\text{Tr}}) \rightarrow \mathbb{L}_1(\mathcal{N}, \mathbb{L}_\infty(\sigma_{\text{Tr}}))\| \leq C.$$

We see here that it is central that the norms we use form an interpolating family of norms.

The last proposition is not optimal, which indicates that point (ii) in Theorem 8.4.1 may also not be, even for a non-trivial evolution. To see this, consider the situation where $\mathcal{N}$ is the algebra of diagonal operator in some orthonormal basis. In this case $|I|$ is equal to the dimension $d_{\mathcal{H}}$ of the Hilbert space $\mathcal{H}$ (the converse is also true: if $|I| = d_{\mathcal{H}}$ then $\mathcal{N}$ is commutative maximal). In this case $D(\rho \| E_{\mathcal{N}*}(\rho)) \leq \ln d_{\mathcal{H}}$, with equality for the maximally coherent state Ω :

$$\Omega = \frac{1}{d_{\mathcal{H}}} \sum_{i,j=1}^{d_{\mathcal{H}}} |e_i\rangle\langle e_j|,$$

where $(e_i)_{i=1,\dots,d_{\mathcal{H}}}$ is the orthonormal basis with respect to which the operators in $\mathcal{N}$ are diagonal. It means that Equation (1.28) saturates and that the implication in Proposition 8.4.6 (i) becomes an equivalence.

8.4.2. A universal upper bound on the weak log-Sobolev constant

Here and in the next section, we show how to get a DF log-Sobolev inequality with universal constants in terms of the spectral gap of the QMS. The first step is to prove that the weak log-Sobolev inequality together with the DF-Poincaré's inequality imply a universal weak log-Sobolev constant.

Theorem 8.4.8. Assume that $\text{LSI}_{2,\mathcal{N}}(c, d)$ holds and denote by $\lambda(\mathcal{L})$ the spectral gap of $\mathcal{L}$. Then $\text{LSI}_{2,\mathcal{N}}(c + 2\frac{d+1}{\lambda(\mathcal{L})}, d' = \ln \sqrt{2})$ holds.

One can obtain from the uniform convexity (8.14) the following inequality

$$\text{Ent}_{2,\sigma}(X) \leq \text{Ent}_{2,\sigma}(|X - \text{Tr}[\sigma X]|_2) + 2 \text{Var}_{\sigma}(X),$$

where $\text{Var}_{\sigma}(X) = \|X - \text{Tr}(\sigma X)\|_{\mathbb{L}_2(\sigma)}^2$ is the variance of $X \in \mathcal{B}_{sa}(\mathcal{H})$ under the state σ , and for any $Z \in \mathcal{B}_{sa}(\mathcal{H})$,

$$|Z|_2 := \Gamma_{\sigma}^{-\frac{1}{2}} |\Gamma_{\sigma}^{\frac{1}{2}}(Z)|.$$

From this we can derive the analogue result of Theorem 8.4.8 in the primitive case (see Theorem 4.2 of [Olkiewicz and Zegarlinski, 1999]). The extension of this result is the subject of the next proposition.

Proposition 8.4.9. For all $X \in \mathcal{B}_{sa}^+(\mathcal{H})$,

$$\text{Ent}_{2,\mathcal{N}}(X) \leq \text{Ent}_{2,\mathcal{N}}(|X - E_{\mathcal{N}}[X]|_2) + 2 \text{Var}_{\mathcal{N}}(X) + 2 \ln \sqrt{2} \|X\|_{\mathbb{L}_2(\sigma_{\text{Tr}})}^2. \quad (8.24)$$

Proof. We shall adopt the notations introduced in Section 8.3.2 and write for $Z \in \mathcal{B}_{sa}(\mathcal{H})$:

$$Z_{\mathcal{N}} = \frac{E_{\mathcal{N}}[I_{1,2}(Z)]}{\|Z\|_{\mathbb{L}_2(\sigma_{\text{Tr}})}^2}.$$

Using Equation (8.42) with $q = 2$ as well as Lemma 8.A.2 and Equation (8.8), we find that

$$\left. \frac{\partial}{\partial p} \Phi(\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{p}}(|Z|_2), A, p)^2 \right|_{p=2} = \frac{1}{2} \text{Ent}_{2,\mathcal{N}}(|Z|_2) + \frac{1}{2} \|Z\|_{\mathbb{L}_2(\sigma_{\text{Tr}})}^2 D(\Gamma_{\sigma_{\text{Tr}}}(Z_{\mathcal{N}}) \| \Gamma_{\sigma_{\text{Tr}}}(A)),$$

where Φ is defined in Equation (8.15), and where we used $I_{1,2}(Z) = I_{1,2}(|Z|_2)$ and $\| |Z|_2 \|_{\mathbb{L}_2(\sigma)} = \|Z\|_{\mathbb{L}_2(\sigma)}$. Consequently, by Equation (8.17) we get that for all $A \in \mathcal{N}(\mathcal{P}) \cap \mathcal{S}_{\mathbb{L}_1(\sigma_{\text{Tr}})}^+$ and for $p = \frac{\|E_{\mathcal{N}}[X]\|_{\mathbb{L}_2(\sigma_{\text{Tr}})}^2}{\|X\|_{\mathbb{L}_2(\sigma_{\text{Tr}})}^2}$,

$$\begin{aligned} \text{Ent}_{2,\mathcal{N}}(X) &\leq \text{Ent}_{2,\mathcal{N}}(|X - E_{\mathcal{N}}[X]|_2) + 2 \text{Var}_{\mathcal{N}}(X) \\ &\quad + \|X - E_{\mathcal{N}}[X]\|_{\mathbb{L}_2(\sigma_{\text{Tr}})}^2 D\left(\sigma_{\text{Tr}}^{\frac{1}{2}}(X - E_{\mathcal{N}}[X])_{\mathcal{N}} \sigma_{\text{Tr}}^{\frac{1}{2}} \| \sigma_{\text{Tr}}^{\frac{1}{2}} A \sigma_{\text{Tr}}^{\frac{1}{2}}\right) \\ &\quad + \|E_{\mathcal{N}}[X]\|_{\mathbb{L}_2(\sigma_{\text{Tr}})}^2 D\left(\sigma_{\text{Tr}}^{\frac{1}{2}}(E_{\mathcal{N}}[X])_{\mathcal{N}} \sigma_{\text{Tr}}^{\frac{1}{2}} \| \sigma_{\text{Tr}}^{\frac{1}{2}} A \sigma_{\text{Tr}}^{\frac{1}{2}}\right) \\ &\quad - \|X\|_{\mathbb{L}_2(\sigma_{\text{Tr}})}^2 D\left(\sigma_{\text{Tr}}^{\frac{1}{2}} X_{\mathcal{N}} \sigma_{\text{Tr}}^{\frac{1}{2}} \| \sigma_{\text{Tr}}^{\frac{1}{2}} A \sigma_{\text{Tr}}^{\frac{1}{2}}\right) \\ &= \text{Ent}_{2,\mathcal{N}}(|X - E_{\mathcal{N}}[X]|_2) + 2 \text{Var}_{\mathcal{N}}(X) \\ &\quad + \|X\|_{\mathbb{L}_2(\sigma_{\text{Tr}})}^2 \times \\ &\quad \times \left\{ p D\left(\sigma_{\text{Tr}}^{\frac{1}{2}}(E_{\mathcal{N}}[X])_{\mathcal{N}} \sigma_{\text{Tr}}^{\frac{1}{2}} \| \sigma_{\text{Tr}}^{\frac{1}{2}} A \sigma_{\text{Tr}}^{\frac{1}{2}}\right) + (1-p) D\left(\sigma_{\text{Tr}}^{\frac{1}{2}}(X - E_{\mathcal{N}}[X])_{\mathcal{N}} \sigma_{\text{Tr}}^{\frac{1}{2}} \| \sigma_{\text{Tr}}^{\frac{1}{2}} A \sigma_{\text{Tr}}^{\frac{1}{2}}\right) \right. \\ &\quad \left. - D\left(\sigma_{\text{Tr}}^{\frac{1}{2}} X_{\mathcal{N}} \sigma_{\text{Tr}}^{\frac{1}{2}} \| \sigma_{\text{Tr}}^{\frac{1}{2}} A \sigma_{\text{Tr}}^{\frac{1}{2}}\right), \right\} \end{aligned} \quad (8.25)$$

since by definition $\text{Var}_{\mathcal{N}}(X) = \|X - E_{\mathcal{N}}[X]\|_{\mathbb{L}_2(\sigma_{\text{Tr}})}^2$ and $\text{Ent}_{2,\mathcal{N}}(E_{\mathcal{N}}[X]) = 0$, and where we also used the fact that for any $Z \in \mathcal{B}_{sa}(\mathcal{H})$, $Z_{\mathcal{N}} = (|Z|_2)_{\mathcal{N}}$. Remark that, since $(E_{\mathcal{N}}[X])_{\mathcal{N}} = E_{\mathcal{N}}[X]^2 / \|E_{\mathcal{N}}[X]\|_{\mathbb{L}_2(\sigma_{\text{Tr}})}^2$,

$$\begin{aligned} X_{\mathcal{N}} &= \frac{E_{\mathcal{N}}[\sigma_{\text{Tr}}^{-1/2} | \sigma_{\text{Tr}}^{1/4} X \sigma_{\text{Tr}}^{1/4} |^2 \sigma_{\text{Tr}}^{-1/2}]}{\|X\|_{\mathbb{L}_2(\sigma_{\text{Tr}})}^2} \\ &= \frac{E_{\mathcal{N}}[\sigma_{\text{Tr}}^{-1/2} (\sigma_{\text{Tr}}^{1/4} X \sigma_{\text{Tr}}^{1/4})^2 \sigma_{\text{Tr}}^{-1/2}]}{\|X\|_{\mathbb{L}_2(\sigma_{\text{Tr}})}^2} \\ &= \frac{E_{\mathcal{N}}[\sigma_{\text{Tr}}^{-1/2} (\sigma_{\text{Tr}}^{1/4} (X - E_{\mathcal{N}}[X] + E_{\mathcal{N}}[X]) \sigma_{\text{Tr}}^{1/4})^2 \sigma_{\text{Tr}}^{-1/2}]}{\|X\|_{\mathbb{L}_2(\sigma_{\text{Tr}})}^2} \\ &= \frac{E_{\mathcal{N}}[\sigma_{\text{Tr}}^{-1/2} (\sigma_{\text{Tr}}^{1/4} E_{\mathcal{N}}[X] \sigma_{\text{Tr}}^{1/4})^2 \sigma_{\text{Tr}}^{-1/2}]}{\|X\|_{\mathbb{L}_2(\sigma_{\text{Tr}})}^2} + \frac{E_{\mathcal{N}}[\sigma_{\text{Tr}}^{-1/2} (\sigma_{\text{Tr}}^{1/4} (X - E_{\mathcal{N}}[X]) \sigma_{\text{Tr}}^{1/4})^2 \sigma_{\text{Tr}}^{-1/2}]}{\|X\|_{\mathbb{L}_2(\sigma_{\text{Tr}})}^2} \\ &\quad + \frac{E_{\mathcal{N}}[\sigma_{\text{Tr}}^{-1/4} (X - E_{\mathcal{N}}[X]) \sigma_{\text{Tr}}^{1/2} E_{\mathcal{N}}[X] \sigma_{\text{Tr}}^{-1/4}]}{\|X\|_{\mathbb{L}_2(\sigma_{\text{Tr}})}^2} + \frac{E_{\mathcal{N}}[\sigma_{\text{Tr}}^{-1/4} E_{\mathcal{N}}[X] \sigma_{\text{Tr}}^{1/2} (X - E_{\mathcal{N}}[X]) \sigma_{\text{Tr}}^{-1/4}]}{\|X\|_{\mathbb{L}_2(\sigma_{\text{Tr}})}^2} \quad (8.26) \\ &= p (E_{\mathcal{N}}[X])_{\mathcal{N}} + (1-p) (X - E_{\mathcal{N}}[X])_{\mathcal{N}}, \end{aligned}$$

where we used Pythagoras theorem $\|X\|_{\mathbb{L}_2(\sigma_{\text{Tr}})}^2 = \|X - E_{\mathcal{N}}[X]\|_{\mathbb{L}_2(\sigma_{\text{Tr}})}^2 + \|E_{\mathcal{N}}[X]\|_{\mathbb{L}_2(\sigma_{\text{Tr}})}^2$, and where the last two terms in Equation (8.26) can be shown to be equal to zero using Equations (0.5) and (6.18),

since

$$\begin{aligned}
E_{\mathcal{N}} \left[\sigma_{\text{Tr}}^{-1/4} (X - E_{\mathcal{N}}[X]) \sigma_{\text{Tr}}^{1/2} E_{\mathcal{N}}[X] \sigma_{\text{Tr}}^{-1/4} \right] &= E_{\mathcal{N}} \left[\sigma_{\text{Tr}}^{-1/4} (X - E_{\mathcal{N}}[X]) \sigma_{\text{Tr}}^{1/2} E_{\mathcal{N}}[X] \sigma_{\text{Tr}}^{-1/2} \sigma_{\text{Tr}}^{1/4} \right] \\
&= E_{\mathcal{N}} \left[\sigma_{\text{Tr}}^{-1/4} (X - E_{\mathcal{N}}[X]) E_{\mathcal{N}}[\sigma_{\text{Tr}}^{1/2} X \sigma_{\text{Tr}}^{-1/2}] \sigma_{\text{Tr}}^{1/4} \right] \\
&= \sigma_{\text{Tr}}^{-1/4} E_{\mathcal{N}} \left[(X - E_{\mathcal{N}}[X]) E_{\mathcal{N}}[\sigma_{\text{Tr}}^{1/2} X \sigma_{\text{Tr}}^{-1/2}] \right] \sigma_{\text{Tr}}^{1/4} \\
&= \sigma_{\text{Tr}}^{-1/4} E_{\mathcal{N}} [(X - E_{\mathcal{N}}[X])] E_{\mathcal{N}}[\sigma_{\text{Tr}}^{1/2} X \sigma_{\text{Tr}}^{-1/2}] \sigma_{\text{Tr}}^{1/4} \\
&= 0,
\end{aligned}$$

and similarly for the second term. Consequently, since $(X - E_{\mathcal{N}}[X])_{\mathcal{N}} = (|X - E_{\mathcal{N}}[X]|_2)_{\mathcal{N}}$, and by a use of the almost convexity of the von Neumann entropy (see Theorem 11.10 of [Nielsen and Chuang, 2010]), the term between brackets in (8.25) can be upper bounded by $H((p, 1-p))$, where H denotes the binary Shannon entropy. This is itself upper bounded by $\ln 2$, from which we get the result. $\square$

We can now easily prove Theorem 8.4.8.

Proof of Theorem 8.4.8 : This is a simple corollary of Proposition 8.4.9. Indeed, $\text{LSI}_{2,\mathcal{N}}(c, d)$ applied to $|X - E_{\mathcal{N}}[X]|_2$ gives

$$\begin{aligned}
\text{Ent}_{2,\mathcal{N}}(|X - E_{\mathcal{N}}[X]|_2) &\leq c \mathcal{E}_{2,\mathcal{L}}(|X - E_{\mathcal{N}}[X]|_2) + 2d \text{Var}_{\mathcal{N}}(X) \\
&\leq c \mathcal{E}_{2,\mathcal{L}}(X) + 2d \text{Var}_{\mathcal{N}}(X),
\end{aligned}$$

where we used that $\mathcal{E}_{2,\mathcal{L}}(|X - E_{\mathcal{N}}[X]|_2) \leq \mathcal{E}_{2,\mathcal{L}}(X - E_{\mathcal{N}}[X]) = \mathcal{E}_{2,\mathcal{L}}(X)$ (see Theorem 4.7 of [Cipriani, 1997]). Besides, the DF-Poincaré inequality $\text{PI}_{\mathcal{N}}(\lambda)$ implies $\lambda(\mathcal{L}) \text{Var}_{\mathcal{N}}(X) \leq \mathcal{E}_{2,\mathcal{L}}(X)$. Consequently, we get by (8.24):

$$\text{Ent}_{2,\mathcal{N}}(X) \leq \left(c + 2 \frac{d+1}{\lambda(\mathcal{L})} \right) \mathcal{E}_{2,\mathcal{L}}(X) + 2 \ln \sqrt{2} \|X\|_{\mathbb{L}_2(\sigma_{\text{Tr}})}^2,$$

which is the desired result. $\square$

8.4.3. Bounding log-Sobolev constants via interpolation

The idea to use interpolation in order to obtain estimates on the log-Sobolev constants goes back to [Gross, 1975a]. The strategy can be summarised as follows: assume a bound of the form $\|P_{t_p} : \mathbb{L}_2(\sigma) \rightarrow \mathbb{L}_p(\sigma)\| \leq M$ is known for some fixed $t_p \geq 0$ and $p > 2$, with $M \geq 1$. Then can one show by extrapolation from this bound that hypercontractivity holds for all $t \geq 0$? The answer is yes and its proof uses the crucial fact that the $\mathbb{L}_p$ norms used for the definition of hypercontractivity form an interpolating family of norms.

Theorem 8.4.10. *Let $(\mathcal{P}_t)_{t \geq 0}$ be a faithful KMS-symmetric QMS on $\mathcal{B}(\mathcal{H})$ and assume that for some $2 < p \leq +\infty$, there exist $t_p, M_p > 0$ such that for all X positive semidefinite, $\|\mathcal{P}_{t_p}(X)\|_{\mathbb{L}_2(\mathcal{N}, \mathbb{L}_p(\sigma_{\text{Tr}}))} \leq M_p \|X\|_{\mathbb{L}_2(\sigma_{\text{Tr}})}$. Then $\text{LSI}_{2,\mathcal{N}}\left(\frac{2p t_p}{p-2}, \frac{p}{p-2} \ln M_p\right)$ holds.*

Proof. The proof follows closely the analogous statement for classical Markov chains [Diaconis and Saloff-Coste, 1996a] and primitive QMS [Temme et al., 2014]. The complex time semigroup

$$\mathcal{P}_z := e^{z\mathcal{L}} = \sum_{n=0}^{\infty} \frac{z^n}{n!} \mathcal{L}^n, \quad z \in \mathbb{C},$$

defines an analytic family of operators. Define the time dilated complex semigroup $\tilde{\mathcal{P}}_z := \mathcal{P}_{t_p z}$. Since $(\mathcal{P}_t)_{t \geq 0}$ is KMS-symmetric, its spectral radius does not change upon the replacement $x \mapsto ix$, and therefore, for any $a > 0$ and X positive semidefinite:

$$\|\tilde{\mathcal{P}}_{ia}(X)\|_{\mathbb{L}_2(\sigma_{\text{Tr}})} \leq \|X\|_{\mathbb{L}_2(\sigma_{\text{Tr}})}.$$

Therefore,

$$\|\tilde{\mathcal{P}}_{1+ia}(X)\|_{\mathbb{L}_2(\mathcal{N}, \mathbb{L}_p(\sigma_{\text{Tr}}))} = \|\tilde{\mathcal{P}}_1 \circ \tilde{\mathcal{P}}_{ia}(X)\|_{\mathbb{L}_2(\mathcal{N}, \mathbb{L}_p(\sigma_{\text{Tr}}))} \leq M_p \|\tilde{\mathcal{P}}_{ia}(X)\|_{\mathbb{L}_2(\sigma_{\text{Tr}})} \leq M_p \|X\|_{\mathbb{L}_2(\sigma_{\text{Tr}})}.$$

Hence, by Stein-Weiss' interpolation Theorem ([Bergh and Löfström, 2012, Stein and Weiss, 2016]), for all $0 \leq s \leq 1$, and any $X \in \mathcal{B}(\mathcal{H})$:

$$\|\tilde{\mathcal{P}}_s(X)\|_{\mathbb{L}_2(\mathcal{N}, \mathbb{L}_{p_s}(\sigma_{\text{Tr}}))} \leq M_p^s \|X\|_{\mathbb{L}_2(\sigma_{\text{Tr}})},$$

for p_s such that

$$\frac{1}{p_s} = \frac{s}{p} + \frac{1-s}{2}.$$

Taking $t = st_p$ and $p(t) := p_s$, we get

$$\|\mathcal{P}_t(X)\|_{\mathbb{L}_2(\mathcal{N}, \mathbb{L}_{p(t)}(\sigma_{\text{Tr}}))} \leq e^{\frac{t}{t_p} \ln M_p} \|X\|_{\mathbb{L}_2(\sigma_{\text{Tr}})}, \quad (8.27)$$

with equality at $t = 0$, where

$$p(t) = \frac{2pt_p}{(2-p)t + pt_p}.$$

Taking derivatives on both sides of (8.27) at 0,

$$-\frac{\ln M_p}{t_p} \|X\|_{\mathbb{L}_2(\sigma_{\text{Tr}})} + \frac{d}{dt} \|\mathcal{P}_t(X)\|_{\mathbb{L}_2(\mathcal{N}, \mathbb{L}_{p(t)}(\sigma_{\text{Tr}}))} \Big|_{t=0} \leq 0. \quad (8.28)$$

Using Corollary 8.3.4, with $p(0) = 2$ and $p'(0) = \frac{2(p-2)}{pt_p}$,

$$\frac{d}{dt} \|\mathcal{P}_t(X)\|_{\mathbb{L}_2(\mathcal{N}, \mathbb{L}_{p(t)}(\sigma_{\text{Tr}}))} \Big|_{t=0} = \frac{p-2}{pt_p \|X\|_{\mathbb{L}_2(\sigma_{\text{Tr}})}} \left[\frac{1}{2} \text{Ent}_{2, \mathcal{N}}(X) - \frac{pt_p}{p-2} \mathcal{E}_{2, \mathcal{L}}(X) \right].$$

Hence, (8.28) can be rewritten as

$$\frac{p-2}{2pt_p} \text{Ent}_{2, \mathcal{N}}(X) \leq \mathcal{E}_{2, \mathcal{L}}(X) + \frac{\ln M_p}{t_p} \|X\|_{\mathbb{L}_2(\sigma_{\text{Tr}})}^2, \quad (8.29)$$

which leads to the desired result. $\square$

In the following corollary, we combine Theorem 8.2.5 and Theorem 8.4.10 to further provide upper bounds on the log-Sobolev constants in terms of the spectral gap of the QMS $(\mathcal{P}_t)_{t \geq 0}$. As such, it can be seen as an extension of Theorem 5 of [Temme et al., 2014] to the case of faithful KMS-symmetric QMS.

Corollary 8.4.11. *Given a KMS-symmetric QMS $(\mathcal{P}_t)_{t \geq 0}$ with spectral gap $\lambda(\mathcal{L})$, $\text{LSI}_{2, \mathcal{N}}(c, \ln \sqrt{2})$*

holds, with

$$c \leq \frac{\ln(\|\sigma_{\text{Tr}}^{-1}\|_{\infty}) + 2}{\lambda(\mathcal{L})}.$$

Proof. From (1.20), we get that for any $X \geq 0$,

$$\|X\|_{\mathbb{L}_2(\mathcal{N}, \mathbb{L}_4(\sigma_{\text{Tr}}))} \leq \|X\|_{\mathbb{L}_4(\sigma_{\text{Tr}})} \leq \|\sigma_{\text{Tr}}^{-1}\|_{\infty}^{1/4} \|X\|_{\mathbb{L}_2(\sigma_{\text{Tr}})},$$

where the last inequality is a well-known property of $\mathbb{L}_p$ norms. Together with the contractivity of $(\mathcal{P}_t)_{t \geq 0}$ (cf. (i) of Proposition 8.3.1), we find

$$\|\mathcal{P}_t(X)\|_{\mathbb{L}_2(\mathcal{N}, \mathbb{L}_4(\sigma_{\text{Tr}}))} \leq \|X\|_{\mathbb{L}_2(\mathcal{N}, \mathbb{L}_4(\sigma_{\text{Tr}}))} \leq \|\sigma_{\text{Tr}}^{-1}\|_{\infty}^{1/4} \|X\|_{\mathbb{L}_2(\sigma_{\text{Tr}})}.$$

We conclude with successive applications of Theorem 8.4.10 and Theorem 8.2.5, taking the limit $t_4 \rightarrow 0$ and $M_4 = \|\sigma_{\text{Tr}}^{-1}\|_{\infty}^{1/4}$. $\square$

8.5. Non-positivity of the strong LSI constant

In this section, we show that a strong DF-log-Sobolev inequality does not hold for a non-trivially faithful QMS, that is a QMS that is neither primitive nor unitary. We deduce from this that the amalgamated $\mathbb{L}_p$ norms do not satisfy uniform convexity for $1 \leq p \leq 2$ as soon as $\mathcal{N}$ is non-trivial.

By comparison of Dirichlet forms, it is enough to consider the case of the $\mathcal{N}$ -decoherent QMS defined by $\mathcal{L}_{\mathcal{N}} := E_{\mathcal{N}} - \text{id}$, where $\mathcal{N}$ is any $*$ -subalgebra of $\mathcal{B}(\mathcal{H})$ and $E_{\mathcal{N}}$ is a conditional expectation on it. Indeed, if $(\mathcal{P}_t)_{t \geq 0}$ is any faithful QMS with DF-algebra $\mathcal{N}$ and the same conditional expectation $E_{\mathcal{N}}$, then the following inequality holds [Müller-Hermes et al., 2016]²:

$$\lambda(\mathcal{L}) \mathcal{E}_{2, \mathcal{L}_{\mathcal{N}}}(X) \leq \mathcal{E}_{2, \mathcal{L}}(X) \leq \left\| \frac{\mathcal{L} + \hat{\mathcal{L}}}{2} : \mathbb{L}_2(\sigma_{\text{Tr}}) \rightarrow \mathbb{L}_2(\sigma_{\text{Tr}}) \right\| \mathcal{E}_{2, \mathcal{L}_{\mathcal{N}}}(X),$$

where $\hat{\mathcal{L}}$ is the conjugate of $\mathcal{L}$ with respect to $\langle \cdot, \cdot \rangle_{\sigma_{\text{Tr}}}$, and $\lambda(\mathcal{L})$ is the spectral gap of $(\mathcal{P}_t)_{t \geq 0}$. From this inequality we directly obtain that if $\text{LSI}_{2, \mathcal{N}}(c_{\mathcal{N}}, 0)$ holds for $\mathcal{L}_{\mathcal{N}}$, then $\text{LSI}_{2, \mathcal{N}}(c, 0)$ holds for $\mathcal{L}$ with:

$$0 < \frac{\lambda(\mathcal{L})}{c_{\mathcal{N}}} \leq \frac{1}{c} \leq \left\| \frac{\mathcal{L} + \hat{\mathcal{L}}}{2} : \mathbb{L}_2(\sigma_{\text{Tr}}) \rightarrow \mathbb{L}_2(\sigma_{\text{Tr}}) \right\| \frac{1}{c_{\mathcal{N}}}.$$

Our goal is thus to show that if $\mathcal{N}$ is non-trivial and $\text{LSI}_{2, \mathcal{N}}(c_{\mathcal{N}}, 0)$ holds for $\mathcal{L}_{\mathcal{N}}$, then $c_{\mathcal{N}} = +\infty$.

Theorem 8.5.1. *Let $\mathcal{N}$ be any non-trivial $*$ -subalgebra of $\mathcal{B}(\mathcal{H})$ (that is, $\mathcal{N} \neq \mathbb{C}\mathbb{1}$ and $\mathcal{N} \neq \mathcal{B}(\mathcal{H})$) and consider the Lindbladian $\mathcal{L}_{\mathcal{N}} := E_{\mathcal{N}} - \text{id}$, where $E_{\mathcal{N}}$ is any conditional expectation on $\mathcal{N}$. Define $\sigma_{\text{Tr}} := E_{\mathcal{N}*}(d_{\mathcal{H}}^{-1} \mathbb{1}_{\mathcal{H}})$. Assume that there exists $\alpha \geq 0$ such that for all positive semi-definite $X \in \mathcal{B}(\mathcal{H})$,*

$$\alpha \text{Ent}_{2, \mathcal{N}}(X) \leq \mathcal{E}_{2, \mathcal{L}_{\mathcal{N}}}(X). \quad (8.30)$$

Then $\alpha = 0$.

Proof. Let $\alpha \geq 0$ be such that inequality (8.30) holds for all positive semi-definite $X \in \mathcal{B}(\mathcal{H})$. We

²A proof of the lower bound is provided in the proof of Corollary 10.1.8

shall construct a sequence $(Z_k)_{k \in \mathbb{N}}$ such that

$$\frac{\mathcal{E}_{2, \mathcal{L}_{\mathcal{N}}}(Z_k)}{\text{Ent}_{2, \mathcal{N}}(Z_k)} \xrightarrow{k \rightarrow \infty} 0,$$

which directly implies that $\alpha = 0$. More precisely, we shall construct a sequence of density matrices $(\rho_k)_{k \geq 1}$ such that $Z_k = \Gamma_{\sigma_{\text{Tr}}}^{-\frac{1}{2}}(\sqrt{\rho_k})$ and

$$\frac{\mathcal{E}_{2, \mathcal{L}_{\mathcal{N}}}(\sigma_{\text{Tr}}^{-1/4} \sqrt{\rho_k} \sigma_{\text{Tr}}^{-1/4})}{D(\rho_k \| \rho_{\mathcal{N}, k})} \xrightarrow{k \rightarrow \infty} 0, \quad (8.31)$$

where $\rho_{\mathcal{N}, k} := E_{\mathcal{N}*}(\rho_k)$. Now assume that $\mathcal{H}$ and $\mathcal{N}$ admit the decomposition given by Equation (0.8) and Equation (0.9). As $\mathcal{N}$ is non-trivial, we can assume that either there exists $i \in I$ such that $\dim \mathcal{H}_i > 1$ and $\dim \mathcal{K}_i > 1$, or $|I| > 1$. We shall construct a sequence $(\rho_k)_{k \geq 1}$ in each case and then treat them simultaneously to prove the limit in (8.31).

We start by considering the first case and, without loss of generality, we assume that $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$ and that $\mathcal{N} = \mathcal{B}(\mathcal{H}_A) \otimes \mathbb{1}_{\mathcal{H}_B}$, with $\dim \mathcal{H}_B := d_B > 1$ and $\dim \mathcal{H}_A = 2$. One can recover the general case by adding zeros in the corresponding entries of ρ_k . Then, it means that there exists a density matrix $\tau \in \mathcal{D}(\mathcal{H}_B)$ such that for all $\omega \in \mathcal{S}_1(\mathcal{H})$,

$$E_{\mathcal{N}*}(\omega) = \text{Tr}_{\mathcal{H}_B}(\omega) \otimes \tau.$$

We define, in an orthonormal basis in which τ is diagonal and in any orthonormal basis of $\mathcal{H}_A$,

$$\Delta := \underbrace{\begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 1 & 0 & & \cdots & 0 \\ 0 & & \ddots & & \vdots \\ \vdots & & & & \\ 0 & \cdots & & & 0 \end{pmatrix}}_{d_B} \otimes \begin{pmatrix} \frac{1}{k} & 0 \\ 0 & 1 - \frac{1}{k} \end{pmatrix} \otimes \tau.$$

It is clear that $E_{\mathcal{N}*}(\Delta) = 0$. Next, define

$$e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix},$$

so that $\langle e_i, \Delta e_j \rangle = 1 - \delta_{ij}$. We also define $\lambda_1 := k \langle e_1, \rho_{\mathcal{N}, k} e_1 \rangle$ and $\lambda_2 := \frac{k}{k-1} \langle e_2, \rho_{\mathcal{N}, k} e_2 \rangle$, which clearly do not depend on k . We now set, for $\varepsilon \geq 0$,

$$\rho_{k, \varepsilon} := \rho_{\mathcal{N}, k} + \varepsilon \Delta,$$

so that $E_{\mathcal{N}*}(\rho_{k, \varepsilon}) = \rho_{\mathcal{N}, k}$. Since the $\rho_{\mathcal{N}, k}$ are full-rank, the $\rho_{k, \varepsilon}$ are well-defined density matrices for ε small enough.

We now turn to the case where $|I| > 1$. Up to adding zero entries in the matrices defining ρ_k , we can assume that $|I| = 2$. Denote by P_i the orthogonal projection on $\mathcal{H}_i \otimes \mathcal{K}_i$ for $i \in I$, and consider $\eta_i = \frac{\mathbb{1}_{\mathcal{H}_i}}{\dim \mathcal{H}_i} \otimes \tau_i$. We also denote by $e_i \in \mathcal{H}_i \otimes \mathcal{K}_i$ an eigenvector of η_i of associated eigenvalue $\lambda_i > 0$.

We then set

$$\Delta = |e_1\rangle\langle e_2| + |e_2\rangle\langle e_1|, \quad \rho_{\mathcal{N},k} = \frac{1}{k} \eta_1 + \left(1 - \frac{1}{k}\right) \eta_2,$$

so that again $E_{\mathcal{N}*}(\Delta) = 0$ and $\langle e_i, \Delta e_j \rangle = 1 - \delta_{ij}$. As before, we define $\rho_{k,\varepsilon} := \rho_{\mathcal{N},k} + \varepsilon \Delta$.

Remark that in both cases, we have $E_{\mathcal{N}*}(\Delta) = 0$ and

$$\begin{aligned} \Delta &= |e_1\rangle\langle e_2| + |e_2\rangle\langle e_1|, & \langle e_i, \Delta e_j \rangle &= 1 - \delta_{ij}, \\ \lambda_1 &:= k \langle e_1, \rho_{\mathcal{N},k} e_1 \rangle, & \lambda_2 &:= \frac{k}{k-1} \langle e_2, \rho_{\mathcal{N},k} e_2 \rangle. \end{aligned} \quad (8.32)$$

This will be enough to treat both cases simultaneously. We shall now prove that the limit in (8.31) holds with $\rho_k = \lim_{\varepsilon \rightarrow 0} \rho_{k,\varepsilon}$. The first step is to obtain a limit for a fixed $k \geq 1$ and $\varepsilon \rightarrow 0$, that is, to obtain a continuous extension of the quotient appearing in the limit at $\rho_{\mathcal{N},k}$. For this purpose, we compute the Taylor expansion of both the numerator and the denominator. A simple calculation using the integral representations of the logarithm and of the square root functions [Hiai et al., 2011] shows that (see also the proofs of Theorem 16 in [Kastoryano and Temme, 2013] and Lemma 3.5 in [Bardet, 2017]):

$$\begin{aligned} D(\rho_{k,\varepsilon} \| \rho_{\mathcal{N},k}) &= \varepsilon^2 \int_0^\infty \text{Tr} \left[\Delta \frac{1}{t - \rho_{\mathcal{N},k}} \Delta \frac{1}{t - \rho_{\mathcal{N},k}} \right] dt + \mathcal{O}(\varepsilon^3) \\ \mathcal{E}_{2,\mathcal{N}}(\sigma_{\text{Tr}}^{-1/4} \sqrt{\rho_{k,\varepsilon}} \sigma_{\text{Tr}}^{-1/4}) &= \pi^2 \varepsilon^2 \iint_{[0,\infty)^2} \sqrt{st} \text{Tr} \left[\frac{1}{t + \rho_{\mathcal{N},k}} \Delta \frac{1}{t + \rho_{\mathcal{N},k}} \Delta \right] \\ &\quad - \pi^2 \varepsilon^2 \iint_{[0,\infty)^2} \sqrt{st} \text{Tr} \left[\frac{\sigma_{\text{Tr}}^{1/4}}{t + \rho_{\mathcal{N},k}} \Delta \frac{\sigma_{\text{Tr}}^{1/4}}{t + \rho_{\mathcal{N},k}} E_{\mathcal{N}} \left[\frac{\sigma_{\text{Tr}}^{-1/4}}{s + \rho_{\mathcal{N},k}} \Delta \frac{\sigma_{\text{Tr}}^{-1/4}}{s + \rho_{\mathcal{N},k}} \right] \right] + \mathcal{O}(\varepsilon^3). \end{aligned}$$

Using Equation (8.32) we can compute explicitly these integrals. For instance, the second integral in the second equation is null, since $E_{\mathcal{N}} \left[\frac{\sigma_{\text{Tr}}^{-1/4}}{s + \rho_{\mathcal{N},k}} \Delta \frac{\sigma_{\text{Tr}}^{-1/4}}{s + \rho_{\mathcal{N},k}} \right] = 0$. This can be checked directly using the fact that both e_1 and e_2 are eigenvectors of $\frac{\sigma_{\text{Tr}}^{-1/4}}{s + \rho_{\mathcal{N},k}}$ and that $E_{\mathcal{N}}[|e_1\rangle\langle e_2|] = E_{\mathcal{N}}[|e_2\rangle\langle e_1|] = 0$. We thus obtain:

$$D(\rho_{k,\varepsilon} \| \rho_{\mathcal{N},k}) = \varepsilon^2 g \left(\frac{1}{k} \lambda_1, \left(1 - \frac{1}{k}\right) \lambda_2 \right) |\langle e_1 | \Delta | e_2 \rangle|^2 + \mathcal{O}(\varepsilon^3), \quad (8.33)$$

$$\mathcal{E}_{2,\mathcal{N}}(\sigma_{\text{Tr}}^{-1/4} \sqrt{\rho_{k,\varepsilon}} \sigma_{\text{Tr}}^{-1/4}) = 2\pi^2 \varepsilon^2 f \left(\frac{1}{k} \lambda_1, \left(1 - \frac{1}{k}\right) \lambda_2 \right) |\langle e_1 | \Delta | e_2 \rangle|^2 + \mathcal{O}(\varepsilon^3), \quad (8.34)$$

where

$$f(x, y) := \begin{cases} \frac{(\sqrt{x} - \sqrt{y})^2}{(x - y)^2} & \text{if } x \neq y \\ \frac{1}{4x} & \text{else} \end{cases}, \quad g(x, y) := \begin{cases} \frac{\ln(x) - \ln(y)}{x - y} & \text{if } x \neq y \\ \frac{1}{x} & \text{else} \end{cases}. \quad (8.35)$$

For a fixed $k \geq 1$, we thus obtain that

$$\frac{\mathcal{E}_{2,\mathcal{N}}(\sigma_{\text{Tr}}^{-1/4} \sqrt{\rho_{k,\varepsilon}} \sigma_{\text{Tr}}^{-1/4})}{D(\rho_{k,\varepsilon} \| \rho_{\mathcal{N},k})} \xrightarrow{\varepsilon \rightarrow 0} 2\pi^2 \frac{f \left(\frac{1}{k} \lambda_1, \left(1 - \frac{1}{k}\right) \lambda_2 \right)}{g \left(\frac{1}{k} \lambda_1, \left(1 - \frac{1}{k}\right) \lambda_2 \right)}.$$

We just have to take the limit $k \rightarrow +\infty$ to conclude. Indeed,

$$\begin{aligned} f\left(\frac{1}{k}\lambda_1, \left(1 - \frac{1}{k}\right)\lambda_2\right) &\xrightarrow{k \rightarrow +\infty} 1/\lambda_2, \\ g\left(\frac{1}{k}\lambda_1, \left(1 - \frac{1}{k}\right)\lambda_2\right) &\xrightarrow{k \rightarrow +\infty} +\infty. \end{aligned}$$

□

The above result implies the following straightforward corollary:

Corollary 8.5.2. *The $\mathbb{L}_2(\mathcal{N}, \mathbb{L}_p(\sigma_{\text{Tr}}))$ spaces do not satisfy the uniform convexity property.*

8.6. Application to decoherence times

In this section, we apply the framework of DF-log-Sobolev inequalities in order to find bounds on the decoherence rates of a non-primitive quantum Markov semigroup. We recall that, for $0 < \varepsilon < 1$, the decoherence time of a KMS-symmetric QMS $(\mathcal{P}_t)_{t \geq 0}$ is defined as

$$\tau_{\text{deco}}(\varepsilon) := \inf \{t \geq 0 : \|\mathcal{P}_{t*}(\rho - \rho_{\mathcal{N}})\|_1 \leq \varepsilon\},$$

where $\rho_{\mathcal{N}} \equiv E_{\mathcal{N}*}(\rho)$. A classical technique to get rapid decoherence for all times comes from looking at the spectral gap of a KMS-symmetric QMS:

$$\begin{aligned} \|\mathcal{P}_t(X - E_{\mathcal{N}}[X])\|_{\infty} &\leq \|\sigma_{\text{Tr}}^{-1}\|_{\infty}^{\frac{1}{2}} \|\mathcal{P}_t(X - E_{\mathcal{N}}[X])\|_{\mathbb{L}_2(\sigma_{\text{Tr}})} \\ &\leq \|\sigma_{\text{Tr}}^{-1}\|_{\infty}^{\frac{1}{2}} e^{-\lambda(\mathcal{L})t} \|X - E_{\mathcal{N}}(X)\|_{\mathbb{L}_2(\sigma_{\text{Tr}})}, \end{aligned}$$

where the second inequality follows from (8.3). In the dual Schrödinger picture, such a bound translates into

$$\|\mathcal{P}_{*t}(\rho - \rho_{\mathcal{N}})\|_1 \leq \|\sigma_{\text{Tr}}^{-1}\|_{\infty}^{1/2} e^{-\lambda(\mathcal{L})t}. \quad (8.36)$$

However, as discussed in Section 3.2, already in the classical case the spectral gap does not usually provide tight enough bounds on the decoherence time of a Markov semigroup [Diaconis and Saloff-Coste, 1996a]. Moreover, in practice, the coefficient $\|\sigma_{\text{Tr}}^{-1}\|_{\infty}^{\frac{1}{2}}$ explodes exponentially fast as the dimension of the system grows. If $\text{LSI}_{2,\mathcal{N}}(c, 0)$ held with $c < \infty$, the original techniques of [Temme et al., 2014] could be adapted to yield

$$\|\mathcal{P}_{*t}(\rho - \rho_{\mathcal{N}})\|_1 \leq (2 \ln \|\sigma_{\text{Tr}}^{-1}\|_{\infty})^{1/2} e^{-\frac{4t}{c}},$$

improving significantly the bound (8.36) derived from the spectral gap method. However, as discussed in the last section, a strong LSI never holds for non-primitive QMS. This motivates the search for a technique that would deal with the weak version of the log-Sobolev inequality. Fortunately, such a technique already exists in the classical literature [Zegarlinski, 1995, Martinelli, 1999, Diaconis and Saloff-Coste, 1996a]: it consists in combining hypercontractivity bounds at short times with the spectral gap at long times (see estimate (3.13)). Using such a method, we can prove the exponential convergence in terms of the ∞ -norm.

Proposition 8.6.1. *Assume that a faithful QMS $(\mathcal{P}_t)_{t \geq 0}$ of corresponding decoherence-free algebra $\mathcal{N}$*

satisfies $\text{HC}_{2,\mathcal{N}}(c, d)$, and that $\|\sigma_{\text{Tr}}^{-1}\|_\infty \geq e$. Then for $t = \frac{c}{4} \ln \ln \|\sigma_{\text{Tr}}^{-1}\|_\infty + \frac{\kappa}{\lambda(\mathcal{L})}$, $\kappa > 0$, and all $X \in \mathcal{B}(\mathcal{H})$

$$\|\mathcal{P}_t(X - E_{\mathcal{N}}[X])\|_\infty \leq \left(\max_{i \in I} \sqrt{d_{\mathcal{H}_i}} \right) e^{1+d-\kappa} \|X\|_\infty, \quad (8.37)$$

where $d_{\mathcal{H}_i}$ denote the dimensions of the spaces $\mathcal{H}_i$ appearing in the decomposition of $\mathcal{N}$ given by (0.9). By duality, we get the following similar bound:

$$\forall \rho \in \mathcal{D}(\mathcal{H}), \quad \|\mathcal{P}_{*t}(\rho - E_{\mathcal{N}*}(\rho))\|_1 \leq \max_{i \in I} \sqrt{d_{\mathcal{H}_i}} e^{1+d-\kappa}. \quad (8.38)$$

The above inequality provides a bound on the decoherence time of the QMS:

$$\tau_{\text{deco}}(\varepsilon) \leq \frac{\ln(\max_{i \in I} \sqrt{d_{\mathcal{H}_i}} \varepsilon^{-1}) + 1 + d}{\lambda(\mathcal{L})} + \frac{c}{4} \ln \ln \|\sigma_{\text{Tr}}^{-1}\|_\infty.$$

Proof. Let $t, s > 0$. Then:

$$\begin{aligned} \|\mathcal{P}_{t+s}(X - E_{\mathcal{N}}[X])\|_{\mathbb{L}_2(\mathcal{N}, \mathbb{L}_\infty(\sigma_{\text{Tr}}))} &\leq \|\sigma_{\text{Tr}}^{-1}\|_\infty^{\frac{1}{p}} \|\mathcal{P}_{t+s}(X - E_{\mathcal{N}}[X])\|_{\mathbb{L}_2(\mathcal{N}, \mathbb{L}_p(\sigma_{\text{Tr}}))} \\ &\leq \|\sigma_{\text{Tr}}^{-1}\|_\infty^{\frac{1}{p}} \exp\left(2d\left(\frac{1}{2} - \frac{1}{p}\right)\right) \|\mathcal{P}_t(X - E_{\mathcal{N}}[X])\|_{\mathbb{L}_2(\sigma_{\text{Tr}})} \\ &\leq \|\sigma_{\text{Tr}}^{-1}\|_\infty^{\frac{1}{p}} \exp\left(2d\left(\frac{1}{2} - \frac{1}{p}\right)\right) \|X - E_{\mathcal{N}}[X]\|_{\mathbb{L}_2(\sigma_{\text{Tr}})} e^{-\lambda(\mathcal{L})t} \\ &\leq e^d \|\sigma_{\text{Tr}}^{-1}\|_\infty^{\frac{1}{p}} \|X - E_{\mathcal{N}}[X]\|_{\mathbb{L}_2(\sigma_{\text{Tr}})} e^{-\lambda(\mathcal{L})t}, \end{aligned}$$

where the first inequality follows from (1.28) in Section 1.1.2 applied to $\mathcal{P}_{t+s}(X - E_{\mathcal{N}}[X])$, the second inequality from $\text{HC}_{2,\mathcal{N}}(c, d)$, and the third one by definition of the spectral gap. Since $\|\sigma_{\text{Tr}}^{-1}\|_\infty \geq e$, one can choose $s := \frac{c}{4} \ln \ln \|\sigma_{\text{Tr}}^{-1}\|_\infty$, and $p \equiv p(s) = 1 + \ln \|\sigma_{\text{Tr}}^{-1}\|_\infty$, so that

$$\|\mathcal{P}_{t+s}(X - E_{\mathcal{N}}[X])\|_{\mathbb{L}_2(\mathcal{N}, \mathbb{L}_\infty(\sigma_{\text{Tr}}))} \leq \|X\|_\infty e^{1+d-\lambda(\mathcal{L})t},$$

where we use that $\|X - E_{\mathcal{N}}[X]\|_{\mathbb{L}_2(\sigma_{\text{Tr}})} \leq \|X\|_{\mathbb{L}_2(\sigma_{\text{Tr}})} \leq \|X\|_\infty$. The result follows by applying the following norm estimate proved in Proposition 1.1.13:

$$\|\text{id} : \mathbb{L}_2(\mathcal{N}, \mathbb{L}_\infty(\sigma_{\text{Tr}})) \rightarrow \mathcal{B}(\mathcal{H})\| \leq \max_{i \in I} \sqrt{d_{\mathcal{H}_i}}.$$

By duality, we get,

$$\begin{aligned} \|\mathcal{P}_{(t+s)*}(\rho - \rho_{\mathcal{N}})\|_1 &= \sup_{\|X\|_\infty \leq 1} \text{Tr}(\mathcal{P}_{(t+s)*}(\rho - E_{\mathcal{N}*}(\rho)) X) \\ &= \sup_{\|X\|_\infty \leq 1} \text{Tr}(\rho \mathcal{P}_{(t+s)}(X - E_{\mathcal{N}}[X])) \\ &\leq \max_{i \in I} \sqrt{d_{\mathcal{H}_i}} e^{1+d-\lambda(\mathcal{L})t}. \end{aligned}$$

□

8.A. Proof of Theorem 8.3.3

Let $t \mapsto X(t) \in \mathcal{B}(\mathcal{H})$ be an operator-valued twice continuously differentiable function, where $X(t) > 0$ for all $t \in [-\eta, \eta]$, for some $\eta > 0$, as well as an increasing twice continuously differentiable function

$\mathbb{R} \ni t \mapsto p(t)$ with $p(0) = q \geq 1$. Define

$$s(t) := \frac{1}{q} - \frac{1}{p(t)},$$

and for a positive definite operator $A \in \mathcal{N}$, such that $\|A\|_{\mathbb{L}_1(\sigma_{\text{Tr}})} = 1$,

$$M(t, A) := A^{-s(t)/2} X(t) A^{-s(t)/2}.$$

Thus $M(t, A)$ is positive definite for any $t \in [-\eta, \eta]$. Define moreover

$$\Phi(X(t), A, p(t)) := \|M(t, A)\|_{p(t)}. \quad (8.39)$$

The following proposition gathers straightforward generalization of results proved in [Beigi and King, 2016] which were used to prove the relation between hypercontractivity and the log-Sobolev inequality for the completely bounded norm. (cf. lemmas 8, 9 of [Beigi and King, 2016]). We recall that $\mathcal{S}_{\mathbb{L}_1(\sigma_{\text{Tr}})}^+$ denotes the set of positive definite operators on the sphere of radius one in $\mathbb{L}_1(\sigma_{\text{Tr}})$.

Proposition 8.A.1. *For a fixed $t \in (-\eta, \eta)$, $A \mapsto \Phi(X(t), A, p(t))^{p(t)}$ is convex for $1 \leq q \leq p(t) \leq 2q$ and concave for $1 \leq p(t) \leq q$. Moreover, the following assertions hold true:*

- 1 The function $(t, A) \mapsto \frac{\partial^2}{\partial t^2} \Phi(X(t), A, p(t))$ is continuous on $(-\eta, \eta) \times \mathcal{N}(\mathcal{P}) \cap \mathcal{S}_{\mathbb{L}_1(\sigma_{\text{Tr}})}^+$.
- 2 The function $A \mapsto \Phi(X(t), A, p(t))$ is continuously differentiable for all $A \in \mathcal{N}(\mathcal{P}) \cap \mathcal{S}_{\mathbb{L}_1(\sigma_{\text{Tr}})}^+$.
- 3 For all $A \in \mathcal{N}(\mathcal{P}) \cap \mathcal{S}_{\mathbb{L}_1(\sigma_{\text{Tr}})}^+$ and $t \in (-\eta, \eta)$,

$$\begin{aligned} \frac{\partial}{\partial t} \Phi(X(t), A, p(t)) &= \frac{p'(t) \Phi(X(t), A, p(t))}{p(t)^2 \text{Tr}[M(t, A)^{p(t)}]} \left(-\text{Tr}[M(t, A)^{p(t)}] \ln \text{Tr}[M(t, A)^{p(t)}] \right. \\ &\quad \left. + \text{Tr}[M(t, A)^{p(t)} \ln M(t, A)^{p(t)}] - \text{Tr}[M(t, A)^{p(t)} \ln A] \right. \\ &\quad \left. + \frac{p(t)^2}{p'(t)} \text{Tr}[M(t, A)^{p(t)-1} A^{-s(t)/2} X'(t) A^{-s(t)/2}] \right). \end{aligned} \quad (8.40)$$

In what follows, we fix a positive definite $Y \in \mathcal{B}(\mathcal{H})$ and set $X(t) = \Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{p(t)}}(Y(t))$, where $t \mapsto Y(t)$ is some twice continuously differentiable matrix-valued function with $Y(0) = Y$. Therefore,

$$\left. \frac{d}{dt} X(t) \right|_{t=0} = \left. \frac{d}{dt} \Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{p(t)}}(Y(t)) \right|_{t=0} = -\frac{p'(0)}{2q^2} \{ \ln \sigma_{\text{Tr}}, \Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y(0)) \} + \Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y'(0)),$$

where we used that $p(0) = q$ and where $\{\cdot, \cdot\}$ is the anticommutator. Thus, using that $M(0, A) = \Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y(0))$ and that $\Phi(X(0), A, q) = \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}$, Equation (8.40) reduces to

$$\begin{aligned} \left. \frac{\partial}{\partial t} \Phi(X(t), A, p(t)) \right|_{t=0} &= \frac{p'(0)}{q^2 \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^{q-1}} \left[-\|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^q \ln \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^q + \text{Tr} \left(\left[\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y) \right]^q \ln \left[\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y) \right]^q \right) \right. \\ &\quad \left. - \text{Tr} \left(\left[\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y) \right]^q \ln A \right) - \text{Tr} \left(\left[\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y) \right]^q \ln \sigma_{\text{Tr}} \right) + \frac{q^2}{p'(0)} \text{Tr} \left(\left[\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y) \right]^{q-1} \Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y'(0)) \right) \right]. \end{aligned} \quad (8.41)$$

In fact, in the case when $Y(t) = Y \in \mathcal{B}_{sa}(\mathcal{H})$, and $p(t) = q + t$, one can similarly show the following

$$\begin{aligned} \left. \frac{\partial}{\partial p} \Phi(\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{p}}(Y), A, p) \right|_{p=q} &= \frac{1}{q^2 \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^{q-1}} \left[-\|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^q \ln \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^q + \text{Tr} \left(\left| \Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y) \right|^q \ln \left| \Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y) \right|^q \right) \right. \\ &\quad \left. - \text{Tr} \left(\left| \Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y) \right|^q \ln A \right) - \text{Tr} \left(\left| \Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y) \right|^q \ln \sigma_{\text{Tr}} \right) \right]. \end{aligned} \quad (8.42)$$

Now, define $G(A)$ as the part in the parenthesis:

$$\begin{aligned} G(A) &:= -\|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^q \ln \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^q + \text{Tr} \left(\left[\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y) \right]^q \ln \left[\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y) \right]^q \right) \\ &\quad - \text{Tr} \left(\left[\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y) \right]^q \ln A \right) - \text{Tr} \left(\left[\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y) \right]^q \ln \sigma_{\text{Tr}} \right) \\ &\quad + \frac{q^2}{p'(0)} \text{Tr} \left(\left[\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y) \right]^{q-1} \Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y'(0)) \right), \end{aligned} \quad (8.43)$$

and let, for a given $Y \in \mathcal{B}_{sa}(\mathcal{H})$,

$$Y_{\mathcal{N}} := \frac{E_{\mathcal{N}}[\mathbb{I}_{1,q}(Y)]}{\|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^q}. \quad (8.44)$$

Next, we derive a formula that will be useful in what follows.

Lemma 8.A.2. *With the above notations and for positive semidefinite $Y \in \mathcal{B}(\mathcal{H})$,*

$$G(A) - G(Y_{\mathcal{N}}) = \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^q D(\Gamma_{\sigma_{\text{Tr}}}(Y_{\mathcal{N}}) \| \Gamma_{\sigma_{\text{Tr}}}(A)). \quad (8.45)$$

Remark that $G(A) - G(Y_{\mathcal{N}})$ does not depend on $Y'(0)$ and therefore one can check that the same result holds for $Y \in \mathcal{B}_{sa}(\mathcal{H})$.

Proof. First note that

$$G(A) - G(Y_{\mathcal{N}}) = \text{Tr} \left(\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y) (\ln Y_{\mathcal{N}} - \ln A) \right).$$

As $Y_{\mathcal{N}}$ and A are in $\mathcal{N}$, they commute with σ_{Tr} and therefore we get

$$G(A) - G(Y_{\mathcal{N}}) = \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^q \text{Tr} \left(\frac{\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y)^q}{\|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^q} (\ln \Gamma_{\sigma_{\text{Tr}}}(Y_{\mathcal{N}}) - \ln \Gamma_{\sigma_{\text{Tr}}}(A)) \right).$$

Now, as again $Y_{\mathcal{N}}, A \in \mathcal{N}(\mathcal{P})$, $\ln Y_{\mathcal{N}}$ and $\ln A$ also belong to $\mathcal{N}(\mathcal{P})$ and we get

$$\begin{aligned} G(A) - G(Y_{\mathcal{N}}) &= \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^q \text{Tr} \left(\frac{\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y)^q}{\|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^q} E_{\mathcal{N}} [\ln \Gamma_{\sigma_{\text{Tr}}}(Y_{\mathcal{N}}) - \ln \Gamma_{\sigma_{\text{Tr}}}(A)] \right) \\ &= \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^q \text{Tr} \left(E_{\mathcal{N}*} \left(\frac{\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y)^q}{\|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^q} \right) (\ln \Gamma_{\sigma_{\text{Tr}}}(Y_{\mathcal{N}}) - \ln \Gamma_{\sigma_{\text{Tr}}}(A)) \right) \\ &= \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^q \text{Tr} (\Gamma_{\sigma_{\text{Tr}}}(Y_{\mathcal{N}}) (\ln \Gamma_{\sigma_{\text{Tr}}}(Y_{\mathcal{N}}) - \ln \Gamma_{\sigma_{\text{Tr}}}(A))), \end{aligned}$$

which is the desired result. $\square$

Theorem 8.3.3 follows from a direct adaptation of the proof of Theorem 7 of [Beigi and King, 2016]. In a nutshell, all the lemmas used in [Beigi and King, 2016] to prove it can be generalized to

our framework, when replacing the equation (25) of [Beigi and King, 2016] by Equation (8.45). In particular, one can prove that

$$\Delta(t) := \frac{1}{t} \left(\|Y(t)\|_{\mathbb{L}_q(\mathcal{N}, \mathbb{L}_{p(t)}(\sigma_{\text{Tr}}))} - \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})} \right) - \frac{G(Y_{\mathcal{N}})p'(0)}{q^2 \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^{q-1}}.$$

converges to 0, which leads to the desired result. The details are provided for sake of completeness.

Lemma 8.A.3. *There exist $\kappa > 0$ and $K < \infty$, such that for all $t \in [-\eta/2, \eta/2]$ and $A \in \mathcal{S}(\kappa) := \mathcal{N}(\mathcal{P}) \cap \{B > 0, \|B\|_{\mathbb{L}_1(\sigma_{\text{Tr}})} = 1, \|B - Y_{\mathcal{N}}\|_{\mathbb{L}_1(\sigma_{\text{Tr}})} \leq \kappa\}$,*

$$\left| \Phi(X(t), A, p(t)) - \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})} - t \frac{p'(0)G(A)}{q^2 \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^{q-1}} \right| \leq Kt^2,$$

where $X(t) = \Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{p(t)}}(Y(t))$.

Proof. The proof is similar to the one of Lemma 10 of [Beigi and King, 2016]. Let $t \in [-\eta/2, \eta/2]$. Since the set $\mathcal{N}(\mathcal{P}) \cap \{B > 0, \|B\|_{\mathbb{L}_1(\sigma_{\text{Tr}})} = 1\}$ is open, there exists $\kappa > 0$ such that $\mathcal{S}(\kappa)$ is a compact and a subset of $\mathcal{N}(\mathcal{P}) \cap \{B > 0, \|B\|_{\mathbb{L}_1(\sigma_{\text{Tr}})} = 1\}$. By Proposition 8.A.1, the function $\frac{\partial^2}{\partial t^2} \Phi(X(t), A, p(t))$ is continuous on $(-\eta, \eta) \times \mathcal{N}(\mathcal{P}) \cap \mathcal{S}_{\mathbb{L}_1(\sigma_{\text{Tr}})}^+$. Hence, there exists $K > \infty$ such that

$$-2K \leq \frac{\partial^2 \Phi(X(t), A, p(t))}{\partial t^2} \leq 2K,$$

for all $t \in [-\eta/2, \eta/2]$ and all $A \in \mathcal{S}(\kappa)$. Therefore, for any $t \in [-\eta/2, \eta/2]$ and $A \in \mathcal{S}(\kappa)$:

$$\begin{aligned} \left| \Phi(X(t), A, p(t)) - \Phi(X(0), A, q) - t \frac{\partial \Phi(X(u), A, p(u))}{\partial u} \Big|_{u=0} \right| \\ = \left| \int_0^t (t-v) \frac{\partial^2 \Phi(X(v), A, p(v))}{\partial v^2} dv \right| \leq Kt^2. \end{aligned}$$

Noting that $\Phi(X(0), A, q) = \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}$ and using the definition of $G(A)$, we find that

$$\left| \Phi(X(t), A, p(t)) - \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})} - t \frac{p'(0)G(A)}{q^2 \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^{q-1}} \right| \leq Kt^2,$$

for all $t \in [-\eta/2, \eta/2]$ and $A \in \mathcal{S}(\kappa)$. □

Lemma 8.A.4. *With the notations of Lemma 8.A.3, for any $0 < \varepsilon \leq \kappa$, there exists $\delta > 0$ such that for all $t \in [-\delta, \delta]$ there is $A(t) \in \mathcal{N}(\mathcal{P}) \cap \mathcal{S}_{\mathbb{L}_1(\sigma_{\text{Tr}})}^+$ satisfying*

$$\|Y(t)\|_{\mathbb{L}_q(\mathcal{N}, \mathbb{L}_{p(t)}(\sigma_{\text{Tr}}))} = \Phi(X(t), A(t), p(t)), \quad \|Y_{\mathcal{N}} - A(t)\|_{\mathbb{L}_1(\sigma_{\text{Tr}})} \leq \varepsilon.$$

Proof. The proof is similar to the one of Lemma 11 of [Beigi and King, 2016]. Given $\varepsilon \leq \kappa$, choose $\delta' > 0$ satisfying

$$\delta' < \min \left\{ \frac{\eta}{2}, \frac{\varepsilon^2 p'(0) \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}}{4Kq^2} \right\}$$

where K is defined in Lemma 8.A.3. We have

$$\mathcal{S}(\varepsilon) \subset \mathcal{S}(\kappa) \subset \mathcal{N}(\mathcal{P}) \cap \mathcal{S}_{\mathbb{L}_1(\sigma_{\text{Tr}})}^+$$

and so the boundary of $\mathcal{S}(\varepsilon)$ is contained in $\mathcal{N}(\mathcal{P}) \cap \mathcal{S}_{\mathbb{L}_1(\sigma_{\text{Tr}})}^+$. Suppose that A is on the boundary of $\mathcal{S}(\varepsilon)$, so that

$$\|Y_{\mathcal{N}} - A\|_{\mathbb{L}_1(\sigma_{\text{Tr}})} = \varepsilon.$$

By the quantum Pinsker inequality,

$$D(\Gamma_{\sigma_{\text{Tr}}}(Y_{\mathcal{N}}) \|\Gamma_{\sigma_{\text{Tr}}}(A)) \geq \frac{1}{2} \|Y_{\mathcal{N}} - A\|_{\mathbb{L}_1(\sigma_{\text{Tr}})}^2 = \frac{\varepsilon^2}{2}.$$

From Equation (8.45) we deduce

$$G(A) \geq G(Y_{\mathcal{N}}) + \frac{\varepsilon^2 \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^q}{2}. \quad (8.46)$$

Let us first consider the case where $t \geq 0$. From Lemma 8.A.3, we deduce that

$$\Phi(X(t), A, p(t)) \geq \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})} + t \frac{p'(0) G(A)}{q^2 \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^{q-1}} - Kt^2. \quad (8.47)$$

Our choice of δ' implies that for all $0 \leq t \leq \delta'$,

$$t \frac{\varepsilon^2 p'(0) \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}}{2q^2} - Kt^2 > Kt^2,$$

and hence, combining this with 8.46 and 8.47,

$$\Phi(X(t), A, p(t)) > \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})} + t \frac{p'(0) G(Y_{\mathcal{N}})}{q^2 \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^{q-1}} + Kt^2. \quad (8.48)$$

Furthermore, from Lemma 8.A.3, we also deduce that

$$\Phi(X(t), Y_{\mathcal{N}}, p(t)) \leq \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})} + t \frac{p'(0) G(Y_{\mathcal{N}})}{q^2 \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^{q-1}} + Kt^2. \quad (8.49)$$

Combining 8.48 and 8.49, we find that

$$\Phi(X(t), Y_{\mathcal{N}}, p(t)) < \Phi(X(t), A, p(t)).$$

Since this inequality holds for any A on the boundary of $\mathcal{S}(\varepsilon)$, we conclude that for all $0 \leq t \leq \delta'$, the function $A \mapsto \Phi(X(t), A, p(t))$ has a local minimum $A(t)$ in the interior of $\mathcal{S}(\varepsilon)$. We now choose $0 < \delta_+ \leq \delta'$ so that $q \leq p(t) \leq 2q$ for all $0 \leq t \leq \delta_+$ (the existence of $\delta_+ > 0$ is guaranteed by the assumptions that $p(0) = q \geq 1$ and that $t \mapsto p(t)$ is increasing and differentiable). Applying Proposition 8.A.1, we conclude that, for all $0 \leq t \leq \delta_+$, the local minimum of the convex function $A \mapsto \Phi(X(t), A, p(t))^{p(t)}$ in the interior of $\mathcal{S}(\varepsilon)$ is in fact a global minimum. Since $A \mapsto \Phi(X(t), A, p(t))$ and $A \mapsto \Phi(X(t), A, p(t))^{p(t)}$ share the same minimum $A(t) \in \mathcal{S}(\varepsilon)$, we conclude that

$$\|Y\|_{\mathbb{L}_q(\mathcal{N}, \mathbb{L}_{p(t)}(\sigma_{\text{Tr}}))} = \Phi(X(t), A(t), p(t)), \quad \|Y_{\mathcal{N}} - A(t)\|_{\mathbb{L}_1(\sigma_{\text{Tr}})} \leq \varepsilon.$$

We consider now the case $t \leq 0$. Using Lemma 8.A.3 as well as inequality 8.46, we deduce that

$$\begin{aligned} \Phi(X(t), A, p(t)) &\leq \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})} + t \frac{p'(0)G(A)}{q^2 \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^{q-1}} + Kt^2 \\ &\leq \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})} + t \frac{p'(0)G(Y_{\mathcal{N}})}{q^2 \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^{q-1}} + t \frac{\varepsilon^2 p'(0) \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}}{2q^2} + Kt^2, \end{aligned}$$

where the second inequality follows from the fact that $t \leq 0$. Now, for $-\delta' \leq t \leq 0$,

$$t \frac{\varepsilon^2 p'(0) \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}}{2q^2} + Kt^2 < -Kt^2,$$

and thus

$$\Phi(X(t), A, p(t)) < \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})} + t \frac{G(Y_{\mathcal{N}})p'(0)}{q^2 \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^{q-1}} - Kt^2. \quad (8.50)$$

Combining with the lower bound for F obtained from Lemma 8.A.3 we deduce that

$$\Phi(X(t), Y_{\mathcal{N}}, p(t)) > \Phi(X(t), A, p(t)),$$

for all A on the boundary of $\mathcal{S}(\varepsilon)$. Thus, we conclude that for all $\delta' \leq t \leq 0$, the function $A \mapsto \Phi(X(t), A, p(t))$ has a local maximum in the interior of $\mathcal{S}(\varepsilon)$. Choose $0 < \delta_- \leq \delta'$ so that $1 \leq p(t) \leq 2$ for all $-\delta_- \leq t \leq 0$. Applying Proposition 8.A.1, we conclude that the local maximum of the concave function $A \mapsto \Phi(X(t), A, p(t))^{p(t)}$ in the interior of $\mathcal{S}(\varepsilon)$ is in fact a global maximum for all $-\delta_- \leq t \leq 0$. Finally, take $\delta := \min\{\delta_+, \delta_-\}$ to deduce that for all $t \in [-\delta, \delta]$ there exists $A(t) \in \mathcal{N}(\mathcal{P}) \cap \mathcal{S}_{\mathbb{L}_1}^+(\sigma_{\text{Tr}})$ satisfying:

$$\|Y\|_{\mathbb{L}_q(\mathcal{N}, \mathbb{L}_{p(t)}(\sigma_{\text{Tr}}))} = \Phi(X(t), A(t), p(t)), \quad \|Y_{\mathcal{N}} - A(t)\|_{\mathbb{L}_1(\sigma_{\text{Tr}})} \leq \varepsilon.$$

□

We are finally ready to state and prove Theorem 8.3.3.

Proof of Theorem 8.3.3 : Recall from (8.43) that

$$\begin{aligned} G(Y_{\mathcal{N}}) &= -\|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^q \ln \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^q + \text{Tr} \left(\left[\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y) \right]^q \ln \left[\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y) \right]^q \right) - \text{Tr} \left(\left[\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y) \right]^q \ln Y_{\mathcal{N}} \right) \\ &\quad - \text{Tr} \left(\left[\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y) \right]^q \ln \sigma_{\text{Tr}} \right) + \frac{q^2}{p'(0)} \text{Tr} \left(\left[\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y) \right]^{q-1} \Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y'(0)) \right). \end{aligned} \quad (8.51)$$

Using the expression (8.44) for $Y_{\mathcal{N}}$, (8.51) reduces to

$$\begin{aligned} G(Y_{\mathcal{N}}) &= \text{Tr} \left(\left[\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y) \right]^q \ln \left[\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y) \right]^q \right) - \text{Tr} \left(\left[\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y) \right]^q \ln E_{\mathcal{N}} \left[\Gamma_{\sigma_{\text{Tr}}}^{-1} \left(\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y) \right)^q \right] \right) \\ &\quad - \text{Tr} \left(\left[\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y) \right]^q \ln \sigma_{\text{Tr}} \right) + \frac{q^2}{p'(0)} \text{Tr} \left(\left[\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y) \right]^{q-1} \Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y'(0)) \right). \end{aligned}$$

Define now

$$\Delta(t) := \frac{1}{t} \left(\|Y(t)\|_{\mathbb{L}_q(\mathcal{N}, \mathbb{L}_{p(t)}(\sigma_{\text{Tr}}))} - \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})} \right) - \frac{p'(0)G(Y_{\mathcal{N}})}{q^2 \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^{q-1}}. \quad (8.52)$$

We next prove that $\Delta(t) \rightarrow 0$ as $t \rightarrow 0$. Let $\varepsilon > 0$ be such that

$$0 < \varepsilon < \min \left\{ \kappa, \eta, \frac{\lambda_{\min}(\Gamma_{\sigma_{\text{Tr}}}(Y_{\mathcal{N}}))}{2} \right\} \quad (8.53)$$

where κ is the parameter introduced in Lemma 8.A.3 and $\lambda_{\min}(\Gamma_{\sigma_{\text{Tr}}}(Y_{\mathcal{N}}))$ is the minimum eigenvalue of $\Gamma_{\sigma_{\text{Tr}}}(Y_{\mathcal{N}})$. According to Lemma 8.A.4, there exists $\delta > 0$ such that for every $0 < t < \delta$ there is an operator $A(t) \in \mathcal{N}(\mathcal{P}) \cap \mathcal{S}_{\mathbb{L}_1(\sigma_{\text{Tr}})}^+$ such that

$$\|A(t) - Y_{\mathcal{N}}\|_{\mathbb{L}_1(\sigma_{\text{Tr}})} \leq \varepsilon \leq \kappa, \quad \|Y(t)\|_{\mathbb{L}_q(\mathcal{N}, \mathbb{L}_{p(t)}(\sigma_{\text{Tr}}))} = \Phi(X(t), A(t), p(t)).$$

Then

$$\begin{aligned} \Delta(t) &= \frac{1}{t} \left(\Phi(X(t), A(t), p(t)) - \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})} \right) - \frac{p'(0) G(Y_{\mathcal{N}})}{q^2 \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^{q-1}} \\ &= \frac{1}{t} \left(\Phi(X(t), A(t), p(t)) - \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})} - t \frac{p'(0) G(A(t))}{q^2 \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^{q-1}} \right) + \frac{p'(0) (G(A(t)) - G(Y_{\mathcal{N}}))}{q^2 \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^{q-1}}. \end{aligned}$$

Since $A(t) \in \mathcal{S}(\varepsilon)$, Lemma 8.A.3 implies that

$$\left| \Phi(X(t), A(t), p(t)) - \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})} - t \frac{p'(0) G(A(t))}{q^2 \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^{q-1}} \right| \leq K t^2. \quad (8.54)$$

Furthermore, from (Equation (8.45)) and using Lemma 14 of [Beigi and King, 2016]:

$$\begin{aligned} |G(A(t)) - G(Y_{\mathcal{N}})| &= \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^q D(\Gamma_{\sigma_{\text{Tr}}}(Y_{\mathcal{N}}) \Gamma_{\sigma_{\text{Tr}}}(A(t))) \leq \frac{2 \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^q}{\lambda_{\min}(\Gamma_{\sigma_{\text{Tr}}}(Y_{\mathcal{N}}))} \|Y_{\mathcal{N}} - A(t)\|_{\mathbb{L}_1(\sigma_{\text{Tr}})} \\ &\leq \frac{2 \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^q}{\lambda_{\min}(\Gamma_{\sigma_{\text{Tr}}}(Y_{\mathcal{N}}))} \varepsilon. \end{aligned} \quad (8.55)$$

Using (8.54) and (8.55), we obtain the bound

$$|\Delta(t)| \leq K t + \frac{2 p'(0) \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}}{\lambda_{\min}(\Gamma_{\sigma_{\text{Tr}}}(Y_{\mathcal{N}})) q^2} \varepsilon, \quad (8.56)$$

for all ε satisfying (8.53) and all $0 < t < \delta$. Therefore,

$$\limsup_{t \rightarrow 0} |\Delta(t)| \leq \frac{2 p'(0) \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}}{\lambda_{\min}(\Gamma_{\sigma_{\text{Tr}}}(Y_{\mathcal{N}})) q^2} \varepsilon, \quad (8.57)$$

and since ε may be arbitrarily small, we deduce that

$$\limsup_{t \rightarrow 0} |\Delta(t)| = \lim_{t \rightarrow 0} |\Delta(t)| = 0.$$

□

8.B. Towards the proof of Theorem 8.4.1(ii)

In this appendix, we define and study the properties of an object that turns out to be useful in the derivation of Theorem 8.4.1(ii): first define the following norm on operators $A \in \mathcal{B}(\mathcal{H})$:

$$\|A\|_{\mathbb{L}_1(\sigma_{\text{Tr}})} := |I| \max_{i \in I} \|P_i A P_i\|_{\mathbb{L}_1(\sigma_{\text{Tr}})}.$$

In what follows, we also denote by $\tilde{\mathcal{S}}_{\mathbb{L}_1(\sigma_{\text{Tr}})}^+$ the set of positive definite operators A of norm $\|A\|_{\mathbb{L}_1(\sigma_{\text{Tr}})} = 1$. Now, given a positive semidefinite operator X and $1 \leq q < p \leq \infty$, let

$$\|X\|_{\mathbb{L}_q(\mathcal{N}, \mathbb{L}_p(\sigma_{\text{Tr}}))} := \inf_{A \in \mathcal{N}(\mathcal{P}) \cap \tilde{\mathcal{S}}_{\mathbb{L}_1(\sigma_{\text{Tr}})}^+} \|A^{-1/2r} X A^{-1/2r}\|_{\mathbb{L}_p(\sigma_{\text{Tr}})}. \quad (8.58)$$

The following lemma is straightforward:

Lemma 8.B.1. *For all X positive semidefinite, and any $1 \leq q < p \leq \infty$, $\|X\|_{\mathbb{L}_q(\mathcal{N}, \mathbb{L}_p(\sigma_{\text{Tr}}))} \geq \|X\|_{\mathbb{L}_q(\mathcal{N}, \mathbb{L}_p(\sigma_{\text{Tr}}))}$, and equality holds whenever $|I| = 1$. Moreover, the optimum in Equation (8.58) is attained on the subset of positive definite operators $A \in \mathcal{N}(\mathcal{P})$ such that $\|P_i A P_i\|_{\mathbb{L}_1(\sigma_{\text{Tr}})} = \frac{1}{|I|}$ for all $i \in I$.*

Proof. The second part of the lemma follows from the observation that for any two positive semidefinite operators $A \in \mathcal{N}(\mathcal{P})$ and $X \in \mathcal{B}(\mathcal{H})$,

$$\|A^{-1/2r} X A^{-1/2r}\|_{\mathbb{L}_p(\sigma_{\text{Tr}})} = \|A^{-1/2r} \Gamma_{\sigma_{\text{Tr}}}^{1/p}(X) A^{-1/2r}\|_p = \|[\Gamma_{\sigma_{\text{Tr}}}^{1/p}(X)]^{1/2} A^{-1/r} [\Gamma_{\sigma_{\text{Tr}}}^{1/p}(X)]^{1/2}\|_p.$$

Since $\frac{1}{r} = \frac{1}{q} - \frac{1}{p} \leq 1$, $x \mapsto x^{1/r}$ is operators monotone and therefore the optimization in Equation (8.58) occurs at the boundary of $\tilde{\mathcal{S}}_{\mathbb{L}_1(\sigma_{\text{Tr}})}^+$, that is for $\|P_i A P_i\|_{\mathbb{L}_1(\sigma_{\text{Tr}})} = \frac{1}{|I|}$ for all $i \in I$. The first part follows directly from the latter fact, since it implies that $\|A\|_{\mathbb{L}_1(\sigma_{\text{Tr}})} = 1$. $\square$

Theorem 8.4.1(ii) relies crucially on the below Lemmas 8.B.3 and 8.B.4, which respectively generalize Lemmas 12 and 13 of [Beigi and King, 2016] to the non unital case and for $|I| \geq 1$. In order to prove these results, we first need to extend Lemmas 8.A.3 and 8.A.4 to the quantity defined in Equation (8.58).

Proposition 8.B.2. *Let $q \geq 1$, $[0, \infty) \ni t \mapsto p(t)$ by a twice continuously differentiable increasing function with $p(0) = q$ and $[0, \infty) \ni Y(t)$ be a twice continuous differentiable positive semidefinite matrix-valued function with $Y(0) = Y$, and for any $\kappa > 0$, define $\tilde{\mathcal{S}}(\kappa) := \mathcal{N}(\mathcal{P}) \cap \{B > 0, \|B\|_{\mathbb{L}_1(\sigma_{\text{Tr}})} = 1, \|B - \tilde{Y}_{\mathcal{N}}\|_{\mathbb{L}_1(\sigma_{\text{Tr}})} \leq \kappa\}$, where*

$$\tilde{Y}_{\mathcal{N}} := \sum_{i \in I} \frac{P_i E_{\mathcal{N}}[\Gamma_{1,q}(Y)] P_i}{|I| \text{Tr}[P_i (\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y))^q P_i]}.$$

Then, there exists $\tilde{\kappa} > 0$ and $\tilde{K} > 0$ such that for all $t \geq 0$ and $A \in \tilde{\mathcal{S}}(\tilde{\kappa})$,

$$\left| \Phi(\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{p(t)}}(Y(t)), A, p(t)) - \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})} - t \frac{p'(0)G(A)}{q^2 \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^{q-1}} \right| \leq \tilde{K} t^2. \quad (8.59)$$

Moreover, for any $\tilde{\varepsilon} \leq \tilde{\kappa}$, there exists $\tilde{\delta} > 0$ such that for all $t \in [0, \tilde{\delta}]$ there is $A(t) \in \mathcal{N}(\mathcal{P}) \cap \tilde{\mathcal{S}}_{\mathbb{L}_1(\sigma_{\text{Tr}})}^+$ satisfying

$$\|Y(t)\|_{\mathbb{L}_q(\mathcal{N}, \mathbb{L}_p(\sigma_{\text{Tr}}))} = \Phi(\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{p(t)}}(Y(t)), A(t), p(t)), \quad \|\tilde{Y}_{\mathcal{N}} - A(t)\|_{\mathbb{L}_1(\sigma_{\text{Tr}})} \leq \tilde{\varepsilon}. \quad (8.60)$$

Proof. The proof of (8.59) follows the exact same lines as the proof of Lemma 8.A.3. Now, let $X(t) := \Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{p(t)}}(Y(t))^{p(t)}$ and given $\tilde{\varepsilon} \leq \tilde{\kappa}$, choose $\tilde{\delta}' > 0$ satisfying

$$\tilde{\delta}' < \frac{\tilde{\varepsilon}^2 \min_{j \in I} \text{Tr}(P_j \Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y)^q P_j) p'(0)}{4 \tilde{K} q^2 \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^{q-1}}.$$

Then, we have

$$\tilde{\mathcal{S}}(\tilde{\varepsilon}) \subset \tilde{\mathcal{S}}(\tilde{\kappa}) \subset \mathcal{N}(\mathcal{P}) \cap \tilde{\mathcal{S}}_{\mathbb{L}_1(\sigma_{\text{Tr}})}^+.$$

Suppose that A belongs to the boundary of $\tilde{\mathcal{S}}(\tilde{\varepsilon})$, so that

$$\|\tilde{Y}_{\mathcal{N}} - A\|_{\mathbb{L}_1(\sigma_{\text{Tr}})} = \tilde{\varepsilon}.$$

Hence, as in the proof of Lemma 8.A.2, we can show that

$$\begin{aligned} G(A) - G(\tilde{Y}_{\mathcal{N}}) &= \text{Tr} \left(\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y)^q (\ln \tilde{Y}_{\mathcal{N}} - \ln A) \right) \\ &= \sum_{i \in I} \text{Tr} \left(E_{\mathcal{N}*} [P_i \Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y)^q P_i] (\ln \tilde{Y}_{\mathcal{N}} - \ln A) \right). \end{aligned}$$

Now, define for any $i \in I$ the states $\sigma_i := |I| P_i \sigma_{\text{Tr}}^{1/2} A \sigma_{\text{Tr}}^{1/2} P_i$ and $\eta_i := |I| P_i \sigma_{\text{Tr}}^{1/2} \tilde{Y}_{\mathcal{N}} \sigma_{\text{Tr}}^{1/2} P_i$, one can easily verify that $E_{\mathcal{N}*} [P_i \Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y)^q P_i] = \text{Tr}(P_i \Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y)^q P_i) \eta_i$, so that

$$\begin{aligned} G(A) - G(\tilde{Y}_{\mathcal{N}}) &= \sum_{i \in I} \text{Tr}(E_{\mathcal{N}*} [P_i \Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y)^q P_i] (\ln \eta_i - \ln \sigma_i)) \\ &= \sum_{i \in I} \text{Tr}(P_i \Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y)^q P_i) D(\eta_i \| \sigma_i) \\ &\geq \frac{1}{2} \sum_{i \in I} \text{Tr}(P_i \Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y)^q P_i) \|\eta_i - \sigma_i\|_1^2 \\ &= \frac{|I|^2}{2} \sum_{i \in I} \text{Tr}(P_i \Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y)^q P_i) \|P_i A P_i - P_i \tilde{Y}_{\mathcal{N}} P_i\|_{\mathbb{L}_1(\sigma_{\text{Tr}})}^2 \\ &\geq \frac{|I|^2}{2} \min_{j \in I} \text{Tr}(P_j \Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y)^q P_j) \sum_{i \in I} \|P_i A P_i - P_i \tilde{Y}_{\mathcal{N}} P_i\|_{\mathbb{L}_1(\sigma_{\text{Tr}})}^2 \\ &\geq \frac{1}{2} \min_{j \in I} \text{Tr}(P_j \Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y)^q P_j) \|A - \tilde{Y}_{\mathcal{N}}\|_{\mathbb{L}_1(\sigma_{\text{Tr}})}^2 \\ &\geq \frac{\tilde{\varepsilon}^2}{2} \min_{j \in I} \text{Tr}(P_j \Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{q}}(Y)^q P_j), \end{aligned}$$

where we used Pinsker's inequality on the third line above. Following the steps of the proof of (8.48), we can show from (8.59) that for all $0 \leq t \leq \tilde{\delta}'$,

$$\Phi(X(t), A, p(t)) > \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})} + t \frac{p'(0) G(\tilde{Y}_{\mathcal{N}})}{q^2 \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^{q-1}} + \tilde{K} t^2.$$

This, together with another use of (8.59) applied to $A = \tilde{Y}_{\mathcal{N}}$ implies that

$$\Phi(X(t), \tilde{Y}_{\mathcal{N}}, p(t)) < \Phi(X(t), A, p(t)).$$

The rest of the proof follows similarly to the proof of Lemma 8.A.4. □

Lemma 8.B.3. Let $Y \in \mathcal{B}(\mathcal{H})$ positive definite and for $1 \leq q < p \leq \infty$, let $\frac{1}{r} := \frac{1}{q} - \frac{1}{p}$. Then the function $\Psi(Y, \cdot, p) : A \mapsto \|A^{-1/2r} Y A^{-1/2r}\|_{\mathbb{L}_p(\sigma_{\text{Tr}})}$ is strictly convex. Moreover, there exists a unique $\tilde{A} \in \mathcal{N}(\mathcal{P}) \cap \tilde{\mathcal{L}}_{\mathbb{L}_1(\sigma_{\text{Tr}})}^+$ such that

$$\Psi(Y, \tilde{A}, p) = \|Y\|_{\mathbb{L}_q(\mathcal{N}, \mathbb{L}_p(\sigma_{\text{Tr}}))}. \quad (8.61)$$

Moreover, the optimizer $\tilde{A}$ of Equation (8.61) satisfies the following constraint

$$P_i \tilde{A} P_i = \frac{P_i E_{\mathcal{N}} [\Gamma_{1,p}(\tilde{A}^{-1/2r} Y \tilde{A}^{-1/2r})] P_i}{|I| \text{Tr} \left[P_i \left(\Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{p}} (\tilde{A}^{-1/2r} Y \tilde{A}^{-1/2r}) \right)^p P_i \right]}. \quad (8.62)$$

Proof. Following the exact same steps as in the proof of Lemma 12 of [Beigi and King, 2016], one can show that the function

$$\Phi(X, \cdot, p) : A \mapsto \|A^{-1/2r} X A^{-1/2r}\|_p$$

is strictly convex. Let $X = \Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{p}}(Y)$. The first point then follows from the observation that $[A, \sigma_{\text{Tr}}] = 0$ for $A \in \mathcal{N}$, so that $\Psi(Y, A, p) = \Phi(X, A, p)$. The fact that the infimum is achieved at a unique point $\tilde{A}$ also follows from the same lemma. Now, we prove Equation (8.62). Let $A \in \mathcal{N}$ such that for all $i \in I$, $\text{Tr}(P_i A P_i) = \frac{1}{|I|}$. Moreover, let $D \in \mathcal{N}(\mathcal{P})$ be a self-adjoint operator such that $\text{Tr}(\sigma_{\text{Tr}} P_i D P_i) = 0$ for all $i \in I$. Then, it follows that for any $x \in \mathbb{R}$ sufficiently small, $A(x) := A + x D$ satisfies the same constraints as A . Let $B(x) := X^{\frac{1}{2}} A(x)^{-s/2}$ and $C(x) := A(x)^{s/2} \frac{d}{dx} A(x)^{-s/2} \in \mathcal{N}(\mathcal{P})$, where $s = 1/r$. Then the minimum is achieved at A if for any such D ,

$$\begin{aligned} 0 &= \frac{d}{dx} \Big|_{x=0} \Phi(X, A(x), p)^p = \frac{d}{dx} \Big|_{x=0} \text{Tr}[(B(x)^* B(x))^p] \\ &= p \text{Tr}[(B(0)^* B(0))^{p-1} (B(0)^* B(0) C(0) + C(0)^* B(0)^* B(0))] \\ &= p \langle \Gamma_{\sigma_{\text{Tr}}}^{-1} ((B(0)^* B(0))^p), (C(0) + C(0)^*) \rangle_{\sigma_{\text{Tr}}} \\ &= p \langle E_{\mathcal{N}} [\Gamma_{\sigma_{\text{Tr}}}^{-1} ((B(0)^* B(0))^p)], (C(0) + C(0)^*) \rangle_{\sigma_{\text{Tr}}} \\ &= p \langle A^{-1/2} E_{\mathcal{N}} [\Gamma_{\sigma_{\text{Tr}}}^{-1} ((B(0)^* B(0))^p)] A^{-1/2}, \Lambda_A(D) \rangle_{\sigma_{\text{Tr}}}, \end{aligned} \quad (8.63)$$

where $D \mapsto \Lambda_A(D) := A^{\frac{1}{2}} (C(0) + C(0)^*) A^{\frac{1}{2}}$ maps the space of Hermitian operators D in $\mathcal{N}(\mathcal{P})$ such that $\text{Tr}[\sigma_{\text{Tr}} P_i D P_i] = 0$ for all block $i \in I$ onto itself. Indeed, for any such D ,

$$\begin{aligned} \text{Tr}[\sigma_{\text{Tr}} P_i \Lambda_A(D) P_i] &= 2 \text{Tr} \sigma_{\text{Tr}} P_i A^{s/2+1} \frac{d}{dx} \Big|_{x=0} (A(x)^{-s/2}) P_i \\ &= 2 \text{Tr} \sigma_{\text{Tr}} P_i A^{s/2+1} (-s/2) A^{-s/2-1} D P_i \\ &= -s \text{Tr}[\sigma_{\text{Tr}} P_i D P_i] = 0. \end{aligned}$$

Moreover, the map $D \mapsto \Lambda_A(D)$ is onto. To show this, we extend the definition of this map to a linear operator $\tilde{\Lambda}_A$ on the whole space of self-adjoint operators in $\mathcal{N}(\mathcal{P})$ and prove that $\tilde{\Lambda}_A$ is onto. First, notice that $D \mapsto D^{-s/2}$ is one-to-one on the set of positive definite matrices in $\mathcal{N}(\mathcal{P})$, and hence its differential at A

$$D \mapsto \frac{d}{dx} \Big|_{x=0} (A + x D)^{-s/2} \quad (8.64)$$

is onto on $\mathcal{N}(\mathcal{P}) \cap \mathcal{B}_{sa}(\mathcal{H})$. This directly implies that $\tilde{\Lambda}_A$ is onto, since it derives from the map

defined in Equation (8.64) by multiplication with positive definite operators. Hence, Λ_A is onto, which together with Equation (8.63) implies that for any $D \in \mathcal{N}(\mathcal{P}) \cap \mathcal{B}_{sa}(\mathcal{H})$ satisfying $\text{Tr}[\sigma_{\text{Tr}} P_i D P_i] = 0$ for all $i \in I$,

$$\langle A^{-1/2} E_{\mathcal{N}}[\Gamma_{\sigma_{\text{Tr}}}^{-1}((B(0)^* B(0))^p)] A^{-1/2}, D \rangle_{\sigma_{\text{Tr}}} = 0. \quad (8.65)$$

Thus, in each block $i \in I$, $P_i A^{-1/2} E_{\mathcal{N}}[\Gamma_{\sigma_{\text{Tr}}}^{-1}((B(0)^* B(0))^p)] A^{-1/2} P_i$ is a multiple of the identity:

$$P_i E_{\mathcal{N}}[\Gamma_{\sigma_{\text{Tr}}}^{-1}((B(0)^* B(0))^p)] P_i = c_i P_i A P_i, \quad c_i \in \mathbb{R}.$$

Replacing $B(0)$ by its definition we find

$$P_i E_{\mathcal{N}}[\mathbb{I}_{1,p}(\tilde{A}^{-1/2r} Y \tilde{A}^{-1/2r})] P_i = c_i P_i A P_i. \quad (8.66)$$

Finally, the multiplicative factors c_i are found after tracing Equation (8.66) against σ_{Tr} , using the fact that $\text{Tr}(\sigma_{\text{Tr}} P_i A P_i) = \frac{1}{|I|}$ for all $i \in I$, and Equation (8.62) follows after rearranging the terms in Equation (8.66). $\square$

Lemma 8.B.4. *Given $X \in \mathcal{B}(\mathcal{H})$ positive definite and $q \geq 1$, the function*

$$[0, \infty) \ni t \mapsto \varphi(t) := \|\mathcal{P}_t(Y)\|_{\mathbb{L}_q(\mathcal{N}, \mathbb{L}_{p(t)}(\sigma_{\text{Tr}}))} \equiv \Phi(X(t), \tilde{A}(t), p(t))$$

is continuous on $[0, \infty)$, for $p(t) := 1 + (q-1) e^{4t/c}$, where Φ is the map defined in Equation (8.39), $X(t) \equiv \Gamma_{\sigma_{\text{Tr}}}^{\frac{1}{p(t)}}(\mathcal{P}_t(Y))$ and $\tilde{A}(t)$ is the optimizer obtained in Lemma 8.B.3.

Proof. From (8.59), there exist $\tilde{\kappa} > 0$ and $\tilde{K} < \infty$, such that for all $t \in [0, \infty)$ and $A \in \tilde{\mathcal{S}}(\tilde{\kappa})$,

$$|\Phi(X(t), A, p(t)) - \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}| \leq t \frac{p'(0)G(A)}{q^2 \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^{q-1}} + \tilde{K} t^2.$$

Moreover, from the second part of Proposition 8.B.2 we know that, for sufficiently small t , the optimizer $\tilde{A}(t)$ is in $\tilde{\mathcal{S}}(\tilde{\kappa})$. Since $\varphi(0) = \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}$, the above inequality implies

$$|\varphi(t) - \varphi(0)| \leq t \frac{p'(0)G(\tilde{A}(t))}{q^2 \|Y\|_{\mathbb{L}_q(\sigma_{\text{Tr}})}^{q-1}} + K t^2.$$

By definition, the map $A \mapsto G(A)$ defined in Equation (8.43) is continuous, and hence uniformly bounded on $\tilde{\mathcal{S}}(\tilde{\kappa})$. Hence, the continuity of φ at 0 follows. We now prove the continuity of φ at $t_0 > 0$. For any $0 < a < t_0 < b$, $t \in [a, b]$ and $s(t) = \frac{1}{q} - \frac{1}{p(t)}$,

$$\begin{aligned} \varphi(t) &= \Phi(X(t), \tilde{A}(t), p(t)) \\ &= \|\tilde{A}(t)^{-s(t)/2} X(t) \tilde{A}(t)^{-s(t)/2}\|_{p(t)} \\ &\geq \|\tilde{A}(t)^{-s(t)}\|_{p(t), \sigma_{\text{Tr}}} \|\mathcal{P}_t(Y)^{-1/2}\|_{\infty}^{-2} \\ &\geq \lambda_{\min}(\sigma_{\text{Tr}}) \lambda_{\min}(\tilde{A}(t))^{-s(t)} \|\mathcal{P}_t(Y)^{-1/2}\|_{\infty}^{-2}, \end{aligned}$$

where $\lambda_{\min}(\tilde{A}(t))$ is the minimum eigenvalue of $\tilde{A}(t)$. On the other hand,

$$\varphi(t) = \inf_A \Phi(X(t), A, p(t)) \leq \Phi(X(t), \mathbb{1}, p(t)) = \|X(t)\|_{p(t)} \leq \|\mathcal{P}_t(Y)\|_{p(t), \sigma_{\text{Tr}}} \leq \|\mathcal{P}_t(Y)\|_{\infty}.$$

Together with the previous bound, we arrive at

$$\lambda_{\min}(\tilde{A}(t))^{-s(a)} \leq \lambda_{\min}(\tilde{A}(t))^{-s(t)} \leq \lambda_{\min}(\sigma_{\text{Tr}}) \|\mathcal{P}_t(Y)^{-1/2}\|_{\infty}^2 \|\mathcal{P}_t(Y)\|_{\infty}.$$

Above, we used that $t \mapsto s(t)$ increases, as well as the fact that $\lambda_{\min}(\tilde{A}(t)) \leq 1$, since $\|\tilde{A}(t)\|_{\mathbb{L}_1(\sigma_{\text{Tr}})} = 1$. By continuity of $t \mapsto \mathcal{P}_t(Y)$, the right hand side of the above chain of inequalities is uniformly bounded by some positive constant $C > 0$ over the interval $[a, b]$. Therefore, $\tilde{A}(t)$ belongs to the compact set $\mathcal{R} := \mathcal{N}(\mathcal{P}) \cap \{B > 0, \|B\|_{\mathbb{L}_1(\sigma_{\text{Tr}})} = 1, \lambda_{\min}(B) \geq C^{-1/s(a)}\}$. The function $(t, A) \mapsto \Phi(X(t), A, p(t))$ restricted to the compact set $[a, b] \times \mathcal{R}$ is uniformly continuous, which means that for any $\varepsilon > 0$, there exists $\delta > 0$ such that for all $t, t' \in [a, b]$ such that $|t - t'| \leq \delta$, and any $A \in \mathcal{R}$,

$$|\Phi(X(t), A, p(t)) - \Phi(X(t'), A, p(t'))| \leq \varepsilon.$$

Therefore,

$$\varphi(t) = \Phi(X(t), \tilde{A}(t), p(t)) \leq \Phi(X(t), \tilde{A}(t'), p(t)) \leq \Phi(X(t'), \tilde{A}(t'), p(t')) + \varepsilon = \varphi(t') + \varepsilon.$$

Conversely, $\varphi(t') \leq \varphi(t) + \varepsilon$. Thus, $|\varphi(t) - \varphi(t')| \leq \varepsilon$ for all $|t - t'| \leq \delta$. We established the continuity of φ on the interval $[a, b]$, and hence at the point $t = t_0 \in [a, b]$. $\square$

Chapter 9.

The transference method

In Chapter 8, we introduced an extension of the theory of quantum hypercontractivity and its associated notion of a logarithmic Sobolev inequality to quantum Markov semigroups that are not necessarily primitive. However, computing the exact constants turns out to be a difficult problem, even for classical evolutions [Saloff-Coste, 1994, Diaconis et al., 1993], and requires a good knowledge of entropic and/or functional analytic methods. In this chapter, we show how one can get estimates on decoherence times for a particular class of QMS from a simple transfer of already known classical mixing times. In substance we show that, given a QMS in this class, there exists a classical Markov semigroup whose contractivity properties, and hence decoherence times, control the ones of the original QMS. On the other hand, given a projective representation of a (discrete or compact Lie) group G on a Hilbert space $\mathcal{H}$, any classical right-invariant transition kernel $(k_t)_{t \geq 0}$ on G gives rise to a convolution QMS on $\mathcal{B}(\mathcal{H})$, the convergence properties of which it controls. The "output" QMS can be primitive or nonprimitive, irrespective of the ergodic properties of the "input" classical Markov chain. The method however is at the moment restricted to the case doubly stochastic QMS, that is evolutions for which the completely mixed state is invariant.

This so-called *transference method* was introduced in [Gao et al., 2018b] who were exclusively concerned with the transfer of classical diffusions. Here, we broaden their scope to incorporate evolutions on finite groups. As we will see, given a quantum Markov semigroup $(\mathcal{P}_t)_{t \geq 0}$, one can find different classical semigroups acting on either a finite or a compact Lie group, both giving rise to $(\mathcal{P}_t)_{t \geq 0}$. The question of the optimality of the decoherence times that one gets from these different choices is then posed. We will provide examples of such situations and compare the results found from these different inputs. One important class of QMS that fits the present discussion is the one of collective decoherence introduced in Section 6.4.2. In particular, we show that given any right-invariant transition kernel on G , the decoherence times we get on the transferred QMS do not depend on the representation of the group that we choose. In the case of collective decoherence, this means that the estimate that we get is independent of the number of qubits constituting the system being studied.

Before moving to a more thorough analysis, we sketch the transference method on a simple example for sake of illustration: consider the Lindblad generator of the weak collective decoherence semigroup on $\mathcal{B}((\mathbb{C}^2)^{\otimes m})$, already introduced in Section 6.4.2:

$$\mathcal{L}_n^{\text{wcd}}(X) := \Sigma_z^n X \Sigma_z^n - \frac{1}{2}((\Sigma_z^n)^2 X + X(\Sigma_z^n)^2), \quad \text{where} \quad \Sigma_z^n := \sum_{i=1}^n \mathbb{1}_{\mathbb{C}^2}^{\otimes i-1} \otimes \sigma_z \otimes \mathbb{1}_{\mathbb{C}^2}^{n-i}.$$

In this example, since the corresponding non-primitive semigroup $(\mathcal{P}_t^{\text{wcd}})_{t \geq 0}$ is KMS-symmetric with respect to the completely mixed state, the decoherence-free subalgebra $\mathcal{N}(\mathcal{P}^{\text{wcd}})$ coincides with the

fixed point algebra

$$\mathcal{N}(\mathcal{P}^{\text{wcd}}) \equiv \mathcal{F}(\mathcal{P}^{\text{wcd}}) = \{X : \forall t \geq 0, \mathcal{P}_t^{\text{wcd}}(X) = X\},$$

and is given by the commutator of Σ_n^{wcd} . As seen in Chapter 6, the asymptotic behavior of the QMS is encoded into a conditional expectation $E_{\mathcal{F}} = E_{\mathcal{F}*}$:

$$E_{\mathcal{F}}(\rho) = \lim_{t \rightarrow \infty} \mathcal{P}_t^{\text{wcd}}(\rho),$$

which is both a completely positive trace preserving map and the orthogonal projection onto the fixpoint algebra with respect to $\langle \cdot, \cdot \rangle_{1/d_{\mathcal{H}}}$. Here, the main quantity of interest is the decoherence time (cf. Section 8.6):

$$\tau_{\text{deco}}(\varepsilon) = \inf \{t \geq 0; \|\mathcal{P}_t(\rho) - E_{\mathcal{F}}(\rho)\|_1 \leq \varepsilon \quad \forall \rho\}.$$

Layout of the chapter: In Section 9.1, we explain the connection of QMS to classical diffusions and jump processes on groups via the so-called *transference technique*. In Section 9.2, we explain the technical tools that allow us to bound various norm estimates of a quantum Markov semigroup in terms of the kernel of an associated classical process. Section 9.3 consists of applications of the techniques developed in the previous sections to the derivation of bounds on decoherence times of some simple QMS.

9.1. Quantum Markov semigroups via group transference

In this chapter, we are exclusively concerned with quantum Markov semigroups that are self adjoint with respect to the Hilbert Schmidt inner product (or, equivalently, that are KMS symmetric with respect to the completely mixed state) on the algebra $\mathcal{B}(\mathcal{H})$ of linear operators on a finite dimensional Hilbert space $\mathcal{H}$. Here, we recall a general construction from [Gao et al., 2018b] to obtain quantum Markov semigroups (QMS) from classical ones, and apply it to form two classes of examples: QMS of diffusive and jump type. This construction can be interpreted as a simplified version of the characterization of quantum convolution semigroups of [Kümmerer and Maassen, 1987] provided in Section 5.5.3.

The starting point is a compact group G , either Lie or finite, with Haar measure μ_G . Let $(P_t)_{t \geq 0}$ be a Markov semigroup on the algebra $\mathbb{L}_{\infty}(G)$ of bounded, measurable functions on G . We will always assume that $(P_t)_{t \geq 0}$ admits a kernel representation (see Section 2.1):

$$P_t(f)(g) = \int_G k_t(g, h) f(h) d\mu_G(h). \quad (9.1)$$

We also assume that $(P_t)_{t \geq 0}$ is right-invariant, which means that the probability to visit h from g only depends on gh^{-1} . This implies that μ_G is an invariant probability distribution and that $k_t(g, h) = k_t(gh^{-1}, e)$ where e is the neutral element of the group. We keep the same notation $k_t(g)$ for $k_t(g, e)$. Let $g \mapsto U_g$ be a projective representation of G on some finite dimensional Hilbert space $\mathcal{H}$. We define the following convolution QMS on $\mathcal{B}(\mathcal{H})$ which we call a *transferred QMS*:

$$\mathcal{P}_t(X) = \int_G k_t(g^{-1}) U_g^* X U_g d\mu_G(g). \quad (9.2)$$

At the root of the transference technique that we study in this chapter is the factorization property between $(P_t)_{t \geq 0}$ and $(\mathcal{P}_t)_{t \geq 0}$, involving the standard co-representation

$$\pi : \mathcal{B}(\mathcal{H}) \rightarrow \mathbb{L}_{\infty}(G, \mathcal{B}(\mathcal{H})) , \quad \pi(X)(g) = U_g^* X U_g .$$

Indeed, it can be checked that the following relation holds for all $t \geq 0$ (Lemma 4.6 in [Gao et al., 2018b]):

$$\pi \circ \mathcal{P}_t = (P_t \otimes \text{id}_{\mathcal{B}(\mathcal{H})}) \circ \pi. \quad (9.3)$$

Indeed, for any $X \in \mathcal{B}(\mathcal{H})$,

$$\begin{aligned} \pi \circ \mathcal{P}_t(X)(g) &= U_g^* \int_G k_t(h^{-1}) U_{h^{-1}} X U_h d\mu_G(h) U_g \\ &= \int_G k_t(gg^{-1}h^{-1}) U_{(hg)^{-1}} X U_{hg} d\mu_G(h) \\ &= \int_G k_t(gh^{-1}) U_{h^{-1}} X U_h d\mu_G(h) \\ &= (P_t \otimes \text{id}_{\mathcal{B}(\mathcal{H})})(\pi(X))(g). \end{aligned}$$

From the invariance of μ_G , one can easily verify that any QMS $(\mathcal{P}_t)_{t \geq 0}$ transferred from $(P_t)_{t \geq 0}$ is doubly stochastic: $\mathcal{P}_{t*}(d_{\mathcal{H}}^{-1} \mathbb{1}) = d_{\mathcal{H}}^{-1} \mathbb{1}$ for any $t \geq 0$. In practice, we only consider situations where the classical Markov semigroup $(P_t)_{t \geq 0}$ is *primitive*, that is, μ_G is the unique invariant distribution and furthermore

$$P_t f \xrightarrow{t \rightarrow +\infty} \mathbb{E}_{\mu_G}[f] = \int_G f(g) d\mu_G(g).$$

This does not imply that the transferred QMS $(\mathcal{P}_t)_{t \geq 0}$ is also primitive. On the other hand, the reversibility of $(P_t)_{t \geq 0}$ is transferred to the QMS $(\mathcal{P}_t)_{t \geq 0}$.

Lemma 9.1.1. *Assume that the Markov semigroup $(P_t)_{t \geq 0}$ is reversible, or equivalently that $k_t(g) = k_t(g^{-1})$ for any $g \in G$. Then any QMS $(\mathcal{P}_t)_{t \geq 0}$ transferred from $(P_t)_{t \geq 0}$ is KMS-symmetric with respect to $d_{\mathcal{H}}^{-1} \mathbb{1}_{\mathcal{H}}$.*

Proof. The result follows from the simple calculation:

$$\begin{aligned} \langle X, \mathcal{P}_t(Y) \rangle_{\text{HS}} &= \int_G k_t(g^{-1}) \text{Tr}(X^* U_g^* Y U_g) d\mu_G(g) \\ &= \int_G k_t(g^{-1}) \text{Tr}((U_g X U_g^*)^* Y) d\mu_G(g) \\ &= \int_G k_t(g^{-1}) \text{Tr}((U_g^* X U_g)^* Y) d\mu_G(g) \\ &= \langle \mathcal{P}_t(X), Y \rangle_{\text{HS}}, \end{aligned}$$

where the third line follows from the identity $k_t(g) = k_t(g^{-1})$ for all $g \in G$. □

Since $d_{\mathcal{H}}^{-1} \mathbb{1}_{\mathcal{H}}$ is an invariant state of $(\mathcal{P}_t)_{t \geq 0}$, Theorem 6.1.2 applies and the set of fixed points $\mathcal{F}$ is an algebra. Assuming moreover that $(P_t)_{t \geq 0}$ is self-adjoint, then so is $(\mathcal{P}_t)_{t \geq 0}$, and the decoherence-free algebra $\mathcal{N}$ is equal to $\mathcal{F}$.

Moreover, the fixed-point algebra $\mathcal{F}$ is then characterized as the commutant of the projective representation (see Theorem 6.13 of [Wolf, 2012]):

$$\mathcal{F} = \{U_g; g \in G\}' . \quad (9.4)$$

By definition, it is also the algebra of fixed points of the $*$ -automorphisms $X \mapsto U_g^* X U_g$, $g \in G$. This implies that the following commuting diagram also holds:

$$\begin{array}{ccc} \mathcal{B}(\mathcal{H}) & \xrightarrow{E_{\mathcal{F}}} & \mathcal{F} \\ \downarrow \pi & & \downarrow \pi \\ \mathbb{L}_{\infty}(G, \mathcal{B}(\mathcal{H})) & \xrightarrow{\mathbb{E}_{\mu_G}} & \mathcal{B}(\mathcal{H}). \end{array}$$

We now turn our attention to two special cases of such construction. In both cases, we explicitly construct the Linblad generator of the QMS.

9.1.1. Diffusion

Let $(P_t)_{t \geq 0}$ be the semigroup defined in Section 3.7.3 corresponding to the Hörmander system $V = \{V_1, \dots, V_m\}$ on a given Lie group G , of corresponding generator

$$L_V = \sum_j V_j^2. \quad (9.5)$$

The generator L_V generates a semigroup $P_t = e^{tL_V}$ on $\mathbb{L}_\infty(G)$. Since the semigroup commutes with the right action of the group, it is implemented by a convolution kernel as in Equation (9.1):

$$P_t(f)(g) = \int k_t(gh^{-1}) f(h) d\mu_G(h).$$

Next, considering a projective representation $g \mapsto U_g$ of G on some finite dimensional Hilbert space $\mathcal{H}$, we want to find the Lindblad generator of the QMS defined by Equation (9.2). We first observe that, given the geodesic g_j associated to the vector field V_j , $U_{g_j(t)}$ is a one parameter family of unitaries and hence

$$\left. \frac{d}{dt} U_{g_j(t)} \right|_{t=0} = iA_j \quad (9.6)$$

is given by a self-adjoint matrix in finite dimension. This implies that, for any $X \in \mathcal{B}(\mathcal{H})$,

$$\begin{aligned} (V_j \otimes \text{id}_{\mathcal{B}(\mathcal{H})}) \circ \pi(X)(g) &= \left. \frac{d}{dt} \pi(X)(g_j(t)g) \right|_{t=0} \\ &= \left. \frac{d}{dt} U_g^* U_{g_j(t)}^* X U_{g_j(t)} U_g \right|_{t=0} \\ &= -i\pi([A_j, X])(g). \end{aligned}$$

Therefore we get

$$(L_V \otimes \text{id}_{\mathcal{B}(\mathcal{H})}) \circ \pi(X) = \sum_j i^2 \pi([A_j, [A_j, X]]) = \pi\left(\sum_j 2A_j X A_j - A_j^2 X - X A_j^2\right).$$

By Equation (9.3), it means that the Linblad generator of the transferred QMS is given by

$$\mathcal{L}_V(X) := \sum_j 2A_j X A_j - A_j^2 X - X A_j^2.$$

Remark that conversly, if a Lindblad generator of a QMS on $\mathcal{B}(\mathcal{H})$ has the form given by the previous equation for some self-adjoint elements $A_j \in \mathcal{B}_{sa}(\mathcal{H})$, then we can consider the A_j as tangent elements of the Lie group $\mathcal{U}(\mathcal{H})$ at the identity $\mathbb{1}_{\mathcal{H}}$. Therefore they generate a Hörmander system. Furthermore they span the whole tangent space of the Lie-subgroup of $\mathcal{U}(\mathcal{H})$ induced by this Hörmander system, so that the corresponding generator L_V defined through Equation (9.5) is the one of a primitive Markov semigroup $(P_t)_{t \geq 0}$. We summarize this discussion in the next theorem.

Theorem 9.1.2. *Let $g \mapsto U_g$ be a projective representation of G on some finite dimensional Hilbert space $\mathcal{H}$. Then the Linblad generator of the transferred QMS $(\mathcal{P}_t = e^{t\mathcal{L}_V})_{t \geq 0}$ as defined by Equation (9.2) is given by*

$$\mathcal{L}_V(X) = \sum_j 2A_j X A_j - A_j^2 X - X A_j^2, \quad (9.7)$$

where the A_j 's are defined by Equation (9.6). Conversely, let $\mathcal{L}$ be the Lindblad generator of a QMS on $\mathcal{B}(\mathcal{H})$ which takes the form (9.7) for some self-adjoint elements $A_j \in \mathcal{B}_{sa}(\mathcal{H})$. Then there exists a compact Lie group G , a continuous projective representation $U : G \rightarrow \mathcal{U}(H)$ and a Hörmander system $V = \{V_1, \dots, V_m\}$ in the Lie algebra of G such that $\pi : X \mapsto (g \mapsto U_g^* X U_g)$ satisfies

$$\pi(\mathcal{L}_V(X)) = (L_V \otimes \text{id}_{\mathcal{B}(\mathcal{H})})(\pi(X)).$$

9.1.2. Jumps

Let now G be a finite group and let $(k_t(g, h))_{g, h \in G}$ be a right-invariant density kernel on G (cf. Section 2.4). We write $(g_t)_{t \geq 0}$ the stochastic process on G induced by this kernel. The corresponding Markov semigroup admits a transition matrix L such that $P_t = e^{tL}$ for all $t \geq 0$. Writing $\sigma_h = L(h^{-1}, e) > 0$ for all $h \neq e$, we then have by right invariance that for all $f \in \mathbb{L}_\infty(G)$,

$$L(f)(g) = \sum_{h \in G} \sigma_h (f(hg) - f(g)),$$

where we used that $\sum_h \sigma_h = 0$. Thanks to the right-invariance, we can then define a family of independent Poisson processes $((\tilde{N}_t^h)_{t \geq 0})_{h \in G}$ with intensity σ_h such that for any function $f \in \mathbb{L}_\infty(G)$:

$$f(g_t) - f(g_{t-}) = \sum_{h \in G} (f(hg_t) - f(g_{t-})) (\tilde{N}_t^h - \tilde{N}_{t-}^h).$$

Define the compensated Poisson process with intensity σ_h and jumps $1/\sqrt{\sigma_h}$:

$$N_t^h = \frac{1}{\sqrt{\sigma_h}} (\tilde{N}_t^h - \sigma_h t).$$

Writing $df(g_t) := f(g_t) - f(g_{t-})$ and $dN_t^h := N_t^h - N_{t-}^h$, we can rewrite the previous equation as the stochastic differential equation:

$$df(g_t) = \sum_{h \in G} \sigma_h (f(hg_t) - f(g_{t-})) dt + \sum_{h \in G} \sqrt{\sigma_h} (f(hg_t) - f(g_{t-})) dN_t^h. \quad (9.8)$$

We are now ready to build a QMS from this Markov chain. Let $g \mapsto U_g$ be a projective representation of G on some finite dimensional Hilbert space $\mathcal{H}$. We want to find a stochastic differential equation for $(U_{g_t})_{t \geq 0}$. To this end, take $Y \in \mathcal{B}(\mathcal{H})$ and define $f_Y : h \in G \mapsto \text{Tr}[Y U_h]$. Applying Equation (9.8) to f_Y we find

$$df_Y(g_t) = \sum_{h \in G} \sigma_h \text{Tr}[Y (U_{hg_{t-}} - U_{g_{t-}})] dt + \sum_{h \in G} \sqrt{\sigma_h} \text{Tr}[Y (U_{hg_{t-}} - U_{g_{t-}})] dN_t^h.$$

From this we deduce

$$dU_{g_t} = \sum_{h \in G} \sigma_h (U_g - \mathbb{1}_{\mathcal{H}}) U_{g_{t-}} dt + \sum_{h \in G} \sqrt{\sigma_h} (U_h - \mathbb{1}_{\mathcal{H}}) U_{g_{t-}} dN_t^h. \quad (9.9)$$

This equation is well-known in the theory of quantum stochastic calculus, see [Hudson and Parthasarathy, 1984, Meyer, 1993].

Theorem 9.1.3. *Let $(P_t = e^{tL})_{t \geq 0}$ be a Markov semigroup on a finite group G with right-invariant Markov kernel. Write $\sigma_g = L(g^{-1}, e)$. Then the generator of the QMS $(\mathcal{P}_t)_{t \geq 0}$ defined by Equation (9.2)*

is given for all $X \in \mathcal{B}(\mathcal{H})$ by

$$\mathcal{L}(X) = \sum_{g \in G} \sigma_g (U_g^* X U_g - X). \quad (9.10)$$

Furthermore, $\frac{\mathbb{1}_{\mathcal{H}}}{d_{\mathcal{H}}}$ is an invariant density matrix and if $(P_t)_{t \geq 0}$ is self-adjoint, then so is $(\mathcal{P}_t)_{t \geq 0}$. Conversely, let $\mathcal{L}$ be a Lindblad generator on $\mathcal{B}(\mathcal{H})$ of the form

$$\mathcal{L}(X) = \sum_{k=1}^m \sigma_k (U_k^* X U_k - X),$$

for some unitary operators $U_k \in \mathcal{U}(\mathcal{H})$ and some positive constants σ_k . Assume that the group G generated by $U_1, \dots, U_m$ is finite and define

$$L(f)(g) = \sum_{k=1}^m \sigma_k (f(U_k g) - f(g)).$$

Then L is the generator of a primitive Markov semigroup $(P_t = e^{tL})_{t \geq 0}$ on the oriented graph (G, E) , where

$$E = \{(g, U_k g) \mid k = 1, \dots, m; g \in G\}.$$

Furthermore, the map $\pi : \mathcal{B}(\mathcal{H}) \rightarrow \mathbb{L}_{\infty}(G, \mathcal{B}(\mathcal{H}))$ satisfying the defining property

$$\pi(X)(U_k) = U_k^* X U_k$$

extend to a $*$ -representation of $\mathcal{B}(\mathcal{H})$ on $\mathbb{L}_{\infty}(G, \mathcal{B}(\mathcal{H}))$ such that $(L \otimes \text{id}_{\mathcal{B}(\mathcal{H})}) \circ \pi(X) = \pi(\mathcal{L}(X))$ for all $X \in \mathcal{B}(\mathcal{H})$.

Proof. We begin by proving Equation (9.10). By definition, we have for all $X \in \mathcal{B}(\mathcal{H})$

$$\mathcal{L}(X) = \frac{d}{dt} \mathcal{P}_t(X) \Big|_{t=0} = \frac{d}{dt} \mathbb{E}[U_{g_t}^* X U_{g_t}] \Big|_{t=0}.$$

Equation (9.10) follows from an application of the Itô formula for compensated Poisson processes. The fact that $\frac{\mathbb{1}_{\mathcal{H}}}{d_{\mathcal{H}}}$ is an invariant density matrix is straightforward as clearly

$$\mathcal{L}_* \left(\frac{\mathbb{1}_{\mathcal{H}}}{d_{\mathcal{H}}} \right) = 0.$$

Now, if $(P_t)_{t \geq 0}$ is self-adjoint, then $\sigma_g = \sigma_{g^{-1}}$ for all $g \in G$. We then have for all $x \in \mathcal{B}(\mathcal{H})$:

$$\begin{aligned} \mathcal{L}_*(X) &= \sum_{g \in G} \sigma_g (U_g X U_g^* - X) \\ &= \sum_{g \in G} \sigma_{g^{-1}} (U_{g^{-1}} X (U_{g^{-1}})^* - X) \\ &= \sum_{g \in G} \sigma_g (U_g^* X U_g - X) \\ &= \mathcal{L}(X). \end{aligned}$$

The second part of the proof is straightforward. □

9.1.3. The general situation

The two cases explored above are particular instances of the convolution QMS defined by [Kossakowski, 1972] and fully characterized (in the finite dimensional case) by [Kümmerer and Maassen, 1987] (cf.

Section 5.5.3). In particular, the generators of the form given by Equation (5.51) are the sum of three parts:

- (i) The first part corresponds to a unitary evolution, with generator given by $\mathcal{B}(\mathcal{H}) \ni x \mapsto i[H, X]$ where H is self-adjoint;
- (ii) A diffusion part, given by

$$\mathcal{B}(\mathcal{H}) \ni X \mapsto \sum_{j=1}^k 2A_j X A_j - A_j^2 X - X A_j^2,$$

where the A_j are self-adjoint operators. Any such family $\{A_j\}$ is a Hörmander system for the sub-Lie algebra that they generate, as elements of the unitary group $\mathcal{U}(\mathcal{H})$ of $\mathcal{H}$. Consequently the result of Section 9.1.1 applies.

- (iii) A jump part, given by

$$\mathcal{B}(\mathcal{H}) \ni X \mapsto \sum_{i=1}^l c_i (U_i^* X U_i - X),$$

where the U_i 's are unitary operators on $\mathcal{H}$. Compared to previously, this class is larger than the one presented in Section 9.1.2. Indeed, the family $\{U_i\}$ spans a subgroup of the unitary group $\mathcal{U}(\mathcal{H})$, however in general it will not be a finite group.

Remark 9.1.4. Starting with a Linblad generator, there may be an ambiguity around the choice of the underlying group and classical Markov semigroup leading to it. Indeed, in the jump scenario when the QMS is self-adjoint, it is always possible to write the Linblad generator as in the diffusive case. Then either the group is large, i.e. the commutator is $\mathbb{C} \mathbb{1}_{\mathcal{H}}$, and we can treat it as an Hörmander system, or the group is small (for us finite) and we can treat it as a Markov semigroup with jumps on the Cayley graph of the group. In both cases, estimates on the decoherence time of the corresponding QMS can be found.

9.2. Norm transference and decoherence times

Now, we show how the machinery described above provides a control on the norms and entropies of the output of the quantum Markov convolution semigroups of Section 9.1 in terms of the kernel of their associated classical semigroup. We start by recalling the notations of Section 9.1: $(P_t)_{t \geq 0}$ is a Markov semigroup on the compact group G (either Lie or finite), with right-invariant kernel $(k_t)_{t \geq 0}$. Let the QMS $(\mathcal{P}_t = e^{t\mathcal{L}})_{t \geq 0}$ be the transferred QMS on $\mathcal{B}(\mathcal{H})$ defined by Equation (9.2) through the projective representation $g \mapsto U_g$ of G on a finite dimensional Hilbert space $\mathcal{H}$.

Next theorem regroups all the transference techniques which will be frequently used in the remaining of this chapter (see [Gao et al., 2018b]). We recall that the spectral gap $\lambda(\mathcal{L})$ of the symmetric Lindblad generator $\mathcal{L}$ of $(\mathcal{P}_t)_{t \geq 0}$ (resp. the spectral gap $\lambda(L)$ of the generator L of $(P_t)_{t \geq 0}$) is the smallest non-zero eigenvalue of $-\mathcal{L}$ (resp. of $-L$).

Theorem 9.2.1 (Transference). *Let $\mathcal{P}_t = e^{t\mathcal{L}}$ and $P_t = e^{tL}$ as in Section 9.1. Then*

- (i) $\lambda(\mathcal{L}) \geq \lambda(L)$;
- (ii) $\|\mathcal{P}_t : \mathbb{L}_s^p(\mathcal{F} \subset \mathcal{B}(\mathcal{H})) \rightarrow \mathbb{L}_s^q(\mathcal{F} \subset \mathcal{B}(\mathcal{H}))\|_{\text{cb}} \leq \|P_t : \mathbb{L}_p(\mu_G) \rightarrow \mathbb{L}_q(\mu_G)\|_{\text{cb}}$;
- (iii) $\|\mathcal{P}_t - E_{\mathcal{F}} : \mathbb{L}_s^p(\mathcal{F} \subset \mathcal{B}(\mathcal{H})) \rightarrow \mathbb{L}_s^q(\mathcal{F} \subset \mathcal{B}(\mathcal{H}))\|_{\text{cb}} \leq \|P_t - \mathbb{E}_{\mu_G}[\cdot] : \mathbb{L}_p(\mu_G) \rightarrow \mathbb{L}_q(\mu_G)\|_{\text{cb}}$.

The above theorem is very powerful, since it directly implies the following bound on the decoherence time of the QMS $(\mathcal{P}_t)_{t \geq 0}$ in terms of the mixing time of the corresponding classical Markov chain $(P_t)_{t \geq 0}$.

Corollary 9.2.2. *Let $(\mathcal{P}_t)_{t \geq 0}$ a self-adjoint QMS of the form of Equation (9.2), of associated self-adjoint classical Markov chain $(P_t)_{t \geq 0}$. Then, for any $\varepsilon > 0$:¹*

$$\tau_{\text{deco}}(\varepsilon) < \tau_{\text{mix}}(\varepsilon/2) := \inf \left\{ t \geq 0 : \left\| P_t \left(\frac{d\nu}{d\mu_G} \right) - 1 \right\|_{\mathbb{L}_1(\mu_G)} \leq \varepsilon \quad \forall \nu \ll \mu_G \right\}. \quad (9.11)$$

Proof. This is a direct consequence of point (iii) of Theorem 9.2.1 together with Lemma 1.1.14. $\square$

As seen in Section 3.4, the semigroup $(P_t)_{t \geq 0}$ is called ultracontractive when it is bounded from $\mathbb{L}_1(\mu_G)$ (or $\mathbb{L}_2(\mu_G)$) to $\mathbb{L}_\infty(G)$. One can recast this property in terms of the density kernel as follows:

$$\|P_t : \mathbb{L}_1(\mu_G) \rightarrow \mathbb{L}_\infty(G)\|_{\text{cb}} = \|P_t : \mathbb{L}_1(\mu_G) \rightarrow \mathbb{L}_\infty(G)\| = \sup_{g \in G} |k_t(g)|, \quad (9.12)$$

where the first identity follows from Lemma 1.1.14, whereas the second identity follows by duality. Similarly (see e.g. [Saloff-Coste, 1997]):

$$\|P_t - \mathbb{E}_{\mu_G}[\cdot] \mathbb{1}_G : \mathbb{L}_1(\mu_G) \rightarrow \mathbb{L}_2(\mu_G)\| = \left(\int_G |k_t(g) - 1|^2 d\mu_G(g) \right)^{1/2}. \quad (9.13)$$

This together with Theorem 9.2.1 implies that any estimate on the kernel directly translates into an estimate on the transferred QMS. Such estimates can be found e.g. in [Saloff-Coste, 1994, Diaconis and Saloff-Coste, 1996a]. In other words, ultracontractivity of a classical semigroup translates to ultracontractivity of its transferred QMS.

9.3. Examples

Here, we illustrate the method developed in the previous sections by listing easy examples of known quantum convolution semigroups, as well as the constants that one gets from the transference technique described in Section 9.2.

9.3.1. The depolarizing QMS

Perhaps the simplest QMS that one can think about is the depolarizing semigroup on $\mathcal{B}(\mathbb{C}^n)$:

$$\mathcal{L}^{\text{depol}}(\rho) = \frac{\mathbb{1}_{\mathbb{C}^n}}{n} - \rho, \quad \mathcal{P}_t^{\text{depol}}(\rho) = e^{-t} \rho + (1 - e^{-t}) \frac{\mathbb{1}_{\mathbb{C}^n}}{n}.$$

Using the representation of $\mathbb{Z}_n \times \mathbb{Z}_n$ via the discrete Weyl matrices $\{U_{i,j}\}_{i,j \in [n]}$ (see e.g. [Wolf, 2012]) together with Equation (9.10), we directly find that the QMS transferred in this representation is

$$\mathcal{L}(\rho) = \frac{1}{n^2} \sum_{i,j=1}^n (U_{i,j} \rho U_{i,j}^* - \rho) \quad (9.14)$$

$$= \text{Tr}(\rho) \frac{\mathbb{1}_{\mathbb{C}^n}}{n} - \rho \quad (9.15)$$

$$= \mathcal{L}^{\text{depol}}(\rho), \quad (9.16)$$

¹The factor $\varepsilon/2$ comes from the definition of the mixing time for classical Markov semigroups in terms of the total variation distance.

where we took $\sigma_{i,j} := \frac{1}{n^2}$ for all i, j . This choice implies that the uniform random walk on the complete graph with n^2 vertices transfers to $(\mathcal{P}_t^{\text{depol}})_{t \geq 0}$. Recalling the bound (3.13) and the constant for the complete graph given in (3.30), we find the following upper bound on the mixing time of $(\mathcal{P}_t^{\text{depol}})_{t \geq 0}$:

$$\tau_{\text{deco}}(\varepsilon) \leq \tau_{\text{mix}}(\varepsilon/2) \leq n^2 \frac{1 - \ln \varepsilon}{n^2 - 1} + \frac{n^2 \ln(n^2 - 1)}{2(n^2 - 2)} \ln \ln n^2 \underset{n \rightarrow \infty}{\sim} \frac{\ln(n) \ln \ln(n)}{2}.$$

This can be compared with the tighter bound that one can get from the modified logarithmic Sobolev constant $\alpha_1(\mathcal{L}^{\text{depol}}) \geq \frac{1}{4}$:

$$\|\rho_t - n^{-1} \mathbb{1}_{\mathbb{C}^n}\|_1 \leq \sqrt{2 \ln n} e^{-\frac{t}{2}} \Rightarrow \tau_{\text{deco}}(\varepsilon) \leq 2 \ln \frac{\sqrt{2 \ln n}}{\varepsilon} \sim \ln \ln n.$$

9.3.2. The dephasing QMS

We recall that the *dephasing quantum Markov semigroup* (also called *decoherent QMS* in [Bardet, 2017]) on $\mathcal{B}(\mathbb{C}^n)$, $n \geq 3$, is given by

$$\mathcal{L}^{\text{deph}}(X) = E_{\text{diag}}[X] - X, \quad \mathcal{P}_t^{\text{deph}}(X) = e^{-t} X + (1 - e^{-t}) E_{\text{diag}}[X],$$

where E_{diag} denotes the projection on the space of matrices that are diagonal in some prefixed eigenbasis. Here, we show how simple representations of the discrete and continuous torus both lead to the *dephasing quantum Markov semigroup*.

Dephasing from the discrete torus Choose the uniform random walk of kernel $K(j, k) = 1/n$ for any $j, k \in \mathbb{Z}_n$. A simple unitary representation of $\mathbb{Z}_n$ is given by taking $\mathcal{H} = \mathbb{C}^n$ and

$$U_j := U^j, \quad j \in \mathbb{Z}_n,$$

where U denotes the Weyl unitary operator given by $U = \text{diag}(1, e^{\frac{2i\pi}{n}}, \dots, e^{\frac{2i(n-1)\pi}{n}})$ on $\mathcal{B}(\mathbb{C}^n)$, where the diagonal is chosen to be the one corresponding to E_{diag} . One can easily verify from Equation (3.29) and Equation (9.2) that the QMS $(\mathcal{P}_t^{\text{deph}})_{t \geq 0}$ coincides with the generator of the transferred QMS corresponding to the uniform kernel on $\mathbb{Z}_n$, since by a direct calculation $E_{\text{diag}}[X] = \frac{1}{n} \sum_{j \in \mathbb{Z}_n} U^{-j} X U^j$.

Now, it results from Corollary 9.2.2, (3.13) and (3.30) that

$$\tau_{\text{deco}}(\varepsilon) \leq n \frac{1 - \ln \varepsilon}{n - 1} + \frac{n \ln(n - 1)}{2(n - 2)} \ln \ln n \underset{n \rightarrow \infty}{\sim} \frac{\ln(n) \ln \ln(n)}{2}.$$

Dephasing from the n -dimensional torus Take the representation of the n -dimensional torus that consists of diagonal unitary matrices:

$$\mathbb{T}^n \ni (t_1, \dots, t_n) \mapsto \begin{bmatrix} e^{2it_1\pi} & & 0 \\ & \ddots & \\ 0 & & e^{2it_n\pi} \end{bmatrix}.$$

The QMS associated to the heat semigroup and the above representation corresponds to $(\mathcal{P}_t^{\text{deco}})_{t \geq 0}$. This simply follows from Equation (9.7) by taking the generators $A_j := |j\rangle\langle j|$ of $\mathbb{T}^n$, so that the

generator of the transferred QMS is equal to

$$\begin{aligned}\mathcal{L}(X) &= \frac{1}{2} \sum_{j=1}^n 2|j\rangle\langle j|X|j\rangle\langle j| - |j\rangle\langle j|X - X|j\rangle\langle j| \\ &= E_{\text{diag}}[X] - X.\end{aligned}$$

Then the estimation (3.32) leads to the following bound on the decoherence time of these QMS:

$$\tau_{\text{deco}}(\varepsilon) \leq \frac{1}{2} \ln \left(\frac{1}{2} n \ln n \right) + 6 - \ln \varepsilon \underset{n \rightarrow \infty}{\sim} \frac{\ln(n)}{2}.$$

Hence, the estimate found on the decoherence time from the continuous torus turns out to be sharper than the one found from the discrete torus. Moreover, these two bounds can be compared with the one found via decoherence-free modified logarithmic Sobolev inequality in [Bardet, 2017]. There, the DF-MLSI constant $\alpha_1(\mathcal{L}^{\text{deph}})$ of the dephasing semigroup was found to be equal to $1/4$. Therefore, a simple use of Pinsker inequality provides:

$$\tau_{\text{deco}}(\varepsilon) \leq 2 \ln \frac{\sqrt{2 \ln n}}{\varepsilon} \underset{n \rightarrow \infty}{\sim} \ln \ln n.$$

9.3.3. Collective decoherence

The bounds provided by the transference method for the examples studied in the last two sections, namely the depolarizing and the dephasing semigroups, are worse than the already known ones derived from the modified logarithmic Sobolev inequality. In this section, on the other hand, we show that our method provides an easy way of deriving estimates for collective decoherence on lattice spin systems (cf. Section 6.4.2). The power of the method lies in the fact that the constants derived are independent of the representation chosen. In particular, we get estimates that are independent of the lattice size by choosing tensor product representations.

More precisely, let G be a group and $U : G \rightarrow \mathcal{B}(\mathcal{H})$ a projective representation of G on some finite dimensional Hilbert space $\mathcal{H}$. For all $n \geq 1$, this representation induces a new representation on $\mathcal{H}^{\otimes n}$ given by:

$$g \mapsto U_g^{\otimes n}.$$

Let $(P_t)_{t \geq 0}$, $(\mathcal{P}_t)_{t \geq 0}$ be defined as in Equations (9.1) and (9.2) using the representation $g \mapsto U_g$. We write $(\mathcal{P}_t^{(n)})_{t \geq 0}$ the corresponding QMS on $\mathcal{H}^{\otimes n}$ for the representation $U^{\otimes n}$ and $\mathcal{L}_n$ its generator.

Diffusive case: In the diffusive case presented in Section 9.1.1, the generator $\mathcal{L}$ of $(P_t)_{t \geq 0}$ has the following form:

$$\mathcal{L}(x) = \sum_k 2A_k X A_k - A_k^2 X - X A_k^2, \quad (9.17)$$

where the A_k 's are selfadjoint operators on $\mathcal{H}$. Then the generator $\mathcal{L}_n$ takes the form

$$\mathcal{L}_n(x) = \sum_k 2A_k(n) X A_k(n) - A_k(n)^2 X - X A_k(n)^2, \quad A_k(n) = \sum_{j=1}^n \mathbb{1}_{\mathcal{H}}^{\otimes j-1} \otimes A_k \otimes \mathbb{1}_{\mathcal{H}}^{n-j},$$

where in the j th term of the above sum, A_k acts on the j th copy of $\mathcal{H}$. If we assume that $\mathcal{L}$ is ergodic, then the A_k belong to some unknown compact Lie group, hence the family satisfies the transference principle and the previous results of this chapter can be applied. As a consequence, we obtain dimension free bounds.

Jump case: In the jump case presented in Section 9.1.2, the generator $\mathcal{L}$ of $(\mathcal{P}_t)_{t \geq 0}$ has the following form:

$$\mathcal{L}(X) = \sum_{k=1}^m \sigma_k (X - U_k^* X U_k) , \quad (9.18)$$

where the U_k 's are unitary operators on $\mathcal{H}$. Then the generator $\mathcal{L}_n$ takes the form

$$\mathcal{L}_n(X) = \sum_{k=1}^m \sigma_k (X - V_k^* X V_k) ,$$

where $V_k = U_k^{\otimes n}$ for all k . If the unitary operators U_k generate a finite group G then thanks to Theorem 9.1.3 we can find a Markov semigroup on G and all the estimates we find on this semigroup can be transferred to $(\mathcal{P}_t^{(n)})_{t \geq 0}$ for all n .

Remark 9.3.1. Unfortunately, it is not the decoherence time or any other interesting quantity for $\mathcal{L}$ itself which transfers to all the $\mathcal{L}_n$, but the underlying group which gives the corresponding estimates. Thus, the choice of the group and the classical Markov semigroup on it are particularly important.

We now discuss two particular examples of collective decoherence already introduced in Section 6.4.2, namely the weak and the strong collective decoherences.

Weak collective decoherence We recall the generator of the weak collective decoherence on n qubits:

$$\mathcal{L}_n^{\text{wcd}}(X) := \Sigma_z^n X \Sigma_z^n - \frac{1}{2}((\Sigma_z^n)^2 X + X(\Sigma_z^n)^2), \quad \text{where} \quad \Sigma_z^n := \sum_{i=1}^n \mathbb{1}_{\mathbb{C}^2}^{\otimes i-1} \otimes \sigma_z \otimes \mathbb{1}_{\mathbb{C}^2}^{\otimes n-i}. \quad (9.19)$$

One can easily show that the completely mixed state $2^{-n} \mathbb{1}_{(\mathbb{C}^2)^{\otimes n}}$ is invariant, since $\mathcal{L}_{n*}^{\text{wcd}}(\mathbb{1}_{(\mathbb{C}^2)^{\otimes n}}) = 0$. Moreover, since Σ_z^n is self-adjoint, $\mathcal{L}_n^{\text{wcd}}$ is KMS-symmetric with respect to that state.

Proposition 9.3.2. *For any $n \geq 2$, $\lambda(\mathcal{L}_n^{\text{wcd}}) = 2$.*

Proof. Finding the spectral gap of the generator $\mathcal{L}_n^{\text{wcd}}$ is equivalent to finding the spectral gap of its matrix representation $\tilde{\mathcal{L}}_n^{\text{wcd}}$ (see e.g. [Wolf, 2012]): denote by $\{|i\rangle\}_{i=0,1}$ the eigenbasis of σ_z . Then,

$$\tilde{\mathcal{L}}_n^{\text{wcd}} := \Sigma_z^n \otimes \Sigma_z^n - \frac{1}{2}((\Sigma_z^n)^2 \otimes \mathbb{1} + \mathbb{1} \otimes (\Sigma_z^n)^2),$$

where $\Sigma_z^n |i_1, \dots, i_n\rangle = \sum_{j=1}^n (-1)^{i_j} |i_1 \dots i_n\rangle$ for any $(i_1, \dots, i_n) \in \{0, 1\}^n$, so that

$$\begin{aligned} \tilde{\mathcal{L}}_n^{\text{wcd}} |i_1 \dots i_n\rangle \otimes |j_1 \dots j_n\rangle &= \left[\sum_{k=1}^n (-1)^{i_k} \sum_{l=1}^n (-1)^{j_l} - \frac{1}{2} \sum_{k,l=1}^n (-1)^{i_k + i_l} - \frac{1}{2} \sum_{k,l=1}^n (-1)^{j_k + j_l} \right] |i_1 \dots i_n\rangle \otimes |j_1 \dots j_n\rangle \\ &= -2(|\mathbf{i}| - |\mathbf{j}|)^2 |i_1 \dots i_n\rangle \otimes |j_1 \dots j_n\rangle, \end{aligned}$$

where $|\mathbf{i}|$, resp. $|\mathbf{j}|$, denotes the number of 1's in the string $(i_1, \dots, i_n)$, resp. $(j_1, \dots, j_n)$. Therefore, the spectral gap of $\mathcal{L}_n^{\text{wcd}}$ is equal to 2. $\square$

From this theorem and the universal upper bound found in Corollary 8.4.11, we find that the weak collective decoherence satisfies $\text{HC}(c, \ln \sqrt{2})$ with

$$c \leq \frac{n \ln 2 + 2}{2}. \quad (9.20)$$

Next, let us first consider the heat diffusion on the one dimensional torus $\mathbb{T}^1$, which we represent on $(\mathbb{C}^2)^{\otimes n}$ as follows:

$$\mathbb{T}^1 \ni \theta \mapsto (e^{i\theta\sigma_z})^{\otimes n}.$$

One can easily verify that the QMS transferred via the above representation is the weak collective decoherence semigroup (up to a rescaling of the Lindblad operators by a factor of $\sqrt{2}$) as a direct consequence of Equation (9.17). Then, by Corollary 9.2.2 and the estimate of Section 3.7.2, we find for all $t \geq 0$, and any $\rho \in \mathcal{D}(\mathcal{H})$:

$$\|\mathcal{P}_t^{\text{wcd},n}(\rho - E_{\mathcal{F}*}(\rho))\|_1 \leq \sqrt{2 + \sqrt{\pi/t}} e^{-\frac{t}{2}},$$

which represents a fast convergence independent of the size n of the system.

Strong collective decoherence

We recall the generator of the strong collective decoherence on n qubits:

$$\mathcal{L}_n^{\text{scd}}(X) := \sum_{i \in \{x,y,z\}} \Sigma_i^n X \Sigma_i^n - \frac{1}{2}((\Sigma_i^n)^2 X + X(\Sigma_i^n)^2), \quad \text{where} \quad \Sigma_i^n := \sum_{k=1}^n \mathbb{1}_{\mathbb{C}^2}^{\otimes k-1} \otimes \sigma_i \otimes \mathbb{1}_{\mathbb{C}^2}^{n-k},$$

where the difference with Equation (9.19) arises from the consideration of all three Lindblad operator Σ_x^n , Σ_y^n and Σ_z^n .

We consider the three-dimensional simple Lie group $SU(2)$ of associated generators σ_x , σ_y , and σ_z spanning the Lie algebra $\mathfrak{su}(2)$, as well as the n -fold representation $SU(2) \ni g \mapsto U_g^{\otimes n}$, where U denotes the defining spin 1/2 representation of $SU(2)$: for any $\psi \in \mathbb{C}^2$, and $g \in SU(2)$,

$$U_g \psi = g \psi.$$

Just like previously, an easy use of Equation (9.17) shows that the semigroup transferred from the heat semigroup on $SU(2)$ via the above tensor product representation coincides with the strong decoherence (up to a rescaling of the Lindblad operators by a factor $\sqrt{2}$).

An easy application of Corollary 9.2.2 and the estimate of Theorem 3.7.1 for $n = 3$ provides the following dimension-independent bound for the decoherence time of the strong collective decoherence:

$$\tau_{\text{deco}}(\varepsilon) \leq \frac{64}{3} - \frac{32}{3} \ln \varepsilon + 4 \ln \left(1 + \frac{3}{2} \ln \frac{3}{4} \right).$$

Chapter 10.

Tensorization

The great advantage of classical logarithmic Sobolev inequalities over other methods resides in their tensorization property: the strong log-Sobolev constant of the product of independent Markovian evolutions is equal to the maximum over the set of strong log-Sobolev constants of the individual evolutions. However, this claim is strongly believed to be false in the non-commutative setting.

More precisely, given a set $\{\mathcal{L}_k\}_{k \in \{1, \dots, N\}}$ of Lindblad generators acting on the algebra $\mathcal{B}(\mathcal{H})$ of bounded operators on a finite dimensional Hilbert space $\mathcal{H}$, define

$$\widehat{\mathcal{L}}_k := \text{id}^{\otimes(k-1)} \otimes \mathcal{L}_k \otimes \text{id}^{\otimes(N-k)} \quad (10.1)$$

as an operator acting on $\mathcal{B}(\mathcal{H}^{\otimes N})$. We also set

$$\mathcal{K}_N := \sum_{k=1}^N \widehat{\mathcal{L}}_k. \quad (10.2)$$

Observe that if each $\mathcal{L}_k$ is primitive/satisfies σ_k -DBC/is KMS-symmetric with respect to a state σ_k , then $\mathcal{K}_N$ is primitive/satisfies σ -DBC/is KMS-symmetric with respect to $\sigma^{(N)} := \bigotimes_{k=1}^N \sigma_k$. Moreover, the $\widehat{\mathcal{L}}_k$'s commute with each other and

$$(e^{t\mathcal{K}_N})_{t \geq 0} = \left(\bigotimes_{k=1}^N e^{t\mathcal{L}_k} \right)_{t \geq 0}.$$

Now we can ask how the logarithmic Sobolev constants of $(e^{t\mathcal{K}_N})_{t \geq 0}$ are related to those for each individual $(e^{t\mathcal{L}_k})_{t \geq 0}$. In the commutative case the answer is easy: for instance, $\alpha_2(\mathcal{K}_N)$ equals $\min_{k \in \{1, \dots, N\}} \alpha_2(\mathcal{L}_k)$ for all N . In particular, if each local semigroup satisfies a (reverse) hypercontractivity inequality, then so does $(e^{t\mathcal{K}_N})_{t \geq 0}$. One easy way to understand this is by noticing that, in the commutative case, operator norms are multiplicative, or equivalently, the entropy function satisfies a certain subadditivity property (see e.g., [Mossel et al., 2013]). The aforementioned property that, in the classical case, $\alpha_p(\mathcal{K}_N)$ is independent of N , is usually called the *tensorization property*. The latter turns out to be useful even when allowing the subsystems to interact, e.g. for the Glauber dynamics [Martinelli, 1999, Cesi, 2001, Dai Pra et al., 2002, Caputo et al., 2015, Marton, 2015].

Tensorization property of log-Sobolev constants of quantum Lindblad generators, unlike its classical counterpart, is highly non-trivial. Thus proving (reverse) hypercontractivity inequalities that are independent of N is a difficult problem in the non-commutative setting: [Montanaro and Osborne, 2010] proved such hypercontractivity inequalities for the qubit depolarizing channel (see also [Kastoryano and Temme, 2013]). King [King, 2014] generalized this result for all unital qubit QMS. [Cubitt et al., 2015] developed the theory of quantum reverse hypercontractivity inequalities

in the unital case in and proved some tensorization-type results. Also, [Cubitt et al., 2015, Temme et al., 2014] developed some techniques for proving bounds on log-Sobolev constants $\alpha_p(\mathcal{K}_N)$ that are independent of N . For example, in Theorem 9 of [Temme et al., 2014], a weaker result was shown to hold in finite dimensions, the proof of which relies on interpolation properties of $\mathbb{L}_p$ spaces:

Theorem 10.0.1. *Let $N \in \mathbb{N}^*$ and, for any $k \in \{1, \dots, N\}$, let $(\mathcal{P}_t^k)_{t \geq 0}$ be a primitive reversible quantum Markov semigroup on $\mathcal{B}(\mathbb{C}^d)$ with respective stationary state σ_k and spectral gap λ_k . Then, the log-Sobolev constant $\alpha_2(\mathcal{K}_N)$ of the product QMS $(\otimes_{i=1}^N \mathcal{P}_t^k)_{t \geq 0}$ satisfies*

$$\frac{\min_k \lambda_k}{\ln(d_{\mathcal{H}}^4 \max_k \|\sigma_k^{-1}\|_{\infty}) + 11} \leq \alpha_2(\mathcal{K}_N) \leq \frac{\min_k \lambda_k}{2}. \quad (10.3)$$

The importance of this result lies in the independence of the lower bound of (10.3) in the number N of systems. However, this bound highly depends on the local dimensions d . In particular, it is not well-suited to the study of quantum diffusions.

From there, two approaches can be pursued. A more conservative one consists in showing that tensorization holds in specific examples. Another novel approach, initiated in the unital case by [Beigi and King, 2016], is to replace the use of non-commutative $\mathbb{L}_p$ norms in the definition of hypercontractivity by their completely bounded versions, which are known to be multiplicative [Devetak et al., 2006] (cf. Section 1.1.2). However, since these norms follow from specific amalgamated $\mathbb{L}_p$ norms associated to algebras of the form $\mathcal{N}_k = \mathcal{B}(\mathbb{C}^k) \otimes \mathbb{1}$, we know from Section 8.5 that there is no hope to get the cancellation of the weak hypercontractivity constant d . Since this constant increases linearly with the number of systems under consideration, the problem of the independence of the strong constant c with N is only shifted.

Layout of the chapter: This chapter is divided into two main parts. In Section 10.1, we show the tensorization property of α_1 and α_2 for tensor products of the generalized depolarizing semigroup introduced in Section 5.5.1. Then, we introduce the related concepts of a complete logarithmic Sobolev inequality and of CB hypercontractivity in Section 10.2 for non-unital primitive QMS, which extends the framework of [Beigi and King, 2016] and Chapter 9.

10.1. Tensorizations for the generalized depolarizing semigroup

10.1.1. Quasi-tensorization of α_1

Theorem 10.1.1 provides a uniform bound on α_1 for the *generalized depolarizing semigroups* and their tensor powers. The proof of this result is a generalization of the proof of a similar result in the classical case [Mossel et al., 2013]. This tensorization result together with the quantum Stroock-Varopoulos inequality (Theorem 5.4.2) provides the tensorization of reverse hypercontractivity inequality which we use in Chapter 13.

Theorem 10.1.1. *Let $\sigma_1, \dots, \sigma_N \in \mathcal{D}_+(\mathbb{C}^d)$. Let $\mathcal{L}_{\sigma_k}(X) = \text{Tr}(\sigma_k X) \mathbb{1}_{\mathbb{C}^d} - X$ be the generator of the generalized depolarizing semigroup associated to the state σ_k , let $\widehat{\mathcal{L}}_{\sigma_k} := \text{id}^{\otimes(k-1)} \otimes \mathcal{L}_{\sigma_k} \otimes \text{id}^{\otimes(N-k)}$ and define $\mathcal{K}_N$ by (10.2). Then $\text{MLSI}(\alpha_1)$ holds for $\alpha_1 \geq \frac{1}{4}$. In particular, this bound is independent of N and d .*

Proof. Let $\mathcal{H}_{A^N} = (\mathbb{C}^d)^{\otimes N}$ and $\mathcal{H}_{A_k} \simeq \mathbb{C}^d$ be the k -th local system. We need to show that for all $\rho_{A^N} \in \mathcal{D}_+(\mathcal{H}_{A^N})$:

$$D(\rho_{A^N} \| \sigma_{A^N}) \leq \text{EP}_{\mathcal{K}_N}(\rho_{A^N}),$$

where $\sigma_{A_k} = \sigma_k$ and $\sigma_{A^N} = \sigma_1 \otimes \cdots \otimes \sigma_N$. Using the expression for the generator $\mathcal{K}_N$, we thus need to show that

$$D(\rho_{A^N} \| \sigma_{A^N}) \leq - \sum_{k=1}^N \text{Tr} \left[\widehat{\mathcal{L}}_{\sigma_k}(\rho_{A^N}) \cdot (\ln \rho_{A^N} - \ln(\sigma_{A^N})) \right] \quad (10.4)$$

$$= \sum_{k=1}^N \text{Tr} \left[(\rho_{A^N} - \rho_{A^{\sim k}} \otimes \sigma_{A_k}) \cdot (\ln \rho_{A^N} - \ln(\sigma_{A^N})) \right] \quad (10.5)$$

$$= \sum_{k=1}^N \left[D(\rho_{A^N} \| \sigma_{A^N}) + D(\rho_{A^{\sim k}} \otimes \sigma_{A_k} \| \rho_{A^N}) - D(\rho_{A^{\sim k}} \otimes \sigma_{A_k} \| \sigma_{A^N}) \right]. \quad (10.6)$$

where $A^{\sim k} = (A_1, \dots, A_{k-1}, A_{k+1}, \dots, A_N)$ and $\rho_{A^{\sim k}} = \text{Tr}_{A_k}(\rho_{A^N})$ is the partial trace of ρ_{A^N} with respect to the k -th subsystem. Now since $D(\rho_{A^{\sim k}} \otimes \sigma_{A_k} \| \rho_{A^N}) \geq 0$, it suffices to show that

$$D(\rho_{A^N} \| \sigma_{A^N}) \leq \sum_{k=1}^N \left[D(\rho_{A^N} \| \sigma_{A^N}) - D(\rho_{A^{\sim k}} \otimes \sigma_{A_k} \| \sigma_{A^N}) \right], \quad (10.7)$$

We note that $D(\xi_B \| \tau_B) = -S(B)_\xi - \text{Tr}(\xi \ln \tau)$ where $S(B)_\xi = -\text{Tr}(\xi \ln \xi)$ is the von Neumann entropy. Moreover, $\ln(\xi \otimes \tau) = \ln \xi \otimes \mathbb{1} + \mathbb{1} \otimes \ln \tau$. Therefore, (10.7) is equivalent to

$$\begin{aligned} -S(A^N)_\rho - \sum_{k=1}^N \text{Tr}(\rho_{A_k} \ln \sigma_k) &\leq \sum_{k=1}^N \left[-S(A^N)_\rho - \sum_{j=1}^N \text{Tr}(\rho_{A_j} \ln \sigma_j) + S(A^{\sim k})_\rho + \sum_{j \neq k} \text{Tr}(\rho_{A_j} \ln \sigma_j) \right] \\ &= \sum_{k=1}^N \left[-S(A^N)_\rho - \text{Tr}(\rho_{A_k} \ln \sigma_k) + S(A^{\sim k})_\rho \right] \\ &= \sum_{k=1}^N \left[-S(A_k | A^{\sim k})_\rho - \text{Tr}(\rho_{A_k} \ln \sigma_k) \right]. \end{aligned}$$

This is itself is equivalent to

$$S(A^N)_\rho \geq \sum_{k=1}^N S(A_k | A^{\sim k})_\rho,$$

which is an immediate consequence of the data processing inequality for the partial trace, i.e., $S(B|C)_\xi \geq S(B|CD)_\xi$ (cf. DPI), once we use the chain rule

$$S(A^N)_\rho = S(A_1)_\rho + \sum_{k=2}^N S(A_k | A_1, \dots, A_{k-1})_\rho.$$

This concludes the proof. $\square$

Remark 10.1.2. Letting the σ_k 's to be identical in the above theorem, we obtain the promised tensorization-type result for the modified logarithmic Sobolev constant. Note that this result was independently obtained recently in [Capel et al., 2018] by introducing the notion of a conditional logarithmic Sobolev constant and finding a uniform lower bound on the latter. Moreover, a special case of the above theorem corresponding to σ being the completely mixed state was already proved in [Müller-Hermes et al., 2016].

We can now use Corollary 7.4.10 and the fact that tensor products of generalized depolarizing semigroups satisfy ω -DBC to conclude the following.

Corollary 10.1.3. *Let $\sigma_1, \dots, \sigma_N$ be arbitrary positive definite density matrices. Let $\mathcal{L}_{\sigma_k}(X) = \text{Tr}(\sigma_k X) \mathbb{1} - X$ be the generator associated to the generalized depolarizing semigroup $\mathcal{P}_{k,t}^{\text{depol}}(X) = e^{-t} X + (1 - e^{-t}) \text{Tr}(\sigma_k X) \mathbb{1}$. Define $\sigma^{(N)} = \sigma_1 \otimes \cdots \otimes \sigma_N$ and $\mathcal{P}_t^{\text{depol},(N)} = \mathcal{P}_{1,t}^{\text{depol}} \otimes \cdots \otimes \mathcal{P}_{N,t}^{\text{depol}}$. Then for*

$p \leq q < 1$ and $t \geq \ln \frac{p-1}{q-1}$, we have for all $N \in \mathbb{N}$ and any positive definite operator $X \in \mathcal{B}(\mathcal{H}^{\otimes N})$:

$$\|\mathcal{P}_t^{\text{depol},(N)}(X)\|_{\mathbb{L}_p(\sigma^{(N)})} \geq \|X\|_{\mathbb{L}_q(\sigma^{(N)})}.$$

10.1.2. Tensorization of α_2

The second tensorization result, Theorem 10.1.4, shows that α_2 is independent of N in the case of qubits. In Corollary 10.1.8 we use these results to establish a uniform bound on the 2-logarithmic Sobolev constant of *any* qubit quantum Markov semigroup and its tensor powers. We note that the latter bound improves over the bounds provided in [Temme et al., 2014].

Before stating the tensorization results of this section, let us briefly explain the ideas behind their proof. Previously, Theorem 10.1.4 was known in the doubly stochastic case (the usual depolarizing semigroup), the proof of which was based on an inequality on the norms of a 2×2 block matrix and its submatrices from [King, 2003]. Our proof of Theorem 10.1.4 is based on the same inequality. First in Lemma 10.1.5 we derive an infinitesimal version of that inequality in terms of the entropies of a 2×2 block matrix and its submatrices, and then use it to prove Theorem 10.1.4. Finally, Corollary 10.1.8 is a quantum generalization of a classical result from [Diaconis and Saloff-Coste, 1996a] with an essentially similar proof except that tensorization is taken care of separately.

Theorem 10.1.4. *Let $\dim \mathcal{H} = 2$ and $\mathcal{L}_\sigma(X) = \text{Tr}(\sigma X)\mathbb{1} - X$ for some positive definite density matrix $\sigma \in \mathcal{D}_+(\mathcal{H})$. Then, for any $N \in \mathbb{N}$,*

$$\alpha_2(\mathcal{K}_N) = \alpha_2(\mathcal{L}_\sigma),$$

where $\mathcal{K}_N$ is defined in (10.2) when taking $\mathcal{L}_k = \mathcal{L}_\sigma$ for all k .

In order to prove this theorem, our main tool is the following entropic inequality that is of independent interest and can be useful elsewhere.

Lemma 10.1.5. *Let $\mathcal{H}$ and $\mathcal{H}'$ be Hilbert spaces with $\dim \mathcal{H} = 2$. Let $X \in \mathcal{B}_{sa}(\mathcal{H})^+$ be a positive semidefinite matrix with the block form*

$$X = \begin{pmatrix} A & C \\ C^* & B \end{pmatrix}, \quad (10.8)$$

where $A, B, C \in \mathcal{B}(\mathcal{H}')$. For a full-rank density matrix ρ on $\mathcal{H}'$, the matrix M defined as

$$M = \begin{pmatrix} \|A\|_{\mathbb{L}_2(\rho)} & \|C\|_{\mathbb{L}_2(\rho)} \\ \|C^*\|_{\mathbb{L}_2(\rho)} & \|B\|_{\mathbb{L}_2(\rho)} \end{pmatrix} \quad (10.9)$$

is positive semidefinite. Moreover, let σ be a full-rank density matrix on $\mathcal{H}$ of the form

$$\sigma = \begin{pmatrix} \theta & 0 \\ 0 & 1 - \theta \end{pmatrix}, \quad (10.10)$$

where $\theta \in (0, 1)$. Then we have

$$\begin{aligned} \text{Ent}_{2, \sigma \otimes \rho}(X) &\leq \text{Ent}_{2, \sigma}(M) + \theta \text{Ent}_{2, \rho}(A) + (1 - \theta) \text{Ent}_{2, \rho}(B) \\ &\quad + \sqrt{\theta(1 - \theta)} \text{Ent}_{2, \rho}(\text{I}_{2,2}(C)) + \sqrt{\theta(1 - \theta)} \text{Ent}_{2, \rho}(\text{I}_{2,2}(C^*)). \end{aligned} \quad (10.11)$$

Proof. For any $p \geq 2$ define

$$M_p := \begin{pmatrix} \|A\|_{\mathbb{L}_p(\rho)} & \|C\|_{\mathbb{L}_p(\rho)} \\ \|C^*\|_{\mathbb{L}_p(\rho)} & \|B\|_{\mathbb{L}_p(\rho)} \end{pmatrix},$$

so that $M_2 = M$. Since $X \geq 0$, both A and B are positive semidefinite. Moreover, we have

$$\Gamma_{\mathbb{1} \otimes \rho}^{\frac{1}{p}}(X) = \begin{pmatrix} \Gamma_{\rho}^{\frac{1}{p}}(A) & \Gamma_{\rho}^{\frac{1}{p}}(C) \\ \Gamma_{\rho}^{\frac{1}{p}}(C^*) & \Gamma_{\rho}^{\frac{1}{p}}(B) \end{pmatrix} \geq 0.$$

As a result, according to Theorem IX.5.9 of [Bhatia, 2015] there exists a *contraction* $R \in \mathcal{B}(\mathcal{H}')$ such that $\Gamma_{\rho}^{\frac{1}{p}}(C) = (\Gamma_{\rho}^{\frac{1}{p}}(A))^{\frac{1}{2}} R (\Gamma_{\rho}^{\frac{1}{p}}(B))^{\frac{1}{2}}$. Therefore, by Hölder's inequality we have

$$\begin{aligned} \|\Gamma_{\rho}^{\frac{1}{p}}(C)\|_p &= \|(\Gamma_{\rho}^{\frac{1}{p}}(A))^{\frac{1}{2}} R (\Gamma_{\rho}^{\frac{1}{p}}(B))^{\frac{1}{2}}\|_p \\ &\leq \|(\Gamma_{\rho}^{\frac{1}{p}}(A))^{\frac{1}{2}}\|_{2p} \cdot \|R\|_{\infty} \cdot \|(\Gamma_{\rho}^{\frac{1}{p}}(B))^{\frac{1}{2}}\|_{2p} \\ &\leq \|(\Gamma_{\rho}^{\frac{1}{p}}(A))^{\frac{1}{2}}\|_{2p} \cdot \|(\Gamma_{\rho}^{\frac{1}{p}}(B))^{\frac{1}{2}}\|_{2p} \\ &= \|\Gamma_{\rho}^{\frac{1}{p}}(A)\|_p^{\frac{1}{2}} \cdot \|\Gamma_{\rho}^{\frac{1}{p}}(B)\|_p^{\frac{1}{2}}. \end{aligned}$$

Then using $\|Y\|_{\mathbb{L}_p(\rho)} = \|\Gamma_{\rho}^{\frac{1}{p}}(Y)\|_p$, we find that

$$\|C\|_{\mathbb{L}_p(\rho)} \leq \|A\|_{\mathbb{L}_p(\rho)}^{\frac{1}{2}} \cdot \|B\|_{\mathbb{L}_p(\rho)}^{\frac{1}{2}},$$

and hence $M_p \geq 0$. In particular, $M_2 = M \geq 0$ and $\text{Ent}_{2,\rho}(M)$ is well-defined.

Define $\psi(p) := \|M_p\|_{\mathbb{L}_p(\sigma)} - \|X\|_{\mathbb{L}_p(\sigma \otimes \rho)}$. It is shown by King [King, 2003] that $\psi(p) \geq 0$ for all $p \geq 2$. Indeed, this inequality is proven in [King, 2003] in the special case where σ and ρ are the identity operators on the relevant spaces. Nevertheless, we have

$$\|X\|_{\mathbb{L}_p(\sigma \otimes \rho)} = \left\| \begin{pmatrix} \theta^{\frac{1}{p}} \Gamma_{\rho}^{\frac{1}{p}}(A) & (\theta(1-\theta))^{\frac{1}{2p}} \Gamma_{\rho}^{\frac{1}{p}}(C) \\ (\theta(1-\theta))^{\frac{1}{2p}} \Gamma_{\rho}^{\frac{1}{p}}(C^*) & (1-\theta)^{\frac{1}{p}} \Gamma_{\rho}^{\frac{1}{p}}(B) \end{pmatrix} \right\|_p,$$

and

$$\|M_p\|_{\mathbb{L}_p(\sigma)} = \left\| \begin{pmatrix} \theta^{\frac{1}{p}} \|\Gamma_{\rho}^{\frac{1}{p}}(A)\|_p & (\theta(1-\theta))^{\frac{1}{2p}} \|\Gamma_{\rho}^{\frac{1}{p}}(C)\|_p \\ (\theta(1-\theta))^{\frac{1}{2p}} \|\Gamma_{\rho}^{\frac{1}{p}}(C^*)\|_p & (1-\theta)^{\frac{1}{p}} \|\Gamma_{\rho}^{\frac{1}{p}}(B)\|_p \end{pmatrix} \right\|_p,$$

Thus, King's result holds for arbitrary ρ and diagonal σ as well, and we have $\psi(p) \geq 0$ for all $p \geq 2$. On the other hand, a straightforward computation verifies that $\psi(2) = 0$. This means that $\psi'(2) \geq 0$, i.e.,

$$\left. \frac{d}{dp} (\|M_p\|_{\mathbb{L}_p(\sigma)} - \|X\|_{\mathbb{L}_p(\sigma \otimes \rho)}) \right|_{p=2} \geq 0.$$

The derivatives can be computed using Proposition 7.4.7. We have

$$\left. \frac{d}{dp} \|X\|_{\mathbb{L}_p(\sigma \otimes \rho)} \right|_{p=2} = \frac{1}{4} \|X\|_{\mathbb{L}_2(\sigma \otimes \rho)}^{-1} \cdot \text{Ent}_{2,\sigma \otimes \rho}(X), \quad (10.12)$$

and

$$\left. \frac{d}{dp} \|M_p\|_{\mathbb{L}_p(\sigma)} \right|_{p=2} = \frac{1}{4} \|M\|_{\mathbb{L}_2(\sigma)}^{-1} \cdot \left(\text{Ent}_{2,\sigma}(M) + 4 \text{Tr} \left[\Gamma_{\sigma}^{\frac{1}{2}}(M'_2) \cdot \Gamma_{\sigma}^{\frac{1}{2}}(M) \right] \right),$$

where

$$M'_2 = \frac{d}{dp} M_p \Big|_{p=2} = \frac{1}{4} \begin{pmatrix} \|A\|_{\mathbb{L}_2(\rho)}^{-1} \cdot \text{Ent}_{2,\rho}(A) & w \\ w & \|B\|_{\mathbb{L}_2(\rho)}^{-1} \cdot \text{Ent}_{2,\rho}(B) \end{pmatrix},$$

and $w = \|C\|_{\mathbb{L}_2(\rho)}^{-1} \cdot (\text{Ent}_{2,\rho}(\text{I}_{2,2}(C)) + \text{Ent}_{2,\rho}(\text{I}_{2,2}(C^*)))$. We conclude that

$$\begin{aligned} \frac{d}{dp} \|M_p\|_{\mathbb{L}_p(\sigma)} \Big|_{p=2} &= \frac{1}{4} \|M\|_{\mathbb{L}_2(\sigma)}^{-1} \cdot (\text{Ent}_{2,\sigma}(M) + \theta \text{Ent}_{2,\rho}(A) + (1-\theta) \text{Ent}_{2,\rho}(B) \\ &\quad + \sqrt{\theta(1-\theta)} \text{Ent}_{2,\rho}(\text{I}_{2,2}(C)) + \sqrt{\theta(1-\theta)} \text{Ent}_{2,\rho}(\text{I}_{2,2}(C^*))). \end{aligned}$$

Comparing to (10.12) and using $\|M\|_{\mathbb{L}_2(\sigma)} = \|X\|_{\mathbb{L}_2(\sigma \otimes \rho)}$ the desired inequality follows. $\square$

We need yet another lemma to prove Theorem 10.1.4.

Lemma 10.1.6. *For any Lindblad generator $\mathcal{K}$ that is KMS-symmetric with respect to some positive definite density matrix ρ we have, for any matrix C ,*

$$\mathcal{E}_{2,\mathcal{K}}(\text{I}_{2,2}(C)) + \mathcal{E}_{2,\mathcal{K}}(\text{I}_{2,2}(C^*)) \leq -\langle C, \mathcal{K}(C) \rangle_\rho - \langle C^*, \mathcal{K}(C^*) \rangle_\rho.$$

Proof. Define $D := \Gamma_\rho^{\frac{1}{2}}(C)$. Then for $j \in \{0, 1\}$ [Bhatia, 2015]:

$$Y_j := \begin{pmatrix} |D| & (-1)^j D^* \\ (-1)^j D & |D^*| \end{pmatrix} \geq 0.$$

Since $\Gamma_\rho^{-1/2}$ is completely positive we have

$$Z_j := \text{id} \otimes \Gamma_\rho^{-1/2}(Y_j) = \begin{pmatrix} \text{I}_{2,2}(C) & (-1)^j C^* \\ (-1)^j C & \text{I}_{2,2}(C^*) \end{pmatrix} \geq 0.$$

On the other hand, $\mathcal{P}_t = e^{t\mathcal{K}}$ is completely positive. Therefore,

$$\text{id} \otimes \mathcal{P}_t(Z_0) = \begin{pmatrix} \mathcal{P}_t(\text{I}_{2,2}(C)) & \mathcal{P}_t(C^*) \\ \mathcal{P}_t(C) & \mathcal{P}_t(\text{I}_{2,2}(C^*)) \end{pmatrix} \geq 0.$$

Putting these observations together we find that for any $t \geq 0$:

$$g(t) := \langle Z_1, \text{id} \otimes \mathcal{P}_t(Z_0) \rangle_{\mathbb{1} \otimes \rho} \geq 0.$$

We note that

$$g(t) = \langle \text{I}_{2,2}(C), \mathcal{P}_t(\text{I}_{2,2}(C)) \rangle_\rho + \langle \text{I}_{2,2}(C^*), \mathcal{P}_t(\text{I}_{2,2}(C^*)) \rangle_\rho - \langle C, \mathcal{P}_t(C) \rangle_\rho - \langle C^*, \mathcal{P}_t(C^*) \rangle_\rho.$$

From this expression it is clear that

$$g(0) = \|\text{I}_{2,2}(C)\|_{\mathbb{L}_2(\rho)}^2 + \|\text{I}_{2,2}(C^*)\|_{\mathbb{L}_2(\rho)}^2 - \|C\|_{\mathbb{L}_2(\rho)}^2 - \|C^*\|_{\mathbb{L}_2(\rho)}^2 = 0.$$

Therefore, we must have $g'(0) \geq 0$ which is equivalent to the desired inequality. $\square$

Now we have all the required tools for proving Theorem 10.1.4. We actually prove a stronger statement from which Theorem 10.1.4 is implied by a simple induction procedure:

Theorem 10.1.7. *Let $\dim \mathcal{H} = 2$ and $\mathcal{L}_\sigma(X) = \text{Tr}(\sigma X) \mathbb{1} - X$ for some positive definite density matrix*

σ . Also let $\mathcal{K}$ be a Lindblad generator associated to a primitive QMS that is reversible with respect to some positive definite state $\rho \in \mathcal{D}(\mathcal{H}')$. Then we have

$$\alpha_2(\mathcal{L}_\sigma \otimes \text{id}' + \text{id} \otimes \mathcal{K}) = \min\{\alpha_2(\mathcal{L}_\sigma), \alpha_2(\mathcal{K})\},$$

where id and id' denote the identity superoperators acting on $\mathcal{B}(\mathcal{H})$ and $\mathcal{B}(\mathcal{H}')$ respectively.

Proof. Let $\alpha = \min\{\alpha_2(\mathcal{L}_\sigma), \alpha_2(\mathcal{K})\}$. By restricting X in the log-Sobolev inequality of order 2 to be of the tensor product form and using

$$\text{Ent}_{2,\sigma \otimes \rho}(Y \otimes Y') = \text{Ent}_{2,\sigma}(Y) + \text{Ent}_{2,\rho}(Y'),$$

we conclude that $\alpha_2(\mathcal{L}_\sigma \otimes \text{id}' + \text{id} \otimes \mathcal{K}) \leq \alpha$. To prove the inequality in the other direction we need to show that for any $X \in \mathcal{B}_{sa}(\mathcal{H} \otimes \mathcal{H}')^+$ we have

$$\alpha \text{Ent}_{2,\sigma \otimes \rho}(X) \leq \mathcal{E}_{2,\mathcal{L}_\sigma \otimes \text{id}' + \text{id} \otimes \mathcal{K}}(X). \quad (10.13)$$

Assume, without loss of generality, that σ is diagonal of the form (10.10), and that $X \in \mathcal{B}_{sa}(\mathcal{H} \otimes \mathcal{H}')^+$ has the block form (10.8). Define M by (10.9). Then by Lemma 10.1.5 we have

$$\begin{aligned} \text{Ent}_{2,\sigma \otimes \rho}(X) &\leq \text{Ent}_{2,\sigma}(M) + \theta \text{Ent}_{2,\rho}(A) + (1 - \theta) \text{Ent}_{2,\rho}(B) \\ &\quad + \sqrt{\theta(1 - \theta)} \text{Ent}_{2,\rho}(\text{I}_{2,2}(C)) + \sqrt{\theta(1 - \theta)} \text{Ent}_{2,\rho}(\text{I}_{2,2}(C^*)). \end{aligned}$$

On the other hand by the definition of α we have

$$\alpha \text{Ent}_{2,\sigma}(M) \leq \mathcal{E}_{2,\mathcal{L}_\sigma}(M),$$

and

$$\alpha \text{Ent}_{2,\rho}(Y) \leq \mathcal{E}_{2,\mathcal{K}}(Y),$$

for all $Y \in \{A, B, \text{I}_{2,2}(C), \text{I}_{2,2}(C^*)\}$. Therefore, we have

$$\begin{aligned} \alpha \text{Ent}_{2,\sigma \otimes \rho}(X) &\leq \mathcal{E}_{2,\mathcal{L}_\sigma}(M) + \theta \mathcal{E}_{2,\mathcal{K}}(A) + (1 - \theta) \mathcal{E}_{2,\mathcal{K}}(B) \\ &\quad + \sqrt{\theta(1 - \theta)} \mathcal{E}_{2,\mathcal{K}}(\text{I}_{2,2}(C)) + \sqrt{\theta(1 - \theta)} \mathcal{E}_{2,\mathcal{K}}(\text{I}_{2,2}(C^*)) \\ &\leq \mathcal{E}_{2,\mathcal{L}_\sigma}(M) + \theta \mathcal{E}_{2,\mathcal{K}}(A) + (1 - \theta) \mathcal{E}_{2,\mathcal{K}}(B) \\ &\quad - \sqrt{\theta(1 - \theta)} \langle C, \mathcal{K}(C) \rangle_\rho - \sqrt{\theta(1 - \theta)} \langle C^*, \mathcal{K}(C^*) \rangle_\rho, \end{aligned} \quad (10.14)$$

where in the second inequality we use Lemma 10.1.6. We now have

$$\begin{aligned} \mathcal{E}_{2,\mathcal{L}_\sigma \otimes \text{id}' + \text{id} \otimes \mathcal{K}}(X) &= -\langle X, (\mathcal{L}_\sigma \otimes \text{id}' + \text{id} \otimes \mathcal{K})(X) \rangle_{\sigma \otimes \rho} \\ &= -\langle X, \mathcal{L}_\sigma \otimes \text{id}'(X) \rangle_{\sigma \otimes \rho} - \left\langle \begin{pmatrix} A & C \\ C^* & B \end{pmatrix}, \begin{pmatrix} \mathcal{K}(A) & \mathcal{K}(C) \\ \mathcal{K}(C^*) & \mathcal{K}(B) \end{pmatrix} \right\rangle_{\sigma \otimes \rho}. \end{aligned}$$

We compute each term in the above sum separately.

$$\begin{aligned}
& -\langle X, (\mathcal{L}_\sigma \otimes \text{id}')(X) \rangle_{\sigma \otimes \rho} \\
&= \left\langle \begin{pmatrix} A & C \\ C^* & B \end{pmatrix}, \begin{pmatrix} (1-\theta)(A-B) & C \\ C^* & \theta(B-A) \end{pmatrix} \right\rangle_{\sigma \otimes \rho} \\
&= \theta(1-\theta)\langle A, A-B \rangle_\rho + \theta(1-\theta)\langle B, B-A \rangle_\rho + 2\sqrt{\theta(1-\theta)}\langle C, C \rangle_\rho \\
&= \theta(1-\theta)\|A\|_{\mathbb{L}_2(\rho)}^2 + \theta(1-\theta)\|B\|_{\mathbb{L}_2(\rho)}^2 - 2\theta(1-\theta)\langle A, B \rangle_\rho + 2\sqrt{\theta(1-\theta)}\|C\|_{\mathbb{L}_2(\rho)}^2 \\
&\geq \theta(1-\theta)\|A\|_{\mathbb{L}_2(\rho)}^2 + \theta(1-\theta)\|B\|_{\mathbb{L}_2(\rho)}^2 - 2\theta(1-\theta)\|A\|_{\mathbb{L}_2(\rho)} \cdot \|B\|_{\mathbb{L}_2(\rho)} + 2\sqrt{\theta(1-\theta)}\|C\|_{\mathbb{L}_2(\rho)}^2 \\
&= -\langle M, \mathcal{L}_\sigma(M) \rangle_\sigma \\
&= \mathcal{E}_{2, \mathcal{L}_\sigma}(M).
\end{aligned}$$

For the second term we compute

$$\begin{aligned}
& -\left\langle \begin{pmatrix} A & C \\ C^* & B \end{pmatrix}, \begin{pmatrix} \mathcal{K}(A) & \mathcal{K}(C) \\ \mathcal{K}(C^*) & \mathcal{K}(B) \end{pmatrix} \right\rangle_{\sigma \otimes \rho} = -\theta\langle A, \mathcal{K}(A) \rangle_\rho - (1-\theta)\langle B, \mathcal{K}(B) \rangle_\rho \\
& \quad - \sqrt{\theta(1-\theta)}\langle C, \mathcal{K}(C) \rangle_\rho - \sqrt{\theta(1-\theta)}\langle C^*, \mathcal{K}(C^*) \rangle_\rho \\
&= \theta\mathcal{E}_{2, \mathcal{K}}(A) + (1-\theta)\mathcal{E}_{2, \mathcal{K}}(B) \\
& \quad - \sqrt{\theta(1-\theta)}\langle C, \mathcal{K}(C) \rangle_\rho - \sqrt{\theta(1-\theta)}\langle C^*, \mathcal{K}(C^*) \rangle_\rho.
\end{aligned}$$

Therefore, we have

$$\begin{aligned}
\mathcal{E}_{2, \mathcal{L}_\sigma \otimes \text{id}' + \text{id} \otimes \mathcal{K}}(X) &\geq \mathcal{E}_{2, \mathcal{L}_\sigma}(M) + \theta\mathcal{E}_{2, \mathcal{K}}(A) + (1-\theta)\mathcal{E}_{2, \mathcal{K}}(B) \\
&\quad - \sqrt{\theta(1-\theta)}\langle C, \mathcal{K}(C) \rangle_\rho - \sqrt{\theta(1-\theta)}\langle C^*, \mathcal{K}(C^*) \rangle_\rho.
\end{aligned}$$

Comparing this to (10.14) we arrive at the desired inequality (10.13). $\square$

We can now derive a tensorization-type result for a wide class of Lindblad generators. Let $\mathcal{L}$ be a σ -reversible and primitive Lindblad generator. Recall that the *spectral gap* of $\mathcal{L}$ is defined by

$$\lambda(\mathcal{L}) = \inf_X \frac{\mathcal{E}_{2, \mathcal{L}}(X)}{\text{Var}_\sigma(X)},$$

where $\text{Var}_\sigma(X) = \langle X, X \rangle_\sigma - \langle X, \mathbb{1} \rangle_\sigma^2 = \|X\|_{\mathbb{L}_2(\sigma)}^2 - \langle X, \mathbb{1} \rangle_\sigma^2$. Observe that $\text{Var}_\sigma(X)$ is the squared length of the projection of X onto the subspace orthogonal to $\mathbb{1} \in \mathcal{B}(\mathcal{H})$ with respect to the inner product $\langle \cdot, \cdot \rangle_\sigma$. On the other hand, $\mathbb{1}$ is the sole¹ 0-eigenvector of $\mathcal{L}$ which is self-adjoint with respect to this inner product. Therefore, $-\lambda(\mathcal{L})$ is the minimum non-zero eigenvalue of $\mathcal{L}$.

The spectral gap satisfies the tensorization property, as shown below. Observe that

$$\mathcal{K}_N = \sum_{k=1}^N \widehat{\mathcal{L}}_k$$

is a sum of mutually commuting operators. Then the eigenvalues of $\mathcal{K}_N$ are summations of eigenvalues of individual $\widehat{\mathcal{L}}_k$'s. Since each $\widehat{\mathcal{L}}_k$ is a tensor product of $\mathcal{L}$ with some identity superoperator, the set of its eigenvalues is the same as that of $\mathcal{L}$. Using these we conclude that for all $N \in \mathbb{N}$,

$$\lambda(\mathcal{K}_N) = \lambda(\mathcal{L}). \quad (10.15)$$

¹This 0-eigenvector is unique since $\mathcal{L}$ is assumed to be primitive.

It is well-known that $\lambda(\mathcal{L}) \geq 2\alpha_2(\mathcal{L})$ (see [Carbone and Martinelli, 2015]). The following corollary gives a lower bound on $\alpha_2(\mathcal{L})$ in terms of $\lambda(\mathcal{L})$.

Corollary 10.1.8. *Let $\dim \mathcal{H} = 2$ and $\sigma \in \mathcal{D}(\mathcal{H})$ be full-rank. For any KMS-symmetric primitive Lindblad generator $\mathcal{L}$ with respect to σ we have*

$$\frac{1 - 2\|\sigma^{-1}\|_{\infty}^{-1}}{\ln(\|\sigma^{-1}\|_{\infty} - 1)} \lambda(\mathcal{L}) \leq \alpha_2(\mathcal{K}_n).$$

Proof. Let $\mathcal{L}_{\sigma}$ be the generalized depolarizing Lindblad generator that is σ -reversible, and let $X \in \mathcal{B}_{sa}(\mathcal{H}^{\otimes N})^+$ be arbitrary. Then by Theorem 10.1.4 and Theorem 7.2.4 we have

$$\frac{1 - 2\|\sigma^{-1}\|_{\infty}^{-1}}{\ln(\|\sigma^{-1}\|_{\infty} - 1)} \text{Ent}_{2, \sigma^{\otimes N}}(X) \leq - \sum_{k=1}^N \langle X, \widehat{\mathcal{L}}_{\sigma_k}(X) \rangle_{\sigma^{\otimes N}}, \quad (10.16)$$

where $\sigma_k = \sigma$ for all k . Next, let $\mathcal{W}_k \subset \mathcal{B}(\mathcal{H}^{\otimes N})$ be the subspace spanned by operators of the form $A_1 \otimes \cdots \otimes A_N \in \mathcal{B}(\mathcal{H}^{\otimes N})$ with $A_k = \mathbb{1} \in \mathcal{B}(\mathcal{H})$. In other words, $\mathcal{W}_k = \ker(\widehat{\mathcal{L}}_{\sigma_k})$. Then $-\langle X, \widehat{\mathcal{L}}_{\sigma_k}(X) \rangle_{\sigma^{\otimes N}}$ equals the squared length of the projection of X onto $\mathcal{W}_k^{\perp}$. On the other hand, since $\mathcal{L}$ is primitive and σ -reversible, we also have $\mathcal{W}_k = \ker \widehat{\mathcal{L}}_k$ and $\mathcal{W}_k^{\perp}$ is invariant under $\widehat{\mathcal{L}}_k$. Moreover, by definition $\lambda(\widehat{\mathcal{L}}_k)$ is the minimum eigenvalue of $\widehat{\mathcal{L}}_k$ restricted to $\mathcal{W}_k^{\perp}$ (i.e., the maximum non-zero eigenvalue). We conclude that

$$-\lambda(\widehat{\mathcal{L}}_k) \langle X, \widehat{\mathcal{L}}_{\sigma_k}(X) \rangle_{\sigma^{\otimes N}} \leq -\langle X, \widehat{\mathcal{L}}_k(X) \rangle_{\sigma^{\otimes N}}.$$

On the other hand since $\widehat{\mathcal{L}}_k$ equals the tensor product of $\mathcal{L}$ with some identity superoperators, $\lambda(\widehat{\mathcal{L}}_k) = \lambda(\mathcal{L})$. Therefore²,

$$-\lambda(\mathcal{L}) \langle X, \widehat{\mathcal{L}}_{\sigma_k}(X) \rangle_{\sigma^{\otimes N}} \leq -\langle X, \widehat{\mathcal{L}}_k(X) \rangle_{\sigma^{\otimes N}}.$$

Using this in (10.16) we arrive at

$$\lambda(\mathcal{L}) \frac{1 - 2\|\sigma^{-1}\|_{\infty}^{-1}}{\ln(\|\sigma^{-1}\|_{\infty} - 1)} \text{Ent}_{2, \sigma^{\otimes N}}(X) \leq - \sum_{k=1}^N \langle X, \widehat{\mathcal{L}}_k(X) \rangle_{\sigma^{\otimes N}} = -\langle X, \mathcal{K}_N(X) \rangle_{\sigma^{\otimes N}}.$$

This gives the desired bound on $\alpha_2(\mathcal{K}_N)$. $\square$

Remark 10.1.9. This corollary is a non-commutative version of Corollary A.4 of [Diaconis and Saloff-Coste, 1996a] and gives a stronger bound compared to Corollary 6 and Theorem 9 of [Temme et al., 2014]. It would be interesting to compare this corollary with the result of [King, 2014] who generalized the hypercontractivity inequalities of [Montanaro and Osborne, 2010] for the qubit depolarizing channel to all doubly stochastic qubit quantum Markov semigroups. Here, having a bound on the 2-logarithmic Sobolev constant of the KMS-symmetric generalized qubit depolarizing channel (and its tensorization property), we derive a bound on the 2-logarithmic Sobolev constant of all qubit σ -reversible QMS.

Corollary 10.1.10. *Let $\dim \mathcal{H} = 2$ and $\sigma \in \mathcal{D}(\mathcal{H})$ be full-rank. Let $\mathcal{L}$ be the generator of a KMS-symmetric QMS $(\mathcal{P}_t)_{t \geq 0}$. Then for any $1 \leq q \leq p$ and $t \geq 0$ satisfying*

$$t \geq \frac{\ln(\|\sigma^{-1}\|_{\infty} - 1)}{4\lambda(\mathcal{L})(1 - 2\|\sigma^{-1}\|_{\infty}^{-1})} \ln \frac{p-1}{q-1},$$

²This comparison of Dirichlet forms was already used in Section 8.5 in order to show the non-positivity of the strong decoherence free log-Sobolev constant by reducing the analysis to the case of the $\mathcal{N}$ -decoherent semigroup.

we have $\|\mathcal{P}_t^{\otimes N}(X)\|_{\mathbb{L}_p(\sigma)} \leq \|X\|_{q,\sigma}$ for all $X > 0$ and any $N \in \mathbb{N}$.

10.2. CB hypercontractivity and complete log-Sobolev inequalities

As already discussed in the introduction of this chapter, in the classical case, log-Sobolev inequalities satisfy the very useful *tensorization property*, that is, given N primitive Markov semigroups $(P_t^{(i)})_{t \geq 0}$ with generators L_i , $i = 1, \dots, N$, if for each i , the semigroup $(P_t^{(i)})_{t \geq 0}$ satisfies the log-Sobolev inequality $\text{LSI}_2(c_i, d_i)$, then the product semigroup $(P_t)_{t \geq 0}$ with $P_t = P_t^1 \otimes \dots \otimes P_t^N$, satisfies the log-Sobolev inequality $\text{LSI}_2(\max_i c_i, \sum_i d_i)$. This can be seen as a consequence of the multiplicativity of the classical weighted $\mathbb{L}_p$ norms. It is strongly believed that this latter property no longer holds true in the quantum case, since quantum weighted $\mathbb{L}_p$ norms are not multiplicative. In [Beigi and King, 2016], the authors proposed to define the hypercontractivity property with respect to the completely bounded norm, which is known to be multiplicative even in the noncommutative framework, and proved that it is equivalent to the notion of a complete logarithmic Sobolev inequality for primitive doubly stochastic QMS. This provides a way to recover the tensorization property in the noncommutative framework. Here, we generalize the framework of [Beigi and King, 2016] to the case of any primitive (not necessarily doubly stochastic) QMS in finite dimensions. Recall that, given an operator $\Phi : \mathcal{B}(\mathcal{H}) \rightarrow \mathcal{B}(\mathcal{H})$, its *weighted completely bounded norm* is defined as follows (cf. Section 1.1.2):

$$\|\Phi : \mathbb{L}_q(\sigma) \rightarrow \mathbb{L}_p(\sigma)\|_{\text{cb}} := \sup_{n \in \mathbb{N}} \|\text{id}_{\mathbb{M}_n(\mathbb{C})} \otimes \Phi : \mathbb{L}_q(n^{-1} \mathbb{1}_{\mathbb{C}^n}, \mathbb{L}_q(\sigma)) \rightarrow \mathbb{L}_q(n^{-1} \mathbb{1}_{\mathbb{C}^n}, \mathbb{L}_p(\sigma))\|, \quad (10.17)$$

where the supremum in 10.17 is over all dimensions n . These norms are known to be multiplicative, as proved in [Devetak et al., 2006]. As a result, in order to define a notion of hypercontractivity and its associated log-Sobolev inequality that satisfy the tensorization property, we embed a primitive QMS $(\mathcal{P}_t)_{t \geq 0}$ on $\mathcal{B}(\mathcal{H})$ into the QMS $(\text{id}_k \otimes \mathcal{P}_t)_{t \geq 0}$ on $\mathcal{B}(\mathbb{C}^k \otimes \mathcal{H})$, and study the latter's DF-hypercontractivity properties, for each integer $k \geq 1$. Let σ be the unique invariant state of $(\mathcal{P}_t)_{t \geq 0}$. Then $\mathcal{N}_k := \mathcal{B}(\mathbb{C}^k) \otimes \mathbb{1}_{\mathcal{H}}$ and $\sigma_{\text{Tr}} = \frac{\mathbb{1}_k}{k} \otimes \sigma$. We are lead to the following definitions.

Definition 10.2.1. We say that $(\mathcal{P}_t)_{t \geq 0}$:

- (i) satisfies a *q-complete logarithmic Sobolev inequality* with positive *strong logarithmic Sobolev constant* $c > 0$ and *weak logarithmic Sobolev constant* $d \geq 0$, which we denote by $\text{cLSI}_q(c, d)$, if for all integer $k \geq 1$, $\text{LSI}_{q, \mathcal{N}_k}(c, d)$ holds.
- (ii) is *q-CB-hypercontractive* for positive constants $c > 0$ and $d \geq 0$, condition denoted by $\text{cHC}_q(c, d)$, if for all $t \geq 0$,

$$\|\mathcal{P}_t : \mathbb{L}_q(\sigma) \rightarrow \mathbb{L}_{p(t)}(\sigma)\|_{\text{cb}} \leq \exp \left(2d \left(\frac{1}{q} - \frac{1}{p(t)} \right) \right),$$

for any function $p : [0, +\infty) \rightarrow \mathbb{R}$ such that for any $t \geq 0$, $q \leq p(t) \leq 1 + (q-1)e^{\frac{4t}{c}}$. In the case $q = 2$, we simply denote the previous property by $\text{cHC}(c, d)$.

The above definitions extend the ones in [Beigi and King, 2016] to non-unital primitive QMS and to weak LSI and weak HC. In the next theorem, we establish the equivalence between the complete logarithmic Sobolev inequality and CB-hypercontractivity, hence extending Theorem 4 of [Beigi and King, 2016] to the cases mentioned above.

Theorem 10.2.2. *Let $(\mathcal{P}_t)_{t \geq 0}$ be a primitive QMS on $\mathcal{B}(\mathcal{H})$ with associated generator $\mathcal{L}$, and let $q \geq 1$, $d \geq 0$ and $p(t) = 1 + (q-1)e^{\frac{4t}{c}}$ for some constant $c > 0$. Then*

- (i) *If $\text{cHC}_q(c, d)$ holds, then $\text{cLSI}_q(c, d)$ holds.*
- (ii) *If $\text{cLSI}_{p(t)}(c, d)$ holds for all $t \geq 0$, then $\text{cHC}_q(c, d)$ holds.*

Proof. We first prove (i). If $\text{cHC}_q(c, d)$ holds, then for any k and any $X \in \mathcal{B}(\mathbb{C}^k \otimes \mathcal{H})$,

$$\|\text{id}_k \otimes \mathcal{P}_t(X)\|_{\mathbb{L}_q(\mathcal{N}_k, \mathbb{L}_{p(t)}(\sigma_{\text{Tr}}))} \leq \exp\left(2d\left(\frac{1}{q} - \frac{1}{p(t)}\right)\right) \|X\|_{\mathbb{L}_q(\frac{1}{k} \otimes \sigma)},$$

that is $\text{HC}_{q, \mathcal{N}}(c, d)$ holds for the QMS $(\text{id}_k \otimes \mathcal{P}_t)_{t \geq 0}$ of corresponding decoherence-free algebra $\mathcal{N}_k = \mathcal{B}(\mathbb{C}^k) \otimes \mathbb{1}_{\mathcal{H}}$ and $\sigma_{\text{Tr}} = \frac{1}{k} \otimes \sigma$. The result then follows from a direct application of 8.2.2(i). (ii) follows similarly from 8.2.2(ii). $\square$

A direct application of the definitions for $\mathbb{L}_p$ regularity of Dirichlet forms then leads to the following:

Theorem 10.2.3. *Assume that $\text{cLSI}_2(c, d)$ holds. Then, if the generator $\mathcal{L}$ is $\mathbb{L}_p$ -regular for some $d_0 \geq 0$, then $\text{cLSI}_q(c, d + c d_0/2)$ holds for all $q \geq 1$, so that $\text{cHC}_2(c, d + c d_0/2)$ holds.*

As in the decoherence-free case, an application of Proposition 5.2 of [Olkiewicz and Zegarliński, 1999] together with Theorem 4 of [Watrous, 2005] leads to the following corollary:

Corollary 10.2.4. *Assume that $\mathcal{L}$ is the generator of a primitive QMS with unique invariant state σ , and that $\text{cLSI}_2(c, d)$ holds.*

- (i) *If $\mathcal{L}$ is KMS-symmetric, then $\text{cLSI}_q(c, d + c(\|\mathcal{L} : \mathbb{L}_2(\sigma) \rightarrow \mathbb{L}_2(\sigma)\| + 1))$ holds for all $q \geq 1$ and consequently $\text{cHC}_2(c, d + c(\|\mathcal{L} : \mathbb{L}_2(\sigma) \rightarrow \mathbb{L}_2(\sigma)\| + 1))$ holds.*
- (ii) *If $\mathcal{L}$ satisfies σ -DBC, then $\text{cLSI}_q(c, d)$ holds for all $q \geq 1$ and consequently $\text{cHC}_2(c, d)$ holds.*

Proof. The result follows directly from the fact that reversibility of $\mathcal{L}$ w.r.t. σ implies reversibility of $\text{id}_k \otimes \mathcal{L}$ w.r.t. σ_{Tr} , for any $k \in \mathbb{N}$, so that Corollary 8.2.4 applies. We conclude by noticing that for any $k \in \mathbb{N}$,

$$\begin{aligned} \|\text{id}_k \otimes \mathcal{L} : \mathbb{L}_2(\sigma) \rightarrow \mathbb{L}_2(k^{-1} \mathbb{1}_k \otimes \sigma)\| &= \|\Gamma_{\frac{1}{k} \otimes \sigma} \circ (\text{id}_k \otimes \mathcal{L}) \circ \Gamma_{\frac{1}{k} \otimes \sigma}^{-1} : \mathcal{T}_2(\mathcal{H}) \rightarrow \mathcal{T}_2(\mathcal{H})\| \\ &= \|\Gamma_{\sigma} \circ \mathcal{L} \circ \Gamma_{\sigma}^{-1} : \mathcal{T}_2(\mathcal{H}) \rightarrow \mathcal{T}_2(\mathcal{H})\| \\ &= \|\mathcal{L} : \mathbb{L}_2(\sigma) \rightarrow \mathbb{L}_2(\sigma)\|, \end{aligned} \tag{10.18}$$

where we used Theorem 4 of [Watrous, 2005] in (10.18). The second part follows similarly by the second part of Corollary 8.2.4. In both cases, hypercontractivity follows from Theorem 10.2.2. $\square$

Moreover, we derive universal bounds on the complete-log-Sobolev constants:

Theorem 10.2.5 (Universal bounds on the complete logarithmic Sobolev constants). *Let $(\mathcal{P}_t)_{t \geq 0}$ be a primitive reversible QMS, with unique invariant state σ and spectral gap $\lambda(\mathcal{L})$. Then, $\text{cLSI}_2(c, \ln \sqrt{2})$ holds, with*

$$c \leq \frac{\ln \|\sigma^{-1}\|_{\infty} + 2}{\lambda(\mathcal{L})}. \tag{10.19}$$

Proof. First notice that for all $k \in \mathbb{N}$, and any $X \in \mathcal{B}(\mathbb{C}^k \otimes \mathcal{H})$,

$$\|\text{id}_k \otimes \mathcal{P}_t(X)\|_{\mathbb{L}_2(\mathcal{N}_k, \mathbb{L}_4(\sigma_{\text{Tr}}))} \leq \|X\|_{\mathbb{L}_2(\mathcal{N}_k, \mathbb{L}_4(\sigma_{\text{Tr}}))} \leq \|\text{id} : \mathbb{L}_2(\sigma) \rightarrow \mathbb{L}_4(\sigma)\|_{\text{cb}} \|X\|_{\mathbb{L}_2(k^{-1} \mathbb{1}_k \otimes \sigma)} \tag{10.20}$$

where the first inequality follows from (i) of Proposition 8.3.1. Then, for any $1 \leq q \leq p \leq \infty$:

$$\|\text{id} : \mathbb{L}_q(\sigma) \rightarrow \mathbb{L}_p(\sigma)\|_{\text{cb}} = \|\Gamma_{\sigma}^{\frac{1}{p} - \frac{1}{q}} : \mathcal{T}_q(\mathcal{H}) \rightarrow \mathcal{T}_p(\mathcal{H})\|_{\text{cb}} = \|\sigma^{-1}\|_{\infty}^{\frac{1}{q} - \frac{1}{p}}. \quad (10.21)$$

Now, an application of Theorem 8.4.10 to the QMS $(\text{id}_k \otimes \mathcal{P}_t)_{t \geq 0}$ together with the fact that $\lambda(\mathcal{L}) = \lambda(\text{id}_k \otimes \mathcal{L})$ for any $k \in \mathbb{N}$ allow us to conclude. $\square$

Using the multiplicativity of CB norms [Devetak et al., 2006], we directly get the tensorization property of the complete logarithmic Sobolev inequality, hence extending Theorem 6 of [Beigi and King, 2016] to any primitive QMS.

Theorem 10.2.6. *Suppose that for all $i = 1, \dots, N$ the primitive QMS $(\mathcal{P}^{(i)})_{t \geq 0}$ on $\mathcal{B}(\mathcal{H}_i)$ generated by $\mathcal{L}_i$ with invariant state σ_i satisfies $\text{cLSI}_q(c_i, d_i)$. Then the QMS $(\mathcal{P}_t)_{t \geq 0}$ on $\mathcal{B}(\otimes_{i=1}^N \mathcal{H}_i)$ generated by $\mathcal{L}^{(N)} := \sum_{i=1}^N \otimes_{k=1}^{i-1} \text{id}_{\mathcal{B}(\mathcal{H}_k)} \otimes \mathcal{L}_i \otimes \otimes_{k=i+1}^N \text{id}_{\mathcal{B}(\mathcal{H}_k)}$ with invariant state $\otimes_{i=1}^N \sigma_i$ satisfies $\text{cLSI}_q(c, d)$ with $c := \max_i c_i$ and $d = \sum_{i=1}^N d_i$.*

The additivity of the weak complete logarithmic Sobolev constant prevents one from obtaining relevant estimates for a large number of tensorized primitive QMS. In particular, estimating both constants separately as in Theorem 10.2.6 leads to weaker bounds than the ones found in [Temme et al., 2014, Müller-Hermes et al., 2016]. This however does not exclude the possibility of better controlling both constants simultaneously when considering tensor products of QMS. One way to achieve this is via the transference method introduced in Chapter 9. Before moving to the next section, we simply mention that a similar notion of a complete modified logarithmic Sobolev inequality was introduced in [Bardet, 2017] (see also [Gao et al., 2018b]). Contrary to the 2-complete logarithmic Sobolev inequality studied here, there exist examples of semigroups for which the “weak MLSI constant” vanishes. The question of the validity of this claim for any quantum Markov semigroup is still open. Another interesting problem is the one of finding the range of parameters q for which the weak complete logarithmic Sobolev constant becomes non-zero.

Chapter 11.

Quantum geometric and information theoretic inequalities

In Section 4.5, we recalled an information theoretical proof of the logarithmic Sobolev inequality for the Ornstein Uhlenbeck semigroup. At the core of the proof lies the entropy power inequality (cf. Theorem 4.5.2): for any two independent random variables X and Y on $\mathbb{R}^n$,

$$N(X + Y) \geq N(X) + N(Y), \quad \text{where} \quad N(X) := e^{2S(X)/n} \quad (11.1)$$

is the entropy power of signal X . Equivalently, the following entropy convex combination inequality holds: $S(\sqrt{\lambda}X + \sqrt{1-\lambda}Y) \geq \lambda S(X) + (1-\lambda)S(Y)$. In words, the latter inequality expresses the fact that the entropy of the sum of two signals is greater than the sum of the entropies of the signals taken individually. The entropy power inequality is also closely related to the idea of isoperimetry and has important applications in channel coding theory [Shannon, 1948].

Recently, the entropy power inequality (EPI) has been a subject of focus in the quantum community [Koenig and Smith, 2014, Audenaert et al., 2016, Carlen et al., 2016]. In this chapter, we aim at extending the results mentioned in Section 4.5 to the quantum phase space setting. In this chapter, we consider a classical-quantum entropy power inequality, where the sum between the two independent random variables X and Y in Theorem 4.5.2 is replaced by the classical-quantum convolution between a quantum state ρ and a classical probability distribution function f as introduced by [Werner, 1984]. We show how this inequality provides convergence times for the quantum heat semigroup defined in Section 5.5.2. Some of the results stated and proved in this section were found independently in [Huber et al., 2017, Datta et al., 2017].

However, a generalization of the proof of the logarithmic Sobolev inequality via EPI for the quantum Ornstein Uhlenbeck semigroup (see Section 5.5.2) seems to require a fully quantum-quantum convolution as introduced in [Koenig and Smith, 2014]¹, which is a channel that models the physical process corresponding to photons coming out of a beamsplitter. The proof was recently found in [De Palma and Trevisan, 2018] and relies on a series of papers [G. De Palma et al., 2014, De Palma et al., 2015b, Koenig, 2015, De Palma and Trevisan, 2018]. We summarize and compare their main results to the ones of Section 11.1 in Section 11.3.

¹We also mention that a finite dimensional version of the quantum-quantum convolution was later introduced and a quantum-quantum entropy power inequality for that convolution was derived in [Audenaert et al., 2016], see also [Carlen et al., 2016] for a more recent proof and [Jeong et al., 2018] for a generalization in the presence of quantum memory.

Layout of the chapter: In Section 11.1, we state and prove some classical-quantum versions of the information theoretic inequalities of Section 4.5. These results, and in particular the isoperimetric inequality of Theorem 11.1.7 are used to prove some convergence bounds on the quantum heat semigroup in Section 11.2. Connection with the fully quantum entropy power inequality of [Koenig and Smith, 2014] is established, and the link to the entropy decay for the quantum Ornstein Uhlenbeck semigroup found in [De Palma and Huber, 2018] is recalled in Section 11.3.

11.1. A quantum-classical entropy power inequality

Given a random variable Z , its entropy power $N(Z)$ only depends on its probability density function f_Z . Therefore, (11.1) only depends on f_X , f_Y and their convolution

$$f_{X+Y}(z) = f_X * f_Y(z) = \int_{\mathbb{R}^n} f_X(x) f_Y(z-x) dx.$$

Any quantum extension of the (11.1) requires a non-commutative notion of convolution. In [Werner, 1984], the classical-quantum convolution between a quantum state $\rho \in \mathcal{D}(\mathcal{H})$, $\mathcal{H} = \mathbb{L}_2(\mathbb{R}^n)$, and a probability density function f on $\mathbb{R}^{2n}$ was defined as follows:

$$\rho * f := \int_{\mathbb{R}^{2n}} f(z) W_z \rho W_{-z} dz,$$

where the integral is defined in the Bochner sense. Here, the Weyl displacement operators, introduced in Section 0.2, play the role of translations in phase space. Next, we define the *quantum entropy power* of a state $\rho \in \mathcal{D}(\mathcal{H})$ as follows

$$N(\rho) := e^{S(\rho)/n}. \quad (11.2)$$

Once again, the difference in the normalization of the exponent in comparison with the classical entropy power (11.1) comes from the fact that the quantum phase space is $2n$ dimensional. The following classical-quantum entropy power inequality was first derived in [Huber et al., 2017]:

Theorem 11.1.1. *For any $\rho \in \mathcal{D}(\mathcal{H})$ and any probability density function f on $\mathbb{R}^{2n}$,*

$$N(\rho * f) \geq N(\rho) + N(f). \quad (\text{cq-EPI})$$

In this section, we provide a proof of Theorem 11.1.1 using tools developed independently in [Datta et al., 2017]. Along the way, we derive other entropic inequalities on the quantum phase space such as a quantum Blachman-Stam inequality, the concavity of the entropy power along the quantum heat semigroup, as well as an entropic isoperimetric inequality, that may be of independent interest.

11.1.1. Fisher informations on phase space

Divergence based quantum Fisher information Classically, the proof of the entropy power inequality relies on the Fisher information inequality of Lemma 4.5.4. The crucial step is the so-called de Bruijn identity (Lemma 1.3.2) relating the Fisher informations to the derivative of the Shannon entropy along the heat semigroup, which allows to interpret it as its corresponding entropy production. In the quantum setting, one is then tempted to define a quantum Fisher information that can similarly be related to the von Neumann entropy.

We first briefly recall the classical setting: given a smooth family $(f_\theta)_{\theta \in \Theta}$ of probability density functions on $\mathbb{R}^{2n}$, its classical Fisher information is defined by Equation (1.70) as the negative of the Laplacian of the function $(\alpha, \beta) \mapsto D(f_\alpha \| f_\beta)$, or equivalently:

$$I(\theta) = \Delta_\alpha D(f_\theta \| f_\alpha)|_{\alpha=\theta}. \quad (11.3)$$

In the case when the parameter θ simply consists of shifts from a fixed distribution f , the expression can be simplified as in Equation (1.71):

$$I(f) := I(0) = - \int f(x) \Delta(\ln f)(x) dx. \quad (11.4)$$

Up to regularity issues, this is nothing but the entropy production associated to the heat semigroup (cf. Section 3.3).

In the quantum case, we saw in Section 1.3 that, given a smooth family $(\rho_\theta)_{\theta \in \Theta}$ of quantum states on a finite dimensional Hilbert space, their Fisher information was defined, similarly, as the Laplacian of $(\alpha, \beta) \mapsto D(\rho_\alpha \| \rho_\beta)$. In analogy with (11.4), we are interested in the case when $(\rho_\theta)_{\theta \in \mathbb{R}^{2n}}$ can be interpreted as a family of shifts in the quantum phase space. This can naturally be done via the Weyl operators introduced in Section 0.2.

We assume from now on that $\mathcal{H} = \mathbb{L}_2(\mathbb{R}^n)$. Then, given a state $\rho \in \mathcal{D}(\mathcal{H})$, define the shifted operator along the direction of R_j as

$$\rho_{\theta, R_j} := e^{i\theta R_j} \rho e^{-i\theta R_j} := W_{z_{j, \theta}} \rho W_{-z_{j, \theta}}, \quad (11.5)$$

where $z_{j, \theta}$ is the vector in $\mathbb{R}^{2n}$ with entries

$$(z_{j, \theta})_r := \begin{cases} \theta & \text{if } r = j, \text{ } j \text{ even,} \\ -\theta & \text{if } r = j, \text{ } j \text{ odd,} \\ 0 & \text{otherwise.} \end{cases} \quad (11.6)$$

Assuming that $\rho \in \mathcal{D}_+(\mathcal{H})$, its *divergence-based quantum Fisher information* is defined in analogy with (11.3) as [Koenig and Smith, 2014, Koenig, 2015]:

$$J(\rho) := \sum_{i=1}^{2n} \frac{d^2}{d\theta^2} D(\rho \| \rho_{\theta, R_j}) \Big|_{\theta=0}, \quad (11.7)$$

whenever the functions $\theta \mapsto D(\rho \| \rho_{\theta, R_j})$ are twice differentiable. Formally [Koenig and Smith, 2014], this quantity can be shown to be equal to the entropy production of the quantum heat semigroup (cf. Equation (5.44)):

$$\frac{d}{dt} S(\mathcal{P}_{*t}^{\text{heat}}(\rho)) \Big|_{t=0} = - \text{Tr}(\mathcal{L}_*^{\text{heat}}(\rho) \ln \rho) = \frac{1}{4} J(\rho). \quad (11.8)$$

This statement was rigorously proved for Gaussian thermal states in [Koenig, 2015]. The second identity can be made more precise when ρ is a Schwartz operator (cf. Section 0.2):

Lemma 11.1.2. *Let ρ be a Schwartz state such that $\ln \rho$ is polynomially bounded. Then for any $j \in \{1, 2, \dots, 2n\}$, the function $\theta \mapsto D(\rho \| \rho_{\theta, R_j})$ is twice differentiable at 0. Moreover, the operators $\rho[R_j, [R_j, \ln \rho]]$ are trace class, and the divergence-based quantum Fisher information defined through*

Equation (11.7) is given by

$$J(\rho) = \sum_{i=1}^{2n} \text{Tr}(\rho [R_j, [R_j, \ln \rho]]). \quad (11.9)$$

In addition, $\frac{d}{d\theta} D(\rho \| \rho_{\theta, R_j})|_{\theta=0} = 0$ for all j .

Proof. We start from the following relative entropy $D(\rho \| \rho_{\theta, R_j}) = \text{Tr}(\rho (\ln \rho - \ln \rho_{\theta, R_j}))$. Since its first term is constant, we can focus on the second term:

$$\begin{aligned} \text{Tr} \rho \ln \rho_{\theta, R_j} &= \text{Tr}(\rho e^{i\theta R_j} \ln \rho e^{-i\theta R_j}) \\ &= \sum_{k=1}^{\infty} \langle \psi_k, e^{-i\theta R_j} \rho e^{i\theta R_j} \ln \rho \psi_k \rangle, \end{aligned} \quad (11.10)$$

where $\{\psi_k\}_{k=1}^{\infty}$ is an orthonormal basis of Schwartz functions of $\mathcal{H}$ (take e.g. the Hermite polynomials). Recall that $e^{i\theta R_j}$ is equal to $W_{z_{j,\theta}}$, with $z_{j,\theta}$ defined in (11.6). Let's denote $\varphi_k := \ln \rho \psi_k$. It is a Schwartz function, since $\ln \rho$ is polynomially bounded (see Proposition 3.21 of [Keyl et al., 2016]). Therefore, each term of Equation (11.10) is of the form

$$\langle \psi_k, W_{-z_{j,\theta}} \rho W_{z_{j,\theta}} \varphi_k \rangle, \quad \psi_k, \varphi_k \in \mathcal{S}(\mathcal{H}). \quad (11.11)$$

We conclude by a direct use of Proposition 3.25 of [Keyl et al., 2016]. $\square$

The divergence-based Fisher information satisfies the following quantum version of the Blachman-Stam inequality. The result was independently found by [Huber et al., 2017] and [Datta et al., 2017].

Theorem 11.1.3 (Quantum Blachman-Stam inequality). *For any $\alpha, \beta > 0$, $t > 0$, for any state ρ such that $\theta \mapsto D(\rho \| \rho_{\theta, R_j})$ is twice differentiable at 0 for all $j = 1, \dots, 2n$, we have*

$$(\alpha + \beta)^2 J(\rho * f) \leq \alpha^2 J(\rho) + \beta^2 I(f). \quad (11.12)$$

In order to prove the quantum Blachman-Stam inequality, we need the following technical lemma:

Lemma 11.1.4. *For any two quantum states $\rho_1, \rho_2 \in \mathcal{D}(\mathcal{H})$ and probability density functions $f_1, f_2 \in \mathbb{L}_1(\mathbb{R}^{2n})$,*

$$D(\rho_1 * f_1 \| \rho_2 * f_2) \leq D(\rho_1 \| \rho_2) + D(f_1 \| f_2) \quad (11.13)$$

Proof. The proof involves a simple application of Uhlmann's monotonicity of the relative entropy (1.56) under Schwarz mapping in the context of general von Neumann algebras. Indeed, the result would follow if we can prove that for any $\rho \in \mathcal{D}(\mathcal{H})$ and any positive $g \in \mathbb{L}_1(\mathbb{R}^{2n})$, with $\|g\|_{\mathbb{L}_1(\mathbb{R}^{2n})} = 1$,

$$\omega_{\rho * g} = (\omega_{\rho} \otimes \omega_g) \circ \Phi, \quad (11.14)$$

where ω_{ρ} , resp. ω_f , is the state corresponding to the density operator ρ , resp. the density function f , and Φ is a unital, completely positive map from $\mathcal{B}(\mathcal{H}) \otimes \mathbb{L}_{\infty}(\mathbb{R}^{2n})$ to $\mathcal{B}(\mathcal{H})$. Indeed, $\omega_{\rho} \otimes \omega_g$ acts on tensor product elements of $\mathcal{B}(\mathcal{H}) \otimes \mathbb{L}_{\infty}(\mathbb{R}^{2n})$ by

$$(\omega_{\rho} \otimes \omega_g)(A \otimes f) = \text{Tr}(\rho A) \times \int_{\mathbb{R}^{2n}} g(z) f(z) dz, \quad A \in \mathcal{B}(\mathcal{H}), f \in \mathbb{L}_{\infty}(\mathbb{R}^{2n}) \quad (11.15)$$

and extends by continuity [Kadison and Ringrose, 1983]. But $\mathcal{B}(\mathcal{H}) \otimes \mathbb{L}_\infty(\mathbb{R}^{2n})$ can be identified with the set $\mathbb{L}_\infty(\mathbb{R}^{2n}, \mathcal{B}(\mathcal{H}))$ of measurable functions from $\mathbb{R}^{2n}$ to $\mathcal{B}(\mathcal{H})$, in which case an element of it is a map $z \mapsto A_z$ and $\omega_\rho \otimes \omega_g$ acts as

$$(\omega_\rho \otimes \omega_g)(z \mapsto A_z) = \int_{\mathbb{R}^{2n}} \text{Tr}(\rho A_z) g(z) dz \quad (11.16)$$

(which of course is consistent with the above if $A_z = f(z)A$). On the other hand, $\omega_{\rho * g}$ acts as

$$\omega_{\rho * g}(A) = \int_{\mathbb{R}^{2n}} g(z) \text{Tr}(\rho W_{-z} A W_z) dz, \quad (11.17)$$

so to prove our statement it is enough to show that the map Φ from $\mathcal{B}(\mathcal{H})$ to $\mathbb{L}_\infty(\mathbb{R}^{2n}, \mathcal{B}(\mathcal{H}))$ defined as

$$\Phi : A \mapsto (z \mapsto W_{-z} A W_z) \quad (11.18)$$

is unital and completely positive. The first requirement is obvious. To show complete positivity, consider a positive matrix $(A_{i,j})_{i,j=1}^N$ of elements of $\mathcal{B}(\mathcal{H})$, in the sense that for any $(\varphi_i)_{i=1}^N$ with each $\varphi_i \in \mathcal{H}$, one has

$$\sum_{i,j} \langle \varphi_i, A_{i,j} \varphi_j \rangle \geq 0.$$

We then prove that $(\Phi(A_{i,j}))_{i,j=1}^N$ is positive when acting on $\mathbb{L}_2(\mathbb{R}^{2n}, \mathcal{H})$: consider $\psi_i = (z \mapsto \psi_i(z)) \in \mathbb{L}_2(\mathbb{R}^{2n}, \mathcal{H})$, therefore

$$\sum_{i,j} \langle \psi_i, \Phi(A_{i,j}) \psi_j \rangle = \sum_{i,j} \int \langle \psi_i(z), W_z^* A_{i,j} W_z \psi_j(z) \rangle dz = \int \sum_{i,j} \langle W_z \psi_i(z), A_{i,j} W_z \psi_j(z) \rangle dz \geq 0.$$

This together with the additivity (1.54) of the relative entropy completes the proof of (11.13). $\square$

We are now ready to prove Theorem 11.1.3. The proof we give can be seen as a quantum analogue of the proof given by Stam in [a.J. Stam, 1959].

Proof of Theorem 11.1.3 : In the following, we define a vector $z = (q_1, p_1, \dots, q_n, p_n) \in \mathcal{Z}$ simply as (q, p) , and for any $a \in \mathbb{R}$ we define

$$(q, p; q_j - a) := (q_1, p_1, \dots, q_{j-1}, p_{j-1}, q_j - a, p_j, q_{j+1}, p_{j+1}, \dots, q_n, p_n), \quad (11.19)$$

$$(q, p; p_j - a) := (q_1, p_1, \dots, q_{j-1}, p_{j-1}, q_j, p_j - a, q_{j+1}, p_{j+1}, \dots, q_n, p_n). \quad (11.20)$$

Then for any $a \in \mathbb{R}$ we define the following functions:

$$f_{a, P_j} : (q, p) \mapsto f(q, p; q_j - a); \quad f_{a, Q_j} : (q, p) \mapsto f(q, p; p_j + a) \quad (11.21)$$

Further, accordingly denoting the Weyl operator W_z as $W_{(q,p)}$, we have for any $\theta \in \mathbb{R}$ and $\alpha, \beta > 0$,

$$\begin{aligned} \rho_{\theta\alpha, P_j} * f_{\theta\beta, P_j} &= \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} f(q, p; q_j - \theta\beta) W_{(q,p)} \rho_{\theta\alpha, P_j} W_{(-q, -p)} dq dp \\ &= \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} f(q, p) W_{(q, p; q_j + \theta\beta)} \rho_{\theta\alpha, P_j} W_{(q, p; q_j + \theta\beta)}^* dq dp, \end{aligned} \quad (11.22)$$

where we made a change of variable $q_j - \theta\beta \rightarrow q_j$. Then using the Weyl-Segal CCR, we obtain

$$W_{(q, p; q_j + \theta\beta)} e^{i\theta\alpha P_j} = e^{ip_j \theta\alpha/2} W_{(q, p; q_j + \theta(\alpha + \beta))} \quad (11.23)$$

Hence,

$$\begin{aligned}
 \rho_{\theta\alpha, P_j} * f_{\theta\beta, P_j} &= \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} f(q, p) W_{(q, p; q_j + \theta(\alpha + \beta))} \rho W_{-(q, p; q_j + \theta(\alpha + \beta))} dq dp \\
 &= e^{i\theta(\alpha + \beta)P_j} \left(\int_{\mathbb{R}^n} \int_{\mathbb{R}^n} f(q, p) W_{(q, p)} \rho W_{-(q, p)} dq dp \right) e^{-i\theta(\alpha + \beta)P_j} \\
 &= \rho_{\theta(\alpha + \beta), P_j}.
 \end{aligned} \tag{11.24}$$

This implies that

$$D(\rho \| \rho_{\theta\alpha, P_j} * f_{\theta\beta, P_j}) = D(\rho \| \rho_{\theta(\alpha + \beta), P_j}). \tag{11.25}$$

Similarly, we can show that

$$D(\rho \| \rho_{\theta\alpha, Q_j} * f_{\theta\beta, Q_j}) = D(\rho \| \rho_{\theta(\alpha + \beta), Q_j}). \tag{11.26}$$

A direct application of Lemma 11.1.4 leads to:

$$D(\rho \| \rho_{\theta\alpha, R_j} * f_{\theta\beta, R_j}) \leq D(\rho \| \rho_{\theta\alpha, R_j}) + D(f \| f_{\theta\beta, R_j}), \tag{11.27}$$

The result then follows by differentiating the above inequality twice, since both sides of the above are functions of θ that are equal up to first order. $\square$

The quantum integral Fisher information The proofs of the entropic inequalities derived in [Koenig and Smith, 2014, Huber et al., 2017, Datta et al., 2017] all rely on the assumption that the state ρ satisfies the de Bruijn identity (11.8). Classically, the de Bruijn identity holds with mild assumptions on the distribution f (cf. Lemma 1.3.2). However, it is not clear which are the minimal conditions that the state ρ should satisfy for the identity to hold. In [De Palma and Trevisan, 2018], the authors took a different approach by showing that a particular entropic quantity is differentiable and defining the Fisher information as its derivative. In this approach, the de Bruijn identity becomes true by definition. Here, we review the approach of [De Palma and Trevisan, 2018] and relate it to the one we described in the previous paragraph. In the next section, we use this better behaved definition for the quantum Fisher to prove the entropic inequalities already appearing in [Koenig and Smith, 2014, Huber et al., 2017, Datta et al., 2017] without requiring any regularity on the quantum state other than finiteness of its entropy and of its first moments.

Given a quantum state $\rho \in \mathcal{D}(\mathcal{H})$ of finite entropy, its *quantum integral Fisher information* $\tilde{J}(\rho)$ is defined as follows: for $t \geq 0$ ²:

$$\tilde{J}(\rho)(t) := \text{Ent} \left(\omega_{\text{QC}}(t) \| \omega_{\mathcal{P}_{*t}^{\text{Heat}}(\rho)} \otimes \omega_{g_{t/2}} \right),$$

where $\omega_{\text{QC}}(t)$ is a state defined on the algebra $\mathcal{B}(\mathcal{H}) \otimes \mathbb{L}_{\infty}(\mathbb{R}^{2n})$ whose action on tensor product elements $X \otimes f$, $X \in \mathcal{B}(\mathcal{H})$, $f \in \mathbb{L}_{\infty}(\mathbb{R}^{2n})$, is as follows,:

$$\omega_{\text{QC}}(t)(X \otimes f) := \int_{\mathbb{R}^{2n}} \text{Tr}(W_z \rho W_{-z} X) f(z) g_{t/2}(z) dz \equiv \omega_{\rho} \otimes \omega_{g_{t/2}} \circ \Phi(X \otimes f). \tag{11.28}$$

In the case when ρ has finite second moments and entropy, one shows that $t \mapsto \tilde{J}(\rho)(t)$ is an increasing, concave continuous function of time (cf. Theorem 2 of [De Palma and Trevisan, 2018]). This provides

²In fact, [De Palma and Trevisan, 2018] more generally defined a conditional quantum integral Fisher information $\tilde{J}$ that can be expressed as a conditional mutual information.

the existence of the following limit:

$$J(\rho) := 4 \lim_{t \rightarrow 0^+} \frac{\tilde{J}(\rho)(t)}{t}. \quad (11.29)$$

This new definition of the Fisher information has the advantage that it directly implies the validity of the de Bruijn identity. Indeed, $\tilde{J}(\rho)$ is a mutual information which can be written as [Belavkin and Ohya, 2002]:

$$\tilde{J}(\rho)(t) = S(\mathcal{P}_{*t}^{\text{Heat}}(\rho)) + S(g_{t/2}) - S(\omega_{\text{QC}}(t)),$$

where the entropy $S(\omega_{\text{QC}}(t))$ is defined as [Umegaki, 1962]

$$S(\omega_{\text{QC}}(t)) = -\tau \left(\frac{d\omega_{\text{QC}}(t)}{d\tau} \ln \frac{d\omega_{\text{QC}}(t)}{d\tau} \right),$$

where τ is the trace defined on tensor elements of the semi-finite von Neumann algebra $\mathcal{B}(\mathcal{H}) \otimes \mathbb{L}_\infty(\mathbb{R}^{2n})$ as follows: for $X \in \mathcal{T}_1(\mathcal{H})$ and $f \in \mathbb{L}_1(\mathcal{H})$, $\tau(X \otimes f) = \text{Tr}(X) \int_{\mathbb{R}^{2n}} f(z) dz$. The operator $\frac{d\omega_{\text{QC}}(t)}{d\tau}$ is the density operator associated to the state $\omega_{\text{QC}}(t)$. From Equation (11.28), it can be identified as $z \mapsto W_z \rho W_{-z} g_{t/2}(z)$. Therefore [De Palma and Trevisan, 2018],

$$\begin{aligned} \tilde{J}(\rho)(t) &= S(\mathcal{P}_{*t}^{\text{Heat}}(\rho)) + S(g_{t/2}) - S(\omega_{\text{QC}}(t)) \\ &= S(\mathcal{P}_{*t}^{\text{Heat}}(\rho)) + S(g_{t/2}) + \int_{\mathbb{R}^{2n}} \text{Tr}(W_z \rho W_{-z} \ln(g_{t/2}(z) W_z \rho W_{-z})) g_{t/2}(z) dz \\ &= S(\mathcal{P}_{*t}^{\text{Heat}}(\rho)) + S(g_{t/2}) - S(g_{t/2}) + \int_{\mathbb{R}^{2n}} S(W_z \rho W_{-z} g_{t/2}(z)) dz \\ &= S(\mathcal{P}_{*t}^{\text{Heat}}(\rho)) - S(\rho). \end{aligned}$$

Even if formally, the quantities defined in (11.29) and (11.7) coincide (cf. Equation (11.7)), the problem of making this intuition rigorous is still open. Fortunately, the new definition (11.8) can still be shown to satisfy Equation (11.12) for $0 \leq \alpha = 1 - \beta \leq 1$ as long as ρ and f both have finite second moments. The proof of this shares some similarities with the one of Theorem 11.1.3 and can be found in Theorem 3 of [De Palma and Trevisan, 2018]. In the next subsection, we assume this fact and give a brief summary of the information theoretic inequalities that it implies.

11.1.2. Information theoretic inequalities on the quantum phase space

The classical-quantum entropy power inequality is now a simple consequence of the Blachman-Stam inequality and the de Bruijn identity. It was originally derived in [Huber et al., 2017] using the divergence based Fisher information (11.7). A similar proof which uses the Fisher information defined in Equation (11.29), which also generalizes to the case of conditional entropies, is provided in Theorem 5 of [De Palma and Huber, 2018]:

Theorem 11.1.5 (Classical-quantum EPI). *For any state $\rho \in \mathcal{D}(\mathcal{H})$, any density function $f \in \mathbb{L}_1(\mathbb{R}^{2n})$, $\int f = 1$, of finite entropy and first and second moments, and any $0 \leq \lambda \leq 1$, the following holds:*

$$\frac{S(\rho * f)}{n} \geq \lambda \frac{S(\rho)}{n} + (1 - \lambda) \frac{S(f)}{n} - \lambda \ln \lambda - (1 - \lambda) \ln(1 - \lambda).$$

After optimization over λ this inequality implies

$$N(\rho * f) \geq N(\rho) + N(f). \quad (\text{CQ-EPI})$$

As expected, the Blachman-Stam inequality also provides quantum generalizations of the concavity of the entropy power, which itself implies a quantum entropic isoperimetric inequality. Our proof of Theorem 11.1.6 can hence be interpreted as a quantum version of the proof of [Dembo, 1989]:

Theorem 11.1.6 (Concavity of the quantum entropy power). *For any state $\rho \in \mathcal{D}(\mathcal{H})$ of finite second moments and finite Fisher information, the entropy power of $\rho_t := \mathcal{P}_{*t}^{\text{qHeat}}(\rho)$ is twice differentiable and satisfies*

$$\left. \frac{d^2}{dt^2} \right|_{t=0^+} N(\rho_t) \leq 0. \quad (11.30)$$

Proof. Since $J(\rho) < \infty$ by assumption, and since $t \mapsto \tilde{J}(\rho_t)$ is concave, $t \mapsto J(\rho_t)$ is a bounded, decreasing function. Letting $\varepsilon > 0$ and setting $\alpha = \frac{1}{J(\rho_t)}$ and $\beta = \frac{\varepsilon}{4n}$ in (11.12), we get

$$\left(\frac{1}{J(\rho_t)} + \frac{\varepsilon}{4n} \right)^2 J(\rho_{t+\varepsilon}) \leq \frac{1}{J(\rho_t)} + \frac{\varepsilon}{4n} \Rightarrow \frac{J(\rho_{t+\varepsilon}) - J(\rho_t)}{\varepsilon} \leq \frac{-J(\rho_t)J(\rho_{t+\varepsilon})}{4n}. \quad (11.31)$$

Since $t \mapsto J(\rho_t)$ is decreasing, both sides of Equation (11.31) converge as $\varepsilon \rightarrow 0^+$, so that $t \mapsto J(\rho_t)$ is differentiable at 0 and:

$$\left. \frac{d}{dt} \right|_{t=0^+} J(\rho_t) \leq -\frac{J(\rho_t)^2}{4n}.$$

It is then an easy calculation to verify that the above inequality together with the definition of J implies (11.30). $\square$

Theorem 11.1.6 has a simple corollary that can be interpreted as a quantum entropic isoperimetric inequality e -Isop. The proof that we provide here only relies on the concavity and asymptotic properties of the entropy power and asymptotics. In contrast, the proof of [De Palma and Trevisan, 2018] makes use of the entropy power inequality.

Theorem 11.1.7 (Isoperimetric inequality for the quantum entropy). *For any state ρ of finite second moments and Fisher information:*

$$J(\rho) N(\rho) \geq 2en. \quad (\text{qe-Isop})$$

Proof. From Equation (11.8), for any such ρ evolving under the action of the quantum heat semigroup,

$$\frac{d}{ds} N(\rho_s) = \frac{1}{4n} J(\rho_s) N(\rho_s). \quad (11.32)$$

Moreover by the concavity of the entropy power (Theorem 11.1.6),

$$\left. \frac{d}{ds} N(\rho_s) \right|_{s=0} \geq \frac{N(\rho_t) - N(\rho)}{t}, \quad \forall t > 0. \quad (11.33)$$

However, by Corollary III.4 of [Koenig and Smith, 2014] (see also Theorem 5 of [De Palma and Trevisan, 2018]), $N(\rho_t) = \frac{et}{2} + \mathcal{O}(1)$. Combining these facts, we get the following inequality:

$$J(\rho) \geq e^{-\frac{1}{n}S(\rho)} 2en, \quad (11.34)$$

which is the statement of Theorem 11.1.7. $\square$

11.2. Convergence properties of the quantum heat semigroup

In this section, we use the quantum isoperimetric inequality (Theorem 11.1.7) to find a bound on the relative entropy $D(\rho_t \| \rho_t^G)$ for any $t > 0$, where $\rho_t = \mathcal{P}_{*t}^{\text{qHeat}}(\rho)$ and for any state ρ of finite first and second moments, ρ_t^G denotes its ‘Gaussification’, i.e. the Gaussian state of same mean and covariance as ρ_t . We begin by recalling the following result of [Koenig and Smith, 2014].

Lemma 11.2.1. *For any state $\rho \in \mathcal{D}(\mathcal{H})$ of finite first and second moments, finite entropy and with covariance matrix satisfying the conditions of Theorem 0.2.4, denote $\rho_t := \mathcal{P}_{*t}^{\text{qHeat}}(\rho)$. Then*

$$D(\rho_t \| \rho_t^G) = -S(\rho_t) + S(\rho_t^G). \quad (11.35)$$

Proof. Without loss of generality we can reduce the proof to the case of a centered state ρ . Then:

$$-S(\rho_t) + S(\rho_t^G) = \text{Tr}(\rho_t \ln \rho_t) - \text{Tr}(\rho_t^G \ln \rho_t^G) \quad (11.36)$$

$$= \text{Tr}(\rho_t \ln \rho_t) - \ln C + \text{Tr}(\rho_t^G \mathbf{R}^T \Gamma \mathbf{R}) \quad (11.37)$$

$$= \text{Tr}(\rho_t \ln \rho_t) - \ln C + \text{Tr}(\rho_t \mathbf{R}^T \Gamma \mathbf{R}) \quad (11.38)$$

$$= \text{Tr}(\rho_t \ln \rho_t) - \text{Tr}(\rho_t \ln \rho_t^G) \quad (11.39)$$

$$= D(\rho_t \| \rho_t^G), \quad (11.40)$$

where in the second line we used the form (0.41) for ρ_t^G , and in the third line we used the fact that ρ_t has same covariance matrix as ρ_t^G for each time t . $\square$

This result enables us to prove that $D(\rho_t \| \rho_t^G)$ converges to 0 as $t \rightarrow \infty$, as pointed out By König and Smith in [Koenig and Smith, 2014]. Indeed for any initial state ρ evolving according to the heat semigroup $(\mathcal{P}_t^{\text{qHeat}})_{t \geq 0}$ they proved that the large time behavior of $S(\rho_t)$ is independent of ρ (see Corollary III.4). Therefore $-S(\rho_t)$ and $S(\rho_t^G)$ asymptotically cancel each other, and a direct application of Lemma 11.2.1 provides the convergence claimed. Below, we provide a refined analysis of the behavior of $D(\rho_t \| \rho_t^G)$ as t goes to infinity using the isoperimetric inequality derived in Theorem 11.1.7. We assume that $n = 1$ for sake of simplicity.

Lemma 11.2.2. *Let $\mathcal{H} := \mathbb{L}_2(\mathbb{R})$. Then for any initial state $\rho \in \mathcal{D}(\mathcal{H})$ of finite first and second moments evolving according to the heat semigroup $(\mathcal{P}_t^{\text{qHeat}})_{t \geq 0}$, the following inequality holds for any $t \geq 0$:*

$$D(\rho_t \| \rho_t^G) \leq e^{-\kappa(t)} D(\rho \| \rho^G) + e^{-\kappa(t)} \int_0^t \frac{e^{\kappa(s)}}{N(\rho_s^G)} \left(\frac{1}{4} J(\rho_s^G) N(\rho_s^G) - \frac{e}{2} \right) ds, \quad (11.41)$$

where:

$$\kappa(t) := \frac{e}{2} \int_0^t N(\rho_s^G)^{-1} ds.$$

Proof. We start by differentiating Equation (11.35):

$$\begin{aligned}
 \frac{dD(\rho_t \| \rho_t^G)}{dt} &= -\frac{dS(\rho_t)}{dt} + \frac{dS(\rho_t^G)}{dt} \\
 &= -\frac{1}{4}J(\rho_t) + \frac{1}{4}J(\rho_t^G) \\
 &\leq -\frac{e}{2}N(\rho_t)^{-1} + \frac{1}{4}J(\rho_t^G) \\
 &= -\frac{e}{2}e^{-(S(\rho_t)-S(\rho_t^G))}N(\rho_t^G)^{-1} + \frac{1}{4}J(\rho_t^G) \\
 &\leq -\frac{e}{2}N(\rho_t^G)^{-1}D(\rho_t \| \rho_t^G) - \frac{e}{2}N(\rho_t^G)^{-1} + \frac{1}{4}J(\rho_t^G), \tag{11.42}
 \end{aligned}$$

where we used Theorem 11.1.7 in the third inequality, as well as the basic inequality $e^x \geq 1 + x$, $x \in \mathbb{R}$ in the fifth one. Now $t \mapsto e^{e \int_0^t N(\rho_s^G)^{-1} ds} D(\rho_t \| \rho_t^G)$ is differentiable on $[0, \infty)$, and for all $t \geq 0$:

$$\frac{d}{dt} \left(e^{\frac{e}{2} \int_0^t N(\rho_s^G)^{-1} ds} D(\rho_t \| \rho_t^G) \right) = \left(\frac{e}{2} N(\rho_t^G)^{-1} D(\rho_t \| \rho_t^G) + \frac{dD(\rho_t \| \rho_t^G)}{dt} \right) e^{\frac{e}{2} \int_0^t N(\rho_s^G)^{-1} ds} \tag{11.43}$$

$$\leq \left(-\frac{e}{2} N(\rho_t^G)^{-1} + \frac{1}{4} J(\rho_t^G) \right) e^{\frac{e}{2} \int_0^t N(\rho_s^G)^{-1} ds}, \tag{11.44}$$

where we used (11.42) in the last line. Integrating the left hand side and the right hand side of the last inequality, we get

$$D(\rho_t \| \rho_t^G) e^{\frac{e}{2} \int_0^t N(\rho_s^G)^{-1} ds} - D(\rho \| \rho^G) \leq \int_0^t \frac{e^{\frac{e}{2} \int_0^s N(\rho_u^G)^{-1} du}}{N(\rho_s^G)} \left(\frac{1}{4} J(\rho_s^G) N(\rho_s^G) - \frac{e}{2} \right) ds, \tag{11.45}$$

which when rearranged gives (11.41). $\square$

The advantage of the upper bound of (11.41) is that it depends on the initial state ρ only through its covariance matrix. In order to interpret the last result as a large time behavior for the quantum heat semigroup, one however still needs to show that this upper bound converges to 0 as t goes to infinity. Fortunately, useful expressions for the Fisher information and the entropy power of a Gaussian state exist, and can be used to prove this claim (see [Koenig and Smith, 2014, Koenig, 2015]).

Lemma 11.2.3. *Let ρ be a one mode Gaussian state of mean photon number $\nu := \text{Tr}(\rho a^* a)$. Then*

$$J(\rho) = 2 \ln \frac{\nu + 1}{\nu} \tag{11.46}$$

$$S(\rho) = (\nu + 1) \ln(\nu + 1) - \nu \ln \nu. \tag{11.47}$$

Lemma 11.2.4. *For any initial Gaussian state of mean photon number ν , and for any time $t \geq 0$, the mean photon number of $\mathcal{P}_{*t}^{\text{qHeat}}(\rho)$ is such that*

$$\nu(t) = \frac{t}{2} + \mathcal{O}(1). \tag{11.48}$$

Combining the two above Lemmas, we prove the following proposition.

Proposition 11.2.5. *Let $\mathcal{H} := \mathbb{L}_2(\mathbb{R})$. Then for any state ρ of finite first and second moments, finite entropy and covariance matrix satisfying the conditions of Theorem 0.2.4. Then, given $\rho_t := \mathcal{P}_{*t}^{\text{qHeat}}(\rho)$, for any $0 < \varepsilon < 1$ there exists $t_\varepsilon > 0$ as well as $\alpha_\varepsilon > 0$ such that for any $t \geq t_\varepsilon$,*

$$D(\rho_t \| \rho_t^G) \leq \alpha_\varepsilon t^{\varepsilon-1}. \tag{11.49}$$

Proof. We first study the large time behavior of the function κ :

$$\kappa(t) = \frac{e}{2} \int_0^t N(\rho_s^G)^{-1} ds = \frac{e}{2} \int_0^t \frac{\nu(s)^{\nu(s)}}{(\nu(s)+1)^{\nu(s)+1}} ds \quad (11.50)$$

$$= \frac{e}{2} \int_0^t \frac{1}{\nu(s)+1} \left(1 + \frac{1}{\nu(s)}\right)^{-\nu(s)} ds, \quad (11.51)$$

where we used the expressions given by Equation (11.46) for $N(\rho_s^G)$. Using Lemma 11.2.4, we know that

$$\left(1 + \frac{1}{\nu(s)}\right)^{-\nu(s)} \xrightarrow{s \rightarrow \infty} e^{-1},$$

which implies that the integrand in Equation (11.51) is equivalent to $2/(e s)$. Therefore $\kappa(t)$ diverges as $t \rightarrow \infty$. Therefore, the first term in the right hand side of (11.41) converges to 0 as t goes to infinity. We now proceed to a more refined analysis of each term. Let's first define the C^1 function $f : (0, \infty) \rightarrow \mathbb{R}$ by $f(x) := (1+x)^{1/x}$. One can easily check that f and its derivative f' can be continuously extended to 0, with $f(0) = e$ and $f'(0) = -e/2$, so that the following Taylor expansion up to order 1 around 0 holds

$$f(x) = e(1 - x/2) + o(x). \quad (11.52)$$

Now for any $u \geq 0$, $N(\rho_u^G) = (\nu(u)+1)f(1/\nu(u))$, so that

$$N(\rho_u^G)^{-1} = \frac{1}{e\nu(u)} + o\left(\frac{1}{\nu(u)}\right). \quad (11.53)$$

Inserting the expression 11.48 for $\nu(u)$, we find that for any $\varepsilon > 0$ there exists $s_0 > 0$ such that for any $u \geq s_0$,

$$\frac{2-2\varepsilon}{eu} \leq N(\rho_u^G)^{-1} \leq \frac{2+2\varepsilon}{eu}. \quad (11.54)$$

Integrating from 0 to $s > s_0$, we get

$$e^{\kappa(s_0)}(s/s_0)^{(1-\varepsilon)} \leq e^{\kappa(s)} \leq e^{\kappa(s_0)}(s/s_0)^{(1+\varepsilon)}. \quad (11.55)$$

Finally,

$$\begin{aligned} J(\rho_s^G) N(\rho_s^G) &= 2 \ln \left(\frac{\nu(s)+1}{\nu(s)} \right) \frac{(\nu(s)+1)^{\nu(s)+1}}{\nu(s)^{\nu(s)}} \\ &= 2 \ln \left((1+1/\nu(s))^{\nu(s)+1} \right) (1+1/\nu(s))^{\nu(s)} \\ &= 2(\ln(f(1/\nu(s))) + \ln(1+1/\nu(s))) f(1/\nu(s)) \\ &= 2e(1+1/(2\nu(s)) + o(1/\nu(s))) \left(1 - \frac{1}{2\nu(s)} + o(1/\nu(s))\right) \\ &= 2e(1+o(1/\nu(s))). \end{aligned}$$

so that

$$J(\rho_s^G) N(\rho_s^G) - 2e = o(1/\nu(s)). \quad (11.56)$$

This, together with the expression (11.48) for $\nu(s)$, implies that for any $\delta > 0$ there exists a $t_0 > 0$ such

that for any $s \geq t_0$,

$$J(\rho_s^G) N(\rho_s^G) - 2e \leq \frac{2\delta + 2}{s} \quad (11.57)$$

Combining (11.57) with (11.55) and (11.54) we end up with the following: for any $t > \max\{t_0, s_0\}$,

$$\begin{aligned} \int_0^t \frac{e^{\kappa(s)}}{N(\rho_s^G)} \left(\frac{1}{4} J(\rho_s^G) N(\rho_s^G) - \frac{e}{2} \right) ds &\leq \int_0^{\max\{s_0, t_0\}} \frac{e^{\kappa(s)}}{N(\rho_s^G)} \left(\frac{1}{4} J(\rho_s^G) N(\rho_s^G) - \frac{e}{2} \right) ds \\ &\quad + e^{\kappa(s_0)} \frac{(1+\varepsilon)(1+\delta)}{e s_0^{1+\varepsilon}} \int_{\max\{s_0, t_0\}}^t s^{\varepsilon-1} ds \\ &= \int_0^{\max\{s_0, t_0\}} \frac{e^{\kappa(s)}}{N(\rho_s^G)} \left(\frac{1}{4} J(\rho_s^G) N(\rho_s^G) - \frac{e}{2} \right) ds \\ &\quad + e^{\kappa(s_0)} \frac{(1+\delta)(1+1/\varepsilon)}{e s_0^{1+\varepsilon}} (t^\varepsilon - \max\{s_0, t_0\}^\varepsilon). \end{aligned}$$

The claim follows for any $0 < \varepsilon < 1$ by combining this last inequality as well as (11.55) with (11.41). $\square$

Remark 11.2.6. Equation (11.56) implies that the isoperimetric inequality is saturated for Gaussian states in the limit of an infinite mean photon number. This fact was already noticed in [Huber et al., 2017]. One could similarly carry out the same kind of analysis for systems of a finite number of modes.

11.3. EPI and LSI for the quantum Ornstein Uhlenbeck semigroup

As we saw in Section 4.5, the classical entropic isoperimetric inequality can be used to derive the logarithmic Sobolev inequality for the Ornstein Uhlenbeck semigroup. This is a manifestation of the strong link existing between the latter and the heat semigroup (cf. Section 4.5). The same type of argument seems to fail in the non-commutative regime. However, another entropy power inequality corresponding to a convolution between quantum states does the job, as observed in [De Palma and Huber, 2018]. Here, we briefly review the argument for sake of comparison with Section 4.5.

A quantum convolution via beamsplitters The *quantum-quantum convolution* that we mentioned in the previous paragraph was first introduced in [Koenig and Smith, 2014]. It is motivated from physics since defined in terms of the beamsplitter (cf Section 0.2): let ρ_1 and ρ_2 be states in $\mathcal{D}(\mathcal{H})$, $\mathcal{H} := \mathbb{L}_2(\mathbb{R}^n)$, and $0 < \lambda < 1$, the λ -*quantum convolution* between ρ_1 and ρ_2 is defined as

$$\rho_1 \boxplus_\lambda \rho_2 := \text{Tr}_2 \left(U_\lambda (\rho_1 \otimes \rho_2) U_\lambda^{-1} \right),$$

where U_λ is defined through the symplectic matrix (0.47). The quantum entropy power inequality corresponding to the above convolution was proved in [Koenig and Smith, 2014, De Palma and Trevisan, 2018]:

Theorem 11.3.1 (Quantum EPI). *For any two quantum states $\rho_1, \rho_2 \in \mathcal{D}(\mathcal{H})$ of finite second moments, and any $0 \leq \lambda, \eta \leq 1$,*

$$S(\rho_1 \boxplus_\eta \rho_2) \geq \lambda S(\rho_1) + (1-\lambda) S(\rho_2) + \lambda \ln \frac{\eta}{\lambda} + (1-\lambda) \ln \frac{|1-\eta|}{1-\lambda}.$$

After maximization over λ , we end up with the following quantum entropy power inequality:

$$N(\rho_1 \boxplus_\lambda \rho_2) \geq \lambda N(\rho_1) + (1 - \lambda)N(\rho_2). \quad (\text{qq-EPI})$$

The quantum entropy power inequality provides a two-lines proof of the exponential convergence in relative entropy of the quantum Ornstein Uhlenbeck semigroup of Section 5.5.2 towards its invariant state σ_ν . This observation was made in Theorem 9 of [De Palma and Huber, 2018]. Indeed, for each $t \geq 0$,

$$\mathcal{P}_{*t}^{\text{qOU}}(\rho) = \text{Tr}_2 \left(U_{\lambda(t)} (\rho \otimes \sigma_\nu) U_{\lambda(t)}^* \right),$$

where $\lambda(t) := e^{-(\mu^2 - \lambda^2)t}$, which can be easily checked by looking at the action of the semigroup on the creation and annihilation operators. Then

Theorem 11.3.2. *The quantum Ornstein Uhlenbeck semigroup $(\mathcal{P}_t^{\text{qOU}})_{t \geq 0}$ satisfies the following identity: for any initial state ρ of finite first and second moments:*

$$D(\mathcal{P}_{*t}^{\text{qOU}}(\rho) \| \sigma_\nu) \leq e^{-(\mu^2 - \lambda^2)t} D(\rho \| \sigma_\nu).$$

Proof. Let $\rho_t := \mathcal{P}_{*t}^{\text{qOU}}(\rho)$ and $\lambda(t) := e^{-(\mu^2 - \lambda^2)t}$. First, observe that

$$\begin{aligned} D(\rho_t \| \sigma_\nu) &= -S(\rho_t) - \text{Tr}(\rho_t \ln \sigma_\nu) \\ &= -S(\rho_t) - \ln \nu \text{Tr}(\rho_t N) - \ln(1 - \nu), \end{aligned}$$

where N is the total number operator. Now, a simple calculation made e.g. p. 23 of [Huber et al., 2017] implies

$$\text{Tr}(\rho_t N) = \lambda(t) \text{Tr}(\rho N) + (1 - \lambda(t)) \text{Tr}(\sigma_\nu N).$$

Therefore

$$\begin{aligned} D(\rho_t \| \sigma_\nu) &= -S(\rho_t) - \ln \nu (\lambda(t) \text{Tr}(\rho N) + (1 - \lambda(t)) \text{Tr}(\sigma_\nu N)) - \ln(1 - \nu) \\ &\leq -\lambda(t)S(\rho) - (1 - \lambda(t))S(\sigma_\nu) - \lambda(t) \ln \nu \text{Tr}(\rho N) - \ln \nu (1 - \lambda(t)) \text{Tr}(\sigma_\nu N) - \ln(1 - \nu) \\ &= -\lambda(t)D(\rho \| \sigma_\nu) - (1 - \lambda(t))S(\sigma_\nu) - (1 - \lambda(t)) \ln \nu \text{Tr}(\sigma_\nu N) - (1 - \lambda(t)) \ln(1 - \nu) \\ &= -\lambda(t)D(\rho \| \sigma_\nu), \end{aligned}$$

where the second line follows from Theorem 11.3.1. □

Chapter 12.

Quantum optimal transport

In Chapter 4, classical functional inequalities were shown to arise from a single geometric inequality called the Ricci curvature lower bound. In particular, we saw in Section 4.2.2 that the Wasserstein distance of order 2 can be interpreted as a metric on the weak Riemannian manifold of probability measures $\mathcal{P}(\mathcal{M})$. This led to a rethinking of the geometry of $\mathcal{M}$ in terms of the convexity properties of a family of functionals on $\mathcal{P}(\mathcal{M})$. From the formal calculus introduced by Otto, the generator of a Markov process naturally arises from the differential structure of $\mathcal{M}$. In the case of Markov chains on non-geodesic, finite sample spaces, the problem is reversed: from the generator of the chain, and its associated transition kernel, [Maas, 2011, Erbar and Maas, 2012] introduced a formal differential calculus and a modified Wasserstein distance that led to the definition of a Ricci curvature lower bound in terms of the displacement convexity of Boltzmann's H functional. Recently, Carlen and Maas introduced an extension of the latter in the context of quantum Markov semigroups acting on finite dimensional C^* -algebras [Carlen and Maas, 2014, Carlen and Maas, 2017].

Layout of the chapter: In this chapter, we further analyse the quantum Ricci lower bound ($\text{Ric}(\mathcal{L}) \geq \kappa$) introduced by Carlen and Maas [Carlen and Maas, 2014, Carlen and Maas, 2017], and derive various equivalent formulations of it. In Section 12.1, we introduce the quantum Wasserstein distance. The quantum version of κ -displacement convexity is studied in Section 12.3. The connection between the Ricci curvature lower bound and the quantum functional and transportation cost inequalities is fleshed out in Section 12.5. This is done by showing that $\text{Ric}(\mathcal{L}) \geq \kappa$ implies a quantum version of the celebrated HWI(κ) inequality which interpolates between the modified logarithmic Sobolev inequality and the a quantum version of the transportation cost inequality of order 2 ($\text{TC}_2(c_2)$). In particular, we show that, in the case of $\kappa > 0$, $\text{HWI}(\kappa) \Rightarrow \text{MLSI}(\kappa)$, recovering the result of [Carlen and Maas, 2017]. On the other hand, we establish that in the case when $\kappa \in \mathbb{R}$, $\text{Ric}(\mathcal{L}) \geq \kappa$ together with $\text{TC}_2(c_2)$ imply $\text{MLSI}(\alpha)$. When $\kappa = 0$, we show that, under the assumption of boundedness of the diameter D of the set of states with respect to the quantum Wasserstein distance $W_{2,\mathcal{L}}$, $\text{Ric}(\mathcal{L}) \geq 0$ implies $\text{PI}(c_1 D^{-2})$ for some universal constant c_1 . When the QMS is doubly stochastic (i.e. one which has the completely mixed state as its unique invariant state), we show that $\text{Ric}(\mathcal{L}) \geq 0$ also implies $\text{MLSI}(c_2 D^{-2})$ for some universal positive constant c_2 . Quantum concentration inequalities are derived from the Poincaré and the transportation cost inequalities in Section 12.7. Finally, we briefly mention extensions of the theory to infinite dimensional systems in Section 12.8.

12.1. The Wasserstein distance $W_{2,\mathcal{L}}$

In this section, we recall the Benamou-Brenier-like construction of the Wasserstein metric $W_{2,\mathcal{L}}$ first defined in [Carlen and Maas, 2017]. Our goal is to define a Riemmanian manifold structure on the set $\mathcal{D}(\mathcal{H})$ of quantum states. Let $\mathcal{L}$ be the generator of a primitive QMS $(\mathcal{P}_t)_{t \geq 0}$ on $\mathcal{B}(\mathcal{H})$, $d_{\mathcal{H}} < \infty$, with invariant state σ . We further assume that $(\mathcal{P}_t)_{t \geq 0}$ satisfies σ -DBC. We recall from Section 5.2.2 that $\mathcal{L}$ can then be written in the following form (cf. Equation (5.20)):

$$\mathcal{L}(X) = \sum_{j \in \mathcal{J}} e^{-\omega_j/2} \left(\tilde{L}_j^* [X, \tilde{L}_j] + [\tilde{L}_j^*, X] \tilde{L}_j \right), \quad (12.1)$$

where $\Delta_{\sigma}(\tilde{L}_j) = e^{-\omega_j} \tilde{L}_j$ for some $\omega_j \in \mathbb{R}$.

Remark 12.1.1. Given a primitive quantum Markov semigroup with generator which is self-adjoint with respect to $\langle \cdot, \cdot \rangle_{1,\sigma}$, the operators $\tilde{L}_j$ in Equation (12.1) are uniquely defined up to a unitary transformation, i.e. for any two representations $(\{\tilde{L}_j\}, \{\omega_j\})$ and $(\{\tilde{L}'_j\}, \{\omega'_j\})$ of $\mathcal{L}$, there exists a $|\mathcal{J}| \times |\mathcal{J}|$ unitary matrix U such that, unless $\omega_j = \omega'_k$, $U_{jk} = 0$ and

$$\sum_{k \in \mathcal{J}} \bar{U}_{kj} U_{kl} = \delta_{jl}, \quad \tilde{L}'_l = \sum_{k \in \mathcal{J}} U_{lk} \tilde{L}_k, \quad (12.2)$$

We also recall from Section 5.3 that, given an operator $X \in \mathcal{B}(\mathcal{H})$, its noncommutative gradient and divergence are respectively defined as:

$$\nabla_{\tilde{\mathbf{L}}} X := (\nabla_{\tilde{L}_j} X)_{j \in \mathcal{J}}, \quad \text{div}_{\tilde{\mathbf{L}}}(\mathbf{A}) := \sum_{j \in \mathcal{J}} [A_j, \tilde{L}_j^*] \equiv - \sum_{j \in \mathcal{J}} \nabla_{\tilde{L}_j}^* A_j,$$

for $X \in \mathcal{B}(\mathcal{H})$ and $\mathbf{A} \equiv (A_j)_{j \in \mathcal{J}} \in \oplus_{j \in \mathcal{J}} \mathcal{B}(\mathcal{H})$, where $\nabla_{\tilde{L}_j} X = [\tilde{L}_j, X]$ and $\nabla_{\tilde{L}_j}^* X := [\tilde{L}_j^*, X]$ for all $j \in \mathcal{J}$. Then, the noncommutative Laplacian takes the following simple form: for $X \in \mathcal{B}(\mathcal{H})$:

$$\Delta(X) := \text{div}_{\tilde{\mathbf{L}}} \circ \nabla_{\tilde{\mathbf{L}}}(X) = - \sum_{j \in \mathcal{J}} \nabla_{\tilde{L}_j}^* \circ \nabla_{\tilde{L}_j}(X).$$

In its Benamou-Brenier formulation, the definition of the commutative Wasserstein distance relies on a continuity equation which admits a unique minimal solution (cf. Section 4.2.1). Similarly, in their finite dimensional quantum setting, [Carlen and Maas, 2017] introduced the following noncommutative version of the continuity equation and showed the existence of a minimal solution. First, since $(\mathcal{P}_t)_{t \geq 0}$ is primitive, $\ker(\Delta) = \ker(\nabla_{\tilde{\mathbf{L}}}) = \mathbb{C}\mathbb{1}_{\mathcal{H}}$, and hence the *noncommutative Poisson equation*

$$\Delta(X) = B \quad (\text{Poisson equation})$$

has a solution if and only if B is orthogonal to $\mathbb{1}_{\mathcal{H}}$ with respect to $\langle \cdot, \cdot \rangle_{\text{HS}}$, i.e. $\text{Tr}(B) = \langle \mathbb{1}, B \rangle_{\text{HS}} = 0$. As we will see later, given a full-rank state ρ and family φ of functions $\varphi_k : [0, \infty) \rightarrow [0, \infty)$, $k \in \mathcal{J}$, it is convenient to introduce the following operator of multiplication by ρ on $\oplus_{j \in \mathcal{J}} \mathcal{B}(\mathcal{H})$:

$$[\rho]_{\varphi} = \{R_{\rho} \circ \varphi_k(\Delta_{\rho})\}_{k \in \mathcal{J}}. \quad (12.3)$$

where $R_{\rho}(X) := X\rho$. The choice of the functions φ_k will be fixed by the requirement that the evolution induced by $(\mathcal{P}_t)_{t \geq 0}$ is the gradient-flow associated with the relative entropy functional $D(\cdot \| \sigma)$. For the time being, we keep this degree of freedom untouched, and simply observe that, in the commuting

case,

$$[\rho]_\varphi(\mathbf{A}) = (\varphi_k(1) A_k \rho)_{k \in \mathcal{J}}.$$

Next, given a continuously differentiable path $(\gamma(s))_{s \in [-\varepsilon, \varepsilon]}$ of full-rank states, with $\partial_s \gamma(s) = \dot{\gamma}(s) \perp_{\mathcal{T}_2(\mathcal{H})} \mathbb{C} \mathbb{1}_{\mathcal{H}}$, there exists a path $\mathbf{V} : [-\varepsilon, \varepsilon] \rightarrow \oplus_{j \in \mathcal{J}} \mathcal{B}(\mathcal{H})$ such that

$$-\operatorname{div}_{\tilde{\mathbf{L}}}([\gamma(s)]_\varphi \mathbf{V}(s)) = \dot{\gamma}(s). \quad (12.4)$$

Indeed, the above identity is satisfied by the vector field $\mathbf{V}(s) = [\gamma(s)]_\varphi^{-1} \nabla_{\tilde{\mathbf{L}}}(A)(s)$, where $A : [-\varepsilon, \varepsilon] \rightarrow \mathcal{B}(\mathcal{H})$ denotes the solution to the Poisson equation. However, the field $\mathbf{V}$ is not unique in general, since any solution of the form $\mathbf{V} + \mathbf{W}$, where $\mathbf{W} = [\gamma]_\varphi^{-1}(\mathbf{F})$, $\operatorname{div}_{\tilde{\mathbf{L}}} \mathbf{F} = 0$, would satisfy Equation (12.4). However, just like in the classical setting, there exists a minimal such solution, which is constructed as follows: for any $\rho \in \mathcal{D}_+(\mathcal{H})$, define the following *modified weighted $\mathbb{L}_{2,\varphi}(\rho)$ inner product* on $\oplus_{j \in \mathcal{J}} \mathcal{B}(\mathcal{H})$: for any $\mathbf{A} = (A_j)_{j \in \mathcal{J}}$ and $\mathbf{B} = (B_j)_{j \in \mathcal{J}}$:

$$\langle \mathbf{A}, \mathbf{B} \rangle_{\mathbb{L}_{2,\varphi}(\rho)} := \sum_{j \in \mathcal{J}} \langle A_j, [\rho]_{\varphi,j}(B_j) \rangle_{\text{HS}}. \quad (12.5)$$

Since the solutions to Equation (12.4) span a closed affine subspace of $\oplus_{j \in \mathcal{J}} \mathcal{B}(\mathcal{H})$, there exists a unique solution $\mathbf{V}(s)$ of minimal $\mathbb{L}_{2,\varphi}(\gamma(s))$ norm, such that for any $\mathbf{W} = [\gamma]_\varphi^{-1}(\mathbf{F})$, $\operatorname{div}_{\tilde{\mathbf{L}}} \mathbf{F} = 0$,

$$\langle \mathbf{V}, \mathbf{F} \rangle_{\oplus_{j \in \mathcal{J}} \mathcal{T}_2(\mathcal{H})} = \langle \mathbf{V}, \mathbf{W} \rangle_{\mathbb{L}_{2,\varphi}(\gamma)} = 0.$$

In other words, $\mathbf{V} \perp_{\oplus_{j \in \mathcal{J}} \mathcal{T}_2(\mathcal{H})} \ker(\operatorname{div}_{\tilde{\mathbf{L}}})$, which implies that $\mathbf{V} \in \operatorname{im}(\nabla_{\tilde{\mathbf{L}}})$. Since $\ker(\nabla_{\tilde{\mathbf{L}}}) = \mathbb{C} \mathbb{1}_{\mathcal{H}}$ by primitivity, there exists a unique traceless operator valued function $\Phi : [-\varepsilon, \varepsilon] \rightarrow \mathcal{B}(\mathcal{H})$ such that $\mathbf{V} = \nabla_{\tilde{\mathbf{L}}} \Phi$, so that

$$-\operatorname{div}_{\tilde{\mathbf{L}}}([\gamma(s)]_\varphi (\nabla_{\tilde{\mathbf{L}}}(\Phi(s)))) = \dot{\gamma}(s). \quad (\text{n-c continuity equation})$$

The vector $\nabla_{\tilde{\mathbf{L}}}(\Phi(0))$ should be interpreted as the tangent vector at the origin associated with the path γ in the manifold $\mathcal{D}_+(\mathcal{H})$ of full-rank states.

Remark 12.1.2. If $-\operatorname{div}_{\tilde{\mathbf{L}}}([\rho]_\varphi (\nabla_{\tilde{\mathbf{L}}}(X^*))) = -\operatorname{div}_{\tilde{\mathbf{L}}}([\rho]_\varphi (\nabla_{\tilde{\mathbf{L}}}(X)))^*$ for any $\rho \in \mathcal{D}_+(\mathcal{H})$ and $X \in \mathcal{B}(\mathcal{H})$, $\Phi(s)$ is moreover self-adjoint for any $s \in [-\varepsilon, \varepsilon]$, by uniqueness of the solution and self-adjointness of $\dot{\gamma}$. This is the case in particular for the following choice for φ : Given $\omega := (\omega_j)_{j \in \mathcal{J}}$, define the linear operator $[\rho]_\varphi \equiv [\rho]_{\varphi^{\log, \omega}}$ on $\oplus_{j \in \mathcal{J}} \mathcal{B}(\mathcal{H})$ through

$$[\rho]_{\varphi^{\log, \omega}}(\mathbf{A}) := ([\rho]_{\varphi^{\log, \omega_j}} A_j)_{j \in \mathcal{J}}, \quad \mathbf{A} \equiv (A_j)_{j \in \mathcal{J}},$$

where for any $\omega \in \mathbb{R}$,

$$[\rho]_{\varphi^{\log, \omega}} = R_\rho \circ \varphi_\omega^{\log}(\Delta_\rho), \quad \varphi_\omega^{\log}(t) := e^{\omega/2} \frac{t - e^{-\omega}}{\ln t + \omega}, \quad t > 0. \quad (12.6)$$

The acronym log stands for relative entropy since the following choice of multiplication by ρ will lead to the gradient-flowness of $(\mathcal{P}_t)_{t \geq 0}$ for the relative entropy functional $D(\cdot \| \sigma)$. The following lemma can be used in order to prove that $\Phi = \Phi^*$ in this case (see Lemma 5.8 of [Carlen and Maas, 2017]):

Lemma 12.1.3. *For any $\omega \in \mathbb{R}$, $\rho \in \mathcal{D}_+(\mathcal{H})$ and $A \in \mathcal{B}(\mathcal{H})$,*

$$\begin{aligned} [\rho]_{\varphi^{\log}, \omega}(A) &= \int_0^1 e^{\omega(1/2-s)} \rho^s A \rho^{1-s} ds, \\ [\rho]_{\varphi^{\log}, \omega}^{-1}(A) &= \int_0^\infty (s \text{id} + e^{-\omega/2} L_\rho)^{-1} (s \text{id} + e^{\omega/2} R_\rho)^{-1}(A) ds. \end{aligned}$$

Now, given two states $\gamma_0, \gamma_1 \in \mathcal{D}_+(\mathcal{H})$, let $\mathcal{P}(\gamma_0, \gamma_1)$ be the collection of smooth paths $\gamma : [0, 1] \rightarrow \mathcal{D}_+(\mathcal{H})$ satisfying $\gamma(0) = \gamma_0$ and $\gamma(1) = \gamma_1$. Then, the *quantum Wasserstein distance* associated with the Lindblad generator $\mathcal{L}$ is defined as

$$W_{2, \mathcal{L}}^\varphi(\gamma_0, \gamma_1) := \inf_{\gamma \in \mathcal{P}(\gamma_0, \gamma_1)} \left\{ \int_0^1 \|\nabla_{\tilde{\mathbf{L}}} \Phi(s)\|_{\mathbb{L}_{2, \varphi}(\gamma(s))}^2 ds \right\}^{\frac{1}{2}},$$

where, for any path γ , $\Phi : [0, 1] \rightarrow \mathcal{B}(\mathcal{H})$ is the unique corresponding (traceless) solution to the n-c continuity equation. The Wasserstein distance endows the manifold $\mathcal{D}_+(\mathcal{H})$ with a smooth Riemannian structure. We also denote by $g_{\mathcal{L}, \varphi}$ the *Wasserstein metric* associated with $W_{2, \mathcal{L}}^\varphi$. More precisely, for any two smooth paths $\gamma, \gamma' : [-\varepsilon, \varepsilon] \rightarrow \mathcal{D}_+(\mathcal{H})$ such that $\gamma(0) = \gamma'(0) = \rho$, with associated tangent vectors $\nabla_{\tilde{\mathbf{L}}} \Phi, \nabla_{\tilde{\mathbf{L}}} \Phi'$, at the origin

$$g_{\mathcal{L}, \rho}^\varphi(\dot{\gamma}, \dot{\gamma}') = \langle \nabla_{\tilde{\mathbf{L}}} \Phi, \nabla_{\tilde{\mathbf{L}}} \Phi' \rangle_{\mathbb{L}_{2, \varphi}(\rho)}. \quad (12.7)$$

When $\varphi = \varphi^{\log}$, we simply write $W_{2, \mathcal{L}}^{\varphi^{\log}} \equiv W_{2, \mathcal{L}}$, $\mathbb{L}_{2, \varphi^{\log}}(\rho) \equiv \mathbb{L}_{2, \log}(\rho)$ and $g_{\mathcal{L}, \rho}^{\varphi^{\log}} \equiv g_{\mathcal{L}, \rho}$. The following lemma provides an alternative expression for $W_{2, \mathcal{L}}^\varphi$:

Lemma 12.1.4. *With the above notations, the Wasserstein distance $W_{2, \mathcal{L}}^\varphi$ between two full-rank states γ_0, γ_1 is equal to the minimal length over the smooth paths joining γ_0 and γ_1 :*

$$W_{2, \mathcal{L}}^\varphi(\gamma_0, \gamma_1) = \inf_{\gamma \text{ const. speed}} \left\{ \int_0^1 \sqrt{g_{\mathcal{L}, \gamma(s)}^\varphi(\dot{\gamma}(s), \dot{\gamma}(s))} ds : \gamma(0) = \gamma_0, \gamma(1) = \gamma_1 \right\}, \quad (12.8)$$

where the infimum is taken over the set of constant speed curves γ , i.e. such that $s \mapsto g_{\mathcal{L}, \gamma(s)}^\varphi(\dot{\gamma}(s), \dot{\gamma}(s))$ is constant on $[0, 1]$.

Proof. By the proof of Lemma 1.1.4 (b) of [Ambrosio et al., 2008], for any smooth path $\gamma : [0, 1] \rightarrow \mathcal{D}_+(\mathcal{H})$, with $\gamma(0) = \gamma_0$ and $\gamma(1) = \gamma_1$ there exists a constant speed path $\eta : [0, 1] \rightarrow \mathcal{D}_+(\mathcal{H})$ with the same boundary conditions: for any $s \in [0, 1]$,

$$\sqrt{g_{\mathcal{L}, \eta(s)}^\varphi(\dot{\eta}(s), \dot{\eta}(s))} = \int_0^1 \sqrt{g_{\mathcal{L}, \gamma(u)}^\varphi(\dot{\gamma}(u), \dot{\gamma}(u))} du.$$

Fix such a path γ . By the Cauchy-Schwarz inequality, for any $s \in [0, 1]$

$$\sqrt{g_{\mathcal{L}, \eta(s)}^\varphi(\dot{\eta}(s), \dot{\eta}(s))} = \int_0^1 \sqrt{g_{\mathcal{L}, \gamma(u)}^\varphi(\dot{\gamma}(u), \dot{\gamma}(u))} du \leq \left(\int_0^1 g_{\mathcal{L}, \gamma(u)}^\varphi(\dot{\gamma}(u), \dot{\gamma}(u)) du \right)^{1/2}. \quad (12.9)$$

with equality if $\gamma = \eta$. The statement follows after taking the infimum over all smooth paths γ on the right hand side of (12.9). $\square$

Extension of the metric to $\mathcal{D}(\mathcal{H})$: Here, we provide a procedure to extend the definition of the Wasserstein distance to the set $\mathcal{D}(\mathcal{H})$ of all states on $\mathcal{H}$. The following technical lemma is going to play a crucial role:

Lemma 12.1.5. *For any $\rho \in \mathcal{D}_+(\mathcal{H})$, the map $D_{\varphi^{\log}}(\rho) : \Phi \mapsto -\operatorname{div}_{\tilde{\mathbf{L}}}([\rho]_{\varphi^{\log},\omega} \nabla_{\tilde{\mathbf{L}}} \Phi)$ is invertible and positive on the space of traceless operators. Moreover, if $\rho \geq \varepsilon \mathbb{1}$ for some $\varepsilon > 0$, then:*

$$D_{\varphi^{\log}}(\rho)^{-1} \leq K_{\mathcal{L}} \varepsilon^{-1} \operatorname{id}, \quad (12.10)$$

where $K_{\mathcal{L}} := \max_{j \in \mathcal{J}} \frac{\omega_j}{e^{\omega_j/2} - e^{-\omega_j/2}} \|(-\operatorname{div}_{\tilde{\mathbf{L}}} \circ \nabla_{\tilde{\mathbf{L}}})(\cdot)^{-1} : \mathcal{T}_2(\mathcal{H}) \rightarrow \mathcal{T}_2(\mathcal{H})\|$, and where the inequality is in the sense of the partial order induced by the convex cone of positive semidefinite operators on $\mathcal{T}_2(\mathcal{H})$.

Proof. Let $\mathcal{W}$ be the space of traceless operators on $\mathcal{H}$. We already saw that, for any $B \in \mathcal{W}$, there exists a unique $\Phi \in \mathcal{W}$ such that $B = -\operatorname{div}_{\tilde{\mathbf{L}}}([\rho]_{\varphi^{\log},\omega} (\nabla_{\tilde{\mathbf{L}}} \Phi))$. We also defined $-\operatorname{div}_{\tilde{\mathbf{L}}}$ as $\nabla_{\tilde{\mathbf{L}}}^*$, where the adjoint is taken with respect to the inner products on $\mathcal{T}_2(\mathcal{H})$ and $\bigoplus_{j \in \mathcal{J}} \mathcal{T}_2(\mathcal{H})$. The result follows from Lemma 12.1.3, since for any $j \in \mathcal{J}$, $[\rho]_{\varphi^{\log},\omega_j} \geq \varepsilon \frac{e^{\omega_j/2} - e^{-\omega_j/2}}{\omega_j} \operatorname{id}$. $\square$

The above proposition allows us to extend the definition of the Wasserstein distance to non-faithful states:

Proposition 12.1.6 (Extension of the metric to $\mathcal{D}(\mathcal{H})$). *Let $\rho, \omega \in \mathcal{D}(\mathcal{H})$ and let $\{\rho_n\}_{n \in \mathbb{N}}$ and $\{\omega_n\}_{n \in \mathbb{N}}$ be sequences of full-rank states satisfying*

$$\operatorname{Tr}[(\rho - \rho_n)^2] \rightarrow 0, \quad \operatorname{Tr}[(\omega - \omega_n)^2] \rightarrow 0, \quad (12.11)$$

as $n \rightarrow \infty$. Then the sequence $\{W_{2,\mathcal{L}}(\rho_n, \omega_n)\}_{n \in \mathbb{N}}$ converges.

Proof. The proof is similar to the one given in Proposition 4.5 of [Carlen and Maas, 2014]. It is enough to show that $\{W_{2,\mathcal{L}}(\rho_n, \omega_n)\}_{n \in \mathbb{N}}$ is Cauchy. By the triangle inequality, it is even enough to prove that $W_{2,\mathcal{L}}(\rho_n, \rho_m) \rightarrow 0$ as $m, n \rightarrow \infty$. Let $\varepsilon \in (0, 1)$ and set $\bar{\rho} := (1 - \varepsilon)\rho + \varepsilon \frac{\mathbb{1}}{d_{\mathcal{H}}}$. Let $N \in \mathbb{N}$ be such that for any $n \geq N$, $\operatorname{Tr}[(\rho - \rho_n)^2] \leq \varepsilon^2$. For $n \geq N$, consider the convex interpolation $\gamma(s) := (1 - s)\rho_n + s\bar{\rho}$. Since $\gamma(s) \geq \varepsilon s \frac{\mathbb{1}}{d_{\mathcal{H}}}$ for $s \in [0, 1]$, we find from Equation (12.8) that

$$\begin{aligned} & W_{2,\mathcal{L}}(\rho_n, \bar{\rho}) \\ & \leq \int_0^1 \sqrt{g_{\mathcal{L},\gamma(s)}(\dot{\gamma}(s), \dot{\gamma}(s))} ds \\ & = \int_0^1 \left[\sum_{j \in \mathcal{J}} \langle \nabla_{\tilde{L}_j} (-\operatorname{div}_{\tilde{\mathbf{L}}} \circ [\gamma]_{\varphi^{\log},\omega} \circ \nabla_{\tilde{\mathbf{L}}})^{-1}(\dot{\gamma}), [\gamma]_{\varphi^{\log},\omega_j} \nabla_{\tilde{L}_j} (-\operatorname{div}_{\tilde{\mathbf{L}}} \circ [\gamma]_{\varphi^{\log},\omega} \circ \nabla_{\tilde{\mathbf{L}}})^{-1}(\dot{\gamma}) \rangle_{\text{HS}} \right]^{1/2} ds \\ & = \int_0^1 \sqrt{\langle \dot{\gamma}(s), (-\operatorname{div}_{\tilde{\mathbf{L}}}([\gamma(s)]_{\varphi^{\log},\omega} \nabla_{\tilde{\mathbf{L}}}(\cdot)))^{-1} \dot{\gamma}(s) \rangle_{\text{HS}}} ds \\ & \leq \sqrt{\frac{K_{\mathcal{L}} d_{\mathcal{H}}}{\varepsilon}} \int_0^1 s^{-1/2} \sqrt{\operatorname{Tr}[(\dot{\gamma}(s))^2]} ds, \end{aligned}$$

where we used Lemma 12.1.5 in the second, third and fourth above lines. Now

$$\begin{aligned} \operatorname{Tr}[(\dot{\gamma}(s))^2] &= \operatorname{Tr}[(\rho - \rho_n + \varepsilon(\mathbb{1}/d_{\mathcal{H}} - \rho))^2] \\ &\leq 2 \operatorname{Tr}[(\rho - \rho_n)^2] + 2\varepsilon^2 \operatorname{Tr}[(\mathbb{1}/d_{\mathcal{H}} - \rho)^2] \\ &\leq 2(1 + \operatorname{Tr}[(\mathbb{1}/d_{\mathcal{H}} - \rho)^2]) \varepsilon^2. \end{aligned}$$

Hence, $W_{2,\mathcal{L}}(\rho_n, \bar{\rho}) \leq C_{\rho} \sqrt{\varepsilon}$, for some constant C_{ρ} depending on ρ . Since ε was arbitrary, we conclude by triangle inequality that $W_{2,\mathcal{L}}(\rho_m, \rho_n) \leq W_{2,\mathcal{L}}(\rho_m, \bar{\rho}) + W_{2,\mathcal{L}}(\bar{\rho}, \rho_n) \rightarrow 0$. $\square$

The above proposition justifies the following extension: The quantum Wasserstein distance $W_{2,\mathcal{L}}$

between two states $\rho, \omega \in \mathcal{D}(\mathcal{H})$ is defined as

$$W_{2,\mathcal{L}}(\rho, \omega) := \lim_{n \rightarrow \infty} W_{2,\mathcal{L}}(\rho_n, \omega_n),$$

where $\{\rho_n\}_{n \in \mathbb{N}}$ and $\{\omega_n\}_{n \in \mathbb{N}}$ are arbitrary sequences in $\mathcal{D}_+(\mathcal{H})$ satisfying (12.11). Next, we recall from Section 1.3 that the functions $k^{\log} : \mathbb{R}_+ \ni t \mapsto (t-1)^{-1} \ln(t)$ belongs to the class

$$\text{OMM} := \{k : \mathbb{R}_+ \rightarrow \mathbb{R} \mid -k \text{ is operator monotone, } k(x^{-1}) = xk(x), k(1) = 1\}. \quad (12.12)$$

Then, since $(\varphi_{\omega_j}^{\log}(t))^{-1} = e^{-\omega_j/2}(\ln t + \omega_j)(t - e^{-\omega_j})^{-1} = e^{\omega_j/2} k^{\log}(e^{\omega_j} t)$, inequality (1.88) implies that, for any $j \in \mathcal{J}$,

$$[\rho]_{\varphi^{\log}, \omega_j} \leq \frac{L_\rho e^{\omega_j/2} + R_\rho e^{-\omega_j/2}}{2}. \quad (12.13)$$

In the next theorem, we show that $(\mathcal{D}(\mathcal{H}), W_{2,\mathcal{L}})$ forms a complete metric space:

Lemma 12.1.7. *For any $\rho, \omega \in \mathcal{D}(\mathcal{H})$,*

$$\|\rho - \omega\|_1 \leq \sqrt{2} \left(\sum_{j \in \mathcal{J}} (e^{-\omega_j/2} + e^{\omega_j/2}) \|\tilde{L}_j\|_\infty^2 \right)^{1/2} W_{2,\mathcal{L}}(\rho, \omega).$$

Proof. The proof is inspired by the proof of Proposition 2.12 of [Erbar and Maas, 2012]. Fix $\delta > 0$, and $\rho, \omega \in \mathcal{D}_+(\mathcal{H})$ without loss of generality. There exists a smooth path $\gamma : [0, 1] \rightarrow \mathcal{D}_+(\mathcal{H})$ such that $\gamma(0) = \rho$, $\gamma(1) = \omega$, and by definition of $W_{2,\mathcal{L}}$:

$$\left(\int_0^1 \sum_{j \in \mathcal{J}} \langle V(s)_j, [\gamma(s)]_{\varphi^{\log}, \omega_j} V(s)_j \rangle_{\text{HS}} ds \right)^{1/2} \equiv \left(\int_0^1 \|\mathbf{V}(s)\|_{\mathbb{L}_{2, \varphi^{\log}}(\gamma(s))}^2 ds \right)^{1/2} \leq W_{2,\mathcal{L}}(\rho, \omega) + \delta, \quad (12.14)$$

where for each $s \in [0, 1]$, $\mathbf{V}(s) \equiv (V(s)_j)_{j \in \mathcal{J}}$ is related to $\gamma(s)$ through the continuity equation:

$$\dot{\gamma}(s) + \text{div}_{\tilde{\mathbf{L}}}([\gamma(s)]_{\varphi^{\log}, \omega} \mathbf{V}(s)) = 0,$$

Hence, for any $X \in \mathcal{B}(\mathcal{H})$,

$$|\text{Tr}(X(\rho - \omega))| \quad (12.15)$$

$$\begin{aligned} &= \left| \text{Tr} \left(X \int_0^1 \frac{d}{ds} \gamma(s) ds \right) \right| \\ &= \left| \int_0^1 \text{Tr} (X \text{div}_{\tilde{\mathbf{L}}}([\gamma(s)]_{\varphi^{\log}, \omega} \mathbf{V}(s))) ds \right| \\ &= \left| \int_0^1 \sum_{j \in \mathcal{J}} \langle \nabla_{\tilde{L}_j} X, [\gamma(s)]_{\varphi^{\log}, \omega_j} V(s)_j \rangle_{\text{HS}} ds \right| \\ &\leq \left(\int_0^1 \sum_{j \in \mathcal{J}} \langle \nabla_{\tilde{L}_j} X, [\gamma(s)]_{\varphi^{\log}, \omega_j} \nabla_{\tilde{L}_j} X \rangle_{\text{HS}} ds \right)^{1/2} \left(\int_0^1 \sum_{j \in \mathcal{J}} \langle V(s)_j, [\gamma(s)]_{\varphi^{\log}, \omega_j} V(s)_j \rangle_{\text{HS}} ds \right)^{1/2} \\ &\leq \left(\int_0^1 \sum_{j \in \mathcal{J}} \langle \nabla_{\tilde{L}_j} X, [\gamma(s)]_{\varphi^{\log}, \omega_j} \nabla_{\tilde{L}_j} X \rangle_{\text{HS}} ds \right)^{1/2} (W_{2,\mathcal{L}}(\rho, \omega) + \delta), \end{aligned} \quad (12.16)$$

where in the fourth line we used the Cauchy-Schwarz inequality with respect to the inner product $\sum_{j \in \mathcal{J}} \int_0^1 \langle \cdot, [\gamma(s)]_{\varphi^{\log}, \omega_j} \cdot \rangle_{\text{HS}} ds$, and the last line comes from (12.14). Now, for each $j \in \mathcal{J}$, and any

$s \in [0, 1]$, (12.13) for $\rho = \gamma(s)$ leads to

$$\begin{aligned} \langle \nabla_{\tilde{L}_j} X, [\gamma(s)]_{\varphi^{\log}, \omega_j} \nabla_{\tilde{L}_j} X \rangle_{\text{HS}} &\leq \frac{1}{2} \left[e^{-\omega_j/2} \text{Tr}(\gamma(s)(\nabla_{\tilde{L}_j} X)^* \nabla_{\tilde{L}_j} X) + e^{\omega_j/2} \text{Tr}((\nabla_{\tilde{L}_j} X)^* \gamma(s) \nabla_{\tilde{L}_j} X) \right] \\ &\leq \frac{e^{-\omega_j/2} + e^{\omega_j/2}}{2} \|\nabla_{\tilde{L}_j} X\|_{\infty}^2, \end{aligned} \quad (12.17)$$

where the last line follows by Hölder's inequality. Substituting this bound into (12.16), we end up with

$$\begin{aligned} |\text{Tr}(X(\rho - \omega))| &\leq 2^{-1/2} \left(\sum_{j \in \mathcal{J}} (e^{-\omega_j/2} + e^{\omega_j/2}) \|\nabla_{\tilde{L}_j}(X)\|_{\infty}^2 \right)^{1/2} (W_{2,\mathcal{L}}(\rho, \omega) + \delta) \\ &\leq \sqrt{2} \|X\|_{\infty} \left(\sum_{j \in \mathcal{J}} (e^{-\omega_j/2} + e^{\omega_j/2}) \|\tilde{L}_j\|_{\infty}^2 \right)^{1/2} (W_{2,\mathcal{L}}(\rho, \omega) + \delta). \end{aligned} \quad (12.18)$$

The result follows after taking the limit $\delta \rightarrow 0$ and by duality between $\|\cdot\|_1$ and $\|\cdot\|_{\infty}$. $\square$

Proposition 12.1.8. *The metric space $(\mathcal{D}(\mathcal{H}), W_{2,\mathcal{L}})$ is complete.*

Proof. This directly follows from Lemma 12.1.7 and Proposition 12.1.6: assume that $\{\rho_n\}_{n \in \mathbb{N}}$ is a Cauchy sequence in $(\mathcal{D}(\mathcal{H}), W_{2,\mathcal{L}})$, that is $W_{2,\mathcal{L}}(\rho_n, \rho_m) \rightarrow 0$ as $m, n \rightarrow \infty$. Then, by Lemma 12.1.7, $\{\rho_n\}_{n \in \mathbb{N}}$ is also Cauchy with respect to the trace norm $\|\cdot\|_1$. By completeness of the normed vector space $(\mathcal{B}(\mathcal{H}), \|\cdot\|_1)$, this implies existence of $\rho_{\infty} \in \mathcal{D}(\mathcal{H})$ such that $\|\rho_n - \rho_{\infty}\|_1 \rightarrow 0$ as $n \rightarrow \infty$. We conclude that $W_{2,\mathcal{L}}(\rho_n, \rho_{\infty}) \rightarrow W_{2,\mathcal{L}}(\rho_{\infty}, \rho_{\infty}) = 0$ by Proposition 12.1.6. $\square$

12.2. Gradient formula

From now on, we restrict ourselves to the study of $W_{2,\mathcal{L}} \equiv W_{2,\mathcal{L}}^{\varphi^{\log}}$, where the functions $\varphi_{\omega_j}^{\log}$ are given in Equation (12.6). In particular, we show that the evolution $(\mathcal{P}_t)_{t \geq 0}$ is the gradient flow for the relative entropy functional $D(\cdot \| \sigma)$ with respect to $W_{2,\mathcal{L}}$. This means that $\mathcal{L}_* \rho = -\text{grad}_{\mathcal{L}} D(\rho \| \sigma)$, where the gradient $\text{grad}_{\mathcal{L}} \mathcal{F}(\rho)$ of a differentiable functional $\mathcal{F} : \mathcal{D}_+(\mathcal{H}) \rightarrow \mathbb{R}$ at a point ρ is defined as the unique element in the tangent space at ρ so that

$$\left. \frac{d}{dt} \mathcal{F}(\gamma(t)) \right|_{t=0} = g_{\mathcal{L},\rho}(\dot{\gamma}, \text{grad}_{\mathcal{L}} \mathcal{F}(\rho)) \quad (12.19)$$

for all smooth paths $\gamma(t)$ defined on $(-\varepsilon, \varepsilon)$ for some $\varepsilon > 0$ with $\gamma(0) = \rho$. In particular, for $\gamma(t) = \rho_t \equiv \mathcal{P}_{*t}(\rho)$,

$$\left. \frac{d}{dt} D(\rho_t \| \sigma) \right|_{t=0} = -g_{\mathcal{L},\rho}(\mathcal{L}_*(\rho), \mathcal{L}_*(\rho)). \quad (12.20)$$

This result was first proved in [Carlen and Maas, 2017], but we hope our derivation makes the link with the general chain rule of Section 5.3 clearer. First recall that the generator $\mathcal{L}_*$ of $(\mathcal{P}_{*t})_{t \geq 0}$ in the Schrödinger picture takes the form (cf. Theorem 5.10 of [Carlen and Maas, 2017])

$$\mathcal{L}_*(\rho) = - \sum_{j \in \mathcal{J}} \nabla_{\tilde{L}_j}^* (\tilde{L}_j e^{-\omega_j/2} \rho - e^{\omega_j/2} \rho \tilde{L}_j).$$

Moreover, by simple integral representation of the logarithm, one can show that for any smooth path $\gamma : [-\varepsilon, \varepsilon] \rightarrow \mathcal{D}_+(\mathcal{H})$ with $\gamma(0) = \rho$,

$$\left. \frac{d}{ds} D(\gamma(s) \| \sigma) \right|_{s=0} = \text{Tr} [\dot{\gamma}(0) (\ln \rho - \ln \sigma)] \quad (12.21)$$

$$= -\text{Tr} [\text{div}_{\tilde{\mathbf{L}}}([\rho]_{\varphi^{\log, \omega}} \nabla_{\tilde{\mathbf{L}}}(\Phi)) (\ln \rho - \ln \sigma)] \quad (12.22)$$

$$= \langle \nabla_{\tilde{\mathbf{L}}} \Phi, \nabla_{\tilde{\mathbf{L}}} (\ln \rho - \ln \sigma) \rangle_{\mathbb{L}_{2, \log}(\rho)}, \quad (12.23)$$

where the second line comes from the n-c continuity equation for the couple (γ, Φ) at $s = 0$. Equation (12.19) would follow with $\text{grad}_{\mathcal{L}} D(\rho \| \sigma) = -\mathcal{L}_*(\rho)$ if one can show that

$$\mathcal{L}_*(\rho) = \text{div}_{\tilde{\mathbf{L}}}([\rho]_{\varphi^{\log, \omega}} \nabla_{\tilde{\mathbf{L}}} (\ln \rho - \ln \sigma)). \quad (12.24)$$

We prove Equation (12.24) using the general chain rule provided in Theorem 5.3.5. First, notice that $[\rho]_{\varphi^{\log, \omega}}$ can be re-expressed as follows:

$$\begin{aligned} [\rho]_{\varphi^{\log, \omega_j}} &= \frac{e^{\omega_j/2} L_\rho - e^{-\omega_j/2} R_\rho}{\ln(\Delta_\rho) + \omega_j} \\ &= \frac{e^{\omega_j/2} L_\rho - e^{-\omega_j/2} R_\rho}{\ln(e^{\omega_j/2} L_\rho) - \ln(e^{-\omega_j/2} R_\rho)} \\ &= T_{(x,y) \mapsto \frac{x-y}{\ln x - \ln y}}^{e^{\omega_j/2} \rho, e^{-\omega_j/2} \rho}, \end{aligned} \quad (12.25)$$

where the second line follows from the commutativity of L_ρ and R_ρ . Moreover, for any $j \in \mathcal{J}$

$$\begin{aligned} \nabla_{\tilde{L}_j} (\ln \rho - \ln \sigma) &= \nabla_{\tilde{L}_j} \ln \rho + \left. \frac{d}{ds} \right|_{s=0} \Delta_\sigma^s(\tilde{L}_j) \\ &= \nabla_{\tilde{L}_j} \ln \rho + \left. \frac{d}{ds} \right|_{s=0} e^{-\omega_j s} \tilde{L}_j \\ &= \nabla_{\tilde{L}_j} \ln \rho - \omega_j \tilde{L}_j \\ &= \tilde{L}_j \ln(e^{-\omega_j/2} \rho) - \ln(e^{\omega_j/2} \rho) \tilde{L}_j. \end{aligned} \quad (12.26)$$

Therefore, the right hand side of Equation (12.24) reads

$$\begin{aligned} - \sum_{j \in \mathcal{J}} \nabla_{\tilde{L}_j^*} \left(T_{(x,y) \mapsto \frac{x-y}{\ln x - \ln y}}^{e^{\omega_j/2} \rho, e^{-\omega_j/2} \rho} \{ \tilde{L}_j \ln(e^{-\omega_j/2} \rho) - \ln(e^{\omega_j/2} \rho) \tilde{L}_j \} \right) &= - \sum_{j \in \mathcal{J}} \nabla_{\tilde{L}_j^*} (\tilde{L}_j e^{-\omega_j/2} \rho - e^{\omega_j/2} \rho \tilde{L}_j) \\ &\equiv \mathcal{L}_*(\rho), \end{aligned}$$

where the above equality comes from Theorem 5.3.5.

12.3. Quantum displacement convexity

The notion of displacement convexity that we introduced in Section 4.2.3 was extended to the quantum realm by [Carlen and Maas, 2014] for the particular case of the Fermionic Fokker-Planck equation. In [Carlen and Maas, 2017], the same authors took a slightly different path inspired by [Otto and Westdickenberg, 2005, Daneri and Savaré, 2008]. In the latter, the authors showed the geodesic convexity of a function F defined on an abstract metric space under the condition of contractivity of the gradient flow for F for the corresponding distance. Carlen and Maas showed that this approach, when used for a QMS $(\mathcal{P}_{*t})_{t \geq 0}$, that is the gradient flow of the relative entropy functional $D(\cdot \| \sigma)$ in

$(\mathcal{D}_+(\mathcal{H}), W_{2,\mathcal{L}})$, leads to the modified logarithmic Sobolev inequality. In this section, we provide a systematic analysis of the initial notion of displacement convexity in the setting of [Carlen and Maas, 2017], including a study of the geodesic equations on the Riemannian manifold $(\mathcal{D}_+(\mathcal{H}), g_{\mathcal{L}})$.

12.3.1. Geodesic equations

Since $(\mathcal{D}_+(\mathcal{H}), g_{\mathcal{L}})$ defines a valid Riemannian manifold, the local existence and uniqueness of constant speed geodesics is guaranteed by standard Riemannian geometry. We first recall that constant speed geodesics $(\gamma(s))_{s \in [0,1]}$, satisfy a Euler-Lagrange equation that we derive in Theorem 12.3.1. This extends Theorem 5.3 of [Carlen and Maas, 2014].

Let $(\mathcal{V}, \langle \cdot, \cdot \rangle)$ be a finite-dimensional real Hilbert space, $\mathcal{W} \subset \mathcal{V}$ be a subspace of $\mathcal{V}$, and $z \in \mathcal{V} \setminus \mathcal{W}$. Consider the affine subspace $\mathcal{W}_z := z + \mathcal{W}$, and let $\mathcal{M} \subset \mathcal{W}_z$ be a relatively open subset. Let $D : \mathcal{M} \rightarrow \mathcal{B}(\mathcal{W})$ be a smooth function such that $D(x)$ is self-adjoint and invertible for all $x \in \mathcal{M}$. We shall write $C(x) := D(x)^{-1}$. Consider the Lagrangian $L : \mathcal{W} \times \mathcal{M} \rightarrow \mathbb{R}$ defined by $L(p, x) = \langle C(x)p, p \rangle$ and the associated minimization problem:

$$\inf_{u(\cdot) \in C^1([0,1], \mathcal{M})} \left(\int_0^1 L(u'(t), u(t)) dt : u(0) = u_0, u(1) = u_1 \right),$$

where $u_0, u_1 \in \mathcal{M}$ are given boundary values. Then the Euler-Lagrange equations are equivalent to the following system of equations:

$$\begin{cases} u'(t) - D(u(t))v(t) = 0, \\ v'(t) + \frac{1}{2} \langle \partial_x D(u(t))v(t), v(t) \rangle = 0. \end{cases} \quad (12.27)$$

Here, we apply this abstract result to the case where $\mathcal{V} = \mathcal{B}_{sa}(\mathcal{H})$, with inner product $\langle \cdot, \cdot \rangle$ the usual Hilbert-Schmidt inner product, $\mathcal{W} = \{A \in \mathcal{V} : \text{Tr}(A) = 0\}$, $z := \mathbb{1}/\dim(\mathcal{H})$, and $\mathcal{M} = \mathcal{D}_+(\mathcal{H})$. Indeed any density operator ρ can be written as $\rho = \mathbb{1}/\dim \mathcal{H} + K$, for some self-adjoint and traceless operator K . For any $\rho \in \mathcal{D}_+(\mathcal{H})$, we already proved in Lemma 12.1.5 that $D_{\varphi^{\log}}(\rho) : \Phi \mapsto -\text{div}_{\tilde{\mathbf{L}}}([\rho]_{\varphi^{\log}, \omega} \nabla_{\tilde{\mathbf{L}}} \Phi)$ is invertible and self-adjoint. Now we use the following identity (see [Carlen and Maas, 2014] p. 21):

$$\left. \frac{d}{dt} (\rho + tA)^\alpha \right|_{t=0} = \int_0^1 \int_0^\alpha \frac{\rho^{\alpha-\beta}}{(1-s)\mathbb{1} + s\rho} A \frac{\rho^\beta}{(1-s)\mathbb{1} + s\rho} d\beta ds \quad (12.28)$$

for any $0 < \alpha < 1$, $\rho \in \mathcal{D}_+(\mathcal{H})$ and $A \in \mathcal{W}$. Hence for all $A, \Phi \in \mathcal{W}$,

$$\begin{aligned} \left. \frac{d}{dt} \right|_{t=0} \langle D_{\varphi^{\log}}(\rho + tA)[\Phi], \Phi \rangle_{\text{HS}} &= \left. \frac{d}{dt} \right|_{t=0} \sum_{j \in \mathcal{J}} \langle \nabla_{\tilde{L}_j} \Phi, [\rho + tA]_{\varphi^{\log}, \omega_j} \nabla_{\tilde{L}_j} \Phi \rangle_{\text{HS}} \\ &= \left. \frac{d}{dt} \right|_{t=0} \sum_{j \in \mathcal{J}} \langle \nabla_{\tilde{L}_j} \Phi, \int_0^1 e^{\omega_j(1/2-\alpha)} (\rho + tA)^\alpha \nabla_{\tilde{L}_j} \Phi (\rho + tA)^{1-\alpha} \rangle_{\text{HS}} d\alpha \\ &= \langle A, \nabla_{\tilde{\mathbf{L}}} \Phi \cdot \rho \nabla_{\tilde{\mathbf{L}}} \Phi \rangle_{\text{HS}}, \end{aligned} \quad (12.29)$$

where for two vectors $\vec{V}_1, \vec{V}_2$ in $\oplus_{j \in \mathcal{J}} \mathcal{B}(\mathcal{H})$,

$$\vec{V}_1 \cdot \rho \vec{V}_2 := \sum_{j \in \mathcal{J}} \int_0^1 \int_0^1 e^{\omega_j(1/2-\alpha)} (\chi_j(\vec{V}_1, \vec{V}_2^*, \rho, \alpha, s) + \chi_j(\vec{V}_1^*, \vec{V}_2, \rho, 1-\alpha, s)) d\alpha ds,$$

where

$$\chi_j(\vec{V}_1, \vec{V}_2, \rho, \alpha, s) := \int_0^\alpha \frac{\rho^\beta}{(1-s)\mathbb{1} + s\rho} (V_1)_j \rho^{1-\alpha} (V_2)_j \frac{\rho^{\alpha-\beta}}{(1-s)\mathbb{1} + s\rho} d\beta.$$

Therefore, in our context the Euler-Lagrange equations (12.27) reduce to the following:

Theorem 12.3.1. *The geodesic equations in the Riemannian manifold $(\mathcal{D}_+(\mathcal{H}), W_{2,\mathcal{L}})$ are given by*

$$\begin{cases} \frac{d}{ds}\gamma(s) + \operatorname{div}_{\tilde{\mathbf{L}}}([\gamma(s)]_{\varphi^{\log}, \omega} \nabla_{\tilde{\mathbf{L}}} \Phi(s)) = 0, \\ \frac{d\Phi(s)}{ds} + \frac{1}{2} \nabla_{\tilde{\mathbf{L}}} \Phi(s) \cdot_{\gamma(s)} \nabla_{\tilde{\mathbf{L}}} \Phi(s) = 0. \end{cases} \quad (12.30)$$

12.3.2. Equivalent formulations for displacement convexity

In analogy with [Erbar and Maas, 2012], we say that a primitive quantum Markov semigroup $(\mathcal{P}_t)_{t \geq 0}$ with associated invariant state σ and generator $\mathcal{L}$ of the form of Equation (12.1), with $\Delta_\sigma(\tilde{L}_j) = e^{-\omega_j} \tilde{L}_j$, has *Ricci curvature bounded from below* by a constant $\kappa \in \mathbb{R}$ if the following inequality holds:

$$\left. \frac{d^2}{ds^2} \right|_{s=0} D(\gamma(s) \| \sigma) \geq \kappa g_{\mathcal{L}, \rho}(\dot{\gamma}(0), \dot{\gamma}(0)), \quad (\operatorname{Ric}(\mathcal{L}) \geq \kappa)$$

for any constant speed geodesic $(\gamma(s))_{s \in (-\varepsilon, \varepsilon)}$ on $\mathcal{D}_+(\mathcal{H})$ such that $\gamma(0) = \rho$. This inequality will also be referred to as the *quantum Ricci lower bound*. For a classical intuition behind this definition for the quantum Ricci curvature lower bound, we refer to Equation (4.7). Theorem 12.3.1 is useful to derive an expression for the second derivative of the relative entropy $D(\gamma(s) \| \sigma)$ with respect to s . We already know from the gradient flow equation 12.20 that

$$\begin{aligned} \frac{d}{ds} D(\gamma(s) \| \sigma) &= -g_{\mathcal{L}, \gamma(s)}(\dot{\gamma}(s), \mathcal{L}_*(\gamma(s))) \\ &= \sum_{j \in \mathcal{J}} \langle \nabla_{\tilde{L}_j} \Phi(s), [\gamma(s)]_{\varphi^{\log}, \omega_j} \nabla_{\tilde{L}_j} (\ln \gamma(s) - \ln \sigma) \rangle_{\text{HS}} \\ &= \sum_{j \in \mathcal{J}} \langle \nabla_{\tilde{L}_j} \Phi(s), [\gamma(s)]_{\varphi^{\log}, \omega_j} (\tilde{L}_j \ln(e^{-\omega_j/2} \gamma(s)) - \ln(e^{\omega_j/2} \gamma(s)) \tilde{L}_j) \rangle_{\text{HS}}, \end{aligned}$$

where the second line comes from Equation (12.24), and the last identity comes from Equation (12.26). Using the chain rule formula of Theorem 5.3.5 together with Equation (12.25)

$$[\gamma(s)]_{\varphi^{\log}, \omega_j} (\tilde{L}_j \ln(e^{-\omega_j/2} \gamma(s)) - \ln(e^{\omega_j/2} \gamma(s)) \tilde{L}_j) = e^{-\omega_j/2} \tilde{L}_j \gamma(s) - e^{\omega_j/2} \gamma(s) \tilde{L}_j,$$

so that we finally get

$$\frac{d}{ds} D(\gamma(s) \| \sigma) = \sum_{j \in \mathcal{J}} \langle \nabla_{\tilde{L}_j} \Phi(s), e^{-\omega_j/2} \tilde{L}_j \gamma(s) - e^{\omega_j/2} \gamma(s) \tilde{L}_j \rangle_{\text{HS}}.$$

Differentiating once more, we get:

$$\begin{aligned} \left. \frac{d^2}{ds^2} D(\gamma(s) \| \sigma) \right|_{s=0} &= \sum_{j \in \mathcal{J}} \left\{ \left\langle \nabla_{\tilde{L}_j} \frac{d}{ds} \Phi(s) \right|_{s=0}, e^{-\omega_j/2} \tilde{L}_j \rho - e^{\omega_j/2} \rho \tilde{L}_j \right\rangle_{\text{HS}} \\ &\quad + \langle \nabla_{\tilde{L}_j} \Phi(s), e^{-\omega_j/2} \tilde{L}_j \dot{\gamma}(0) - e^{\omega_j/2} \dot{\gamma}(0) \tilde{L}_j \rangle_{\text{HS}} \right\}. \end{aligned} \quad (12.31)$$

We first take care of the second line of Equation (12.31). Using Theorem 12.3.1, we find

$$\begin{aligned}
& \langle \nabla_{\tilde{L}_j} \Phi, e^{-\omega_j/2} \tilde{L}_j \dot{\gamma}(0) - e^{\omega_j/2} \dot{\gamma}(0) \tilde{L}_j \rangle_{\text{HS}} \\
&= -\langle \nabla_{\tilde{L}_j} \Phi, e^{-\omega_j/2} \tilde{L}_j \operatorname{div}_{\tilde{\mathbf{L}}}([\rho]_{\varphi^{\log}, \omega} \nabla \Phi) - e^{\omega_j/2} \operatorname{div}_{\tilde{\mathbf{L}}}([\rho]_{\varphi^{\log}, \omega} \nabla \Phi) \tilde{L}_j \rangle_{\text{HS}} \\
&= -\langle \nabla_{\tilde{L}_j} \Phi, e^{-\omega_j/2} \tilde{L}_j \sum_{k \in \mathcal{J}} [[\rho]_{\varphi^{\log}, \omega_k} \nabla_{\tilde{L}_k} \Phi, \tilde{L}_k^*] - e^{\omega_j/2} \sum_{k \in \mathcal{J}} [[\rho]_{\varphi^{\log}, \omega_k} \nabla_{\tilde{L}_k} \Phi, \tilde{L}_k^*] \tilde{L}_j \rangle_{\text{HS}} \\
&= \sum_{k \in \mathcal{J}} \left(e^{-\omega_j/2} \langle \nabla_{\tilde{L}_k} (\tilde{L}_j^* \nabla_{\tilde{L}_j} \Phi), [\rho]_{\varphi^{\log}, \omega_k} \nabla_{\tilde{L}_k} \Phi \rangle_{\text{HS}} - e^{\omega_j/2} \langle \nabla_{\tilde{L}_k} (\nabla_{\tilde{L}_j} \Phi \tilde{L}_j^*), [\rho]_{\varphi^{\log}, \omega_k} \nabla_{\tilde{L}_k} \Phi \rangle_{\text{HS}} \right) \\
&= \sum_{k \in \mathcal{J}} \langle \nabla_{\tilde{L}_k} \left(e^{-\omega_j/2} \tilde{L}_j^* \nabla_{\tilde{L}_j} \Phi - e^{\omega_j/2} \nabla_{\tilde{L}_j} \Phi \tilde{L}_j^* \right), [\rho]_{\varphi^{\log}, \omega_k} \nabla_{\tilde{L}_k} \Phi \rangle_{\text{HS}}.
\end{aligned}$$

Hence by Equation (12.1),

$$\begin{aligned}
\sum_{j \in \mathcal{J}} \langle \nabla_{\tilde{L}_j} \Phi, e^{-\omega_j/2} \tilde{L}_j \dot{\gamma}(0) - e^{\omega_j/2} \dot{\gamma}(0) \tilde{L}_j \rangle &= - \sum_{k \in \mathcal{J}} \langle \nabla_{\tilde{L}_k} \mathcal{L}(\Phi), [\rho]_{\varphi^{\log}, \omega_k} \nabla_{\tilde{L}_k} \Phi \rangle_{\text{HS}} \\
&= -\langle \nabla_{\tilde{\mathbf{L}}} \mathcal{L}(\Phi), \nabla_{\tilde{\mathbf{L}}} \Phi \rangle_{\mathbb{L}_{2, \log}(\rho)}. \tag{12.32}
\end{aligned}$$

By (12.30), the first line of (12.31) is equal to

$$\frac{1}{2} \sum_{j \in \mathcal{J}} \langle \nabla_{\tilde{L}_j} (\nabla_{\tilde{\mathbf{L}}} \Phi \cdot \rho \nabla_{\tilde{\mathbf{L}}} \Phi), e^{\omega_j/2} \rho \tilde{L}_j - e^{-\omega_j/2} \tilde{L}_j \rho \rangle_{\text{HS}} \tag{12.33}$$

$$\begin{aligned}
&= \frac{1}{2} \sum_{j \in \mathcal{J}} \langle \nabla_{\tilde{\mathbf{L}}} \Phi \cdot \rho \nabla_{\tilde{\mathbf{L}}} \Phi, [\tilde{L}_j^*, \rho \tilde{L}_j] e^{\omega_j/2} - e^{-\omega_j/2} [\tilde{L}_j^*, \tilde{L}_j \rho] \rangle_{\text{HS}} \\
&= \frac{1}{2} \langle \nabla_{\tilde{\mathbf{L}}} \Phi \cdot \rho \nabla_{\tilde{\mathbf{L}}} \Phi, \mathcal{L}_*(\rho) \rangle_{\text{HS}}, \tag{12.34}
\end{aligned}$$

where we used that, replacing $\tilde{L}_j$ by $\tilde{L}_j^*$ so that $\omega_j \rightarrow -\omega_j$,

$$\mathcal{L}_*(\rho) = \sum_{j \in \mathcal{J}} \left(e^{\omega_j/2} [\tilde{L}_j^* \rho, \tilde{L}_j] + e^{-\omega_j/2} [\tilde{L}_j, \rho \tilde{L}_j^*] \right) = \sum_{j \in \mathcal{J}} \left(e^{-\omega_j/2} [\tilde{L}_j \rho, \tilde{L}_j^*] + e^{\omega_j/2} [\tilde{L}_j^*, \rho \tilde{L}_j] \right),$$

Hence, using (12.32) and (12.34), (12.31) reduces to

$$\left. \frac{d^2}{ds^2} D(\gamma(s) \| \sigma) \right|_{s=0} = \frac{1}{2} \langle \nabla_{\tilde{\mathbf{L}}} \Phi \cdot \rho \nabla_{\tilde{\mathbf{L}}} \Phi, \mathcal{L}_*(\rho) \rangle_{\text{HS}} - \langle \nabla_{\tilde{\mathbf{L}}} \mathcal{L}(\Phi), \nabla_{\tilde{\mathbf{L}}} \Phi \rangle_{\mathbb{L}_{2, \log}(\rho)}. \tag{12.35}$$

One can compare this expression with the one derived in Proposition 4.3 of [Erbar and Maas, 2012]. To make this analogy more clear, we denote the quantity on the right hand side of Equation (12.35) by

$$B(\rho, \Phi) \equiv \frac{1}{2} \langle \nabla_{\tilde{\mathbf{L}}} \Phi \cdot \rho \nabla_{\tilde{\mathbf{L}}} \Phi, \mathcal{L}_*(\rho) \rangle_{\text{HS}} - \langle \nabla_{\tilde{\mathbf{L}}} \mathcal{L}(\Phi), \nabla_{\tilde{\mathbf{L}}} \Phi \rangle_{\mathbb{L}_{2, \log}(\rho)}. \tag{12.36}$$

The above *quantum Bochner formula* is analogous to the one derived by the formal Otto calculus in Equation (4.4), and together with Equation (12.35) justifies the interpretation of $\operatorname{Ric}(\mathcal{L}) \geq \kappa$ as a quantum Ricci curvature lower bound. Next, the following lemma extends Lemma 4.6 of [Erbar and Maas, 2012] to the quantum regime, as well as part of the proof of Proposition 5.11 of [Carlen and Maas, 2014], and will prove to be useful in what follows:

Lemma 12.3.2. *Let $(\gamma(s))_{s \in [0,1]}$ be a smooth curve in $\mathcal{D}_+(\mathcal{H})$. For each $t \geq 0$, set $\gamma(s, t) := \mathcal{P}_{*st}(\gamma(s))$, and let $(\Phi(s, t))_{s \in [0,1]}$ be a smooth curve satisfying the continuity equation*

$$\partial_s(\gamma(s, t)) + \operatorname{div}_{\tilde{\mathbf{L}}}([\gamma(s, t)]_{\varphi^{\log}, \omega} \nabla_{\tilde{\mathbf{L}}} \Phi(s, t)) = 0, \quad s \in [0, 1]. \tag{12.37}$$

Therefore,

$$\frac{1}{2}\partial_t g_{\mathcal{L},\gamma(s,t)}(\partial_s\gamma(s,t), \partial_s\gamma(s,t)) + \partial_s D(\gamma(s,t)\|\sigma) = -sB(\gamma(s,t), \Phi(s,t)).$$

Proof. Start by noticing that

$$\begin{aligned}\partial_s D(\gamma(s,t)\|\sigma) &= \partial_s \text{Tr}(\gamma(s,t)(\ln \gamma(s,t) - \ln \sigma)) \\ &= \text{Tr}(\partial_s \gamma(s,t)(\ln \gamma(s,t) - \ln \sigma)) \\ &= -\text{Tr}((\ln \gamma(s,t) - \ln \sigma) \text{div}_{\tilde{\mathbf{L}}}([\gamma(s,t)]_{\varphi^{\log}, \omega} \nabla_{\tilde{\mathbf{L}}} \Phi(s,t))) \\ &= -\langle \ln \gamma(s,t) - \ln \sigma, \text{div}_{\tilde{\mathbf{L}}}([\gamma(s,t)]_{\varphi^{\log}, \omega} \nabla_{\tilde{\mathbf{L}}} \Phi(s,t)) \rangle_{\text{HS}} \\ &= -\sum_{j \in \mathcal{J}} \langle \ln \gamma(s,t) - \ln \sigma, [[\gamma(s,t)]_{\varphi^{\log}, \omega_j} (\nabla_{\tilde{L}_j} \Phi(s,t)), \tilde{L}_j^*] \rangle_{\text{HS}} \\ &= \sum_{j \in \mathcal{J}} \langle \nabla_{\tilde{L}_j} (\log \gamma(s,t) - \log \sigma), [\gamma(s,t)]_{\varphi^{\log}, \omega_j} (\nabla_{\tilde{L}_j} \Phi(s,t)) \rangle_{\text{HS}} \\ &= \sum_{j \in \mathcal{J}} \langle [\gamma(s,t)]_{\varphi^{\log}, \omega_j} (\nabla_{\tilde{L}_j} (\log \gamma(s,t) - \log \sigma)), \nabla_{\tilde{L}_j} \Phi(s,t) \rangle_{\text{HS}} \\ &= \sum_{j \in \mathcal{J}} \langle \nabla_{\tilde{L}_j}^* [\gamma(s,t)]_{\varphi^{\log}, \omega_j} (\nabla_{\tilde{L}_j} (\log \gamma(s,t) - \log \sigma)), \Phi(s,t) \rangle_{\text{HS}} \\ &= -\langle \mathcal{L}_*(\gamma(s,t)), \Phi(s,t) \rangle_{\text{HS}},\end{aligned}$$

where in the third line we used (12.37), in last line Equation (12.24), and in the second line we used the integral representation of the logarithm to prove that $\text{Tr}(\gamma(s,t)\partial_s \log \gamma(s,t)) = 0$. Moreover,

$$\begin{aligned}\frac{1}{2}\partial_t g_{\mathcal{L},\gamma(s,t)}(\partial_s\gamma(s,t), \partial_s\gamma(s,t)) &= \frac{1}{2}\partial_t \sum_{j \in \mathcal{J}} \langle \nabla_{\tilde{L}_j} \Phi(s,t), [\gamma(s,t)]_{\varphi^{\log}, \omega_j} \nabla_{\tilde{L}_j} \Phi(s,t) \rangle_{\text{HS}} \\ &= \sum_{j \in \mathcal{J}} \left(\langle \partial_t (\nabla_{\tilde{L}_j} \Phi(s,t)), [\gamma(s,t)]_{\varphi^{\log}, \omega_j} \nabla_{\tilde{L}_j} \Phi(s,t) \rangle_{\text{HS}} \right. \\ &\quad \left. + \frac{1}{2} \langle \nabla_{\tilde{L}_j} \Phi(s,t), \partial_t ([\gamma(s,t)]_{\varphi^{\log}, \omega_j}) \nabla_{\tilde{L}_j} \Phi(s,t) \rangle_{\text{HS}} \right). \quad (12.38)\end{aligned}$$

From (12.29),

$$\begin{aligned}\sum_{j \in \mathcal{J}} \langle \nabla_{\tilde{L}_j} \Phi(s,t), \partial_t ([\gamma(s,t)]_{\varphi^{\log}, \omega_j}) \nabla_{\tilde{L}_j} \Phi(s,t) \rangle_{\text{HS}} &= \langle \partial_t \gamma(s,t), \nabla \Phi(s,t) \cdot_{\gamma(s,t)} \nabla \Phi(s,t) \rangle_{\text{HS}} \\ &= s \langle \mathcal{L}_*(\gamma(s,t)), \nabla \Phi(s,t) \cdot_{\gamma(s,t)} \nabla \Phi(s,t) \rangle_{\text{HS}}. \quad (12.39)\end{aligned}$$

Moreover,

$$\begin{aligned}\sum_{j \in \mathcal{J}} \langle \partial_t (\nabla_{\tilde{L}_j} \Phi(s,t)), [\gamma(s,t)]_{\varphi^{\log}, \omega_j} \nabla_{\tilde{L}_j} \Phi(s,t) \rangle_{\text{HS}} &\quad (12.40) \\ &= -\sum_{j \in \mathcal{J}} \langle \partial_t \Phi(s,t), [[\gamma(s,t)]_{\varphi^{\log}, \omega_j} (\nabla_{\tilde{L}_j} \Phi(s,t)), \tilde{L}_j^*] \rangle_{\text{HS}} \\ &= -\langle \partial_t \Phi(s,t), \text{div}_{\tilde{\mathbf{L}}}([\gamma(s,t)]_{\varphi^{\log}, \omega} (\nabla \Phi(s,t))) \rangle_{\text{HS}} \\ &= \langle \partial_t \Phi(s,t), \partial_s \gamma(s,t) \rangle_{\text{HS}} \\ &= \partial_t (\langle \Phi(s,t), \partial_s \gamma(s,t) \rangle_{\text{HS}}) - \langle \Phi(s,t), \partial_s \partial_t \gamma(s,t) \rangle_{\text{HS}} \\ &= \partial_t g_{\mathcal{L},\gamma(s,t)}(\partial_s \gamma(s,t), \partial_s \gamma(s,t)) - \langle \Phi(s,t), \partial_s \partial_t \gamma(s,t) \rangle_{\text{HS}} \\ &= \partial_t g_{\mathcal{L},\gamma(s,t)}(\partial_s \gamma(s,t), \partial_s \gamma(s,t)) - \langle \Phi(s,t), \partial_s (s \mathcal{L}_*(\gamma(s,t))) \rangle_{\text{HS}}, \quad (12.41)\end{aligned}$$

where we used once again 12.37 in the third and fifth lines above. Therefore, using 12.39 and 12.41, the right hand side of 12.38 reduces to

$$\begin{aligned} & \frac{1}{2} \partial_t g_{\mathcal{L}, \gamma(s,t)} (\partial_s \gamma(s,t), \partial_s \gamma(s,t)) \\ &= \langle \Phi(s,t), \partial_s (s \mathcal{L}_* (\gamma(s,t))) \rangle_{\text{HS}} - \frac{1}{2} s \langle \mathcal{L}_* (\gamma(s,t)), \nabla \Phi(s,t) \cdot_{\gamma(s,t)} \nabla \Phi(s,t) \rangle_{\text{HS}}. \end{aligned}$$

Hence,

$$\begin{aligned} & \frac{1}{2} \partial_t g_{\mathcal{L}, \gamma(s,t)} (\partial_s \gamma(s,t), \partial_s \gamma(s,t)) + \partial_s D(\gamma(s,t) \| \sigma) \\ &= s \langle \Phi(s,t), \mathcal{L}_* \partial_s (\gamma(s,t)) \rangle_{\text{HS}} - \frac{1}{2} s \langle \mathcal{L}_* \gamma(s,t), \nabla \Phi(s,t) \cdot_{\gamma(s,t)} \nabla \Phi(s,t) \rangle_{\text{HS}} \\ &= -s \langle \mathcal{L}(\Phi(s,t)), \text{div}_{\tilde{\mathbf{L}}}([\gamma(s,t)]_{\varphi^{\log, \omega}} \nabla \Phi(s,t)) \rangle_{\text{HS}} - \frac{1}{2} s \langle \mathcal{L}_* \gamma(s,t), \nabla \Phi(s,t) \cdot_{\gamma(s,t)} \nabla \Phi(s,t) \rangle_{\text{HS}} \\ &= s \langle \nabla \mathcal{L}(\Phi(s,t)), \nabla \Phi(s,t) \rangle_{\mathbb{L}_{2, \log}(\gamma(s,t))} - \frac{1}{2} s \langle \mathcal{L}_* (\gamma(s,t)), \nabla \Phi(s,t) \cdot_{\gamma(s,t)} \nabla \Phi(s,t) \rangle_{\text{HS}} \\ &= -s B(\gamma(s,t), \Phi(s,t)), \end{aligned}$$

which is what needed to be proved. $\square$

Theorem 12.3.3. *Let $\mathcal{L}$ be the generator of a primitive QMS $(\mathcal{P}_t)_{t \geq 0}$, with unique invariant state σ , of the form of Equation (12.1). Then, for $\kappa \in \mathbb{R}$, the following are equivalent:*

- (i) Ricci curvature lower bound: for any constant speed geodesic $(\gamma(s))_{s \in (-\varepsilon, \varepsilon)}$ on $\mathcal{D}_+(\mathcal{H})$ such that $\gamma(0) = \rho$:

$$\left. \frac{d^2}{ds^2} \right|_{s=0} D(\gamma(s) \| \sigma) \geq \kappa g_{\mathcal{L}, \rho}(\dot{\gamma}(0), \dot{\gamma}(0)), \quad (\text{Ric}(\mathcal{L}) \geq \kappa)$$

- (ii) For all $\rho \in \mathcal{D}_+(\mathcal{H})$, and $\Phi \in \mathcal{W}$,

$$B(\rho, \Phi) \geq \kappa \|\nabla \Phi\|_{\mathbb{L}_{2, \log}(\rho)}^2$$

- (iii) Evolution variational inequality: for all $\rho, \omega \in \mathcal{D}_+(\mathcal{H})$ and all $t \geq 0$, writing $\rho_t := \mathcal{P}_{*t}(\rho)$:

$$\left. \frac{1}{2} \frac{d}{dt} \right|_{t^+} (W_{2, \mathcal{L}}(\rho_t, \omega))^2 + \frac{\kappa}{2} W_{2, \mathcal{L}}(\rho_t, \omega)^2 \leq D(\omega \| \sigma) - D(\rho_t \| \sigma). \quad (12.42)$$

- (iv) Equation (12.42) holds for any $\rho, \omega \in \mathcal{D}(\mathcal{H})$.

- (v) κ -displacement convexity of the relative entropy: for any constant speed geodesic $(\gamma(s))_{s \in [0, 1]}$ in $\mathcal{D}(\mathcal{H})$,

$$D(\gamma(s) \| \sigma) \leq (1-s) D(\gamma(0) \| \sigma) + s D(\gamma(1) \| \sigma) - \frac{\kappa}{2} s(1-s) W_{2, \mathcal{L}}(\gamma(0), \gamma(1))^2. \quad (12.43)$$

Proof. The proof is inspired by the one of Theorem 4.5 of [Erbar and Maas, 2012]. That (i) $\Leftrightarrow$ (ii) follows from Equation (12.36). We use Lemma 12.3.2 to show that (ii) $\Rightarrow$ (iii): Take a smooth path $(\gamma(s))_{s \in [0, 1]}$ such that $\gamma(0) = \omega$, $\gamma(1) = \rho$ and

$$\int_0^1 g_{\mathcal{L}, \gamma(s)}(\dot{\gamma}(s), \dot{\gamma}(s)) ds \leq W_{2, \mathcal{L}}(\rho, \omega)^2 + \varepsilon. \quad (12.44)$$

With the notations of Lemma 12.3.2,

$$\frac{1}{2} \partial_t \left(e^{2\kappa st} g_{\mathcal{L}, \gamma(s,t)}(\partial_s \gamma(s,t), \partial_s \gamma(s,t)) \right) + \partial_s \left(e^{2\kappa st} D(\gamma(s,t) \| \sigma) \right) \leq 2\kappa t e^{2\kappa st} D(\gamma(s,t) \| \sigma).$$

Integrating with respect to $t \in [0, h]$, for some $h > 0$, and $s \in [0, 1]$,

$$\begin{aligned} 2\kappa \int_0^1 ds \int_0^h dt \, t e^{2\kappa st} D(\gamma(s,t) \| \sigma) &\geq \int_0^h \left(e^{2\kappa t} D(\gamma(1,t) \| \sigma) - D(\gamma(0,t) \| \sigma) \right) dt \\ &\quad + \frac{1}{2} \int_0^1 e^{2\kappa sh} g_{\mathcal{L}, \gamma(s,h)}(\partial_s \gamma(s,h), \partial_s \gamma(s,h)) - g_{\mathcal{L}, \gamma(s,0)}(\partial_s \gamma(s,0), \partial_s \gamma(s,0)) ds. \end{aligned} \quad (12.45)$$

The following inequality, for which a classical equivalent is given in the proof of Theorem 4.5 of [Erbar and Maas, 2012], can be derived similarly to Lemma 5.1 of [Daneri and Savaré, 2008]:

$$m(\kappa h) W_{2,\mathcal{L}}(\rho_h, \omega)^2 \leq \int_0^1 e^{2\kappa sh} g_{\mathcal{L}, \gamma(s,h)}(\partial_s \gamma(s,h), \partial_s \gamma(s,h)) ds, \quad (12.46)$$

where $m(x) := x e^x / \sinh(x)$. Indeed, define $f : s \mapsto e^{2\kappa sh}$, and denote $L_f := \int_0^1 \frac{1}{f(s)} ds$. Then, let $g : [0, 1] \mapsto [0, 1]$ be the smooth increasing map defined as $g(s) = L_f^{-1} \int_0^s \frac{1}{f(u)} du$, and denote its inverse k such that $k'(g(s)) = L_f f(s)$. Then define the reparametrized curve $(\gamma(k(r), h), k'(r) \Phi(k(r), h))_{r \in [0, 1]}$ which satisfies the continuity equation:

$$\begin{aligned} \partial_r \gamma(k(r), h) &= k'(r) \partial_1 \gamma(k(r), h) \\ &= -k'(r) \operatorname{div}_{\tilde{\mathbf{L}}}([\gamma(k(r), h)]_{\varphi^{\log, \omega}} \nabla \Phi(k(r), h)), \end{aligned}$$

where we used Equation (12.37) in order to established the second line. This curve satisfies $\gamma(k(0), h) = \omega$ and $\gamma(k(1), h) = \rho_h$, so that

$$\begin{aligned} W_{2,\mathcal{L}}(\rho_h, \omega)^2 &\leq \int_0^1 g_{\mathcal{L}, \gamma(k(r), h)}(\partial_r \gamma(k(r), h), \partial_r \gamma(k(r), h)) dr \\ &= \int_0^1 k'(r)^2 \|\nabla_{\tilde{\mathbf{L}}} \Phi(k(r), h)\|_{\mathbb{L}_{2, \log}(\gamma(k(r), h))}^2 dr \\ &= \int_0^1 k'(g(s)) \|\nabla_{\tilde{\mathbf{L}}} \Phi(s, h)\|_{\mathbb{L}_{2, \log}(\gamma(s, h))}^2 ds \\ &= L_f \int_0^1 f(s) g_{\mathcal{L}, \gamma(s, h)}(\partial_s \gamma(s, h), \partial_s \gamma(s, h)) ds, \end{aligned}$$

which directly leads to 12.46. This inequality, together with 12.44, implies

$$\begin{aligned} &\frac{m(h\kappa)}{2} W_{2,\mathcal{L}}(\rho_h, \omega)^2 - \frac{1}{2} W_{2,\mathcal{L}}(\rho, \omega)^2 - \varepsilon + \int_0^h e^{2\kappa t} dt \, D(\rho_h \| \sigma) - h D(\omega \| \sigma) \\ &\leq \frac{1}{2} \int_0^1 e^{2\kappa sh} g_{\mathcal{L}, \gamma(s,t)}(\partial_s \gamma(s,t), \partial_s \gamma(s,t)) ds - \frac{1}{2} \int_0^1 g_{\mathcal{L}, \gamma(s)}(\dot{\gamma}(s), \dot{\gamma}(s)) ds \\ &\quad + \int_0^h e^{2\kappa t} D(\rho_t \| \sigma) dt - h D(\omega \| \sigma) \\ &\leq 2\kappa \int_0^1 ds \int_0^h dt \, t e^{2\kappa st} D(\gamma(s,t) \| \sigma). \end{aligned}$$

where, in the first inequality, we also used the monotonicity of the relative entropy so that $D(\rho_h \| \sigma) = D(\rho_h \| \mathcal{P}_{*h} \sigma) \leq D(\rho_t \| \mathcal{P}_{*t} \sigma) = D(\rho_t \| \sigma)$, and in the second one that for all $t > 0$, $\gamma(1, t) = \rho_t$, $\gamma(0, t) = \omega$,

as well as 12.45. Since for all $s \in [0, 1]$, $t \mapsto D(\gamma(s, t) \| \sigma)$ is bounded,

$$\lim_{h \rightarrow 0} \frac{1}{h} \int_0^1 \int_0^h t e^{2\kappa st} D(\gamma(s, t) \| \sigma) dt ds = 0.$$

Moreover,

$$\lim_{h \rightarrow 0} \frac{1}{h} \left(\int_0^h e^{2\kappa t} dt D(\rho_h \| \sigma) - h D(\omega \| \sigma) \right) = D(\rho \| \sigma) - D(\omega \| \sigma)$$

Since $\varepsilon > 0$ was arbitrary, we arrive at

$$\left. \frac{d}{dh} \right|_{h=0^+} \left(\frac{m(\kappa h)}{2} W_{2, \mathcal{L}}(\rho_h, \omega)^2 \right) + D(\rho \| \sigma) - D(\omega \| \sigma) \leq 0.$$

The result for $t = 0$ follows from the fact that the first term in the left hand side above is equal to $\frac{\kappa}{2} W_{2, \mathcal{L}}(\rho_h, \omega)^2 + \frac{1}{2} \left. \frac{d}{dh} \right|_{h=0^+} W_{2, \mathcal{L}}(\rho_h, \omega)^2$. The case $t \geq 0$ directly follows from the case $t = 0$.

(iii) $\Rightarrow$ (iv) follows from Theorem 3.3 of [Daneri and Savaré, 2008] together with the fact that $(\mathcal{D}(\mathcal{H}), W_{2, \mathcal{L}})$ is complete (cf. Proposition 12.1.8).

(iv) $\Rightarrow$ (v) follows directly from Theorem 3.2 of [Daneri and Savaré, 2008].

(v) $\Rightarrow$ (i) can easily be proved as follows: let $0 < \varepsilon < \varepsilon'$, and without loss of generality, let $\gamma : (-\varepsilon', \varepsilon') \rightarrow \mathcal{D}_+(\mathcal{H})$ be speed 1 geodesic, and that $\gamma(0) = \rho$. Then, construct the following constant speed geodesic $\tilde{\gamma} : [0, 1] \rightarrow \mathcal{D}_+(\mathcal{H})$ as follows: for any $s \in [0, 1]$, $\tilde{\gamma}(s) := \gamma(2\varepsilon s - \varepsilon)$. It then follows that $W_{2, \mathcal{L}}(\tilde{\gamma}(0), \tilde{\gamma}(1)) = 2\varepsilon$. Moreover, by applying (12.43) to $\tilde{\gamma}$, we find, after a suitable rearrangement of the terms:

$$\frac{D(\gamma(\varepsilon) \| \sigma) - 2D(\rho \| \sigma) + D(\gamma(-\varepsilon) \| \sigma)}{\varepsilon^2} \geq \kappa.$$

The result follows after taking the limit $\varepsilon \rightarrow 0$. □

Other equivalent formulations of displacement convexity Here, we provide other characterizations of the Ricci curvature lower bound in terms of some contraction properties of the Wasserstein metric along the semigroup $(\mathcal{P}_t)_{t \geq 0}$. In the next theorem, the characterization of displacement convexity in terms of gradient estimates can be interpreted as a non-commutative version of Bakry-Émery's original gradient bound (see Theorem 4.7.2 of [Bakry et al., 2014]): in particular they showed that the Ricci curvature lower bound is equivalent to the following pointwise inequality for smooth enough functions (e.g. functions satisfying Condition 2.2.1):

$$\Gamma(P_t(f), P_t(f)) \leq e^{-2\kappa t} P_t(\Gamma(f, f)), \quad (12.47)$$

where Γ stands for the carré du champ operator introduced in Section 2.2.

Proposition 12.3.4 (Gradient estimate). *For any $\kappa \in \mathbb{R}$, $\text{Ric}(\mathcal{L}) \geq \kappa$ is equivalent to the following gradient estimate: for any $\rho \in \mathcal{D}_+(\mathcal{H})$, any traceless $\Phi \in \mathcal{B}_{sa}(\mathcal{H})$ and all $t > 0$:*

$$\|\nabla_{\tilde{\mathbf{L}}}(\mathcal{P}_t(\Phi))\|_{\mathbb{L}_{2, \log}(\rho)}^2 \leq e^{-2\kappa t} \|\nabla_{\tilde{\mathbf{L}}} \Phi\|_{\mathbb{L}_{2, \log}(\mathcal{P}_{*t}(\rho))}^2. \quad (12.48)$$

Proof. We first prove the direct implication: define for $u \in [0, t]$ $\rho_u \equiv \mathcal{P}_{*u}(\rho)$ and $\Phi(u) \equiv \mathcal{P}_u(\Phi)$. Next, define

$$\psi(s) := e^{-2\kappa s} \|\nabla_{\tilde{\mathbf{L}}} \Phi(t-s)\|_{\mathbb{L}_{2,\log}(\rho_s)}^2 \equiv e^{-2\kappa s} \sum_{j \in \mathcal{J}} \langle \nabla_{\tilde{L}_j}(\Phi(t-s)), [\rho_s]_{\varphi^{\log}, \omega_j} \nabla_{\tilde{L}_j}(\Phi(t-s)) \rangle_{\text{HS}}.$$

Then, $\psi(0) = \|\nabla_{\tilde{\mathbf{L}}} \Phi(t)\|_{\mathbb{L}_{2,\log}(\rho)}^2$ and $\psi(t) = e^{-2\kappa t} \|\nabla_{\tilde{\mathbf{L}}} \Phi\|_{\mathbb{L}_{2,\log}(\rho_t)}^2$. It is then enough to prove that ψ has non-negative derivative to prove the claim. But:

$$\begin{aligned} \psi'(s) &= 2e^{-2\kappa s} \left[-\kappa \|\nabla_{\tilde{\mathbf{L}}} \Phi(t-s)\|_{\mathbb{L}_{2,\log}(\rho_s)}^2 + \frac{1}{2} \frac{\partial}{\partial s} \|\nabla_{\tilde{\mathbf{L}}} \Phi(t-s)\|_{\mathbb{L}_{2,\log}(\rho_s)}^2 \right] \\ &= 2e^{-2\kappa s} \left[-\kappa \|\nabla_{\tilde{\mathbf{L}}} \Phi(t-s)\|_{\mathbb{L}_{2,\log}(\rho_s)}^2 + \sum_{j \in \mathcal{J}} \langle \nabla_{\tilde{L}_j} \partial_s \Phi(t-s), [\rho_s]_{\varphi^{\log}, \omega_j} \nabla_{\tilde{L}_j} \Phi(t-s) \rangle_{\text{HS}} \right. \\ &\quad \left. + \frac{1}{2} \sum_{j \in \mathcal{J}} \langle \nabla_{\tilde{L}_j} \Phi(t-s), \partial_s([\rho_s]_{\varphi^{\log}, \omega_j}) \nabla_{\tilde{L}_j} \Phi(t-s) \rangle_{\text{HS}} \right] \\ &= 2e^{-2\kappa s} \left[-\kappa \|\nabla_{\tilde{\mathbf{L}}} \Phi(t-s)\|_{\mathbb{L}_{2,\log}(\rho_s)}^2 - \langle \nabla_{\tilde{\mathbf{L}}} \mathcal{L}(\Phi(t-s)), \nabla_{\tilde{\mathbf{L}}} \Phi(t-s) \rangle_{\mathcal{L}, \rho_s} \right. \\ &\quad \left. + \frac{1}{2} \langle \mathcal{L}_*(\rho_s), \nabla_{\tilde{\mathbf{L}}} \Phi(t-s) \cdot_{\rho_s} \nabla_{\tilde{\mathbf{L}}} \Phi(t-s) \rangle_{\text{HS}} \right] \\ &= 2e^{-2\kappa s} \left[-\kappa \|\nabla_{\tilde{\mathbf{L}}} \Phi(t-s)\|_{\mathbb{L}_{2,\log}(\rho_s)}^2 + B(\rho_s, \Phi(t-s)) \right] \end{aligned}$$

where we used Equation (12.29) in the second line. We conclude by a use of (ii) of Theorem 12.3.3. For the reverse implication, assume that Equation (12.48) holds. Then,

$$\begin{aligned} 0 &\leq e^{-2\kappa t} \|\nabla_{\tilde{\mathbf{L}}} \Phi\|_{\mathbb{L}_{2,\log}(\mathcal{P}_{*t}(\rho))}^2 - \|\nabla_{\tilde{\mathbf{L}}}(\mathcal{P}_t(\Phi))\|_{\mathbb{L}_{2,\log}(\rho)}^2 \\ &= (e^{-2\kappa t} - 1) \|\nabla_{\tilde{\mathbf{L}}} \Phi\|_{\mathbb{L}_{2,\log}(\rho)}^2 + e^{-2\kappa t} (\|\nabla_{\tilde{\mathbf{L}}} \Phi\|_{\mathbb{L}_{2,\log}(\mathcal{P}_{*t}(\rho))}^2 - \|\nabla_{\tilde{\mathbf{L}}} \Phi\|_{\mathbb{L}_{2,\log}(\rho)}^2) \\ &\quad - \|\nabla_{\tilde{\mathbf{L}}}(\mathcal{P}_t(\Phi))\|_{\mathbb{L}_{2,\log}(\rho)}^2 + \|\nabla_{\tilde{\mathbf{L}}} \Phi\|_{\mathbb{L}_{2,\log}(\rho)}^2. \end{aligned}$$

By dividing by t and letting $t \rightarrow 0$, we once again obtain that $-\kappa \|\nabla_{\tilde{\mathbf{L}}} \Phi\|_{\mathbb{L}_{2,\log}(\rho)}^2 + B(\rho, \Phi) \geq 0$, which allows us to conclude from Equation (12.29). $\square$

In the commutative smooth setting, the Ricci curvature lower bound is also known to be equivalent to the contraction of the Wasserstein distance along the semigroup $(P_t)_{t \geq 0}$ (see Theorem 9.7.2 of [Bakry et al., 2014]):

$$W_2(P_{t*}(\nu), P_{t*}(\nu')) \leq e^{-\kappa t} W_2(\nu, \nu').$$

This still holds true in the non-commutative, finite dimensional setting:

Proposition 12.3.5. *For any $\kappa \in \mathbb{R}$, $\text{Ric}(\mathcal{L}) \geq \kappa$ is equivalent to the contraction of the Wasserstein distance along the flow generated by $(\mathcal{P}_t)_{t \geq 0}$: for any $\rho, \omega \in \mathcal{D}_+(\mathcal{H})$*

$$W_{2,\mathcal{L}}(\mathcal{P}_{t*}(\rho), \mathcal{P}_{t*}(\omega)) \leq e^{-\kappa t} W_{2,\mathcal{L}}(\rho, \omega). \quad (12.49)$$

Proof. The direct implication follows from Proposition 3.1 of [Daneri and Savaré, 2008], which holds in great generality for gradient flows on metric spaces, and Theorem 12.3.3(iii). The reverse implication is proved as in inequality (2.12) of [Daneri and Savaré, 2008], using the smooth Riemannian structure provided by $(\mathcal{D}_+(\mathcal{H}), W_{2,\mathcal{L}})$ in the finite dimensional case. $\square$

In the commutative diffusive setting, the contraction of (12.47) is actually known to be equivalent

to its “square root” version, usually referred to as the *strong gradient bound*:

$$\sqrt{\Gamma(P_t(f), P_t(f))} \leq e^{-\kappa t} P_t(\sqrt{\Gamma(f, f)}). \quad (12.50)$$

The proof of (12.50) $\Rightarrow$ (12.47) follows by a simple use of Jensen’s inequality, the converse being the content of Theorem 3.3.18 of [Bakry et al., 2014]. The advantage of this formulation arises from the fact that some canonical semigroups (e.g. the quantum Ornstein Uhlenbeck semigroup on $\mathbb{R}^n$) saturate the inequality, or equivalently:

$$[L, \nabla] = \kappa \nabla.$$

Therefore, the Ricci lower bound is equivalent to comparing the commutation of a semigroup with the gradient to the one of a canonical semigroup. This is similar in spirit to the geometric inequalities briefly mentioned in Chapter 11 that involve the comparison of the curvature of a space to the one of a model space of constant curvature (e.g. the sphere). A similar reasoning lead [Johnson, 2017] to formulate a Bakry-Émery condition for birth and death processes on $\mathbb{N}$ in terms of a comparison to the Poisson process. Going back to our non-commutative setting, [Carlen and Maas, 2017] showed that the quantum Ornstein-Uhlenbeck semigroup, as well as its fermionic version on the Clifford algebra, do satisfy such a commutation relation. They used this fact to derive the modified logarithmic Sobolev constant for these QMS via the contraction (12.49). In the next proposition, we recall their argument:

Proposition 12.3.6. *Assume that the following equalities hold: there exists $\kappa \in \mathbb{R}$ such that, for any $j \in \mathcal{J}$ and any $t \geq 0$,*

$$\nabla_{\tilde{L}_j} \circ \mathcal{P}_t = e^{-\kappa t} \mathcal{P}_t \circ \nabla_{\tilde{L}_j}. \quad (12.51)$$

Then, $\text{Ric}(\mathcal{L}) \geq \kappa$ holds.

Proof. From Proposition 12.3.5, it is enough to prove that (12.49) holds. Assume that $(\gamma(s))_{s \in [0,1]}$ is a minimal geodesic relating ρ to σ and denote by $(\mathbf{A}(s))_{s \in [0,1]}$ the unique solution of the continuity equation

$$\dot{\gamma}(s) = \text{div } \mathbf{A}(s).$$

By duality, Equation (12.51) implies that $\mathcal{P}_{t*} \dot{\gamma}(s) = e^{-\kappa t} \text{div } \vec{\mathcal{P}}_{t*} \mathbf{A}(s)$, where $\vec{\mathcal{P}}_{t*} \mathbf{A} := (\mathcal{P}_{t*} A_j)_{j \in \mathcal{J}}$. Then, denoting $\gamma(s, t) := \mathcal{P}_{t*}(\gamma(s))$,

$$\begin{aligned} g_{\mathcal{L}, \gamma(s,t)}(\dot{\gamma}(s, t), \dot{\gamma}(s, t)) &= e^{-2\kappa t} \sum_{j \in \mathcal{J}} \langle \mathcal{P}_{*t}(A_j(s)), [\gamma(s, t)]_{\varphi^{\log}, \omega_j}^{-1}(\mathcal{P}_{*t}(A_j(s))) \rangle_{\text{HS}} \\ &\leq e^{-2\kappa t} \sum_{j \in \mathcal{J}} \langle A_j(s), [\gamma(s)]_{\varphi^{\log}, \omega_j}^{-1} A_j(s) \rangle_{\text{HS}} \\ &= e^{-2\kappa t} g_{\mathcal{L}, \gamma(s)}(\dot{\gamma}(s), \dot{\gamma}(s)), \end{aligned}$$

where the inequality arises from the property of monotonicity of Fisher information metrics (cf. Theorem 1.3.3). The result follows after taking the integral over the geodesic path. $\square$

12.4. Example: the quantum depolarizing semigroup

In this section, we derive a Ricci curvature lower bound for the depolarizing semigroup (cf. Section 5.5.1):

Theorem 12.4.1. *The quantum depolarizing semigroup $(\mathcal{P}_t^{\text{depol}})_{t \geq 0}$ satisfies $\text{Ric}(\mathcal{L}^{\text{depol}}) \geq \frac{1}{2}$.*

Proof. In the Schrödinger picture, the generator $\mathcal{L}_*^{\text{depol}}$ can be written as

$$\mathcal{L}_*^{\text{depol}}(\rho) = \frac{\mathbb{1}}{d_{\mathcal{H}}} - \rho = \frac{1}{d_{\mathcal{H}}^2} \sum_{j=1}^{d_{\mathcal{H}}^2} (U_j \rho U_j^* - \rho), \quad \rho \in \mathcal{D}(\mathcal{H}),$$

where the operators U_j can be chosen to be unitary (e.g. generalized Pauli matrices [Wolf, 2012]). In this case, $\tilde{L}_j = (\sqrt{2}d_{\mathcal{H}})^{-1} U_j$ and $\omega_j = 0$, $j = 1, \dots, d_{\mathcal{H}}^2$. Now, given a vector $\vec{V} = (V_1, \dots, V_{d_{\mathcal{H}}^2}) \in \bigoplus_j \mathcal{B}(\mathcal{H})$ and $\rho \in \mathcal{D}_+(\mathcal{H})$,

$$\langle \vec{V}, \vec{V} \rangle_{\mathbb{L}_{2, \varphi^{\log}}(\rho)} = \sum_{j=1}^{d_{\mathcal{H}}^2} \langle V_j, [\rho]_{\varphi^{\log}}(V_j) \rangle_{\text{HS}} \equiv \sum_{j=1}^{d_{\mathcal{H}}^2} \int_0^1 \text{Tr}(V_j^* \rho^s V_j \rho^{1-s}) ds$$

Now, given $\Phi \in \mathcal{B}_{sa}(\mathcal{H})$ and $\rho \in \mathcal{D}_+(\mathcal{H})$, the following holds:

$$\begin{aligned} \nabla_{\tilde{L}} \Phi \cdot_{\rho} \nabla_{\tilde{L}} \Phi &= \sum_{j=1}^{d_{\mathcal{H}}^2} \int_0^1 \int_0^1 \int_0^{\alpha} \frac{\rho^{\beta}}{(1-s)\mathbb{1} + s\rho} \nabla_{\tilde{L}_j} \Phi \rho^{1-\alpha} (\nabla_{\tilde{L}_j} \Phi)^* \frac{\rho^{\alpha-\beta}}{(1-s)\mathbb{1} + s\rho} d\beta ds d\alpha \\ &\quad + \int_0^1 \int_0^1 \int_0^{1-\alpha} \frac{\rho^{\beta}}{(1-s)\mathbb{1} + s\rho} (\nabla_{\tilde{L}_j} \Phi)^* \rho^{\alpha} \nabla_{\tilde{L}_j} \Phi \frac{\rho^{1-\alpha-\beta}}{(1-s)\mathbb{1} + s\rho} d\beta ds d\alpha, \end{aligned}$$

Then:

$$\begin{aligned} &\frac{1}{2} \langle \nabla_{\tilde{L}} \Phi \cdot_{\rho} \nabla_{\tilde{L}} \Phi, \mathcal{L}_*^{\text{depol}}(\rho) \rangle_{\text{HS}} \\ &= \frac{1}{2} \sum_{j=1}^{d_{\mathcal{H}}^2} \int_0^1 ds \int_0^1 d\alpha \left\{ \int_0^{\alpha} \text{Tr} \left[\left(\frac{\mathbb{1}}{d_{\mathcal{H}}} - \rho \right) \frac{\rho^{\beta}}{(1-s)\mathbb{1} + s\rho} \nabla_{\tilde{L}_j} \Phi \rho^{1-\alpha} (\nabla_{\tilde{L}_j} \Phi)^* \frac{\rho^{\alpha-\beta}}{(1-s)\mathbb{1} + s\rho} \right] d\beta \right. \\ &\quad \left. + \int_0^{1-\alpha} \text{Tr} \left[\left(\frac{\mathbb{1}}{d_{\mathcal{H}}} - \rho \right) \frac{\rho^{\beta}}{(1-s)\mathbb{1} + s\rho} (\nabla_{\tilde{L}_j} \Phi)^* \rho^{\alpha} \nabla_{\tilde{L}_j} \Phi \frac{\rho^{1-\alpha-\beta}}{(1-s)\mathbb{1} + s\rho} \right] d\beta \right\}. \end{aligned}$$

By cyclicity of the trace, and since $\int_0^1 \frac{1}{((1-s)\mathbb{1} + s\rho)^2} ds = \rho^{-1}$, forgetting about the positive contributions coming from the terms in $d_{\mathcal{H}}^{-1} \mathbb{1}$ the above expression can be lower bounded as follows:

$$\begin{aligned} &\frac{1}{2} \langle \nabla_{\tilde{L}} \Phi \cdot_{\rho} \nabla_{\tilde{L}} \Phi, \mathcal{L}_*^{\text{depol}}(\rho) \rangle_{\text{HS}} \\ &\geq -\frac{1}{2} \sum_{j=1}^{d_{\mathcal{H}}^2} \int_0^1 \left\{ \int_0^{\alpha} \text{Tr}(\rho^{\beta} \nabla_{\tilde{L}_j} \Phi \rho^{1-\alpha} (\nabla_{\tilde{L}_j} \Phi)^* \rho^{\alpha-\beta}) + \int_0^{1-\alpha} \text{Tr}(\rho^{\beta} (\nabla_{\tilde{L}_j} \Phi)^* \rho^{\alpha} \nabla_{\tilde{L}_j} \Phi \rho^{1-\alpha-\beta}) d\beta \right\} d\alpha \\ &= -\frac{1}{2} \sum_{j=1}^{d_{\mathcal{H}}^2} \int_0^1 \left\{ \alpha \text{Tr}(\nabla_{\tilde{L}_j} \Phi \rho^{1-\alpha} (\nabla_{\tilde{L}_j} \Phi)^* \rho^{\alpha}) + (1-\alpha) \text{Tr}((\nabla_{\tilde{L}_j} \Phi)^* \rho^{\alpha} \nabla_{\tilde{L}_j} \Phi \rho^{1-\alpha}) \right\} d\alpha \\ &= -\frac{1}{2} \sum_{j=1}^{d_{\mathcal{H}}^2} \int_0^1 \text{Tr}((\nabla_{\tilde{L}_j} \Phi)^* \rho^{\alpha} \nabla_{\tilde{L}_j} \Phi \rho^{1-\alpha}) d\alpha. \end{aligned}$$

On the other hand,

$$\begin{aligned} \langle \nabla_{\tilde{L}} \mathcal{L}^{\text{depol}}(\Phi), \nabla_{\tilde{L}} \Phi \rangle_{\mathbb{L}_{2, \varphi^{\log}}(\rho)} &= \sum_{j=1}^{d_{\mathcal{H}}^2} \int_0^1 \text{Tr} \left[\left\{ \nabla_{\tilde{L}_j} \left(\frac{\mathbb{1}}{d_{\mathcal{H}}} \text{Tr}(\Phi) - \Phi \right) \right\}^* \rho^s \nabla_{\tilde{L}_j} \Phi \rho^{1-s} \right] ds \\ &= -\sum_{j=1}^{d_{\mathcal{H}}^2} \int_0^1 \text{Tr} \left[(\nabla_{\tilde{L}_j} \Phi)^* \rho^s \nabla_{\tilde{L}_j} \Phi \rho^{1-s} \right] ds \end{aligned}$$

Therefore,

$$B(\rho, \Phi) \geq \frac{1}{2} \sum_{j=1}^{d_{\mathcal{H}}^2} \int_0^1 \text{Tr} \left[(\nabla_{\tilde{L}_j} \Phi)^* \rho^s \nabla_{\tilde{L}_j} \Phi \rho^{1-s} \right] ds = \frac{1}{2} \|\nabla_{\tilde{\mathbf{L}}} \Phi\|_{\mathbb{L}_{2, \log}(\rho)}^2,$$

and the result follows. $\square$

12.5. Link to quantum functional inequalities

In this section, we relate the Ricci curvature lower bound to other quantum functional inequalities. In Section 12.5, we show that the above displacement convexity implies a quantum generalization of the HWI inequality of [Erbar and Maas, 2012], which itself implies MLSI in the case of a positive curvature. The quantum transportation cost inequality of order 2 is introduced, and its link to MLSI and PI is strengthened in Section 12.5.2.

12.5.1. The quantum HWI inequality

In [Erbar and Maas, 2012], the authors proved that, in the classical discrete setting, $\text{Ric}(\mathcal{L}) \geq \kappa$ for $\kappa \in \mathbb{R}$ implies an HWI-like inequality (cf. Theorem 4.3.1). Here, we provide a quantum generalization of their result and relate it to the modified logarithmic Sobolev inequality. First of all, we need the following lemma that will also prove its usefulness in Section 12.7:

Lemma 12.5.1. *Let $\rho, \tau \in \mathcal{D}_+(\mathcal{H})$. Then for all $t > 0$, $\rho_t \equiv \mathcal{P}_{*t}(\rho)$ satisfies*

$$\frac{d}{dt} W_{2, \mathcal{L}}(\rho_t, \tau) \leq \sqrt{\text{EP}_{\sigma}(\rho_t)}.$$

Proof. We proceed similarly to the proof of Proposition 7.1 of [Erbar and Maas, 2012]: Firstly, by the triangle inequality,

$$\frac{d}{dt} W_{2, \mathcal{L}}(\rho_t, \tau) = \lim_{s \rightarrow 0} \frac{1}{s} (W_{2, \mathcal{L}}(\rho_{t+s}, \tau) - W_{2, \mathcal{L}}(\rho_t, \tau)) \leq \lim_{s \rightarrow 0} \frac{1}{s} W_{2, \mathcal{L}}(\rho_t, \rho_{t+s}), \quad (12.52)$$

Now, by Lemma 12.1.4,

$$W_{2, \mathcal{L}}(\rho_t, \rho_{t+s}) = \inf_{\gamma(s)} \left\{ \int_0^1 g_{\mathcal{L}, \gamma(u)}(\dot{\gamma}(u), \dot{\gamma}(u)) du : \gamma(0) = \rho_t, \gamma(1) = \rho_{t+s} \right\},$$

This implies by a change of variable $v = t + us$ that for any smooth curve γ such that $\gamma(t) = \rho_t$ and $\gamma(t+s) = \rho_{t+s}$,

$$W_{2, \mathcal{L}}(\rho_t, \rho_{t+s}) \leq \int_t^{t+s} g_{\mathcal{L}, \gamma(v)}(\dot{\gamma}(v), \dot{\gamma}(v)) dv. \quad (12.53)$$

Moreover, from Equation (12.20):

$$g_{\mathcal{L}, \rho_t}(\dot{\rho}_t, \dot{\rho}_t) = -\frac{d}{dt} D(\rho_t \| \sigma) = \text{EP}_{\sigma}(\rho_t), \quad (12.54)$$

where the second identity holds by Theorem 7.3.1. Hence, choosing $\gamma(v) = \rho_v$, we bound the right hand side of (12.52) as follows:

$$\frac{d}{dt} W_{2, \mathcal{L}}(\rho_t, \tau) \leq \lim_{s \rightarrow 0} \frac{1}{s} \int_t^{t+s} \sqrt{\text{EP}_{\sigma}(\rho_v)} dv = \sqrt{\text{EP}_{\sigma}(\rho_t)},$$

where the last equality holds since $t \rightarrow \sqrt{\text{EP}_\sigma(\rho_t)}$ is continuous. $\square$

Theorem 12.5.2. *Assume that $\text{Ric}(\mathcal{L}) \geq \kappa$, for some $\kappa \in \mathbb{R}$. Then $\mathcal{L}$ satisfies the following inequality*

$$\forall \rho \in \mathcal{D}_+(\mathcal{H}), \quad D(\rho \parallel \sigma) \leq W_{2,\mathcal{L}}(\rho, \sigma) \sqrt{\text{EP}_\sigma(\rho)} - \frac{\kappa}{2} W_{2,\mathcal{L}}(\rho, \sigma)^2. \quad (\text{HWI}(\kappa))$$

Proof. By Theorem 12.3.3, for any $\rho, \omega \in \mathcal{D}_+(\mathcal{H})$

$$\frac{1}{2} \frac{d}{dt} \Big|_{t=0^+} W_{2,\mathcal{L}}(\rho_t, \omega)^2 + \frac{\kappa}{2} W_{2,\mathcal{L}}(\rho, \omega)^2 \leq D(\omega \parallel \sigma) - D(\rho \parallel \sigma).$$

Taking $\omega := \sigma$, this implies that

$$D(\rho \parallel \sigma) \leq -\frac{1}{2} \frac{d}{dt} \Big|_{t=0^+} W_{2,\mathcal{L}}(\rho_t \parallel \sigma)^2 - \frac{\kappa}{2} W_{2,\mathcal{L}}(\rho, \sigma)^2. \quad (12.55)$$

Then,

$$\begin{aligned} -\frac{1}{2} \frac{d}{dt} \Big|_{t=0^+} W_{2,\mathcal{L}}(\rho_t, \sigma)^2 &= \liminf_{s \rightarrow 0^+} \frac{1}{2s} (W_{2,\mathcal{L}}(\rho, \sigma)^2 - W_{2,\mathcal{L}}(\rho_s, \sigma)^2) \\ &\leq \limsup_{s \rightarrow 0^+} \frac{1}{2s} (W_{2,\mathcal{L}}(\rho, \rho_s)^2 + 2W_{2,\mathcal{L}}(\rho, \rho_s)W_{2,\mathcal{L}}(\rho_s, \sigma)) \\ &\leq \limsup_{s \rightarrow 0^+} \frac{1}{2s} W_{2,\mathcal{L}}(\rho, \rho_s)^2 + W_{2,\mathcal{L}}(\rho, \sigma) \sqrt{\text{EP}_\sigma(\rho)} \\ &= W_{2,\mathcal{L}}(\rho, \sigma) \sqrt{\text{EP}_\sigma(\rho)}. \end{aligned}$$

where the second inequality follows from Lemma 12.5.1. The result follows from inserting this back into (12.55). $\square$

In the case when $\kappa > 0$, we recover the modified logarithmic Sobolev inequality:

Corollary 12.5.3 (Quantum Bakry-Émery theorem). *Assume that $\text{Ric}(\mathcal{L}) \geq \kappa$, for some $\kappa > 0$. Then $(\mathcal{P}_t)_{t \geq 0}$ satisfies $\text{MLSI}(\alpha_1)$ with $\alpha_1 = \frac{\kappa}{2}$.*

Proof. By Theorem 12.5.2, $\mathcal{L}$ satisfies $\text{HWI}(\kappa)$. $\text{MLSI}(\kappa)$ follows from an application of Young's inequality:

$$xy \leq cx^2 + \frac{1}{4c} y^2, \quad \forall x, y \in \mathbb{R}, c > 0, \quad (12.56)$$

in which we set $x = W_{2,\mathcal{L}}(\rho, \sigma)$, $y = \sqrt{\text{EP}_\sigma(\rho)}$, and $c = \frac{\kappa}{2}$. $\square$

12.5.2. The quantum transportation cost inequality

In analogy with Section 4.6, we introduce the concept of a quantum transportation cost inequality associated to the Wasserstein distance $W_{2,\mathcal{L}}$. More precisely, given a primitive QMS $(\mathcal{P}_t)_{t \geq 0}$ on $\mathcal{B}(\mathcal{H})$ satisfying σ -DBC $(\mathcal{P}_t)_{t \geq 0}$ is said to satisfy a *transportation cost inequality of order 2* with constant $c_2 > 0$ if for all $\rho \in \mathcal{D}_+(\mathcal{H})$,

$$W_{2,\mathcal{L}}(\rho, \sigma) \leq \sqrt{2c_2 D(\rho \parallel \sigma)}. \quad (\text{TC}_2(c_2))$$

The QMS is also said to satisfy the *quantum* $\text{MLSI} + \text{TC}_2(c)$ inequality of constant $c > 0$ if all $\rho \in \mathcal{D}_+(\mathcal{H})$,

$$W_{2,\mathcal{L}}(\rho, \sigma) \leq c \sqrt{\text{EP}_\sigma(\rho)}. \quad (\text{MLSI} + \text{TC}_2(c))$$

In the case $\text{Ric}(\mathcal{L}) \geq \kappa$ for $\kappa \in \mathbb{R}$, $\text{HWI}(\kappa)$ still implies the modified log-Sobolev inequality under the further condition that TC_2 holds. This is a direct quantum generalization of Theorem 7.8 of [Erbar and Maas, 2012] (see also Corollary 3.1 of [Otto and Villani, 2000a]):

Corollary 12.5.4. *Assume that $\text{Ric}(\mathcal{L}) \geq \kappa$, $\kappa \in \mathbb{R}$, and that $\text{TC}_2(c_2)$ holds with $c_2^{-1} \geq \max(0, -\kappa)$. then $\text{MLSI}(\alpha_1)$ holds for*

$$\alpha_1 = \max \left[\frac{1}{8c_2} (1 + c_2\kappa)^2, \frac{\kappa}{2} \right]$$

Proof. The proof is rudimentary and identical to the one of Corollary 3.1 of [Otto and Villani, 2000a]. $\square$

Similarly, we can show that $\text{Ric}(\mathcal{L}) \geq \kappa$ for $\kappa \in \mathbb{R}$ implies $\text{MLSI}(\alpha_1)$ as long as $\text{MLSI} + \text{TC}_2(c)$ holds.

Corollary 12.5.5. *Assume that $\text{Ric}(\mathcal{L}) \geq \kappa$, $\kappa \in \mathbb{R}$, and that $\text{MLSI} + \text{TC}_2(c)$ holds with $c^{-1} \geq \max(\kappa, 0)$, then $\text{MLSI}(\alpha_1)$ holds, with*

$$\alpha_1 = \frac{1}{2c(2 - \kappa)}.$$

Proof. See Corollary 3.2 of [Otto and Villani, 2000a]. $\square$

The following result, namely that $\text{MLSI}(\alpha_1)$ implies $\text{TC}_2(c_2)$ was first proven in the classical, continuous case by Otto and Villani in [Otto and Villani, 2000a] (see also [Bobkov et al., 2001, Gozlan, 2009] for alternative proofs, Theorem 7.5 of [Erbar and Maas, 2012] for the classical, discrete case, and [Carlen and Maas, 2014] for the case of the fermionic Fokker-Planck semigroup).

Theorem 12.5.6. *If $(\mathcal{P}_t)_{t \geq 0}$ satisfies $\text{MLSI}(\alpha_1)$, then $\text{TC}_2(c_2)$ holds with $2c_2 = \alpha_1^{-1}$.*

Proof. Fix $\rho \in \mathcal{D}_+(\mathcal{H})$, and set $\rho_t = \mathcal{P}_{*t}(\rho)$. First note that as $t \rightarrow \infty$,

$$D(\rho_t \| \sigma) \rightarrow 0 \text{ and } W_{2,\mathcal{L}}(\rho, \rho_t) \rightarrow W_{2,\mathcal{L}}(\rho, \sigma) \quad (12.57)$$

Define now the function

$$F(t) := W_{2,\mathcal{L}}(\rho_t, \rho) + \sqrt{\frac{1}{\alpha_1} D(\rho_t \| \sigma)}.$$

Obviously $F(0) = \sqrt{D(\rho \| \sigma) / \alpha_1}$, and by (12.57), $F(t) \rightarrow W_{2,\mathcal{L}}(\rho, \sigma)$ as $t \rightarrow \infty$. Hence it is sufficient to prove that F is non-increasing. In order to do so, we only need to show that its derivative is non-positive. If $\rho_t \neq \sigma$, we know from Lemma 12.5.1 that

$$\frac{d}{dt} F(t) \leq \sqrt{\text{EP}_\sigma(\rho_t)} + \sqrt{\frac{1}{\alpha_1} \frac{d}{dt} D(\rho_t \| \sigma)} = \sqrt{\text{EP}_\sigma(\rho_t)} - \frac{\text{EP}_\sigma(\rho_t)}{\sqrt{4\alpha_1 D(\rho_t \| \sigma)}} \leq 0.$$

where we used $\text{MLSI}(\alpha_1)$ in the last inequality. If $\rho_t = \sigma$, then the relation also holds true, since this implies that $\rho_r = \sigma$ for all $r \geq t$. $\square$

Remark 12.5.7. In the last version of [Carlen and Maas, 2017], the authors independently added a slightly different proof of Theorem 12.5.6.

The following corollary of Theorem 12.5.6 easily follows

Corollary 12.5.8. *Assume that $(\mathcal{P}_t)_{t \geq 0}$ satisfies $\text{MLSI}(\alpha_1)$ for some $\alpha_1 > 0$. Then it also satisfies $\text{MLSI} + \text{TC}_2(c)$ with $2c = \alpha_1^{-1}$.*

A new proof of Pinsker's inequality We show that a refinement of Lemma 12.1.7 as well as Theorem 12.5.6 can be used to provide a new proof of the quantum Pinsker inequality (cf. Equation (1.57)).

Theorem 12.5.9 (Quantum Pinsker's inequality). *For any $\rho, \sigma \in \mathcal{D}_+(\mathcal{H})$,*

$$\|\rho - \sigma\|_1 \leq \sqrt{2D(\rho\|\sigma)}.$$

Proof. Let $\mathcal{L}_{\mathbb{1}/d_{\mathcal{H}}}$ be the generator of the quantum depolarizing semigroup with unique invariant state $\mathbb{1}/d_{\mathcal{H}}$: for any $X \in \mathcal{B}(\mathcal{H})$:

$$\mathcal{L}_{\mathbb{1}/d_{\mathcal{H}}}(X) = \frac{1}{d_{\mathcal{H}}} \text{Tr}(X) \mathbb{1} - X.$$

It is shown in Equation (5.42) of that $\mathcal{L}_{\mathbb{1}/d_{\mathcal{H}}}$ can take the following form:

$$\mathcal{L}_{\mathbb{1}/d_{\mathcal{H}}}(X) = \frac{1}{2d_{\mathcal{H}}} \sum_{k,l=1}^{d_{\mathcal{H}}} |e_k\rangle\langle e_l| [X, |e_l\rangle\langle e_k|] + [|e_k\rangle\langle e_l|, X] |e_l\rangle\langle e_k|,$$

for any orthonormal basis $\{e_i\}$. Recall the last line (12.17) of the proof of Lemma 12.1.7, where we showed that for any $\delta > 0$, any smooth path $(\gamma(s))_{s \in [0,1]}$ such that $\gamma(0) = \rho$, $\gamma(1) = \sigma$, and

$$\left(\int_0^1 \|\dot{\gamma}(s)\|_{g_{\mathcal{L}_{\mathbb{1}/d_{\mathcal{H}}}, \gamma(s)}}^2 ds \right)^{1/2} \leq W_{2, \mathcal{L}_{\mathbb{1}/d_{\mathcal{H}}}}(\rho, \sigma) + \delta,$$

and for any self-adjoint operator X ,

$$\begin{aligned} & |\text{Tr}(X(\rho - \sigma))| \\ & \leq \left(\int_0^1 \left\{ \sum_{j \in \mathcal{J}} \text{Tr}(\gamma(s) (\nabla_{\tilde{L}_j} X)^* \nabla_{\tilde{L}_j} X) + \text{Tr}(\gamma(s) \nabla_{\tilde{L}_j} X (\nabla_{\tilde{L}_j} X)^*) \right\} ds \right)^{\frac{1}{2}} (W_{2, \mathcal{L}_{\frac{1}{d_{\mathcal{H}}}}}(\rho, \sigma) + \delta), \end{aligned} \quad (12.58)$$

where for the depolarizing semigroup, the index $j \in \mathcal{J}$ represents a couple (k, l) , so that $\tilde{L}_{kl} = \frac{1}{\sqrt{2d_{\mathcal{H}}}} |e_k\rangle\langle e_l|$. Choosing the basis $\{e_k\}_{k=1}^{d_{\mathcal{H}}}$ to be the one diagonalizing the operator $X := \sum_{k=1}^{d_{\mathcal{H}}} \varphi(k) |e_k\rangle\langle e_k|$, the term in brackets on the right hand side of (12.58) reduces to

$$\frac{1}{2d_{\mathcal{H}}} \sum_{k,l} (\varphi(l) - \varphi(k))^2 (\langle e_k, \gamma(s)e_k \rangle + \langle e_l, \gamma(s)e_l \rangle) \leq \|\varphi\|_{\text{lip}, \text{H}}^2,$$

where $\|\varphi\|_{\text{lip}, \text{H}} := \sup_{k \neq l} |\varphi(k) - \varphi(l)|$. Therefore, letting δ tend to 0,

$$|\text{Tr}(X(\rho - \sigma))| \leq \|\varphi\|_{\text{lip}, \text{H}} W_{2, \mathcal{L}_{\mathbb{1}/d_{\mathcal{H}}}}(\rho, \sigma).$$

Assuming, moreover, that $0 \leq X \leq \mathbb{1}$, this implies that, for any $k \neq l$, $|\varphi(k) - \varphi(l)| \leq 1$, and thus $\|\varphi\|_{\text{lip}, \mathcal{H}} \leq 1$. By duality,

$$\|\rho - \sigma\|_1 \equiv \sup_{0 \leq X \leq \mathbb{1}} |\text{Tr}(X(\rho - \sigma))| \leq \sup_{X = \sum_j \varphi(j)|e_j\rangle\langle e_j|: \|\varphi\|_{\text{lip}, \mathcal{H}} \leq 1} |\text{Tr} X(\rho - \sigma)| \leq W_{2, \mathcal{L}_{1/d_{\mathcal{H}}}}(\rho, \sigma).$$

We conclude using Theorem 12.5.6 as well as the well-known fact that in the case of the depolarizing semigroup, $\alpha_1 = 1/4$ (see e.g. Lemma 25 of [Kastoryano and Temme, 2013]). $\square$

From transportation cost to Poincaré inequality In the classical, Riemannian case, [Otto and Villani, 2000a] proved that $\text{TC}_2(c_2)$ implies the Poincaré inequality. In the discrete setting, this was proved in Proposition 7.6 of [Erbar and Maas, 2012]. Theorem 12.5.10 below extends these results to the quantum regime.

Theorem 12.5.10. *If $(\mathcal{P}_t)_{t \geq 0}$ satisfies $\text{TC}_2(c_2)$, then $\text{PI}(\lambda)$ holds with respect to $\langle \cdot, \cdot \rangle_\sigma$, with $\lambda = (c_2 \kappa_{\mathcal{L}})^{-1}$, where $\kappa_{\mathcal{L}} = \sup_{j \in \mathcal{J}} \|\sigma\|_{\varphi^{\log, -\omega_j}}^{-\frac{1}{2}} \circ [\sigma]_{\varphi^{\log, \omega_j}} \circ [\sigma]_{\varphi^{\log, -\omega_j}}^{-\frac{1}{2}} : \mathcal{T}_2(\mathcal{H}) \rightarrow \mathcal{T}_2(\mathcal{H})\|$.*

Remark 12.5.11. In the commutative setting, Theorem 12.5.10 reduces to Proposition 7.6 of [Erbar and Maas, 2012]. Indeed, in this case, one can easily verify that for any $j \in \mathcal{J}$, $[\sigma]_{\varphi^{\log, \omega_j}}(X) = \frac{1}{2} \sinh(\omega_j/2) \sigma X$ and $[\sigma]_{\varphi^{\log, -\omega_j}}^{-1}(X) = 2X/(\sigma \sinh(\omega_j/2))$. Therefore $[\sigma]_{\varphi^{\log, \omega_j}} \circ [\sigma]_{\varphi^{\log, -\omega_j}}^{-1}(X) = X$, so that $\kappa_{\mathcal{L}} = 1$ and the result follows.

Proof. Let $X \in \mathcal{B}(\mathcal{H})$ such that $\text{Tr}(\sigma X) = 0$, and for some ε small enough, define $X^\varepsilon := \mathbb{1} + \varepsilon X > 0$. Then, define the completely positive, trace-preserving map Ξ_σ through the following equation: for any $Y \in \mathcal{B}(\mathcal{H})$,

$$\Xi_\sigma(Y) := \int_0^\infty \frac{\sigma^{1/2}}{t\mathbb{1} + \sigma} Y \frac{\sigma^{1/2}}{t\mathbb{1} + \sigma} dt$$

In order to get the result we will need the following two technical lemmas:

Lemma 12.5.12. *With the notations of Equation (12.1),*

$$\tilde{L}_j(t\mathbb{1} + \sigma)^{-1} = \frac{e^{-\omega_j} \sigma^{-1}}{\mathbb{1} + t e^{-\omega_j} \sigma^{-1}} \tilde{L}_j \quad (12.59)$$

$$(t\mathbb{1} + \sigma)^{-1} \tilde{L}_j = \tilde{L}_j \frac{\sigma^{-1} e^{\omega_j}}{\mathbb{1} + t \sigma^{-1} e^{\omega_j}}. \quad (12.60)$$

Proof. These identities follow directly from the fact that $\tilde{L}_j$ is an eigenvector of Δ_σ with associated eigenvalue $e^{-\omega_j}$. $\square$

Lemma 12.5.13. *For $\Gamma_\sigma(X) \equiv \sigma^{\frac{1}{2}} X \sigma^{\frac{1}{2}}$,*

$$\nabla_{\tilde{L}_j}(\Xi_\sigma(X)) = [\sigma]_{\varphi^{\log, -\omega_j}}^{-1} \circ \Gamma_\sigma \circ \nabla_{\tilde{L}_j} X. \quad (12.61)$$

Proof. Using that for each $j \in \mathcal{J}$, $\Delta_\sigma^{\pm \frac{1}{2}}(\tilde{L}_j) = e^{\mp \omega_j/2} \tilde{L}_j$,

$$\begin{aligned} \nabla_{\tilde{L}_j}(\Xi_\sigma(X)) &= e^{\omega_j/2} \sigma^{1/2} \tilde{L}_j \int_0^\infty (t\mathbb{1} + \sigma)^{-1} X (t\mathbb{1} + \sigma)^{-1} \sigma^{1/2} dt \\ &\quad - e^{-\omega_j/2} \sigma^{1/2} \int_0^\infty (t\mathbb{1} + \sigma)^{-1} X (t\mathbb{1} + \sigma)^{-1} \tilde{L}_j \sigma^{1/2} dt \end{aligned} \quad (12.62)$$

Let us first consider the first term on the right hand side of Equation (12.62). By Equation (12.59) it is equal to

$$\begin{aligned} \int_0^\infty \frac{e^{\omega_j/2}}{e^{\omega_j} \sigma + t \mathbb{1}} \sigma^{1/2} \tilde{L}_j X \sigma^{1/2} (t \mathbb{1} + \sigma)^{-1} dt &= \int_0^\infty \frac{1}{e^{\omega_j} \sigma + e^{\omega_j/2} u \mathbb{1}} \Gamma_\sigma(\tilde{L}_j X) \frac{e^{\omega_j}}{e^{\omega_j/2} u \mathbb{1} + \sigma} du \\ &= \int_0^\infty \frac{e^{-\omega_j/2}}{e^{\omega_j/2} \sigma + u \mathbb{1}} \Gamma_\sigma(\tilde{L}_j X) \frac{e^{-\omega_j/2}}{u \mathbb{1} + e^{-\omega_j/2} \sigma} e^{\omega_j} du \\ &= [\sigma]_{\varphi^{\log, -\omega_j}}^{-1} \circ \Gamma_\sigma(\tilde{L}_j X), \end{aligned}$$

where we made the change of variable $e^{\omega_j/2} u = t$ in the first line, and used Lemma 12.1.3 in the last line. Similarly, using (12.60), the second term on the right hand side of (12.62) is equal to

$$\begin{aligned} \int_0^\infty \frac{e^{-\omega_j/2}}{t \mathbb{1} + \sigma} \Gamma_\sigma(X \tilde{L}_j) \frac{1}{e^{-\omega_j} \sigma + t \mathbb{1}} dt &= \int_0^\infty \frac{1}{u + e^{\omega_j/2} \sigma} \Gamma_\sigma(X \tilde{L}_j) \frac{1}{e^{-\omega_j/2} \sigma + u} du \\ &= [\sigma]_{\varphi^{\log, -\omega_j}}^{-1} (\Gamma_\sigma(X \tilde{L}_j)), \end{aligned}$$

where we made the change of variable $t = e^{-\omega_j/2} u$. Hence Equation (12.61) follows. $\square$

We are now ready to prove Theorem 12.5.10. Start by the following:

$$\begin{aligned} \langle X, \Xi_\sigma(X) \rangle_\sigma &= \text{Tr}(\Gamma_\sigma(X) \Xi_\sigma(X)) = \frac{1}{\varepsilon} \text{Tr}(\sigma^{1/2} \Xi_\sigma(X) \sigma^{1/2} (X^\varepsilon - \mathbb{1})) \\ &= \frac{1}{\varepsilon} \text{Tr}(\Xi_\sigma(X) (\sigma^{1/2} X^\varepsilon \sigma^{1/2} - \sigma)). \end{aligned}$$

For any $\delta > 0$, there exists a smooth path $(\gamma^\varepsilon(s))_{s \in [0,1]}$, with associated vector field $(\mathbf{V}^\varepsilon(s) = \nabla_{\tilde{\mathbf{L}}}(\Phi(s)))_{s \in [0,1]}$, interpolating between $\rho^\varepsilon := \Gamma_\sigma(X^\varepsilon)$ and σ , and such that

$$\left(\int_0^1 g_{\mathcal{L}, \gamma^\varepsilon(s)}(\dot{\gamma}^\varepsilon(s), \dot{\gamma}^\varepsilon(s)) ds \right)^{1/2} \leq W_{2, \mathcal{L}}(\rho^\varepsilon, \sigma) + \delta \quad (12.63)$$

This implies that

$$\begin{aligned} \langle X, \Xi_\sigma(X) \rangle_\sigma &= -\frac{1}{\varepsilon} \text{Tr} \left(\Xi_\sigma(X) \int_0^1 \frac{d}{ds} \gamma^\varepsilon(s) ds \right) \\ &= \frac{1}{\varepsilon} \text{Tr} \left(\Xi_\sigma(X) \int_0^1 \text{div}([\gamma^\varepsilon(s)]_{\varphi^{\log, \omega}} \mathbf{V}^\varepsilon(s)) ds \right) \\ &= -\frac{1}{\varepsilon} \int_0^1 \sum_{j \in \mathcal{J}} \langle \nabla_{\tilde{L}_j} \Xi_\sigma(X), [\gamma^\varepsilon(s)]_{\varphi^{\log, \omega_j}} (V^\varepsilon(s))_j \rangle_{\text{HS}} ds \\ &\leq \left(\sum_{j \in \mathcal{J}} \int_0^1 \langle \nabla_{\tilde{L}_j} \Xi_\sigma(X), [\gamma^\varepsilon(s)]_{\varphi^{\log, \omega_j}} \circ \nabla_{\tilde{L}_j} \Xi_\sigma(X) \rangle_{\text{HS}} ds \right)^{1/2} \frac{\left(\int_0^1 \|\mathbf{V}^\varepsilon(s)\|_{\mathcal{L}, \gamma^\varepsilon(s)}^2 ds \right)^{1/2}}{\varepsilon} \\ &\leq \left(\sum_{j \in \mathcal{J}} \int_0^1 \langle \nabla_{\tilde{L}_j} \Xi_\sigma(X), [\gamma^\varepsilon(s)]_{\varphi^{\log, \omega_j}} \circ \nabla_{\tilde{L}_j} \Xi_\sigma(X) \rangle_{\text{HS}} ds \right)^{1/2} \frac{W_{2, \mathcal{L}}(\rho^\varepsilon, \sigma) + \delta}{\varepsilon} \\ &\leq \left(\sum_{j \in \mathcal{J}} \int_0^1 \langle \nabla_{\tilde{L}_j} \Xi_\sigma(f), [\gamma^\varepsilon(s)]_{\varphi^{\log, \omega_j}} \circ \nabla_{\tilde{L}_j} \Xi_\sigma(X) \rangle_{\text{HS}} ds \right)^{1/2} \frac{\sqrt{2c_2 D(\rho^\varepsilon \|\sigma)} + \delta}{\varepsilon}, \end{aligned} \quad (12.64)$$

where the first inequality comes from a use of the Cauchy-Schwarz inequality with respect to the inner product $\sum_{j \in \mathcal{J}} \langle \cdot, \int_0^1 [\gamma^\varepsilon(s)]_{\varphi^{\log, \omega_j}} ds \cdot \rangle_{\text{HS}}$, the second from Equation (12.63) and the last one

from $\text{TC}_2(c_2)$. As $\varepsilon \rightarrow 0$, the term in brackets in (12.64) converges to

$$\begin{aligned}
& \sum_{j \in \mathcal{J}} \langle \nabla_{\tilde{L}_j} \Xi_\sigma(X), [\sigma]_{\varphi^{\log}, \omega_j} \circ \nabla_{\tilde{L}_j} \Xi_\sigma(X) \rangle_{\text{HS}} \\
&= \sum_{j \in \mathcal{J}} \langle \nabla_{\tilde{L}_j} X, \Gamma_\sigma \circ [\sigma]_{\varphi^{\log}, -\omega_j}^{-1} \circ [\sigma]_{\varphi^{\log}, \omega_j} \circ [\sigma]_{\varphi^{\log}, -\omega_j}^{-1} \circ \Gamma_\sigma(\nabla_{\tilde{L}_j} X) \rangle_{\text{HS}} \\
&\leq \sup_{j \in \mathcal{J}} \|[\sigma]_{\varphi^{\log}, -\omega_j}^{-\frac{1}{2}} \circ [\sigma]_{\varphi^{\log}, \omega_j} \circ [\sigma]_{\varphi^{\log}, -\omega_j}^{-\frac{1}{2}} : \mathcal{T}_2(\mathcal{H}) \rightarrow \mathcal{T}_2(\mathcal{H})\| \mathcal{E}_{2, \mathcal{L}}(X, \Xi_\sigma(X)) \\
&\equiv \kappa_{\mathcal{L}} \mathcal{E}_{2, \mathcal{L}}(X, \Xi_\sigma(X))
\end{aligned}$$

where we used Lemma 12.5.13 on the first line. As $\delta > 0$ was chosen arbitrarily, we can now take the limit $\delta \rightarrow 0$. Moreover, following the approach of the proof of Theorem 16 of [Kastoryano and Temme, 2013], one can prove that

$$\frac{D(\rho^\varepsilon \| \sigma)}{\varepsilon^2} \rightarrow \frac{1}{2} \text{Tr}(\Gamma_\sigma(X) \Xi_\sigma(X))$$

Substituting into (12.64), we get

$$\frac{1}{c_2 \kappa_{\mathcal{L}}} \langle X, \Xi_\sigma(X) \rangle_\sigma \leq \mathcal{E}_{2, \mathcal{L}}(X, \Xi_\sigma(X)).$$

This is exactly the form that was derived at the end of the proof of Theorem 16 of [Kastoryano and Temme, 2013] which led to the Poincaré inequality. $\square$

12.6. Diameter estimates

The *diameter* of $\mathcal{D}(\mathcal{H})$ in the Wasserstein distance $W_{2, \mathcal{L}}$ is defined as follows:

$$\text{Diam}_{\mathcal{L}}(\mathcal{D}(\mathcal{H})) := \sup_{\rho, \sigma \in \mathcal{D}(\mathcal{H})} W_{2, \mathcal{L}}(\rho, \sigma).$$

Another straightforward consequence of the κ -displacement convexity of the quantum relative entropy for $\kappa > 0$ is the following estimate on the diameter of $\text{Diam}_{\mathcal{L}}(\mathcal{D}(\mathcal{H}))$, which is a quantum analogue of the Bonnet-Myers theorem (cf. (4.10), see also Proposition 7.3 of [Erbar and Fathi, 2018]).

Proposition 12.6.1. *Assume that $\text{Ric}(\mathcal{L}) \geq \kappa$ holds for $\kappa > 0$. Then for any two states $\rho, \omega \in \mathcal{D}(\mathcal{H})$,*

$$W_{2, \mathcal{L}}(\rho, \omega)^2 \leq \frac{4}{\kappa} (D(\rho \| \sigma) + D(\omega \| \sigma)).$$

Therefore,

$$\text{Diam}_{\mathcal{L}}(\mathcal{D}(\mathcal{H})) \leq \sqrt{\frac{8 \ln \|\sigma^{-1}\|_\infty}{\kappa}}.$$

Proof. The result follows directly from the convexity of the quantum relative entropy (cf. (v) of Theorem 12.3.3):

$$0 \leq D(\gamma(1/2) \| \sigma) \leq \frac{1}{2} D(\rho \| \sigma) + \frac{1}{2} D(\omega \| \sigma) - \frac{\kappa}{8} W_{2, \mathcal{L}}(\rho, \omega)^2.$$

for a given constant speed geodesic $(\gamma(s))_{s \in [0, 1]}$ relating ρ and ω . $\square$

In the next two subsections, we show how one can recover the Poincaré and modified logarithmic

Sobolev inequalities in the case when $\text{Ric}(\mathcal{L}) \geq 0$ by further assuming a bound on the diameter $\text{Diam}_{\mathcal{L}}$.

12.6.1. From diameter bound to the Poincaré inequality

In this section we show that $\text{Ric}(\mathcal{L}) \geq 0$ together with a condition of finiteness of the diameter of $\mathcal{D}(\mathcal{H})$ with respect to the distance $W_{2,\mathcal{L}}$ implies the Poincaré inequality, hence extending Proposition 5.9 of [Erbar and Fathi, 2018] to our non-commutative setting. Throughout this section, we fix $(\mathcal{P}_t)_{t \geq 0}$ to be a primitive QMS on $\mathcal{B}(\mathcal{H})$, $\mathcal{H}$ finite dimensional, with unique invariant state σ and associated generator $\mathcal{L}$, satisfying σ -DBC. The next result is a non-commutative extension of the third equivalent statement in Theorem 4.7.2 of [Bakry et al., 2014]:

Proposition 12.6.2 (Reverse quantum Poincaré inequality). *Assume that $\text{Ric}(\mathcal{L}) \geq \kappa$ holds for some $\kappa \in \mathbb{R}$. Then for any $\rho \in \mathcal{D}_+(\mathcal{H})$, any $\Phi \in \mathcal{B}_{sa}(\mathcal{H})$ and all $t > 0$:*

$$\text{Tr}(\mathcal{P}_{*t}(\rho)\Phi^2) - \text{Tr}(\rho(\mathcal{P}_t(\Phi))^2) \geq \frac{e^{2\kappa t} - 1}{\kappa} \|\nabla_{\tilde{\mathbf{L}}} \mathcal{P}_t(\Phi)\|_{\mathbb{L}_{2,\log}(\rho)}^2 \quad (12.65)$$

Proof. The proof is similar to the one of Theorem 3.5 of [Erbar and Fathi, 2018]. For $u \geq 0$, let $\rho_u \equiv \mathcal{P}_{*u}(\rho)$ and $\Phi(u) \equiv \mathcal{P}_u(\Phi)$. Then, from Proposition 12.3.4,

$$\begin{aligned} & 2 e^{2\kappa s} \|\nabla_{\tilde{\mathbf{L}}} \Phi(t)\|_{\mathbb{L}_{2,\log}(\rho)}^2 \\ &= 2 e^{2\kappa s} \|\nabla_{\tilde{\mathbf{L}}} (\mathcal{P}_s \Phi(t-s))\|_{\mathbb{L}_{2,\log}(\rho)}^2 \\ &\leq 2 \|\nabla_{\tilde{\mathbf{L}}} \Phi(t-s)\|_{\mathbb{L}_{2,\log}(\rho_s)}^2 \\ &= 2 \sum_{j \in \mathcal{J}} \langle \nabla_{\tilde{L}_j} \Phi(t-s), [\rho_s]_{\varphi^{\log}, \omega_j} \nabla_{\tilde{L}_j} \Phi(t-s) \rangle_{\text{HS}} \\ &\leq \sum_{j \in \mathcal{J}} \langle \nabla_{\tilde{L}_j} \Phi(t-s), (e^{-\omega_j/2} R_{\rho_s} + e^{\omega_j/2} L_{\rho_s}) \nabla_{\tilde{L}_j} \Phi(t-s) \rangle_{\text{HS}} \\ &= \sum_{j \in \mathcal{J}} \left(e^{\omega_j/2} \text{Tr}[\rho_s \nabla_{\tilde{L}_j} \Phi(t-s) (\nabla_{\tilde{L}_j} \Phi(t-s))^*] + e^{-\omega_j/2} \text{Tr}[\rho_s (\nabla_{\tilde{L}_j} \Phi(t-s))^* \nabla_{\tilde{L}_j} \Phi(t-s)] \right) \\ &= \sum_{j \in \mathcal{J}} \left(e^{\omega_j/2} \text{Tr}(\rho_s [\tilde{L}_j, \Phi(t-s)] [\Phi(t-s), \tilde{L}_j^*]) + e^{-\omega_j/2} \text{Tr}(\rho_s [\Phi(t-s), \tilde{L}_j^*] [\tilde{L}_j, \Phi(t-s)]) \right) \\ &= \sum_{j \in \mathcal{J}} \left(\text{Tr}(\rho_s (\tilde{L}_j \Phi(t-s)^2 \tilde{L}_j^* e^{\omega_j/2} + e^{-\omega_j/2} \tilde{L}_j^* \Phi(t-s)^2 \tilde{L}_j)) \right. \\ &\quad \left. + e^{\omega_j/2} \text{Tr}(\rho_s (-\tilde{L}_j \Phi(t-s) \tilde{L}_j^* \Phi(t-s) - \Phi(t-s) \tilde{L}_j \Phi(t-s) \tilde{L}_j^* + \Phi(t-s) \tilde{L}_j \tilde{L}_j^* \Phi(t-s))) \right. \\ &\quad \left. + e^{-\omega_j/2} \text{Tr}(\rho_s (\Phi(t-s) \tilde{L}_j^* \tilde{L}_j \Phi(t-s) - \tilde{L}_j^* \Phi(t-s) \tilde{L}_j \Phi(t-s) - \Phi(t-s) \tilde{L}_j^* \Phi(t-s) \tilde{L}_j)) \right) \\ &= \text{Tr}(\rho_s \mathcal{L}(\Phi(t-s)^2)) \\ &\quad + \sum_{j \in \mathcal{J}} e^{\omega_j/2} \text{Tr}[\rho_s (-\tilde{L}_j \Phi(t-s) \tilde{L}_j^* \Phi(t-s) - \Phi(t-s) \tilde{L}_j \Phi(t-s) \tilde{L}_j^* \\ &\quad \quad + \Phi(t-s) \tilde{L}_j \tilde{L}_j^* \Phi(t-s) + \Phi(t-s)^2 \tilde{L}_j \tilde{L}_j^*)] \\ &\quad + e^{-\omega_j/2} \text{Tr}[\rho_s (\Phi(t-s) \tilde{L}_j^* \tilde{L}_j \Phi(t-s) - \tilde{L}_j^* \Phi(t-s) \tilde{L}_j \Phi(t-s) \\ &\quad \quad - \Phi(t-s) \tilde{L}_j^* \Phi(t-s) \tilde{L}_j + \tilde{L}_j^* \tilde{L}_j \Phi(t-s)^2)] \\ &= \text{Tr}(\rho_s \mathcal{L}(\Phi(t-s)^2)) - \text{Tr}(\rho_s \Phi(t-s) \mathcal{L}(\Phi(t-s))) - \text{Tr}(\rho_s \mathcal{L}(\Phi(t-s)) \Phi(t-s)) \\ &= \frac{\partial}{\partial s} \text{Tr}(\rho_s \Phi^2(t-s)) \end{aligned}$$

where we used (12.13) in the fifth line. The claim follows after integrating from 0 to t . $\square$

Theorem 12.6.3. $\text{Ric}(\mathcal{L}) \geq 0 + \text{Diam}_{\mathcal{L}}(\mathcal{D}(\mathcal{H})) \leq D \Rightarrow \text{PI}(\frac{1}{eD^2})$.

Proof. Let $X \in \mathcal{B}_{sa}(\mathcal{H})$ an eigenvector of $\mathcal{L}$ with associated eigenvalue opposite to the spectral gap λ of $\mathcal{L}$. Without loss of generality, $\|X\|_\infty = 1$, and by primitivity of $(\mathcal{P}_t)_{t \geq 0}$, $\text{Tr}(\sigma X) = 0$. Now, note that $\mathcal{P}_t(X) = e^{-\lambda t} X$. Therefore, the reverse Poincaré inequality (12.65) in the case when $\kappa = 0$ implies that for any $\rho \in \mathcal{D}_+(\mathcal{H})$,

$$\|\nabla_{\tilde{\mathbf{L}}} X\|_{\mathbb{L}_{2, \log}(\rho)}^2 \leq \frac{e^{2\lambda t}}{2t} \|X\|_\infty^2.$$

Optimizing in t and using $\|X\|_\infty = 1$, we find

$$\|\nabla_{\tilde{\mathbf{L}}} X\|_{\mathbb{L}_{2, \log}(\rho)}^2 \leq e \lambda.$$

Given the following spectral decomposition of $X = \sum_\mu \mu P_\mu$, since $\text{Tr}(\sigma X) = 0$, the minimum and maximum eigenvalues of X , respectively denoted by $\mu_{\min}$ and $\mu_{\max}$, obey $\mu_{\min} < 0 < \mu_{\max}$. Since we assumed $\|X\|_\infty = 1$, this implies that given a path $(\gamma(s))_{s \in [0,1]}$ in $\mathcal{D}(\mathcal{H})$ joining the states $\gamma(0) = \frac{P_{\mu_{\max}}}{\text{Tr}(P_{\mu_{\max}})}$ and $\gamma(1) = \frac{P_{\mu_{\min}}}{\text{Tr}(P_{\mu_{\min}})}$ such that $\int_0^1 g_{\mathcal{L}, \gamma(s)}(\dot{\gamma}(s), \dot{\gamma}(s)) ds \leq W_{2, \mathcal{L}}(\gamma(0), \gamma(1))^2 + \varepsilon$,

$$\begin{aligned} 1 \leq |\mu_{\max} - \mu_{\min}| &= \left| \text{Tr} X \left(\frac{P_{\mu_{\min}}}{\text{Tr}(P_{\mu_{\min}})} - \frac{P_{\mu_{\max}}}{\text{Tr}(P_{\mu_{\max}})} \right) \right| \\ &= \left| \text{Tr} \left(X \int_0^1 \dot{\gamma}(s) ds \right) \right| \\ &= \left| \int_0^1 \sum_{j \in \mathcal{J}} \langle \nabla_{\tilde{L}_j} X, [\gamma(s)]_{\varphi^{\log, \omega_j}} \nabla_{\tilde{L}_j} \Phi(s) \rangle_{\text{HS}} ds \right| \\ &\leq \sqrt{(D^2 + \varepsilon)} \left(\int_0^1 \|\nabla_{\tilde{\mathbf{L}}} X\|_{\mathbb{L}_{2, \log}(\gamma(s))}^2 ds \right)^{1/2} \leq \sqrt{(D^2 + \varepsilon)} \lambda e, \end{aligned}$$

where in the last line we used the Cauchy-Schwarz inequality with respect to the inner product $\sum_{j \in \mathcal{J}} \langle \cdot, \int_0^1 [\gamma(s)]_{\varphi^{\log, \omega_j}} ds \cdot \rangle_{\text{HS}}$, and the result directly follows. $\square$

12.6.2. From diameter bound to modified log-Sobolev inequality

In [Erbar and Fathi, 2018], a modified logarithmic Sobolev inequality was proved to hold under the conditions that $\text{Ric}(\mathcal{L}) \geq 0$ and of boundedness of the diameter of the underlying space under the modified Wasserstein distance. Here, we extend their results to the quantum regime under the further assumption that the semigroup $(\mathcal{P}_t)_{t \geq 0}$ is unital, leaving the study of the general case to later. The idea of the proof is to get a non-tight logarithmic Sobolev inequality from HWI(0), and then to tighten it using ideas borrowed from [Barthe and Kolesnikov, 2008].

Given two states $\rho, \omega \in \mathcal{D}(\mathcal{H})$, with associated spectral decompositions $\rho = \sum_{i \in E} \lambda_i P_i$, $\omega = \sum_{j \in F} \mu_j Q_j$, where E and F are two finite index sets, a *coupling* of ρ and ω is a probability distribution q on $E \times F$ such that

$$\begin{aligned} \sum_{i \in E} q(i, j) &= \mu_j \text{Tr}(Q_j) \\ \sum_{j \in F} q(i, j) &= \lambda_i \text{Tr}(P_i). \end{aligned}$$

The set of couplings between ρ and ω is denoted by $\Pi(\rho, \omega)$. In analogy with the classical literature (see e.g. [Erbar and Maas, 2012]), given an primitive semigroup $(\mathcal{P}_t)_{t \geq 0}$ with associated generator $\mathcal{L}$,

the coupling Wasserstein distance of order two between ρ and ω is defined as follows:

$$W_{2,\mathcal{L},c}(\rho, \omega)^2 := \inf_{q \in \Pi(\rho, \omega)} \sum_{i \in E, j \in F} q(i, j) W_{2,\mathcal{L}}(\rho_i, \omega_j)^2,$$

where

$$\rho_i := \frac{P_i}{\text{Tr}(P_i)}, \quad \omega_j := \frac{Q_j}{\text{Tr}(Q_j)}, \quad i \in E, j \in F.$$

The following result is a quantum generalization of Proposition 2.14 of [Erbar and Maas, 2012]:

Proposition 12.6.4. *Let $(\mathcal{P}_t)_{t \geq 0}$ be a primitive QMS, with unique invariant state σ and associated generator $\mathcal{L}$, satisfying the detailed balance condition. Then, for any $\rho, \omega \in \mathcal{D}_+(\mathcal{H})$,*

$$W_{2,\mathcal{L}}(\rho, \omega) \leq W_{2,\mathcal{L},c}(\rho, \omega).$$

Proof. Given $\rho = \sum_{i \in E} \lambda_i P_i$, $\omega = \sum_{j \in F} \mu_j Q_j$ the spectral decompositions of the states ρ and ω . For $(i, j) \in E \times F$, define $\rho_i := \frac{P_i}{\text{Tr}(P_i)}$, $\omega_j := \frac{Q_j}{\text{Tr}(Q_j)}$, and let $\varepsilon > 0$. By definition of the Wasserstein distance $W_{2,\mathcal{L}}$, there exists a curve $\gamma_{ij} : [0, 1] \mapsto \mathcal{D}(\mathcal{H})$ from ρ_i to ω_j such that

$$\int_0^1 g_{\mathcal{L}, \gamma_{ij}(s)}(\dot{\gamma}_{ij}(s), \dot{\gamma}_{ij}(s)) ds \leq W_{2,\mathcal{L}}(\rho_i, \omega_j)^2 + \varepsilon.$$

For any coupling $q : E \times F \rightarrow \mathbb{R}_+$ of the states ρ and ω , define the path $(\gamma(s))_{s \in [0, 1]}$ on $\mathcal{D}(\mathcal{H})$ as

$$\gamma(s) = \sum_{i \in E, j \in F} q(i, j) \gamma_{ij}(s),$$

so that $\gamma(0) = \rho$ and $\gamma(1) = \omega$. Now,

$$\begin{aligned} W_{2,\mathcal{L}}(\rho, \omega)^2 &\leq \int_0^1 g_{\mathcal{L}, \gamma(s)}(\dot{\gamma}(s), \dot{\gamma}(s)) ds \\ &\leq \sum_{i \in E, j \in F} q(i, j) \int_0^1 g_{\mathcal{L}, \gamma(s)}(\dot{\gamma}_{ij}(s), \dot{\gamma}_{ij}(s)) ds \\ &\leq \sum_{i \in E, j \in F} q(i, j) W_{2,\mathcal{L}}(\rho_i, \omega_j)^2 + \varepsilon. \end{aligned}$$

where we used the convexity of $g_{\mathcal{L}}$ in the second line (see e.g. equation (8.15) of [Carlen and Maas, 2017]). As ε was arbitrary, the result follows after optimizing over the couplings q . $\square$

In what follows, we assume that the semigroup satisfies $\sigma = \mathbb{1}_{\mathcal{H}}/d_{\mathcal{H}}$ -DBC. In order to prove the main result of this section, we need the following two lemmas that are extensions of Lemmas 6.2 and 6.3 of [Erbar and Fathi, 2018]:

Lemma 12.6.5. *Assume that $\text{Ric}(\mathcal{L}) \geq 0$ and $\text{Diam}_{\mathcal{L}}(\mathcal{D}(\mathcal{H})) \leq D$. Then for any $\delta > 0$ and $X \in \mathcal{B}_{sa}(\mathcal{H})$ such that $\text{Tr}(X^2) = d_{\mathcal{H}}$:*

$$D(X^2/d_{\mathcal{H}} \| \mathbb{1}/d_{\mathcal{H}}) \leq \delta D^2 \text{EP}_{\mathbb{1}/d_{\mathcal{H}}}(X^2/d_{\mathcal{H}}) + \frac{1}{4d_{\mathcal{H}}\delta} \text{Tr}(X^2 \mathbb{1}_{(1, \infty)}(X^2)).$$

Proof. The case when X is not of full support is trivial, as then $\text{EP}_{\sigma}(X^2/d_{\mathcal{H}}) = \infty$. Without loss of generality, we assume that X has full support, so that $X^2/d_{\mathcal{H}} \in \mathcal{D}_+(\mathcal{H})$. Write $X = \sum_{i \in E} \varphi(i) P_i$ the spectral decomposition of X , for some index set E . From HWI(0), and Young's inequality (12.56)

with $c = \delta D^2$, $x = \sqrt{\text{EP}_\sigma(X^2/d_{\mathcal{H}})}$, and $y = W_{2,\mathcal{L}}(X^2/d_{\mathcal{H}}, \mathbb{1}/d_{\mathcal{H}})$:

$$D(X^2/d_{\mathcal{H}} \| \mathbb{1}/d_{\mathcal{H}}) \leq \delta D^2 \text{EP}_{\mathbb{1}/d_{\mathcal{H}}}(X^2/d_{\mathcal{H}}) + \frac{1}{4\delta D^2} W_{2,\mathcal{L}}(X^2/d_{\mathcal{H}}, \mathbb{1}/d_{\mathcal{H}})^2.$$

From Proposition 12.6.4, for any coupling $q : E \times F \rightarrow \mathbb{R}_+$ between $X^2/d_{\mathcal{H}}$ and $\mathbb{1}/d_{\mathcal{H}}$ such that $q(i, j) = 0$ whenever $\varphi(i)^2 \leq 1$,

$$\begin{aligned} D(X^2/d_{\mathcal{H}} \| \mathbb{1}/d_{\mathcal{H}}) &\leq \delta D^2 \text{EP}_{\mathbb{1}/d_{\mathcal{H}}}(X^2/d_{\mathcal{H}}) + \frac{1}{4\delta D^2} \sum_{i,j: \varphi(i)^2 > 1} q(i, j) W_{2,\mathcal{L}}\left(\frac{P_i}{\text{Tr}(P_i)}, \frac{P_j}{\text{Tr}(P_j)}\right)^2 \\ &\leq \delta D^2 \text{EP}_{\mathbb{1}/d_{\mathcal{H}}}(X^2/d_{\mathcal{H}}) + \frac{1}{4d_{\mathcal{H}}\delta} \text{Tr}(X^2 \mathbb{1}_{(1,\infty)}(X^2)), \end{aligned}$$

which is what was needed to be proved. $\square$

Lemma 12.6.6. *For any $A > 1$ there exists $\gamma > 0$ such that for any $X \in \mathcal{B}_{sa}(\mathcal{H})$ with $\text{Tr}(X^2) = d_{\mathcal{H}}$,*

$$\frac{1}{d_{\mathcal{H}}} \text{Tr}(X^2 \mathbb{1}_{[A^2, \infty)}(X^2)) \leq \left(\frac{A}{A-1}\right)^2 \text{Var}_{\mathbb{1}/d_{\mathcal{H}}}(X), \quad (12.66)$$

$$D(X^2/d_{\mathcal{H}} \| \mathbb{1}/d_{\mathcal{H}}) \leq \gamma \text{Var}_{\mathbb{1}/d_{\mathcal{H}}}(X) + \frac{1}{d_{\mathcal{H}}} \text{Tr}(X^2 \ln X^2 \mathbb{1}_{[A^2, \infty)}(X^2)). \quad (12.67)$$

Proof. This is a direct rewriting of Lemma 2.5 of [Barthe and Kolesnikov, 2008]. $\square$

Theorem 12.6.7. *Let $(\mathcal{P}_t)_{t \geq 0}$ be a primitive semigroup that satisfies $\mathbb{1}/d_{\mathcal{H}}$ -DBC with associated generator $\mathcal{L}$. Assume that $\text{Ric}(\mathcal{L}) \geq 0$ and that $\text{Diam}_{\mathcal{L}}(\mathcal{D}(\mathcal{H})) \leq D$. Then $\text{MLSI}(cD^{-2})$ holds, for some universal constant c .*

Proof. Let $A > 1$ and $X \in \mathcal{B}_{sa}(\mathcal{H})$ of spectral decomposition $X = \sum_{i \in E} \varphi(i) P_i$, with $\text{Tr}(X^2) = d_{\mathcal{H}}$. Without loss of generality, we can assume X positive definite. Then, set $X_A := X \vee A \equiv \sum_{i: \varphi(i) \geq A} \varphi(i) P_i + A \mathbb{1}_{(-\infty, A)}(X)$. Define the state $\rho_A = X_A^2 / \text{Tr}(X_A^2)$. By (12.67),

$$D(X^2/d_{\mathcal{H}} \| \mathbb{1}/d_{\mathcal{H}}) \leq \gamma \text{Var}_{\mathbb{1}/d_{\mathcal{H}}}(X) + \frac{1}{d_{\mathcal{H}}} \text{Tr}(X^2 \ln X^2 \mathbb{1}_{[A^2, \infty)}(X^2)). \quad (12.68)$$

By Theorem 12.6.3,

$$\gamma \text{Var}_{\mathbb{1}/d_{\mathcal{H}}}(X) \leq -e D^2 \gamma \frac{1}{d_{\mathcal{H}}} \langle X, \mathcal{L}(X) \rangle_{\text{HS}} \leq \frac{\gamma e D^2}{4} \text{EP}_{\mathbb{1}/d_{\mathcal{H}}}(X^2/d_{\mathcal{H}}), \quad (12.69)$$

where in the last inequality, we used the strong regularity of Dirichlet forms of unital semigroups (see [Kastoryano and Temme, 2013]). Moreover,

$$\begin{aligned} \frac{1}{d_{\mathcal{H}}} \text{Tr}(X^2 \ln X^2 \mathbb{1}_{[A^2, \infty)}(X^2)) &= \frac{1}{d_{\mathcal{H}}} \text{Tr}(X_A^2 \ln X_A^2) - \frac{1}{d_{\mathcal{H}}} A^2 \ln A^2 \text{Tr}(\mathbb{1}_{(-\infty, A)}(X)) \\ &\leq (1 + A^2) D(\rho_A \| \mathbb{1}/d_{\mathcal{H}}) + \frac{\text{Tr}(X_A^2)}{d_{\mathcal{H}}} \ln \left(\frac{\text{Tr}(X_A^2)}{d_{\mathcal{H}}} \right) - \frac{A^2 \ln A^2}{d_{\mathcal{H}}} \text{Tr}(\mathbb{1}_{(-\infty, A)}(X)), \end{aligned} \quad (12.70)$$

where in the last line we used that $\frac{1}{d_{\mathcal{H}}} \text{Tr}(X_A^2) \leq 1 + A^2$. However, from Lemma 12.6.5 applied to X_A , since $\text{EP}_{\mathbb{1}/d_{\mathcal{H}}}(\rho_A) \leq \frac{d}{\text{Tr}(X_A^2)} \text{EP}_{\mathbb{1}/d}(X^2/d)$ by convexity of monotone Riemannian metrics (see e.g.

equation (8.16) of [Carlen and Maas, 2017]),

$$\begin{aligned} D(\rho_A \| \mathbb{1}/d_{\mathcal{H}}) &\leq \frac{d_{\mathcal{H}} \delta D^2}{\text{Tr}(X_A^2)} \text{EP}_{\mathbb{1}/d_{\mathcal{H}}}(X^2/d_{\mathcal{H}}) + \frac{1}{4 \text{Tr}(X_A^2) \delta} \text{Tr}(X_A^2 \mathbb{1}_{[\frac{1}{d_{\mathcal{H}}}, \text{Tr}(X_A^2), \infty)}(X_A^2)) \\ &\leq \frac{\delta D^2}{A^2} \text{EP}_{\mathbb{1}/d_{\mathcal{H}}}(X^2/d_{\mathcal{H}}) + \frac{1}{4d_{\mathcal{H}} \delta A^2} \text{Tr}(X^2 \mathbb{1}_{[A^2, \infty)}(X^2)), \end{aligned} \quad (12.71)$$

where in the last line we used that $A^2 \leq \frac{1}{d_{\mathcal{H}}} \text{Tr}(X_A^2)$. Using (12.66) and (12.69) together with Theorem 12.6.3,

$$\begin{aligned} D(\rho_A \| \mathbb{1}/d_{\mathcal{H}}) &\leq \frac{\delta D^2}{A^2} \text{EP}_{\mathbb{1}/d_{\mathcal{H}}}(X^2/d_{\mathcal{H}}) + \frac{1}{4\delta(A-1)^2} \text{Var}_{\mathbb{1}/d_{\mathcal{H}}}(X) \\ &\leq \frac{\delta D^2}{A^2} \text{EP}_{\mathbb{1}/d_{\mathcal{H}}}(X^2/d_{\mathcal{H}}) - \frac{e D^2}{2\delta(A-1)^2} \frac{1}{d} \langle X, \mathcal{L}(X) \rangle_{\text{HS}} \\ &\leq \left(\frac{\delta D^2}{A^2} + \frac{e D^2}{8\delta(A-1)^2} \right) \text{EP}_{\mathbb{1}/d_{\mathcal{H}}}(X^2/d_{\mathcal{H}}). \end{aligned}$$

Now,

$$\begin{aligned} &\frac{\text{Tr}(X_A^2)}{d_{\mathcal{H}}} \ln \left(\frac{\text{Tr}(X_A^2)}{d_{\mathcal{H}}} \right) - \frac{A^2 \ln(A^2)}{d_{\mathcal{H}}} \text{Tr}(\mathbb{1}_{(-\infty, A)}(X)) \\ &\leq \frac{1}{d_{\mathcal{H}}} (A^2 \text{Tr}(\mathbb{1}_{(-\infty, A)}(X)) + \text{Tr}(X^2 \mathbb{1}_{[A^2, \infty)}(X^2))) \ln \left[\frac{A^2 \text{Tr}(\mathbb{1}_{(-\infty, A)}(X)) + \text{Tr}(X^2 \mathbb{1}_{[A^2, \infty)}(X^2))}{d_{\mathcal{H}}} \right] \\ &\quad - \frac{A^2 \ln(A^2)}{d_{\mathcal{H}}} \text{Tr}(\mathbb{1}_{(-\infty, A)}(X)) \\ &= A^2 \frac{1}{d_{\mathcal{H}}} \text{Tr}(\mathbb{1}_{(-\infty, A)}(f)) \ln \left(\frac{1}{d_{\mathcal{H}}} \text{Tr}(\mathbb{1}_{(-\infty, A)}(X)) + \frac{\text{Tr}(X^2 \mathbb{1}_{[A^2, \infty)}(X^2))}{d_{\mathcal{H}} A^2} \right) \\ &\quad + \frac{1}{d_{\mathcal{H}}} \text{Tr}(X^2 \mathbb{1}_{[A^2, \infty)}(X^2)) \ln \left(A^2 \frac{1}{d_{\mathcal{H}}} \text{Tr}(\mathbb{1}_{(-\infty, A]}(X)) + \frac{1}{d_{\mathcal{H}}} \text{Tr}(X^2 \mathbb{1}_{[A^2, \infty)}(X^2)) \right) \\ &\leq A^2 \ln \left(1 + \frac{\text{Tr}(X^2 \mathbb{1}_{[A^2, \infty)}(X^2))}{d_{\mathcal{H}} A^2} \right) + \frac{1}{d_{\mathcal{H}}} \text{Tr}(X^2 \mathbb{1}_{[A^2, \infty)}(X^2)) \ln(1 + A^2) \\ &\leq (1 + \ln(1 + A^2)) \frac{1}{d_{\mathcal{H}}} \text{Tr}(X^2 \mathbb{1}_{[A^2, \infty)}(X^2)), \end{aligned}$$

where in the fourth line we used that $\frac{1}{d_{\mathcal{H}}} \text{Tr}(X^2) = 1$. Using once more (12.66) and (12.69) together with Theorem 12.6.3, we find

$$\frac{\text{Tr}(X_A^2)}{d_{\mathcal{H}}} \ln \left(\frac{\text{Tr}(X_A^2)}{d_{\mathcal{H}}} \right) - \frac{A^2 \ln(A^2)}{d_{\mathcal{H}}} \text{Tr}(\mathbb{1}_{(-\infty, A]}(X)) \leq \frac{e D^2 A^2 (1 + \ln(1 + A^2))}{2(1 - A)^2} \text{EP}_{\mathbb{1}/d_{\mathcal{H}}}(X^2/d_{\mathcal{H}}). \quad (12.72)$$

The result follows after combining (12.72), (12.68), (12.69), (12.70) and (12.71). $\square$

12.7. Quantum concentration inequalities

12.7.1. The quantum Wasserstein distance of order 1

The Wasserstein distance of order 1 can also be extended to the quantum setting. We recall from Section 4.6 that the classical Wasserstein distance $W_1(\mu, \nu)$ between two probability measures μ, ν on

a metric space has the following Kantorovich-Rubinstein dual representation:

$$W_1(\mu, \nu) = \sup_{\varphi: \|\varphi\|_{\text{Lip}} \leq 1} \left\{ \int \varphi d\mu - \int \varphi d\nu \right\},$$

where $\|\varphi\|_{\text{Lip}}$ denotes the Lipschitz constant of the function φ . In the same spirit, given a primitive quantum Markov semigroup $(\mathcal{P}_t = e^{t\mathcal{L}})_{t \geq 0}$ on $\mathcal{B}(\mathcal{H})$, $\mathcal{H}$ finite dimensional, satisfying σ -DBC, we define the following Lipschitz constant of a operator $X \in \mathcal{B}(\mathcal{H})$:

$$\|X\|_{\text{Lip}} := \left(\sum_{j \in \mathcal{J}} (e^{-\omega_j/2} + e^{\omega_j/2}) \|\nabla_{\tilde{L}_j} X\|_{\infty}^2 \right)^{1/2}. \quad (12.73)$$

Next we define the *non-commutative 1-Wasserstein distance* between two states ρ and σ associated to the generator $\mathcal{L}$ to be

$$W_{1,\mathcal{L}}(\rho, \omega) := \sup_{\|X\|_{\text{Lip}} \leq 1} |\text{Tr}(X(\rho - \omega))|. \quad (12.74)$$

Note that similar distances were recently defined in [Chen et al., 2017a, Chen et al., 2017b]. The difference with our definition (12.74) lies in the definition of the Lipschitz constant.

Lemma 12.7.1. *The non-commutative 1-Wasserstein distance $W_{1,\mathcal{L}}$ defines a distance on $\mathcal{D}(\mathcal{H})$.*

Proof. Symmetry and non-negativity are obvious by definition. If $W_{1,\mathcal{L}}(\rho, \omega) = 0$, then for all $X \in \mathcal{B}(\mathcal{H})$, $\text{Tr}(X(\rho - \omega)) = 0$, which implies $\rho = \omega$. Finally, let $\rho, \omega, \tau \in \mathcal{D}(\mathcal{H})$. Then, for any $X \in \mathcal{B}(\mathcal{H})$ such that $\|X\|_{\text{Lip}} \leq 1$, we have by the triangle inequality

$$|\text{Tr}(X(\rho - \tau))| \leq |\text{Tr}(X(\rho - \omega))| + |\text{Tr}(X(\omega - \tau))| \leq W_{1,\mathcal{L}}(\rho, \omega) + W_{1,\mathcal{L}}(\omega, \tau).$$

The result follows by taking the supremum over such operators X on the right hand side of the above inequality. $\square$

Next proposition justifies the definition of the quantum Lipschitz norm:

Lemma 12.7.2. *For any $\rho, \omega \in \mathcal{D}_+(\mathcal{H})$,*

$$\sqrt{2} W_{1,\mathcal{L}}(\rho, \omega) \leq W_{2,\mathcal{L}}(\rho, \omega).$$

Proof. The proof is a direct consequence of inequality (12.18): for any $X \in \mathcal{B}(\mathcal{H})$:

$$\sqrt{2} |\text{Tr}(X(\rho - \omega))| \leq \left(\sum_{j \in \mathcal{J}} (e^{-\omega_j/2} + e^{\omega_j/2}) \|\nabla_{\tilde{L}_j} X\|_{\infty}^2 \right)^{1/2} W_{2,\mathcal{L}}(\rho, \omega). \quad (12.75)$$

$\square$

Other possible definitions for $W_{1,\mathcal{L}}$ Notice that the expression (12.73) depends on the choice of Lindblad operators. An alternative definition of the quantum Lipschitz constant which is independent of the representation is as follows:

$$\|X\|_{\text{Lip},2} := \left(\sum_{j \in \mathcal{J}} (e^{-\omega_j/2} + e^{\omega_j/2}) \|\nabla_{\tilde{L}_j} X\|_2^2 \right)^{1/2}.$$

$\|X\|_{\text{Lip},2}$ does not depend on the representation $(\{\tilde{L}_j\}, \{\omega_j\})$ in (12.1). Indeed, given two such representations $(\{\tilde{L}_j\}, \{\omega_j\})$ and $(\{\tilde{L}'_j\}, \{\omega'_j\})$,

$$\begin{aligned}
\sum_{j \in \mathcal{J}} (e^{-\omega'_j/2} + e^{\omega'_j/2}) \|\nabla_{\tilde{L}'_j} X\|_2^2 &= \sum_{j \in \mathcal{J}} (e^{-\omega'_j/2} + e^{\omega'_j/2}) \text{Tr}([X^*, (\tilde{L}'_j)^*][\tilde{L}'_j, X]) \\
&= \sum_{j,l,k \in \mathcal{J}} (e^{-\omega'_j/2} + e^{\omega'_j/2}) U_{jk} \bar{U}_{jl} \text{Tr}([X^*, \tilde{L}_l^*][\tilde{L}_k, X]) \\
&= \sum_{j,l,k \in \mathcal{J}} U_{jk} \bar{U}_{jl} (e^{-\omega_k/2} + e^{\omega_k/2}) \text{Tr}([X^*, \tilde{L}_l^*][\tilde{L}_k, X]) \\
&= \sum_{l,k \in \mathcal{J}} \delta_{kl} (e^{-\omega_k/2} + e^{\omega_k/2}) \text{Tr}([X^*, \tilde{L}_l^*][\tilde{L}_k, X]) \\
&= \sum_{l \in \mathcal{J}} (e^{-\omega_l/2} + e^{\omega_l/2}) \text{Tr}([X^*, \tilde{L}_l^*][\tilde{L}_l, X]) \\
&= \sum_{j \in \mathcal{J}} (e^{-\omega_j/2} + e^{\omega_j/2}) \|\nabla_{\tilde{L}_j} X\|_2^2,
\end{aligned}$$

where we used Remark 12.1.1 in the above lines. Moreover, since $\|\cdot\|_\infty \leq \|\cdot\|_2$, it follows that $\|X\|_{\text{Lip}} \leq \|X\|_{\text{Lip},2}$. One can also defined the Wasserstein distance of order 1 corresponding to this Lipschitz norm:

$$W_{1,\mathcal{L},2}(\rho, \omega) := \sup_{\|X\|_{\text{Lip},2} \leq 1} |\text{Tr } X(\rho - \omega)|, \quad (12.76)$$

for which Lemma 12.7.1 extends, i.e. $W_{1,\mathcal{L},2}$ defines a distance on $\mathcal{D}_+(\mathcal{H})$. The following proposition follows directly from the direct observation that $\|X\|_{\text{Lip}} \leq \|X\|_{\text{Lip},2}$.

Proposition 12.7.3. *For any $\rho, \omega \in \mathcal{D}_+(\mathcal{H})$,*

$$W_{1,\mathcal{L},2}(\rho, \omega) \leq W_{1,\mathcal{L}}(\rho, \omega).$$

12.7.2. Transportation cost inequality and Gaussian concentration

A primitive QMS $(\mathcal{P}_t)_{t \geq 0}$ in σ -DBC is said to satisfy a *transportation cost inequality of order 1* with constant $c_1 > 0$ if for all $\rho \in \mathcal{D}_+(\mathcal{H})$

$$W_{1,\mathcal{L}}(\rho, \sigma) \leq \sqrt{2c_1 D(\rho|\sigma)}. \quad (\text{TC}_1(c_1))$$

The following result is a direct consequence of Lemma 12.7.2:

Theorem 12.7.4. *If $(\mathcal{P}_t)_{t \geq 0}$ satisfies $\text{TC}_2(c_2)$, then it satisfies $\text{TC}_1(c_1)$ with $2c_1 = c_2$.*

The first proof that the classical transportation cost inequality of order 1 implies Gaussian concentration is due to Marton [Marton, 1996a]. The following theorem is a quantum generalization of Bobkov-Götze's proof [Bobkov and Goetze, 1999] which relies on the variational representation of the 1 Wasserstein distance (see also Theorem 36 of [Raginsky and Sason, 2014] or Proposition 7.7 of [Erbar and Maas, 2012]):

Theorem 12.7.5. *Let $(\mathcal{P}_t)_{t \geq 0}$ be a primitive QMS on $\mathcal{B}(\mathcal{H})$ in σ -DBC. If $(\mathcal{P}_t)_{t \geq 0}$ satisfies $\text{TC}_1(c_1)$, the following quantum Gaussian concentration inequality holds: for any $X \in \mathcal{B}_{sa}(\mathcal{H})$,*

$$\text{Tr}(\sigma \mathbb{1}_{[r,\infty)}(X - \text{Tr}(\sigma X))) \leq \exp\left(\frac{-r^2}{2c_1 \|(\Delta_\sigma^{-1/2} X)\|_{\text{Lip}}^2}\right). \quad (\text{qGauss})$$

Proof. Here we follow the lines of the proof of Theorem 36 of [Raginsky and Sason, 2014]. Let $Y \in \mathcal{B}(\mathcal{H})$ be such that $\text{Tr}(\sigma Y) = 0$ and $\|Y\|_{\text{Lip}} \leq 1$. From $\text{TC}_1(c_1)$, we know that for any $\rho \in \mathcal{D}_+(\mathcal{H})$,

$$|\text{Tr}(\rho Y)| \leq W_{1,\mathcal{L}}(\rho, \sigma) \leq \sqrt{2c_1 D(\rho\|\sigma)}.$$

Next, from the fact that

$$\inf_{\theta > 0} \left(\frac{a}{\theta} + \frac{b\theta}{2} \right) = \sqrt{2ab}$$

for any $a, b \geq 0$, we see that any such Y must satisfy the following bound for any $\theta > 0$:

$$|\text{Tr}(\rho Y)| \leq \frac{1}{\theta} D(\rho\|\sigma) + \frac{c_1\theta}{2}.$$

Rearranging the terms, we obtain for any $\theta > 0$:

$$2\theta |\text{Tr}(\rho Y)| - c_1\theta^2 \leq 2D(\rho\|\sigma) \leq 2\hat{D}(\rho\|\sigma), \quad (12.77)$$

where we have used (1.66), and $\hat{D}(\rho\|\sigma)$ is the maximal divergence defined through Equation (1.67). Define $\rho := \sigma^{1/2} e^{\theta X} \sigma^{1/2} / (\text{Tr}(\sigma e^{\theta X}))$, where $X \in \mathcal{B}_{sa}(\mathcal{H})$ is to be specified later. Hence Equation (12.77) becomes

$$2\theta |\text{Tr}(\rho Y)| - c_1\theta^2 \leq 2\theta \frac{\text{Tr}(\sigma e^{\theta X} X)}{\text{Tr}(\sigma e^{\theta X})} - 2\ln(\text{Tr}(\sigma e^{\theta X})).$$

Now for $X = \Delta_\sigma^{1/2}(Y)$, the last expression further simplifies into

$$-\frac{c_1\theta^2}{2} \leq -\ln(\text{Tr}(\sigma e^{\theta X})) \quad \Rightarrow \quad M_X(\theta) \equiv \text{Tr}(\sigma e^{\theta X}) \leq e^{c_1\theta^2/2}.$$

We conclude by Markov's inequality:

$$\text{Tr}(\sigma \mathbb{1}_{[r,\infty)}(X)) \leq e^{-r\theta} M_X(\theta) \leq e^{-r\theta} e^{c_1\theta^2/2}. \quad (12.78)$$

Optimizing over all $\theta > 0$,

$$\text{Tr}(\sigma \mathbb{1}_{[r,\infty)}(X)) \leq e^{-\frac{r^2}{2c_1}}.$$

In order to achieve this bound we assumed that $Y = \Delta_\sigma^{-1/2}(X) \in \mathcal{B}(\mathcal{H})$ is such that $\|Y\|_{\text{Lip}} \leq 1$ and $\text{Tr}(\sigma Y) = 0$. This implies that $\text{Tr}(\sigma X) = \text{Tr}(\sigma \Delta_\sigma^{1/2}(Y)) = \text{Tr}(\sigma Y) = 0$. The result follows by rescaling. $\square$

12.7.3. Poincaré inequality and exponential concentration

The following theorem is a generalization of the classical results of [Gromov and Milman, 1983] (see also the review [Milman, 2009b]). It states that the Poincaré inequality implies exponential concentration. For this, we need the following well-known chain rule (see e.g. [Carlen and Maas, 2017] Lemma 5.5):

Lemma 12.7.6. *For all $L \in \mathcal{B}(\mathcal{H})$, and all $X \in \mathcal{B}(\mathcal{H})$,*

$$[L, e^X] = \int_0^1 e^{sX} [L, X] e^{X(1-s)} ds.$$

Proof.

$$\frac{d}{ds} e^{(1-s)X} L e^{sX} = e^{(1-s)X} [L, X] e^{sX}.$$

The result follows by integrating the above equation from 0 to 1. $\square$

Theorem 12.7.7. *Let $(\mathcal{P}_t)_{t \geq 0}$ be a primitive quantum Markov semigroup on $\mathcal{B}(\mathcal{H})$ satisfying σ -DBC. Then for any $X \in \mathcal{B}_{sa}(\mathcal{H})$,*

$$\mathrm{Tr}(\sigma \mathbb{1}_{[r, \infty)}(X - \mathrm{Tr}(\sigma X))) \leq 3 e^{-r\sqrt{\lambda(\mathcal{L})}/(\|X\|_{\mathrm{Lip}} C_{X, \lambda(\mathcal{L})})}.$$

where $\|\cdot\|_{\mathrm{Lip}}$ is defined in Equation (12.73), and $C_{X, \lambda(\mathcal{L})} \equiv \frac{e^{2\sqrt{\lambda(\mathcal{L})}\|X\|_{\infty}/\|X\|_{\mathrm{Lip}}}-1}{\sqrt{2\lambda(\mathcal{L})}\|X\|_{\infty}/\|X\|_{\mathrm{Lip}}}$.

Proof. Assume without loss of generality that $\mathrm{Tr}(\sigma X) = 0$. For $\theta \geq 0$, and $X \neq 0$ self-adjoint, let $M_X(\theta) := \mathrm{Tr}(\sigma e^{\theta X})$. Since $(\mathcal{P}_t)_{t \geq 0}$ is self-adjoint with respect to $\langle \cdot, \cdot \rangle_{1, \sigma}$, $\lambda(\mathcal{L})$ also satisfies the following inequality:

$$\lambda(\mathcal{L})(M_X(\theta) - M_X(\theta/2)^2) \leq -\langle e^{\theta X/2}, \mathcal{L}(e^{\theta X/2}) \rangle_{1, \sigma}. \quad (12.79)$$

Moreover, from Equation (12.1):

$$\begin{aligned} -\langle e^{\theta X/2}, \mathcal{L}(e^{\theta X/2}) \rangle_{1, \sigma} &= \sum_{j \in \mathcal{J}} e^{-\omega_j/2} \mathrm{Tr} \left[\sigma \left(\nabla_{\tilde{L}_j} e^{\theta X/2} \right)^* \nabla_{\tilde{L}_j} e^{\theta X/2} \right] \\ &= \frac{\theta^2}{4} \sum_{j \in \mathcal{J}} e^{-\omega_j/2} \iint_{[0,1]^2} \mathrm{Tr} \left(\sigma e^{\frac{(1-s)\theta X}{2}} (\nabla_{\tilde{L}_j} X)^* e^{\frac{s\theta X}{2}} e^{\frac{u\theta X}{2}} \nabla_{\tilde{L}_j} X e^{\frac{(1-u)\theta X}{2}} \right) duds. \end{aligned} \quad (12.80)$$

where we used Lemma 12.7.6 in the second line. Moreover, for each $u, s \in [0, 1]$, the trace in Equation (12.80) is equal, by cyclicity, to

$$\begin{aligned} \mathrm{Tr} \left[\left(e^{\theta X/2} \sigma e^{\theta X/2} \right)^* \left(e^{-\frac{s\theta X}{2}} (\nabla_{\tilde{L}_j} X)^* e^{\frac{(s+u)\theta X}{2}} \nabla_{\tilde{L}_j} X e^{-\frac{u\theta X}{2}} \right) \right] &\leq \mathrm{Tr} \left(e^{\theta X/2} \sigma e^{\theta X/2} \right) e^{\theta(s+u)\|X\|_{\infty}} \|\nabla_{\tilde{L}_j} X\|_{\infty}^2 \\ &= M_X(\theta) e^{\theta(s+u)\|X\|_{\infty}} \|\nabla_{\tilde{L}_j} X\|_{\infty}^2, \end{aligned}$$

where we used Hölder's inequality as well as the submultiplicativity of the operator norm in the second line. Substituting into (12.80), we thus get:

$$\begin{aligned} -\langle e^{\theta X/2}, \mathcal{L}(e^{\theta X/2}) \rangle_{1, \sigma} &\leq \frac{M_X(\theta)}{4\|X\|_{\infty}^2} \sum_{j \in \mathcal{J}} e^{-\omega_j/2} \|\nabla_{\tilde{L}_j} X\|_{\infty}^2 (e^{\theta\|X\|_{\infty}} - 1)^2 \\ &\leq \|X\|_{\mathrm{Lip}}^2 M_X(\theta) \frac{(e^{\theta\|X\|_{\infty}} - 1)^2}{4\|X\|_{\infty}^2}, \end{aligned}$$

However, for any $0 \leq \theta < 2\sqrt{\lambda(\mathcal{L})}/\|X\|_{\mathrm{Lip}}$,

$$\frac{e^{\theta\|X\|_{\infty}} - 1}{\theta\|X\|_{\infty}} \leq \frac{e^{2\sqrt{\lambda(\mathcal{L})}\|X\|_{\infty}/\|X\|_{\mathrm{Lip}}} - 1}{\sqrt{2\lambda(\mathcal{L})}\|X\|_{\infty}/\|X\|_{\mathrm{Lip}}} \equiv C_{X, \lambda(\mathcal{L})} > 1.$$

Hence, substituting into Equation (12.79):

$$\lambda(\mathcal{L})(M_X(\theta) - M_X(\theta/2)^2) \leq \theta^2 \|X\|_{\mathrm{Lip}}^2 C_{X, \lambda(\mathcal{L})}^2 M_X(\theta)/4.$$

This last inequality implies that

$$M_X(\theta) \leq \frac{1}{1 - \theta^2 \|X\|_{\text{Lip}}^2 C_{X,\lambda(\mathcal{L})}^2 / (4\lambda(\mathcal{L}))} M_X(\theta/2)^2,$$

for every $\theta < 2\sqrt{\lambda(\mathcal{L})}/(C_{X,\lambda(\mathcal{L})}\|X\|_{\text{Lip}})$. A simple iteration procedure yields

$$M_X(\theta) \leq \prod_{k=0}^{n-1} \left(\frac{1}{1 - \theta^2 \|X\|_{\text{Lip}}^2 C_{X,\lambda(\mathcal{L})}^2 / (4^{k+1}\lambda(\mathcal{L}))} \right)^{2^k} M_X(\theta/2^n)^{2^n}.$$

Note that $M_X(\theta) = 1 + \theta \text{Tr}(\sigma X) + \mathcal{O}(\theta^2)$, and we have assumed that $\text{Tr}(\sigma X) = 0$. Thus letting $n \rightarrow \infty$:

$$M_X(\theta) \leq \prod_{k=0}^{\infty} \left(\frac{1}{1 - \theta^2 \|X\|_{\text{Lip}}^2 C_{X,\lambda(\mathcal{L})}^2 / (4^{k+1}\lambda(\mathcal{L}))} \right)^{2^k}.$$

Set $\theta = \sqrt{\lambda(\mathcal{L})}/(\|X\|_{\text{Lip}} C_{X,\lambda(\mathcal{L})})$, then the right hand side is a universal constant contained between e and 3. So we proved that

$$M_X \left(\sqrt{\lambda(\mathcal{L})}/(\|X\|_{\text{Lip}} C_{X,\lambda(\mathcal{L})}) \right) \leq 3.$$

Now by functional calculus, for any $r \in \mathbb{R}$ and $\theta > 0$:

$$\mathbb{1}_{[r,\infty)}(X) = \mathbb{1}_{[\exp(\theta r),\infty)}(\exp(\theta X)) \leq e^{-\theta r} e^{\theta X}.$$

This leads to the following Markov-type inequality:

$$\text{Tr}(\sigma \mathbb{1}_{[r,\infty)}(X)) \leq e^{-r\theta} \text{Tr}(\sigma \exp(\theta X)) = e^{-r\theta} M_X(\theta). \quad (12.81)$$

Therefore

$$\text{Tr}(\sigma \mathbb{1}_{[r,\infty)}(X)) \leq 3 e^{-r\sqrt{\lambda(\mathcal{L})}/(\|X\|_{\text{Lip}} C_{X,\lambda(\mathcal{L})})}.$$

□

12.7.4. Concentration of product states

As already mentioned in (10.3), [Temme et al., 2014] proved that the 2-logarithmic Sobolev constant satisfies

$$\frac{\min_k \lambda_k}{\ln(d_{\mathcal{H}}^4 \max_k \|\sigma_k^{-1}\|_{\infty}) + 11} \leq \alpha_2(\mathcal{K}_N) \leq \frac{\min_k \lambda_k}{2}. \quad (12.82)$$

Moreover, it was shown in Proposition 13 of [Kastoryano and Temme, 2013] that the generator of a primitive semigroup satisfies the following inequality:

$$\alpha_2(\mathcal{L}) \leq \alpha_1(\mathcal{L}) \quad (12.83)$$

provided it is strongly $\mathbb{L}_p$ -regular, which is always the case for semigroups satisfying a detailed balance condition by the Stroock Varopoulos inequality. Therefore, by a joint use of Theorems 12.5.6, 12.7.4 and 12.7.5 as well as 12.82 and 12.83, the following holds true: for any self-adjoint operator X_N on

$\mathcal{H}^{\otimes N},$

$$\mathrm{Tr}(\sigma^{(N)} \mathbb{1}_{[r,\infty)}(X_N - \mathrm{Tr}(\sigma^{(N)} X_N))) \leq \exp \left(\frac{-2 \min_k \lambda_k r^2}{(\ln(d_{\mathcal{H}}^4 \max_k \|\sigma_k^{-1}\|_{\infty}) + 11) \|(\Delta_{\sigma^{(N)}}^{-1/2} X_N)\|_{\mathrm{Lip}}^2} \right). \quad (12.84)$$

where $\sigma^{(N)} := \bigotimes_{k=1}^N \sigma_k$. Assume now that X_N has the following form:

$$X_N := \frac{1}{N} \sum_{k=1}^N \mathbb{1}^{\otimes(k-1)} \otimes X \otimes \mathbb{1}^{\otimes(N-k)}.$$

and that the generators $\mathcal{L}_k$ are all identical, with associated invariant state σ . Then, it is easy to show that

$$\|(\Delta_{\sigma^{(N)}}^{-1/2} X_N)\|_{\mathrm{Lip}} = \frac{1}{\sqrt{N}} \|(\Delta_{\sigma}^{-1/2} X)\|_{\mathrm{Lip}}. \quad (12.85)$$

In this case, (12.84) reduces to

$$\mathrm{Tr}(\sigma^{(N)} \mathbb{1}_{[r,\infty)}(X_N - \mathrm{Tr}(\sigma^{(N)} X_N))) \leq \exp \left(\frac{-2N \min_k \lambda_k r^2}{(\ln(d_{\mathcal{H}}^4 \max_k \|\sigma_k^{-1}\|_{\infty}) + 11) \|(\Delta_{\sigma}^{-1/2} X)\|_{\mathrm{Lip}}^2} \right).$$

In the case of the generalized depolarizing semigroup, the quasi-tensorization of the MLSI constant found in Theorem 10.1.1 provides the simpler bound:

$$\mathrm{Tr}(\sigma^{(N)} \mathbb{1}_{[r,\infty)}(X_N - \mathrm{Tr}(\sigma^{(N)} X_N))) \leq \exp \left(\frac{-r^2}{2 \|(\Delta_{\sigma^{(N)}}^{-1/2} X_N)\|_{\mathrm{Lip}}^2} \right). \quad (12.86)$$

12.8. Excursion to infinite dimensions

Here, we aim at defining a quantum Wasserstein distance $W_{2,\mathcal{L}^{\mathrm{qOU}}}$ associated to the quantum Ornstein Uhlenbeck semigroup $(\mathcal{P}_t^{\mathrm{qOU}})_{t \geq 0}$ of Section 5.5.2. In the general classical setting of Theorem 4.2.1, one assumes that the curves $(\mu_t)_{t \geq 0}$ are absolutely continuous on the Wasserstein space $(\mathcal{P}_2(\mathcal{M}), W_2)$. This assumption is essential in the proof of the existence and uniqueness of a tangent vector $\nabla \varphi \in \mathbb{L}_2(\mu, T_{\mu}(\mathcal{P}_2(\mathcal{M})))$ satisfying the continuity equation. For a compact metric space, a similar proof can be carried out for regular enough paths of absolutely continuous measures with associated densities in $\mathbb{L}_2(\mathcal{M})$, without the need to use any property of the Wasserstein distance. In particular, one can show that the Poisson equation admits a unique weak solution by means of Poincaré estimates. This proof has to be modified in the case when $\mathcal{M} = \mathbb{R}^n$ is unbounded, since such estimates typically diverge. In fact, in this case, there exist multiple (possibly distributional) solutions to the Poisson equation, due to the degeneracy of the Laplace operator. We briefly recall how to construct a solution in the Sobolev space

$$\mathcal{W}^{1,2}(\mathbb{R}^n) := \overline{\{f \in C_c^{\infty}(\mathbb{R}^n)\}}^{\|\cdot\|_{\mathcal{W}^{1,2}(\mathbb{R}^n)}} \subset \mathbb{L}_2(\mathbb{R}^n),$$

that is, the closure of the space $C_c^{\infty}(\mathbb{R})$ of smooth, compactly supported functions on $\mathbb{R}^n$ in the norm $\|\cdot\|_{\mathcal{W}^{1,2}(\mathbb{R}^n)}$ defined as

$$\|f\|_{\mathcal{W}^{1,2}(\mathcal{H})} := \left(\sum_{\substack{\alpha \in \mathbb{N}^n \\ |\alpha| \leq 1}} \|\partial^{\alpha} f\|_{\mathbb{L}_2(\mathbb{R}^n)}^2 \right)^{1/2}.$$